

Particulate Matter Monitoring & Health Effects

September 2026 | Entry window 1–30 September 2026, 30 daily issues | 622 records pooled

622

RECORDS POOLED
(30 DAILY ISSUES)

148

EXPOSURE ASSESS-
MENT
(24%), LARGEST BAND

173

EFFECT ESTIMATES
FROM 93 SOURCES

143

TIER-A
RECORDS (23%)

98

METADATA-ONLY
RECORDS (16%)

Signal of the month

September settled one question at population scale: per unit mass, fire-sourced $\text{PM}_{2.5}$ is more lethal than background $\text{PM}_{2.5}$, and the excess sits in its carbonaceous fraction — the part a mass-reporting optical network does not resolve.

Three large, methodologically independent studies landed inside sixteen days. Su et al. (*Lancet Oncol*, 15 Sep) followed 414,016 SEER–Medicare lung-cancer patients: per $1 \mu\text{g m}^{-3}$, death HR **1.078** (1.064–1.092) for wildfire against **1.017** (1.015–1.019) for non-wildfire $\text{PM}_{2.5}$. Ju et al. (*Lancet Planet Health*, 16 Sep), with instrumental-variable models across 1,366 communities in ten countries, found landscape-fire particles are **19.0 %** of ambient $\text{PM}_{2.5}$ mass but **30.6 %** of the attributable deaths, and that conventional time-series estimates understate the fire effect by more than half. Jin et al. (*ES&T*, 30 Sep) split smoke $\text{PM}_{2.5}$ by composition in 77.5 million Medicare beneficiaries: HR **1.032** per $1 \mu\text{g m}^{-3}$ for smoke mass and **1.087** for smoke carbon, with the carbon fraction carrying most of the 22,420 attributable deaths a year. **The common caveat is arithmetic:** chronic smoke exposure is small and narrow (median $0.28 \mu\text{g m}^{-3}$ in Su et al.), so “per $1 \mu\text{g m}^{-3}$ ” coefficients extrapolate across most of their own range, and on the IQR scale the ranking can reverse. The direction, though, now holds across a cancer cohort, a multi-country IV design and a composition-resolved national cohort. Kaur et al.’s twenty-year western-US source apportionment (*Atmos Pollut Res*, 20 Sep) adds the measurement side: the mass that is disappearing is secondary sulfate and nitrate, which optical sensors see well, while the share from residential-wood and fire combustion is rising. **The month’s verdict: for health-relevant exposure in fire-affected regions, a network that adds an absorption or BC channel buys more than one that adds sensor density.**

What changed in September

- **Calibration moved from “does it fit” to “is the fit honest”.** Sahin et al. (*Atmosphere*, 29 Sep) showed six ML calibrators reaching R^2 0.86–0.89 under random splits and near or below zero under chronological hold-out, with validation design explaining **63–91 %** of the R^2 spread and algorithm choice 3.6–10.6 %. Kaur, Mangin & Kelly (*Aerosol Sci Technol*, 3 Sep) showed Plantower’s PMSX003N fixes dust ($\text{PM}_{2.5}$ slope 1.85–1.91 vs 0.15 for the PMS5003T) by re-binning $>1 \mu\text{m}$ particles, which is the same mechanism that makes it over-read in humid air. Khalili Hollo et al. (*ACS Environ Au*, 25 Sep) put fifty Atmos tubes and five QuantAQ units through four co-locations in two years: unit response **drifted between co-locations**, 5–11 % of wearables were dead at any comparison, and RH correction cut RMSE 42–50 %. Together these define the operating rule: temporally blocked validation, re-co-location rather than a single pre-deployment fit, and slope reported with RMSE.
- **Neither side of the co-location is stationary.** Zhou et al. (*PNAS*, 14 Sep) used the 2025 suspension of the US embassy monitoring programme as a natural experiment: local satellite AOD rose **24–33 %** after the monitors went dark, most where no substitute data existed — monitoring changed behaviour, not just the record. A Korean analysis (25 Sep) is titled on step changes in the regulatory PM record when stations relocate, and a mobile monitor 250 m from a regulatory station published a different AQI colour one time in ten (7 Sep).
- **August’s prediction (1) was met.** A field evaluation of three community-priced condensation particle counters at an urban regulatory station (*Aerosol Sci Technol*, 12 Sep) gave $R^2 > 0.99$ against a research-grade CPC with a 3 % inter-unit CV over four months. The Partector 2 Pro matched SMPS/CPC number concentration within $\pm 10 \%$ but under-sized by up to 28 % (*J Occup Environ Hyg*, 15 Sep), and a six-channel electro-inertial sizer reproduced SMPS distributions within 10 % over a 16-day rooftop deployment (Le et al., *Meas Sci Technol*, 22 Sep). None ran longer than one season, so drift is still unmeasured. **Prediction (2) — an ageing term inside an operational network correction — was not met;** the Denver campaign corrects drift by re-co-location, not by a modelled ageing term.
- **Epidemiological method got sharper.** Leung, Neophytou & Wilson (*Int J Epidemiol*, 13 Sep) proved that zero-filling gestational exposure histories in distributed-lag models *manufactures* the late-pregnancy window it is used to detect. A 115,056-birth analysis returned an informative null for short-term $\text{PM}_{2.5}$ and spontaneous preterm birth (17 Sep). In the Sichuan Basin, excess cardiorespiratory mortality per $10 \mu\text{g m}^{-3}$ rose from 0.4 % to **1.4 %** as concentrations fell, handing back about **19 %** of the Clean Air Action’s avoided deaths (6 Sep) — the

supralinear low-concentration slope showing up as a policy number.

- ▶ **Personal and continuous metrics carried signals area metrics missed.** Time above $35 \mu\text{g m}^{-3}$ indoors, not the period mean, carried the cystic-fibrosis pilot signal (26 Sep); a personal-monitor panel in older adults found a next-day activity deficit of about 8 min per IQR that area PM did not reproduce (27 Sep); and a third of indoor $\text{PM}_{2.5}$ variance sat between homes after full adjustment (8 Sep). Against that, **two African maternal cohorts** (21–22 Sep) rested exposure on uncorrected consumer optical sensors in biomass smoke — the cheapest sensor in the hardest aerosol.
- ▶ **Composition outran mass across the toxicology band.** Brake dust as a catalyst (10 Sep), saturating rather than additive oxidative potential (19 Sep), *p*-phenylenediamine antioxidants in open waste burning at $11.5 \mu\text{g g}^{-1}$ $\text{PM}_{2.5}$ (29 Sep) and tyre-related chemicals in Leipzig roadside PM_{10} (30 Sep). None of these is visible to a mass sensor.
- ▶ **Occupational & indoor nearly doubled** (37 → 72 records) and **Cardiovascular & metabolic doubled** (23 → 46), while **Sensing fell** from 140 to 116 (25% → 19%) as the AMT/ACP Crossref channel delivered 32 records against 48 in August. Exposure assessment (148, 24%) is now the largest band.

Corpus analytics — September 2026

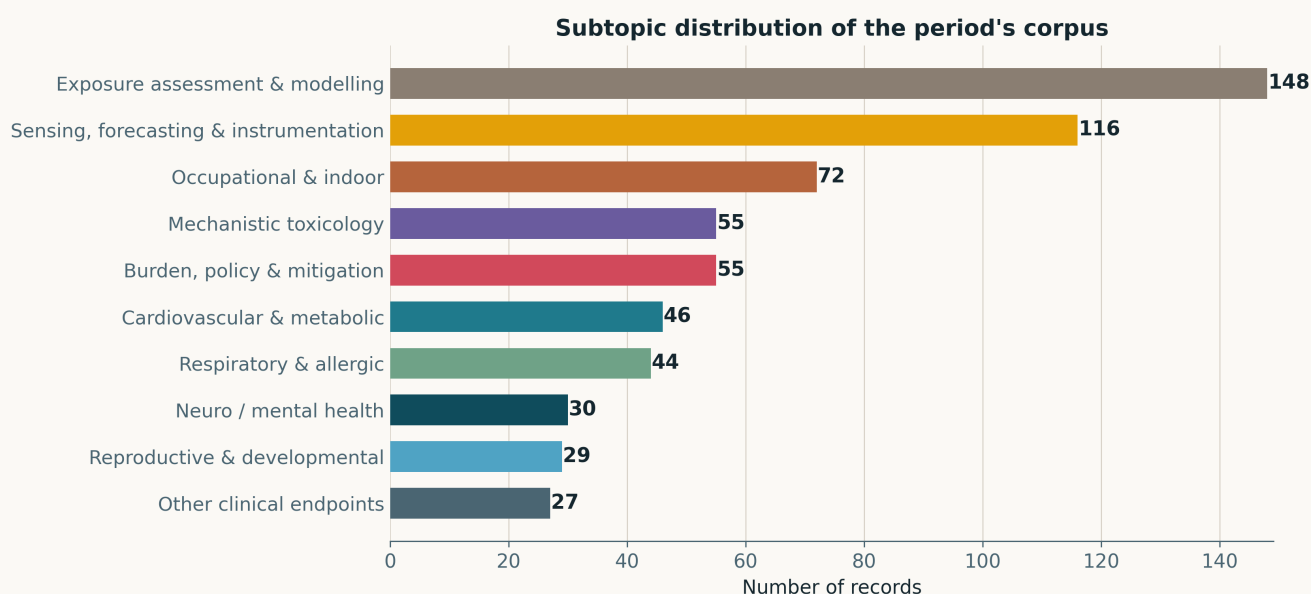


Fig. Subtopic distribution across all 622 pooled records; all ten subtopics populated. *Exposure assessment & modelling* 148 (24%) and *Sensing, forecasting & instrumentation* 116 (19%) together hold 43%, down from August's 48%. *Occupational & indoor* 72, *Mechanistic toxicology* and *Burden, policy & mitigation* 55 each, *Cardiovascular & metabolic* 46, *Respiratory & allergic* 44, *Neuro / mental health* 30, *Reproductive & developmental* 29, *Other clinical* 27. Single-label assignment by dominant contribution.

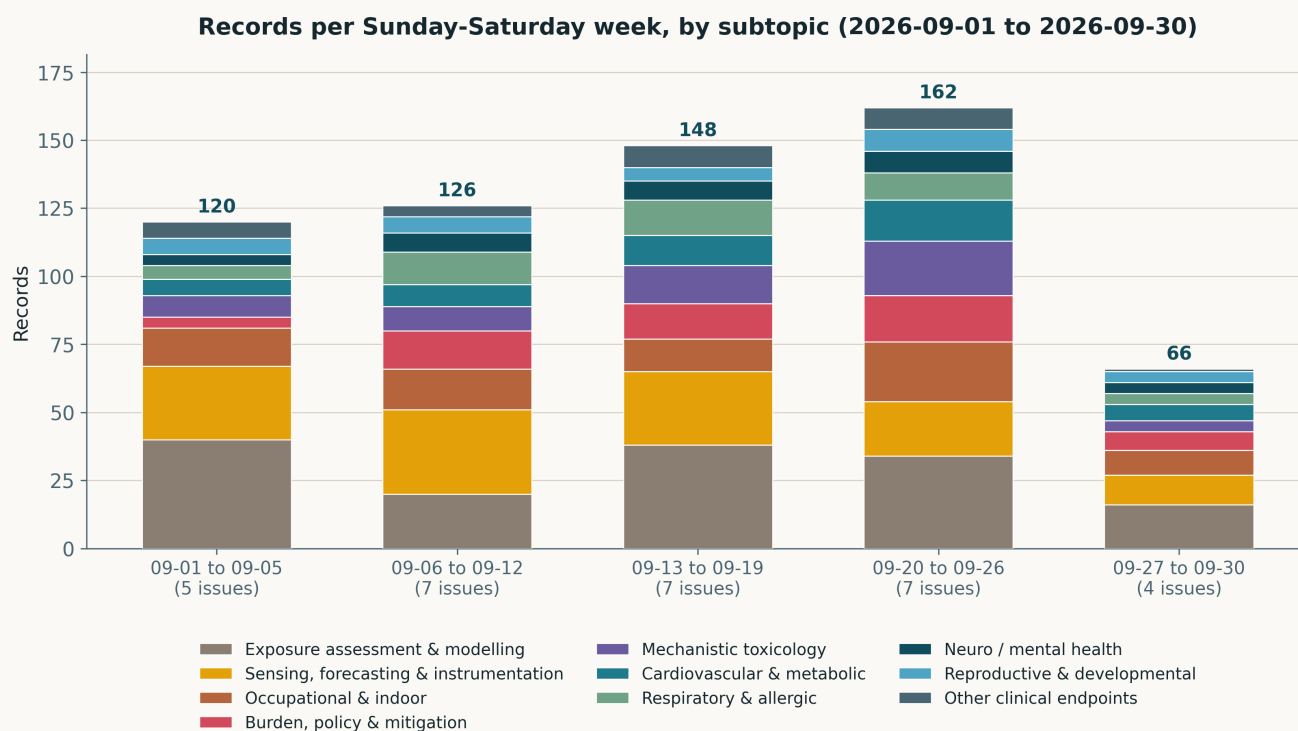


Fig. Records per Sunday-Saturday week by subtopic across the 30 September issues (the first and last weeks are partial): 120 for 1–5 Sep, then 126, 148 and 162, and 66 for 27–30 Sep. Daily volume ranged from 4 (27 Sep, a Sunday) to 38 (3 Sep) and tracks publisher deposit timing, so week totals are the readable unit. Volume rose through the month; the 26–28 Sep issues were built late as a backfill but each kept its own entry-date window. This panel replaces the July-era “what the daily cadence missed” chart, which had no meaning once every day had its own issue and was shipped with a stale July legend in the August monthly.

Study architecture

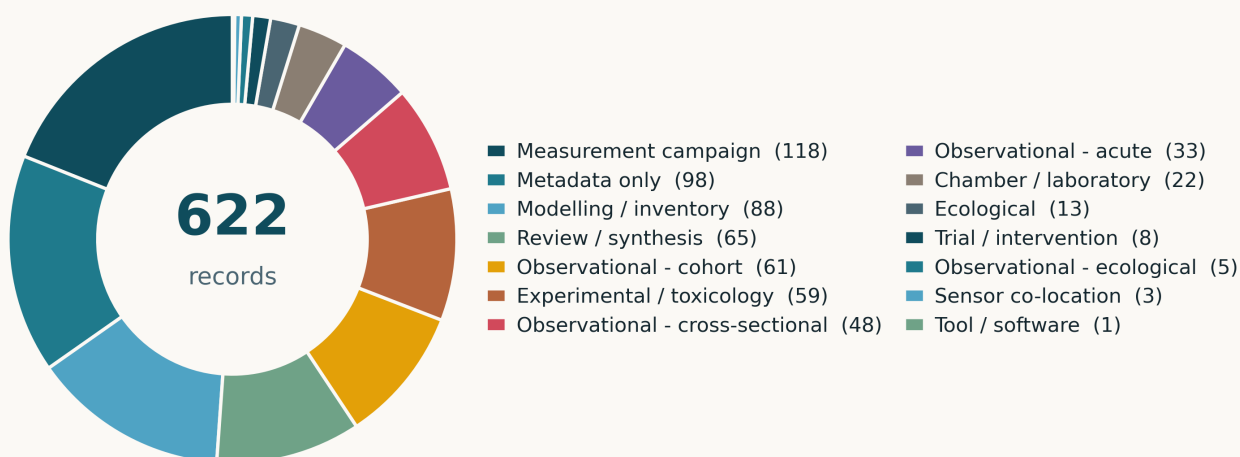


Fig. Study architecture. Measurement campaigns 118, metadata-only 98, modelling/inventory 88, reviews 65, cohorts 61, experimental/toxicology 59, cross-sectional 48, acute designs 33, chamber/laboratory 22, ecological 18. **Metadata-only rose from 12% in August to 16%**, almost all Elsevier deposits without abstracts, while Scholar Gateway (the full-text route) has been disconnected since 29 Sep. **Trials or interventions: 8 records in 622** — double August's four, still about 1%.

Health endpoint distribution — reported as text, not as a figure

ENDPOINT_CANON now exists (August's 144 raw strings collapse to a bounded map), but at month scale it still yields **48 categories**, so the pooled chart would be 48 rows at an unreadable size. **303 records (49%) carry no human health endpoint** — instrument, exposure, emissions and atmospheric-process work plus the metadata-only records. Of the remainder: **respiratory 66, cardiovascular 33, other clinical 23, neuro 28** (cognitive 17, mental health 11), **cancer 26** (still split as "Oncologic" 15 and "Cancer" 11 — the open canonicalisation defect), reproductive 17, mortality 18, metabolic 11, in-vitro toxicological 12, neurodevelopmental 6, cerebrovascular 6, with the balance in one- to five-record categories. **Fix still owed:** merge Cancer/Oncologic and the two neuro labels, and cap the map at ~20 categories so f2b can return at rollout scale.

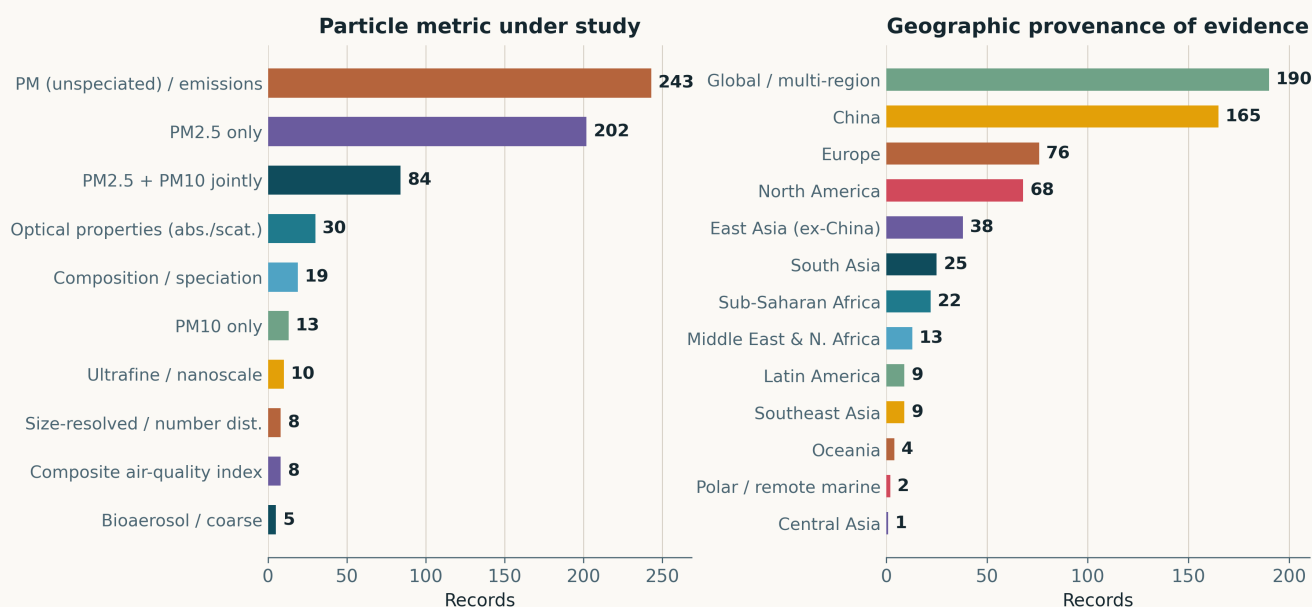


Fig. Left — particle metric. Unspeciated PM or emissions 243, PM_{2.5} alone 202, PM_{2.5} with PM₁₀ 84, optical properties 30, composition/speciation 19, PM₁₀ alone 13, **ultrafine 10 and size-resolved number 8** (3% together, against 4% in August), composite indices 8, bioaerosol 5. Right — geography. Global / multi-region 190 (inflated by metadata-only records with no recoverable site), **China 165 (27%)**, Europe 76, North America 68, East Asia ex-China 38, South Asia 25, **Sub-Saharan Africa 22** (16 in August), Middle East & N. Africa 13, Southeast Asia 9, Latin America 9 (13 in August), Oceania 4, polar/remote 2, Central Asia 1.

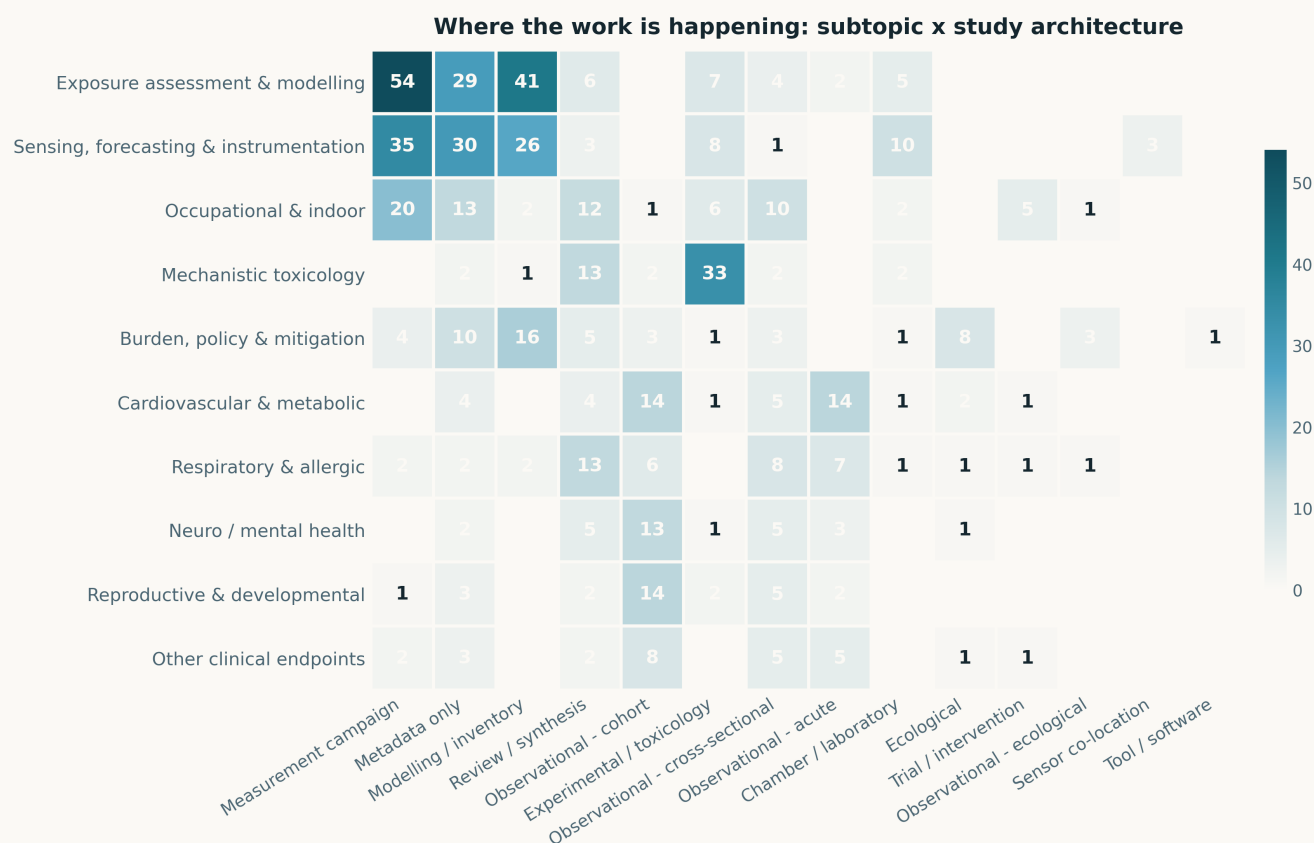


Fig. Subtopic × study-architecture cross-tabulation. The dense cells are the expected ones — sensing × measurement campaign, exposure × modelling and metadata-only. The informative absences are unchanged: the trial column is nearly empty in every health subtopic, and the sensing and exposure rows contribute almost nothing to the effect-estimate pool.

Life-course windows addressed this period (records may span several stages)

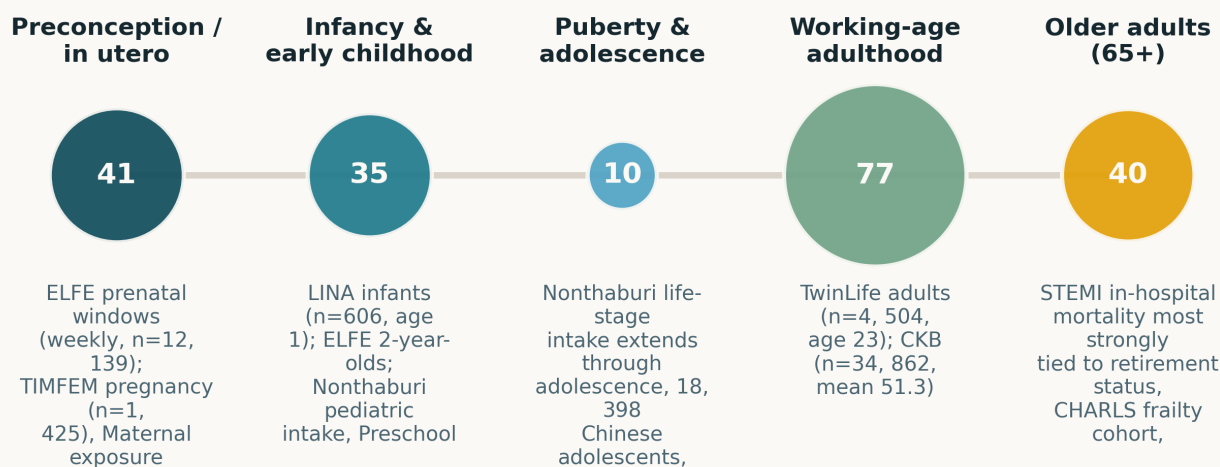


Fig. Life-course windows across the month (non-exclusive; only records with an identifiable human window): in utero 41, infancy/childhood 35, adolescence 10, working age 77, older adults 40. **Adolescence remains the thinnest window**, as in July and August.

Quantitative associations reported in the period's corpus (1 of 6)

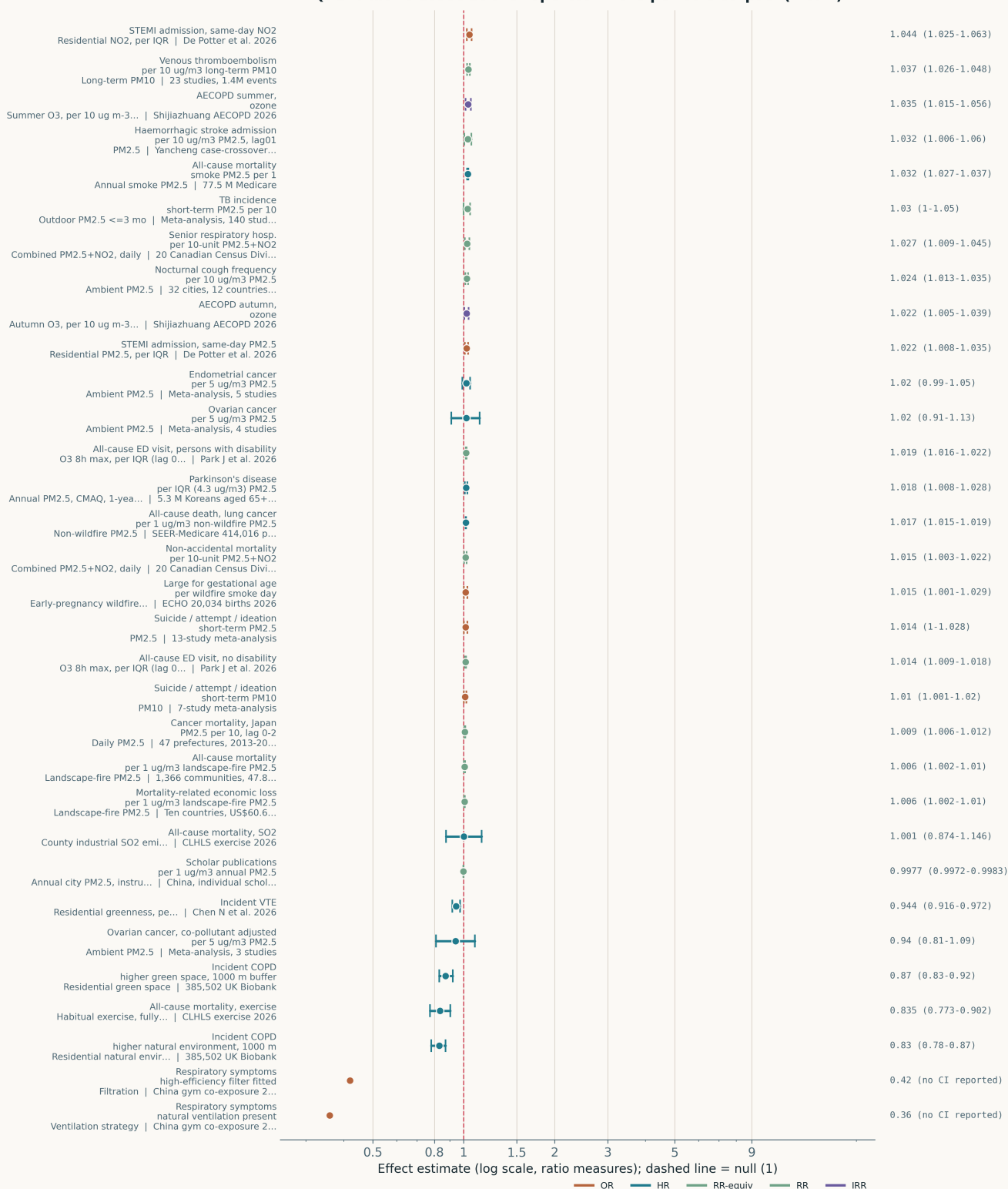


Fig. Quantitative effect estimates extracted across September, panel 1 of 6. **This is not a meta-analysis and no pooled summary is drawn.** Contrasts are not commensurable (per 1, per 10, per IQR, per doubling, threshold, quartile, fuel-type and occupational contrasts) and the outcome scale mixes OR, HR, RR, IRR, PR, sHR and ratio-of-means across unrelated diseases, so **heterogeneity is not estimable and none is reported.** Of 173 estimates, 161 carry a CI on a ratio or difference scale: **132 exclude the null in the harmful direction, 10 in the protective direction and 19 cross it.**

Quantitative associations reported in the period's corpus (2 of 6)

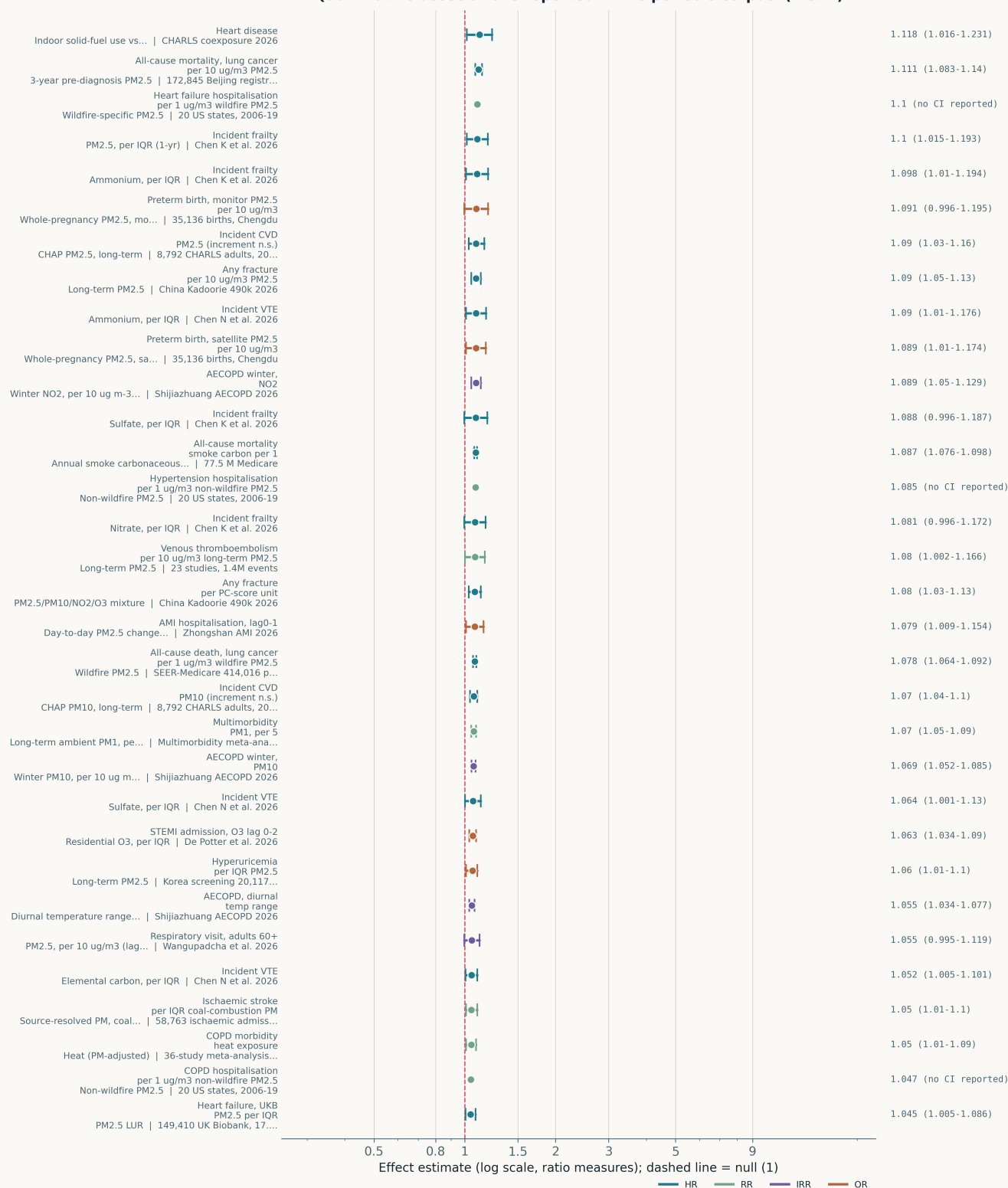


Fig. Effect estimates, panel 2 of 6. Same caveats as panel 1.

Quantitative associations reported in the period's corpus (3 of 6)

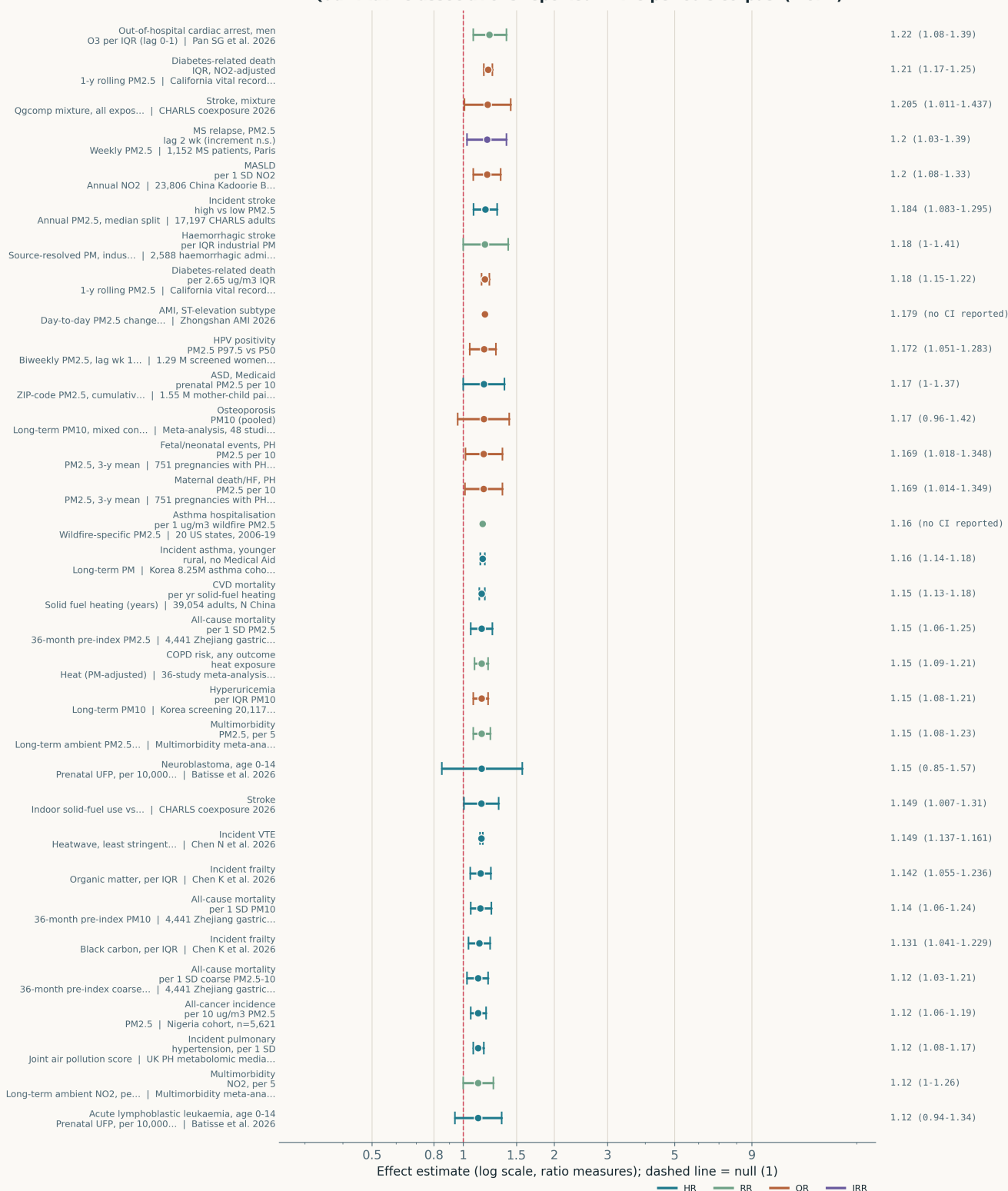


Fig. Effect estimates, panel 3 of 6. Same caveats as panel 1.

Quantitative associations reported in the period's corpus (4 of 6)

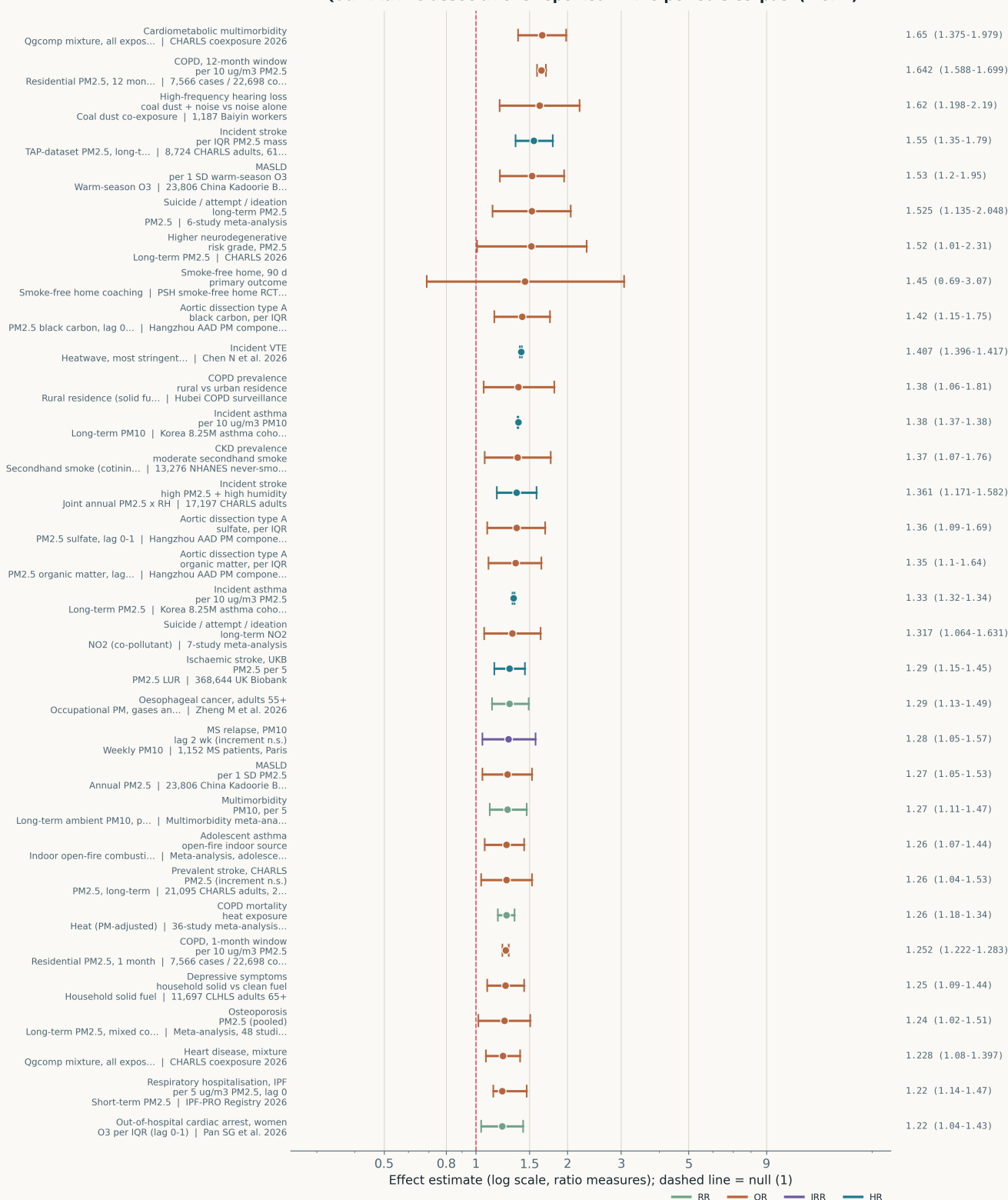


Fig. Effect estimates, panel 4 of 6. Same caveats as panel 1.

Quantitative associations reported in the period's corpus (5 of 6)

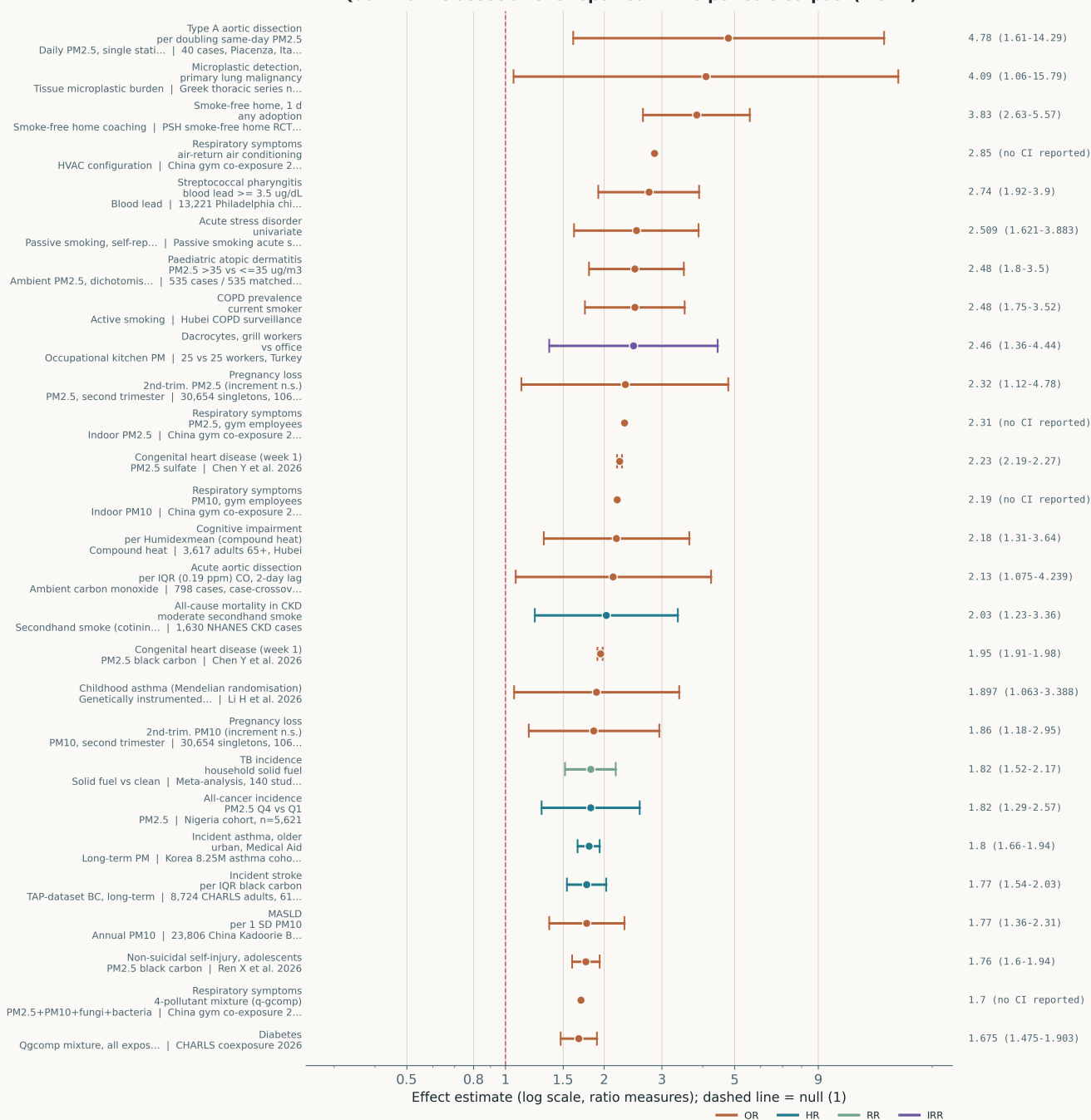


Fig. Effect estimates, panel 5 of 6. Same caveats as panel 1.

Quantitative associations reported in the period's corpus (6 of 6)

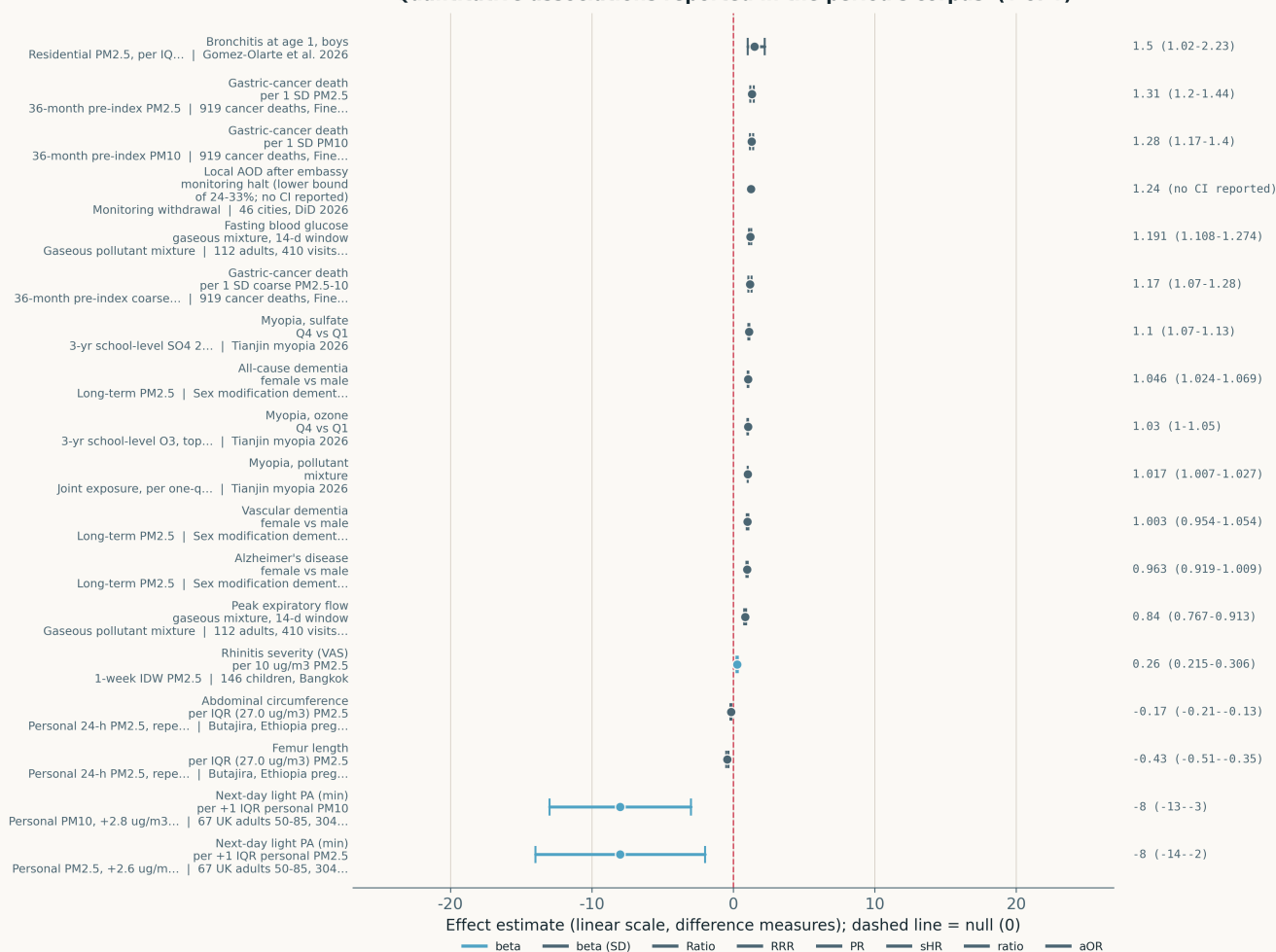


Fig. Effect estimates, panel 6 of 6. None of the ten estimates below the null is a protective PM effect: four are protective *co-exposures* (greenness against VTE, green space and natural environment against COPD, exercise against mortality) and six are harms on a decreasing scale (peak-flow ratio, femur length, abdominal circumference, two next-day activity betas, and scholarly output per $\mu\text{g m}^{-3}$). The largest ratios (OR 4.78 for type A aortic dissection per doubling of same-day $\text{PM}_{2.5}$; OR 4.09 for microplastic detection in lung malignancy) come from small case series with wide intervals.

Emerging, fading, and recurring

Emerging. (i) **Source-specific potency as the default framing for fire PM** — see the signal box; three large designs agree on direction. (ii) **Validation honesty in sensor calibration** — temporally blocked hold-outs (Sahin et al.), stratified error reporting across AQI bins (the Vishakhapatnam co-location, 15 Sep), and repeated co-location to track drift (Khalili Hollo et al.). (iii) **Community-cost particle-number instruments** reaching reference agreement (cCPC, Partector 2 Pro, electro-inertial sizer). (iv) **Monitoring as an intervention** — the embassy-shutdown natural experiment is the first record in this corpus that treats a monitor's existence as the exposure. (v) **Continuous-sensor-only metrics** (time-above-threshold, next-day personal exposure) carrying signals that means and area estimates miss.

Fading. Two complete months are still too few to call anything fading, and no decline is asserted. The observation carried from August stands: **single-pollutant time-series studies** continue to report PM as a weak term once co-pollutants enter the model — the Japanese cancer-mortality series (30 Sep) lost significance after NO₂ or SO₂ adjustment. The share of *Sensing* fell from 25 % to 19 %, but that tracks the Crossref AMT/ACP channel (48 → 32 records), not the literature.

Recurring groups and venues. **No author-cluster analysis is possible** — the corpus has no affiliation field (open since 15 August), so surname repeats in a China-heavy corpus are not investigator clusters. Venue concentration is supportable: *Environ Sci Technol* 46, *Environ Pollut* 36, *Front Public Health* 28, *Atmosphere* 26, *Atmos Environ* 24, *Atmos Chem Phys* 23, *Atmos Pollut Res* 15, *Environ Res* 14, *J Hazard Mater* 13, *Indoor Air* 12, *Atmos Meas Tech* 9. *ES&T* displaced *ACP* as the largest venue. **Most-cited work of the period is not reported:** citation counts on records days to weeks old measure indexing lag, not influence. The month's highest-venue records were Su et al. (*Lancet Oncol*), Ju et al. (*Lancet Planet Health*) and Zhou et al. (*PNAS*).

Trial registry watch

Daily registry sweeps (LastUpdatePostDate windows) ran through September; `state/trials.json` holds **16 trials**, **seven** of them first seen in September (the August rollup reported 24; the difference is not reconciled in the run log — see defect (vi)). The recurring finding of August hardened into a pattern: **registrations framed around air pollution that register no exposure measurement**.

NCT07817836 (IDEA, Pavia, observational, $n=151$): urban outdoor workers vs office controls, personal passive sampling, but no $PM_{2.5}$ outcome measure. **NCT07713264** (VISIONS, Stanford, interventional, $n=50$, outdoor workers): `analyze_endpoints` shows three psychosocial scales and no air-quality measure. **NCT07793123** (interventional, $n=60$, placebo-controlled): the third consecutive topical hit whose PM framing no measured variable carries. **NCT05423665** (observational, $n=360$) lists air pollution as a condition beside fetal growth restriction. **NCT07196436** (AIRVAC, energy-recovery ventilation plus air cleaner, $n=80$) was found through its protocol paper rather than the sweep; registry co-primaries are spirometry and asthma control, the protocol primary is asthma control alone. **NCT06541691** (COATED-AIR, $n=3,000$) has no exposure endpoint and a stale recruiting status. **NCT02944682** (HAPIN, $n=6,704$, LPG intervention) posted a registry update; main results are published.

Published trial-class records rose to **8** in 622, the most instrument-relevant being the protocol for a Dutch primary-school trial (NCT07479420) randomising **180 classrooms in 29 schools** to HEPA, ionisation/plasma or no cleaner (9 Sep) and a smoke-free-home RCT. The 30 Sep sweep returned one observational COPD-exacerbation cohort (CAN-EXACT, NCT07850648), screened out: environmental factors appear only inside a generic secondary measure. **The interventional evidence base for personal and household PM reduction is still thickening in the registry faster than in the literature, and the registry entries increasingly omit the exposure they claim to target.**

Open gaps and read on direction

Gaps in the literature

(i) **Composition is where the health signal is, and networks report mass.** Fire-carbon potency, oxidative-potential saturation and quinone-forming antioxidants all argue for absorption, BC or OP channels; no September record deployed one at network scale with a field calibration. (ii) **Drift is measured but not modelled.** Denver shows drift between co-locations; no record places an ageing term inside an operational correction (August's prediction 2, re-issued). (iii) **Ultrafine instruments now pass co-location but have no long-term record:** every September PN validation ran for a season or less. (iv) **Low-cost sensors in biomass smoke without correction** underpin new maternal-health cohorts in Sub-Saharan Africa — the region whose share of the corpus is growing (22 records) and whose aerosol is the worst case for optical sensing. (v) **Trials without exposure measurement.** Eight trial-class records; registry entries increasingly name air pollution without measuring it.

Gaps and defects this pipeline creates

(i) **No duplicates this month.** 622 records, **622 unique DOIs**; the DUPLICATE-DOI hard-fail added on 1 September held for all 30 issues. (ii) **98 records (16%) are metadata-only**, up from 12%. They are counted but carry no assessed content. The rise tracks Elsevier deposit practice and the loss of Scholar Gateway ("could not resolve user identity" since 29 Sep; needs reconnection). (iii) **Endpoint canonicalisation is incomplete** (48 categories; Cancer/Oncologic split). (iv) **No affiliation field** — no author or institution clusters. (v) **Source legs.** OpenAlex returned 429/403 all month; Europe PMC was down for the whole 25 Sep run (HTTP 503); arXiv returned 406 from mid-September and recovered on 29 Sep; Consensus returned **0 new** records on every September sweep. Three daily runs (26–28 Sep) did not fire and were backfilled day by day on 29 Sep, each with its own window. (vi) **Trial-list continuity.** `trials.json` lists 16 trials against the 24 the August rollup reported; earlier entries appear to have been dropped without a log line. To audit before the October rollup.

Read on direction. August's read was that the two halves of the field were optimising different objective functions — sensing on how far a $PM_{2.5}$ mass correction transfers, epidemiology on toxicity and number. September narrows that gap from both sides. Epidemiology supplied the strongest population-scale evidence yet that source and composition, not mass, set potency; instrumentation supplied the first community-cost particle counters with reference agreement and, more usefully, a set of papers that make calibration claims auditable (blocked validation, re-co-location, stratified error). **What has not moved is deployment:** no network in the corpus yet measures the fraction the epidemiology now points to. **Falsifiable predictions for October:** (1) a low-cost network with an absorption/BC or particle-number channel and a field calibration of at least one full season appears; and (2) at least one sensor-calibration paper reports a temporally blocked or prospective hold-out as its primary metric, citing the validation-leakage argument. If (2) fails, random-split R^2 remains the field's reporting norm and should be discounted by default.

Paper-level digest — the month by subtopic

Every pooled record appears below. Records that carry a full assessed summary have it lifted verbatim from the daily issue that first published it; a title-only or extractive entry is **not an assessment** and the absence of criticism is not approval. Metadata-only records are a pointer to a DOI and nothing more.

Sensing, forecasting & instrumentation

116 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (116 OF 116 RECORDS IN THIS SUBTOPIC)

3D-LSS particle shape 2026 J Aerosol Sci

METADATA ONLY — *enhanced 3D light-scattering sensor for online particle shape*

No abstract retrievable on the entry date. The title commits to two things: an *enhanced* three-dimensional light-scattering sensor for online aerosol shape characterisation, and a validation against test aerosols. Neither the size range, the angular sampling scheme, the sphericity metric nor the validation particle set is recoverable, and none is asserted here. Shape is the property optical low-cost sensors silently assume away when they convert scattering to mass, so an online shape channel is directly relevant; the predecessor 3D-LSS work established the concept at 500 nm–5 μm , and what “enhanced” means against that baseline is precisely what the abstract would settle.

doi: 10.1016/j.jaerosci.2026.106860

AAQR 2026 Aerosol Air Qual Res

A Sensor-Driven State-Space Framework for PM_{2.5}-Informed Natural Ventilation Opportunities Under Coupled Pollution-Thermal-Humidity Constraints.

Four residential communities - Kuala Lumpur, Bogor, Singapore and Palmerston (AU) - over 1 Jan to 1 Feb 2026, 724-744 hourly bins each. Crossing a sustained-PM_{2.5} screen with a T-RH envelope (24-30 C, RH $\leq$ 65%) shows the two constraints are close to anti-correlated: clean-hour fractions were 14.5 / 34.3 / 30.8 / 97.4%, while standalone thermal feasibility ran 97.0 / 71.5 / 98.5 / 20.6%. The site with clean air has almost no comfortable hours and vice versa. Limits are real - one month, one season, four sites, uncalibrated AirGradient data taken as-is with no co-location against reference monitors, and fixed rather than locally derived thresholds. Still the most directly usable result of the issue for LCS network design: it turns a raw sensor feed into an actionable control signal.

doi: 10.1007/s44408-026-00167-6

AIRWISE Delhi 2026 Atmos Chem Phys

Improving aerosol-radiation interactions in the operational forecasting system - AIRWISE.

Replacing globally prescribed refractive indices with Delhi-specific values inside the operational AIRWISE system cut surface downwelling shortwave by up to ~80 W/m² instantaneously (~43 W/m² diurnally) during a severe post-monsoon episode, and monthly mean SWDOWN by 25-37 W/m², a quarter to a third of total clear-sky aerosol-radiation interaction. The meteorological response is coherent and the right sign: surface temperature -0.2 C, near-surface wind -0.4 m/s, boundary layer height -200 m, daytime humidity +3-4%, and consequently higher PM_{2.5} - the shallower, calmer boundary layer traps the aerosol that caused the dimming. Validated against WiFEX surface radiation at Delhi airport with 62% (December) and 35% (January) bias reduction. Limits: one city, one season, refractive indices tuned where they are evaluated, and radiation bias reduction does not demonstrate PM_{2.5} forecast skill improvement, which is the operational claim a reader will assume. Relevant because the feedback runs the wrong way for forecasting: getting optics right makes the pollution episode worse, so a network assimilating into a model with generic refractive indices inherits a systematic low bias during exactly the episodes that matter.

doi: 10.5194/acp-26-13189-2026

Sun et al. 2026 Atmos Chem Phys

Aerosol properties during a transatlantic smoke transport event from Aeolus observations

ALADIN, a spaceborne high-spectral-resolution lidar, retrieves extinction and backscatter directly rather than through an assumed lidar ratio, and is used to follow a September 2020 western-US smoke plume to Europe. Mean extinction near 200 Mm⁻¹ across selected cross-sections; AOD fell 0.54 → 0.33 along the path; the layer rose from ~4.4 km over the US to above 6 km over Europe and arrived above cloud, with lidar ratios first falling then rising. One event constrains generality. For satellite-to-surface PM_{2.5} fusion the relevant point is blunt: this plume was invisible at ground level throughout.

doi: 10.5194/acp-26-12953-2026

Aerosol core/morphology optics 2026 J Aerosol Sci

Coupled Effects of Core Material and Aggregate Morphology on the Optical Properties of Atmospheric Aerosols

METADATA ONLY. Metadata only; no abstract in Crossref, Europe PMC or PubMed on per-DOI retry. Retained on title because core composition and aggregate morphology set the scattering cross-section that every optical PM sensor inverts to mass, which is the physical root of the humidity and composition dependence seen in co-location studies. No finding is asserted. On the retry list.

doi: 10.1016/j.jaerosci.2026.106897

Alderotti et al. 2026 Sci Total Environ

Raman spectroscopy as a non-destructive biomonitoring readout

Mature *Quercus ilex* sampled along a high/intermediate/low NO₂ and PM₁₀ gradient in Tuscany. Leaf Raman spectra show coordinated modulation of pigment (chlorophyll, carotenoid) and inducible flavonoid bands, cross-validated against destructive biochemistry and chlorophyll fluorescence, and consistent with *acclimation* rather than photoinhibitory damage. A composite Raman index tracked site-level NO₂ most closely, which the authors read as nitrogen driving the response. The attraction is a rapid, non-destructive, scalable field readout; the limitation is that a three-site gradient cannot separate NO₂ from PM₁₀ from any co-varying urban factor, so the index is at best a composite urban-exposure indicator and not a pollutant-specific one.

doi: 10.1016/j.scitotenv.2026.182253

Alkylamine IC-MS/MS method 2026 J Chromatogr A*Speciated atmospheric alkylamines at picogram levels in gas and particle phase*

Alkylamines drive new particle formation and early aerosol growth, and are analytically awkward — strongly polar, present at trace level, prone to adsorption and decomposition. This method pairs alkaline purging with acidic absorption for selective enrichment ahead of ion chromatography–tandem MS. Method detection limits are **9–94 pg m⁻³ gaseous and 0.1–1.1 pg m⁻³ in PM**, a substantial gain on existing offline methods, and the approach resolves the dimethylamine/monoethylamine isobaric pair and quantifies tertiary amines that derivatisation-based methods structurally cannot. Field-applied at septic tanks and traffic tunnels in Dalian, it recovered a broader speciation profile than previous work and seasonal variation in each species. It is an offline enrichment method, so time resolution is whatever the sampling interval is — this does not compete with online amine measurement for process studies, it competes on speciation and sensitivity.

doi: 10.1016/j.chroma.2026.467416

Alraddadi et al. 2026 Atmosphere*Explainable one-hour-ahead AQI forecasting across eight Saudi cities*

Attention-based bidirectional recurrent networks against a persistence baseline in eight climatically distinct cities. The deep models **matched persistence on RMSE and R^2** (cross-city mean $R^2 \approx 0.90$) and beat it on MAE by roughly a fifth, with the margin largest in the most variable cities. Event-based evaluation flagged **84–91% of high-AQI exceedances**. Surface meteorology added little on average and its value was city- and architecture-specific. SHAP and LIME put particulate matter as the dominant driver everywhere. The persistence comparison is the part worth copying: at one-hour horizon hourly AQI is so autocorrelated that an unbenchmarked R^2 of 0.90 means nothing, and most papers in this genre do not report the baseline.

doi: 10.3390/atmos17090858

Anwer et al. 2026 Atmos Environ*Artificial Intelligence in Sand and Dust Storm Research: A Systematic and Bibliometric Review (1990-2025)*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 26 Sep with no abstract; Crossref, PubMed fetch, Europe PMC and the publisher landing page returned nothing. The title indicates a systematic and bibliometric review of AI methods in sand and dust storm research over 1990-2025. Of interest only if it tabulates forecast skill against PM₁₀ observations; bibliometric reviews usually do not. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.atmosenv.2026.122387

BC coating optics 2026 J Aerosol Sci*METADATA ONLY — metal-oxide coatings and the optical properties of black carbon*

No abstract retrievable on the entry date. The record establishes that the optical effect of V₂O₅, NiO and Fe₂O₃ surface coatings on black carbon aggregates has been characterised; it establishes no absorption enhancement factor, wavelength dependence, coating thickness or morphology treatment, and none is asserted here. The reason to track it: absorption-based BC measurement — aethalometers, photoacoustic instruments and the filter-based methods most networks run — is systematically biased by the coating enhancement factor, and the metal-oxide coatings specifically are the industrial and brake-wear case rather than the usual organic or sulfate one.

doi: 10.1016/j.jaerosci.2026.106885

Bai et al. 2026 Atmos Meas Tech*Low-level jet evolution and vertical aerosol redistribution: a Doppler wind lidar study*

Two dust episodes over Hefei, East China. April 2021: the northwesterly jet core descended from 2.5 km to 0.3 km then re-ascended to 1.0 km while surface PM₁₀ peaked at **410 µg/m³**, with shear above **0.04 s⁻¹** and increased spectral width at the lower jet interface and a downward shift of enhanced backscatter. March 2022: a stable jet core and persistent lower-tropospheric stability held aerosol aloft, the PM₁₀ peak arriving at 16:00 LT, **~8 h after inferred frontal onset**. Two regimes — rapid downward mixing during jet descent versus delayed surface impact from a decoupled layer. Two cases is mechanism illustration, not climatology, and the inertial-oscillation attribution rests on hodograph rotation alone. The clearest statement yet of why a surface-only network can read clean for hours.

doi: 10.5194/amt-19-5951-2026

Baltic ship plume OFR aging 2026 Environ Sci Technol Lett*Fresh and photochemically aged emission factors from individual ships, measured at sea*

An oxidation flow reactor operated from a moving research vessel, sampling identified ocean-going ships at **0.6–1.6 km** standoff in the Baltic, plus the research vessel's own plume as a control. Gas-phase and potential-aerosol-mass emission factors were measured fresh and after photochemical aging. **Particle mass rose after aging for every vessel sampled**, and secondary aerosol formation potential scaled with SO₂ emission — i.e. with fuel sulfur content, consistent with sulfate-driven particle formation. This is a pilot with a small vessel sample and no size-resolved chemistry reported in the abstract, so the emission factors are demonstrative rather than inventory-grade. The contribution is the framework: stack-level aging experiments and fixed ambient monitoring have both existed for years, and this joins them on a platform that can follow an individual, identifiable source.

doi: 10.1021/acs.estlett.6c00695

Barth et al. 2026 Environ Sci Atmos*Online characterisation of oxidative potential and reactive oxygen species of fresh and aged aircraft turbine and ship engine particulate emissions.*

Only a one-sentence abstract was deposited: fresh and aged transport-sector particles from shipping and aircraft contribute significantly to particle oxidative potential. The title establishes the design - online (not filter-offline) OP and ROS measurement of fresh and aged turbine and marine-engine emissions - but no OP values, ageing conditions, fuels or particle metrics can be stated. Carried at tier B for the method: online OP is the toxicity-relevant metric a sensor network cannot yet measure, and fresh-versus-aged contrasts determine whether near-source mass is a fair proxy for harm. Retry list for the full abstract.

doi: 10.1039/d5ea00174a

Bogota BiLSTM Huber 2026 Atmosphere*Asymmetric-loss BiLSTM for 24h PM_{2.5} forecasting and exceedance detection*

Hourly records from the Puente Aranda industrial-corridor station, Bogotá, 2022–2025 ($n = 31,961$). The motivating statistic is sharp: **98.7%** of days exceeding the WHO 2021 15 µg/m³ guideline contain at least one hour above 25, median peak-to-mean **1.90**, so a model trained on mean error systematically misses the episodes that matter. Three mechanisms are stacked — an asymmetric Huber loss penalising underprediction α -fold, exceedance sample weighting, and Optuna-TPE over 100 trials. The honest result is a reported **regression–detection tension**: no α minimised both global and peak MAE. Optimisation chose $\alpha = 2.80$, giving the best detection (AP **0.724**, 95% CI 0.582–0.821) and lowest peak MAE (10.64 µg/m³) at the cost of $R^2 \approx 0$ globally; $\alpha = 1.5$ is the compromise (AP 0.718, RMSE 6.65, $R^2 = 0.372$). One station, so nothing here is a spatial claim, and the wide bootstrap intervals on AP overlap across all configurations — the ranking is weaker than the point estimates suggest.

doi: 10.3390/atmos17090871

Boston sparse network 2026 Environ Sci Technol*What sparse fixed air-quality networks miss, measured against mobile monitoring*

Fixed-station PM_{2.5} paired with simultaneous mobile measurement at a dense set of locations within a 250 m radius, at three contrasting areas of the Boston metro area across spring and summer 2025. Typical agreement is close — mean difference **2.0 $\mu\text{g}/\text{m}^3$** , median 1.5, IQR 0.6–2.8 — but dynamic events drove differences past **300%** and **800 $\mu\text{g}/\text{m}^3$** , and **10.4% of cases** (21.8% at one site in spring) fell either side of an AQI category boundary. Crucially **Euclidean distance alone could not explain the discrepancies**, which undercuts radius-based representativeness and every inverse-distance product built on a reference network. Relative humidity was the strongest reported correlate ($r = -0.84$), a signed rather than absolute association that the abstract does not fully disambiguate. Two seasons and one metro area, and mobile platforms carry inlet and response-time artefacts of their own; but this is the number the dense-network argument has been missing.

doi: 10.1021/acs.est.6c03923

Broda et al. 2026 Atmos Meas Tech*A new balloon-borne system for measuring the vertical variability of aerosol optical properties using low-cost sensors.*

A balloon-borne AeroSonde carrying calibrated SPS30 optical PM sensors was referenced against an Aurora 4000 nephelometer and an LAS 3340 spectrometer to retrieve aerosol scattering coefficient, scattering Angstrom exponent and effective radius, with Pearson r above 0.88. Agreement was best for the scattering coefficient; SAE and effective radius carried substantially larger uncertainty and are honestly restricted to qualitative fine-versus-coarse discrimination. Soundings in June and August 2025 reached 10–27 km at better than 10 m vertical resolution and captured North American biomass-burning plumes and Saharan dust, agreeing with active remote sensing on layer structure and extensive properties at r 0.74–0.81. Two campaigns and no independent in-situ profile leave the intensive retrievals unvalidated aloft. Still the clearest demonstration this quarter that a commodity OPC can do profiling work.

doi: 10.5194/amt-19-5989-2026

CALIOP Shanghai aerosol stratification 2026 Atmos Res*Long-term CALIOP observations of vertical aerosol stratification over Shanghai: Meteorological modulation and height-resolved transport*

METADATA ONLY. Metadata only - no abstract deposited to Crossref and none in Europe PMC or PubMed on per-DOI retry. Carried on title because height-resolved spaceborne lidar climatology over a megacity is the long-record complement to the single-tower fusion result in this same issue, and because column-to-surface conversion error is the central obstacle for satellite PM_{2.5}. No finding asserted. On the retry list.

doi: 10.1016/j.atmosres.2026.109329

Cai J et al. 2026 Environ Sci Technol*Subway air microbiome is homogeneous at network scale and heterogeneous at platform scale*

Stratified cluster sampling of **43 underground stations across 13 lines** of the Shanghai subway, collecting total suspended particles and PM_{2.5} *simultaneously* at entrances, halls and platforms, then integrating taxonomy, transcriptional potential, PM_{2.5} chemistry, microclimate and network topology. Bacteria dominate both size fractions and most transcriptional signal; eukaryotes are strongly enriched in TSP, **a clean demonstration of particle-size filtering of the biological load**. Dominant taxa (*Staphylococcus*, *Bacillus*) show weak abundance–transcription coupling, so abundance is a poor proxy for activity. Diversity and composition are essentially uniform across the network — consistent with passenger mixing — while finer-scale structure appears within stations. For exposure work this means a single station is a reasonable network sentinel for community composition and a poor one for microenvironment. Weakness: one city, one season not stated, no human outcome.

doi: 10.1021/acs.est.6c00174

Wang et al. 2026 Environ Sci Technol*14-day PM_{2.5} and O₃ forecasting over eastern China with a CNN–Transformer*

Hybrid architecture combining multilevel regional meteorological fields with emission inventories at a 14-day lead. Relative R^2 gains over baseline are **40% for O₃** (0.58 to 0.81) and **18% for PM_{2.5}** (0.44 to 0.52); 850 hPa fields add 5% for O₃ and emissions add 9.8% for PM_{2.5}; 66–83% of O₃ episodes are captured in the north. Read honestly, PM_{2.5} skill remains poor — R^2 0.52 at 14 days is not decision-grade, and the paper's own framing concedes O₃ is the stronger case. No episode-capture rate is reported for PM_{2.5}, which is the number a forecast user would act on, and gridded reanalysis truth hides station-scale error.

doi: 10.1021/acs.est.6c05923

Chaparro et al. 2026 Sci Total Environ*Biological and technological sensors for airborne particles at heritage sites*

Ships on metadata only. The title indicates parallel biomonitoring and instrumental sensing of airborne fine particles for preventive conservation of cultural heritage, but no abstract was deposited and the record is not in PubMed, Europe PMC or Semantic Scholar on this run, so the sensor models, the biomonitor species and any agreement statistics between the two are unavailable. Counted and listed; no numbers claimed. On the retry list — cross-validation of a biological against an instrumental particle sensor is directly relevant to low-cost network validation and rarely published.

doi: 10.1016/j.scitotenv.2026.182285

Chen C et al. 2026 J Hazard Mater*Amine-containing particles from fireworks by single-particle mass spectrometry*

2023 Spring Festival field campaign with single-particle aerosol mass spectrometry plus source samples. Amine-containing particles rose **96.2%** during the firework period, peak size shifted **0.42 → 0.44 μm** , and internal mixing with sulfate increased; fireworks raised metal and elemental-organic-carbon classes while depressing biomass-burning and secondary classes. **N-methylaniline** is proposed as a tracer separating fireworks from firecrackers. A multi-model comparison identifies modified aerosol acidity, not liquid water content, as the dominant control on gas–particle partitioning of amines, with ALWC showing a non-linear threshold. One festival, one city; the tracer claim needs a second site before it is usable.

doi: 10.1016/j.jhazmat.2026.143446

Clean air advocacy data approaches 2026 Curr Environ Health Rep*Examining Data-Driven Approaches for Clean Air Advocacy in a Changing Climate: a Scoping Review.*

Fifty studies from 41 countries and over 100 cities, sorted into five approaches: sensor-based monitoring, modelling, digital platforms and visualisation, community-centred approaches, and lived-experience assessment. 86% combined at least two. The review's operational finding for network builders is that low-cost monitoring has changed data availability far faster than it has changed governance: projects generate evidence but rarely outlive their funding cycle, and open-data formats are not standardised enough for community data to enter policy processes reliably. Scoping reviews do not grade evidence quality, so a single-site pilot and a multi-city network count equally here; no quantitative synthesis of advocacy outcomes is attempted, and publication bias toward successful campaigns is unaddressed.

doi: 10.1007/s40572-026-00563-5

Climate-stratified PM_{2.5} LightGBM 2026 PLoS One*Climate-stratified assessment of PM_{2.5} using machine learning: Geographic controls dominate meteorological factors*

49,997 monthly MERRA-2 samples (2019–2023) across five latitude-band climate zones under Leave-One-Climate-Out cross-validation. Tuned LightGBM reaches a global LOCO R² of 0.367 (MAE 6.43 $\mu\text{g m}^{-3}$), but zone predictability ranges from R² 0.604 subtropical to 0.142 tropical. SHAP puts 64% of importance on geographic coordinates against 23% on meteorology, and a retrain ablation confirms the coordinates are not mere spatial autocorrelation - dropping lat/lon collapses mean R² from 0.367 to -0.099. Temporal split validation gives 0.544, so LOCO is the harder test. The honest limitation is that the target is reanalysis PM_{2.5}, not measured PM_{2.5}, so the model is benchmarked against a model; and coordinates proxying emissions means the framework cannot transfer to a region whose emission geography it never saw.

doi: 10.1371/journal.pone.0347591

Contini et al. 2026 Environ Pollut*10-year trends of particle size distributions and equivalent black carbon at the Environmental-Climate Observatory (ECO) in Lecce (Italy).*

A decade (2015–2024) of co-located online PM_{2.5}, coarse PM, particle number size distribution (10–10,000 nm) and equivalent BC at a southern-Italian observatory, with offline OC/EC to derive the mass absorption cross-section. PM_{2.5} fell -2.58%/yr, eBC -3.17%/yr and total PNC -2.99%/yr (Aitken and accumulation modes), while PM_{10-2.5} was flat, held up by Saharan dust on 13.2% of days. MAC was stable at 12.4 $\pm$ 2.4 m²/g, so falling absorption reflects falling eBC rather than changing optics; new particle formation occurred on about 25% of days. Single site, and the abstract gives neither the trend estimator nor uncertainty on the slopes. Relevance: a stable decade-long MAC is a rare field-validated constant for converting absorption to BC mass - and the fine/coarse decoupling under dust is precisely where low-cost optical sensors misreport.

doi: 10.1016/j.envpol.2026.129234

Cooking aerosol FIGAERO partitioning 2026 Atmos Chem Phys*Unravelling gas-particle partitioning dynamics in cooking aerosol oxidation through FIGAERO-CIMS analysis*

Heated corn oil emissions aged in a Go:PAM oxidation flow reactor, with FIGAERO-I-CIMS and a 2-D VBS parameterisation tracking composition, volatility and partitioning. Kinetic calculations give molar-mass-dependent partitioning with the surface process rate-limiting. Primary cooking aerosol sits biased toward the gas phase relative to equilibrium; in secondary aerosol the crossover is near 120–220 g mol⁻¹, below which compounds favour gas and above which they favour particle. The honest result is the negative one: a substantial unresolved gap remains between FIGAERO-derived effective volatility and equilibrium thermodynamic estimates, which the authors attribute to combined measurement and model uncertainty rather than resolving. Cooking is a rising share of urban OA as combustion sources are controlled, and non-equilibrium partitioning is a direct problem for any mass-based sensor calibrated on ambient mix

doi: 10.5194/acp-26-12793-2026

Coriolis sampler inactivation 2026 Indoor Air*Sampler-induced loss of respiratory virus infectivity in spiked collection liquid*

Rhinovirus, SARS-CoV-2 and influenza A spiked into the collection liquid of a Coriolis μ liquid impinger and run at several airflow rates — deliberately *not* an aerosol or ambient-air collection, which isolates mechanical stress in the liquid from capture efficiency. Culturable virus fell by up to 98.6% for SARS-CoV-2 and influenza A, while RNA copies were largely unchanged for SARS-CoV-2 and rhinovirus and reduced only for influenza A. Rhinovirus infectivity was essentially unaffected, so the effect is virus-specific rather than a general artefact. The design's strength is also its limit: it cannot say how much infectivity is lost during aerosol capture itself, only during the liquid phase, so it bounds one term of the total sampling loss. The consequence for bioaerosol surveillance is direct — a PCR-positive, culture-negative air sample is not evidence of non-infectious aerosol.

doi: 10.1155/ina/5068026

Cui C et al. 2026 Environ Sci Technol*Eighteen years of CCN reconstructed by XGBoost with SHAP attribution*

An observation-driven XGBoost framework reconstructs CCN number concentration at a North China Plain mountain site, 2007–2025. Particle number size distribution carries 55.7–66.3% of predictive importance, with the controlling size range shifting from 100–200 nm at 0.2–0.4% supersaturation to 50–100 nm at 1.0%. No monotonic CCN trend, but summer bulk activation ratio and hygroscopicity parameter both **declined from 2009 to 2025**. Coupling PMF with the ML model attributes that to falling SO₂ influence and rising contributions from new particle formation and growth — a legible consequence of desulfurisation policy. Reconstruction over an 18-year record inherits every instrument change in that record, and SHAP importance is not a causal decomposition.

doi: 10.1021/acs.est.6c04501

Cyprus dust typing + DRE 2026 Atmos Meas Tech*Characterization of Saharan and Middle Eastern dust aerosols and their associated shortwave direct radiative effects over Cyprus*

Seven years (2015–2022) at Agia Marina Xyliatou. Dust is 28.3% of aerosol cases in MAM and 12.7% in SON; 86% Saharan, 14% Middle Eastern. AOD₄₄₀ 0.34 $\pm$ 0.13 Saharan vs 0.40 $\pm$ 0.13 Middle Eastern, SSA₄₄₀ 0.93 for both, i.e. the two source regions are separable by trajectory and loading but not by single-scattering albedo. Modelled GHI within $\pm$ 5% of ground irradiance for 87% of cases. At SZA 60 deg the strongest monthly mean surface forcing is -92 W m⁻² (Saharan) and -86 W m⁻² (Middle Eastern), with +72 and +59 W m⁻² of atmospheric heating. Limitation: DRE is reported at one fixed SZA, so the monthly means are not diurnally integrated and cannot be read as energy budgets. Column optical typing is the reference against which a low-cost PM network's dust-episode flags should be scored

doi: 10.5194/amt-19-5753-2026

Delkash 2026 Environ Pollut*Barometric pressure variability in community air monitoring near a landfill*

Ships on metadata only — the Crossref deposit has no abstract and Semantic Scholar returns the record with a null abstract field. The title describes a receptor-scale multi-method analysis of how barometric pressure variability imprints on community monitoring data around an elevated-temperature landfill, across two mitigation interventions. Counted and listed; no numbers claimed. This is the single most operationally relevant record of the day for network work — pressure-driven emission modulation is a known confounder in fenceline and community monitoring and is almost never quantified — so it goes to the top of the retry list.

doi: 10.1016/j.envpol.2026.129084

Zhou et al. 2026 Proc Natl Acad Sci U S A*Air pollution after the shutdown of US embassy monitoring*

Matched difference-in-differences across 46 cities with same-country controls, using real-time satellite AOD. No differential pre-trend; a prompt **24–33% rise in local pollution** after US reporting stopped, largest where substitutes were poorest. Calibrating AOD to concurrent pre-shutdown embassy PM_{2.5} gives +1.9 $\mu\text{g m}^{-3}$ on the authors' conservative translation, with alternatives spanning 1.5–16.3 — an order-of-magnitude spread that sits in the calibration step, not the DiD, and no CI accompanies the headline. The design is the contribution: monitoring treated as an intervention with a measurable atmospheric effect.

doi: 10.1073/pnas.2605467123

Europe BLH regression kriging 2026 Atmos Meas Tech*Monthly boundary-layer height maps for Europe at 25 km from radiosonde, satellite and ERA5*

Regression kriging over 2010–2020 combining radiosonde profiles with land-surface temperature, topographic covariates and ERA5, at 25 km and monthly means for 00 and 12 UTC. Random forest beat linear regression and gradient boosting at both hours. Against ERA5 the product is consistently sharper: winter spatial variability at 12 UTC gives **RMSE ≤ 170 m versus ≥ 250 m**, summer temporal variability **≤ 260 m versus 340 m**. Performance **declines considerably at 00 UTC**, which is the honest limitation and the awkward one — the nocturnal stable layer is exactly where mixing-height error propagates hardest into surface concentration. Monthly means also cannot serve episode-scale exposure work. Still, boundary-layer height is the dominant meteorological term in almost every surface PM model, and a validated European product that beats reanalysis is directly usable as a covariate.

doi: 10.5194/amt-19-5697-2026

FY-4B/AGRI AOD evaluation 2026 Atmos Environ*Evaluation of the FY-4B/AGRI Land Aerosol Optical Depth Product and Comparison with Geostationary Satellite Products over Asia*

METADATA ONLY. No abstract was available on the entry date from any of the four sources queried - the same same-day Elsevier deposit lag logged repeatedly since 2026-09-08. Title and journal establish a validation of the FY-4B/AGRI land AOD product against other geostationary retrievals over Asia. No validation statistics, no reference network and no spatial or temporal coverage are available, so nothing is asserted here and the record is tier C. Worth re-reading when the abstract deposits: geostationary AOD is the only route to sub-daily satellite-surface PM fusion, and product-versus-product agreement is the term that dominates the error budget of any such fusion

doi: 10.1016/j.atmosenv.2026.122354

Gökozan 2026 Chemosensors*Machine-learning calibration of a low-cost PM_{2.5} sensor with location-specific covariates and feature engineering*

Ablation design separating environmental covariates, auxiliary gases and feature engineering for one outdoor residential sensor, validated on a leakage-safe time-based split. Feature engineering helped in every configuration: raw PM_{2.5} plus engineered features raised Random Forest test R^2 from **0.724 to 0.847** and cut RMSE from 6.96 to **5.19 $\mu\text{g}/\text{m}^3$** ; humidity interactions and cyclic time encodings carried the importance. One sensor at one site is the limit, and 5.19 $\mu\text{g}/\text{m}^3$ is large against typical ambient means — but the operational claim, that useful correction is reachable without continuous co-location, is the one network operators need tested. *Connector-sourced; outside the entry window.*

doi: 10.3390/chemosensors14070154

Front Bioeng 2026 Front Bioeng Biotechnol*Bionic breathing-type airway exposure system for inhalation exposure studies.*

A 3D-printed trachea-to-G3 airway replica with millisecond breathing control (five waveforms, oral/nasal/combined routes), CO₂ exchange and multi-zone exposure interfaces. Validation recovers the expected right-dominant airflow split (right main bronchus 51.6+/-0.1% vs left 48.4+/-0.1%) and separates deposition by breathing pattern (large-loop 78.0+/-1.6% vs small-loop 61.7+/-1.3%). BEAS-2B cells showed dose-dependent cytotoxicity; hiPSC-derived lung models gave selective inflammatory responses. The weakness is that airflow-split fidelity is a low bar - it follows from the geometry, not the dynamics - and no comparison against in vivo human deposition data is offered, so anatomical realism is asserted rather than demonstrated. For sensing work it matters as a dosimetry bench: it converts a measured mass concentration into a branch-resolved deposited dose.

doi: 10.3389/fbioe.2026.1871869

Galioto et al. 2026 Atmosphere*Physics-informed ML for street-canyon NO₂ in Palermo*

Six years of hourly data (2020–2025) on one of Europe's densest traffic corridors, resolved to **~ 11 m**: COPERT-Italy emission factors over the OpenStreetMap network, dispersed with the SIRANE street-network model, then fused with measurements in an ML stage and mapped in GIS. Observation-model scatter over **44,752 hours** centres on the 1:1 line. The genuinely useful move is inverting the chain to recover corridor traffic volume from observed NO₂ — a template for using a fixed monitor as a traffic sensor, which transfers to PM only where the non-exhaust fraction is separately handled. Analyte is gas-phase; single corridor; and a physics-informed hybrid validated against the same stations that trained it needs an out-of-corridor test before the inversion can be trusted.

doi: 10.3390/atmos17090865

Genoa absorption intercomparison 2026 Atmosphere*AE33 aethalometer against two filter-based absorption techniques in a coastal city*

Multi-wavelength field intercomparison at the Multedo urban coastal site, Genoa, where the aerosol mixes traffic, shipping, industry and marine air. The AE33 was run against the Multi-Wavelength Absorption Analyzer and the Broadband Light Analyzer of Complex Aerosols across UV, visible and near-IR. Correlations are excellent — R^2 **0.93–0.99**, tightest between the two offline methods — but AE33 reports absorption coefficients **12–22% higher**, with the offset varying by wavelength. That pattern implicates the instrument's default multiple-scattering correction, which is tuned to a generic aerosol and not to this one. Absorption Ångström exponents from all three sit near unity, consistent with dominant primary combustion. The limitation is that one site and one campaign cannot fix the correction factor generally — which is precisely the point: a site-specific bias is not a constant you can publish once.

doi: 10.3390/atmos17090883

Giang et al. 2026 J Radiol Prot*Fifteen consumer radon monitors, 98 days, one occupied house*

45 units (15 models $\times$ 3) co-located with an AlphaGuard PQ2000 and two alpha-track detectors in the lowest lived-in space of an occupied Ontario home, with PM_{2.5}, CO₂, temperature and RH logged alongside; reference mean **99 Bq/m³** (range 62–141). Cumulative performance: **8 of 15 models** had two or more units within $\pm 25\%$ relative percent error, the best reaching 3–5% after 13 weeks, while others ran from **–47% to +220%**. Intra-model CV was mostly under 15%; real-time RMSE for the best units was 5–6 Bq/m³ with Spearman $\rho=0.70$ –0.77 on two-day averages. Different analyte, and the QA design is directly transferable: this is the between-model spread estimate that consumer PM networks routinely lack. One home, one season, one concentration regime.

doi: 10.1088/1361-6498/aea16a

Guth et al. 2026 Atmosphere*Time-resolved emission factors for 19 prescribed burns, measured with low-cost sensors*

Emission factors for CO, PM_{2.5}, EC, OC and speciated organics across **19 prescribed fires** in Colorado and southeastern Georgia, using multiple low-cost monitors per burn to get temporal resolution that filter-based EF campaigns cannot reach. CO EFs are driven by combustion phase — **+3260%** at low modified combustion efficiency, i.e. smouldering. Broadcast and pile burns are compared directly, which is rare. The methodological point for a distributed network is that the LCS bias that disqualifies a sensor for regulatory-equivalent concentration is tolerable here because the measurand is a ratio within a plume; the risk is that any RH- or size-dependent bias in the PM channel propagates straight into the PM_{2.5}/CO EF ratio, and the abstract does not report the sensor correction applied.

doi: 10.3390/atmos17090867

Hall et al. 2026 Indoor Air*Indoor Air Quality Forecasting With Machine Learning: A Systematic Review*

Systematic review of 99 studies (2010–2026) on machine-learning forecasting of indoor air quality. Output grew sharply after 2020; LSTM-type time-series models dominate and transformers are emerging for longer horizons. Only 13% of studies use occupancy or behavioural inputs, although occupant activity drives indoor peaks. Evaluation is almost entirely RMSE and R2, with little assessment against health-relevant thresholds or by forecast horizon, and Africa and South America are barely represented. No pooled skill metric is given, so the review maps practice rather than grading it. Relevance: high for indoor sensor networks - it is the same validation gap Sahin et al. show today: headline R2 without horizon-specific or peak-event skill says little about operational usefulness.

doi: [10.1155/ina/8524332](https://doi.org/10.1155/ina/8524332)**He J et al. 2026** ACS Sens*Room-temperature ppb NO₂ chemiresistor by molecular packing control*

Alkyl chains of varying length on cyano-substituted distyrylbenzene tune molecular conformation and film microcrystallinity, exposing more surface-accessible functional groups and raising NO₂ sensitivity; the optimised device reaches a **30 ppb** detection limit at room temperature, and response is *enhanced* by humidity through NO₂–H₂O synergy. Demonstrated in a flexible wireless vehicle-exhaust monitor. Carried here on the sensing axis with the analyte stated plainly: this measures a gas, not a particle, and contributes nothing to PM mass or size. It is included because room-temperature operation and positive rather than negative humidity dependence are both unusual for the co-located gas channels that sit in multi-pollutant low-cost nodes. No long-term drift or cross-sensitivity data.

doi: [10.1021/acssensors.6c00977](https://doi.org/10.1021/acssensors.6c00977)**Herman et al. 2026** Atmos Environ*An Advanced, Multifunctional, and Extreme-Condition Test Chamber for Performance Evaluation of Air Quality Sensors*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes a chamber for evaluating air-quality sensors under extreme conditions. Directly relevant to LCS characterisation: humidity, temperature and high-load excursions are where field calibrations fail. First on the retry list. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: [10.1016/j.atmosenv.2026.122401](https://doi.org/10.1016/j.atmosenv.2026.122401)**Hong et al. 2026** Indoor Air*Predicting fungal and bacterial bioaerosol concentrations across building types*

9,377 indoor air samples across 27 multiuse facility types in South Korea, modelled with a generalised additive mixed model to capture non-linear temperature and humidity effects, then a random forest on the resulting sensitivity classes. Fungi respond mainly to relative humidity (16 facility types), bacteria mainly to temperature (7); peak microbial levels occur at **55–60% RH**, supporting 60% as the practical dampness threshold. Stratifying by meteorological sensitivity before prediction raised F1 from 0.53 to **0.65 (bacteria) and 0.75 (mould)**. Culture-based counts, so the non-viable fraction is invisible, and the facility-type classes are administrative rather than physical. Directly relevant to anyone adding T/RH channels to an indoor node and expecting them to carry bioaerosol information.

doi: [10.1155/ina/6724151](https://doi.org/10.1155/ina/6724151)**Hu et al. 2026** Atmos Chem Phys*Biomass burning aerosols during the 2023 Canadian wildfires from dual-site Raman and fluorescence lidar*

Dual-site observation of transported smoke, May–September 2023, at ATOLL (France, multi-wavelength Raman with a 466 nm fluorescence channel) and GPI (Russia, five-channel broadband fluorescence at 355 nm excitation). Upper-troposphere/lower-stratosphere layers showed higher depolarisation, slightly lower lidar ratios, lower Ångström exponents and a redshifted fluorescence peak against free-troposphere layers; fluorescence magnitude and spectral shape agreed across two sites and two instrument designs, the strongest evidence yet that these signatures are transferable rather than instrument-specific. The consequential finding is negative: **near-absence of hygroscopic growth at 90–100% RH**, contradicting the standard assumption that aged biomass-burning aerosol is hygroscopic — with direct consequences for every κ -Köhler humidity correction applied to optical PM sensors during smoke episodes. One fire season, two sites; altitude correlations at r^2 0.61–0.68 that the authors concede proxy for injection height and plume origin.

doi: [10.5194/acp-26-13027-2026](https://doi.org/10.5194/acp-26-13027-2026)**Huang et al. 2026** Environ Sci Technol*Interpretable spatiotemporal coupling network for emission prediction in sludge gasification*

PISTC-Net couples a directed process topology with per-feature optimal time lags, reaching R^2 of **91.18% for SO₂, 80.37% for NO_x and 78.09% for PM** on industrial-scale data. Under simulated abrupt sensor failure it continued tracking emission trajectories via learned process inertia, and hierarchical attention shifted from upstream thermochemical zones for SO₂ to whole-process attention for PM, flagging disturbance propagation **5–8 minutes** before emission peaks. That PM is the worst-predicted species is the informative result: particle formation runs through nucleation, condensation and agglomeration paths a topology-plus-lag representation captures least well, whereas gas-phase sulfur is nearly a mass balance. Single plant, no external validation site, and interpretability is inferred from attention maps, which are a saliency artefact rather than a causal readout.

doi: [10.1021/acs.est.6c09135](https://doi.org/10.1021/acs.est.6c09135)**Ichikawa et al. 2026** Atmos Chem Phys*Drone sampling agrees with a flux tower to within 30%, and that is the useful number*

Multi-year warm-season BVOC measurements at multiple heights on a 30 m flux tower in a *Quercus serrata*-dominated suburban Tokyo forest, with supplemental drone sampling and comparison against MEGAN. Isoprene is **over 90% of measured BVOC composition** at peak summer, monoterpenes stay low with weak vertical gradients, and daily average isoprene emission fluxes span **–0.05 to 15.30 mg m^{–2} h^{–1}**. The methodological result is the one worth carrying: **drone-measured horizontal variability is 10–30% within 30 m of the tower, and tower- and drone-derived flux estimates differ by about 30%**. That is a direct, published estimate of the representativeness error a mobile aerial platform introduces relative to a fixed mast — the same quantity a mobile PM platform needs and rarely reports. Limits: BVOCs are SOA precursors, not particles; intermittent campaigns over multiple years confound season with instrument configuration.

doi: [10.5194/acp-26-12521-2026](https://doi.org/10.5194/acp-26-12521-2026)

India winter PM_{2.5} seasonal forecast 2026 Sci Adv*Winter PM_{2.5} severity over northern India is predictable a season ahead*

Over a decade of quality-controlled Indian government reference PM_{2.5} monitoring, used to establish that **roughly 70% of interannual variance** in northern Indian winter PM_{2.5} is explained by western disturbances — Mediterranean-origin extratropical storms — acting through ventilation and wet removal. The storms in turn track to the position and strength of the subtropical jet, which is traceable to North Atlantic and tropical Indian Ocean sea-surface-temperature anomalies in the preceding autumn. Using only pre-season SST, the authors predict winter WD precipitation and PM_{2.5} severity one season ahead at r^2 **0.52–0.69**. The honest caveat is that this is meteorological predictability, not emissions predictability: it forecasts how well a given emission load will be ventilated, and a decade of records is a short sample for an interannual teleconnection. For network planning it is nonetheless actionable — it says which winters to instrument heavily.

doi: 10.1126/sciadv.aee4549

J Aerosol Sci 2026 J Aerosol Sci*High-efficiency plasma-near-passage unipolar aerosol charger for sub-10 nm particles.*

METADATA ONLY. The only instrumentation record in the window and the one whose missing abstract costs most. Crossref deposit of 19 September with no abstract; not indexed by Semantic Scholar, and the OpenAlex per-work fallback is dead for same-day deposits. No charging efficiency, particle loss fraction or flow rate can be quoted, so none is. Context, not content: sub-10 nm extrinsic charging efficiency is the binding constraint on any mobility-sizing instrument at the ultrafine end, and the published state of the art for this class sits well below unity at 5 nm, so a genuine 'high-efficiency' claim here would matter to anyone trying to put size distribution rather than mass on a distributed node. Priority pickup on the next run.

doi: 10.1016/j.jaerosci.2026.106898

Ji et al. 2026 Atmos Res*Dust aerosol aging tracked by combined Mie, polarization and fluorescence lidar*

Metadata only. Crossref carries the title, author list (8) and 5 September registration date but no abstract, and there is no Europe PMC record; the publisher's full text was not retrievable on this run. From the title the study links optical evolution during dust aging to chemical transformation using a three-channel lidar — a combination relevant to anyone attempting composition inference from optical instruments rather than mass. **No quantities, site, campaign length or retrieval performance are asserted here, because none could be verified.** Added to the abstract retry list.

doi: 10.1016/j.atmosres.2026.109312

Jonsson et al. 2026 Atmos Chem Phys*Tracing biological, anthropogenic, and inorganic sources of coarse aerosols via single-particle fluorescence and optical morphology*

Reference library of fluorescence spectra and morphology for coarse-mode particles (pollen, dust, bacteria, microplastics) generated in controlled experiments on a Multiparameter Bioaerosol Spectrometer, then used to train a supervised classifier tested against chemical tracers at Zeppelin Observatory, Svalbard. Biological and non-biological particles overlapped enough to bias fluorescence-only classes; the classifier confused biomass burning with biological particles and dust with sea spray. Domain adaptation with co-located observations reproduced the published annual bioaerosol cycle and raised summer concentrations toward offline values. No per-class accuracy is given in the abstract, and lab aerosols may not represent ambient ones. Relevance: moderate - an open classifier for real-time coarse-particle typing, the fraction optical LCS mass-convert worst.

doi: 10.5194/acp-26-13617-2026

Kalluri et al. 2026 — metadata only Atmos Environ*Long-term (2008–2024) black carbon over a semi-arid rain-shadow region in southern peninsular India*

The expected same-day Elsevier deposit gap: no abstract in Crossref, Europe PMC or PubMed on per-DOI retry. Carried on title because a 17-year single-site black-carbon record in a semi-arid rain-shadow setting is the kind of long baseline trend detection needs and South Asian networks mostly lack, and because absorption-coefficient records are what constrain the optical-to-mass conversion every filter- and scattering-based instrument depends on. No trend, magnitude or source attribution is asserted — the title names methods, not results. On the retry list.

doi: 10.1016/j.atmosenv.2026.122366

Kaur et al. 2026 Aerosol Sci Technol*Plantower PMSX003N against FEM through a Utah winter*

Three PMSX003N, three PMS5003T and three OPC-N3 co-located with FEM monitors at a Utah DEQ station for five months, plus laboratory size-binning against monodisperse DOS aerosol. The laboratory result drives everything: the PMSX003N **preferentially assigns particles >1 μm to the 2.5–5 μm bin**. In dust events this recovers coarse mass its predecessor loses — **PM_{2.5} slope 1.85–1.91, $R^2 > 0.7$ vs 0.15, $R^2 < 0.37$; PM₁₀ slope 0.91–1.07 vs ~0.02** — and under freezing fog, mist, snow and rain it substantially overestimates both fractions. Three units per model over one site-season is a small basis for a between-unit claim, and a slope near 1.9 is a large residual bias even where the correlation is good. Humidity-dependent correction is not optional for this part.

doi: 10.1080/02786826.2026.2716649

Khalili Hollo et al. 2026 ACS Environ Au*Validation and Correction of PM_{2.5} Mass Concentrations from Consumer-Grade Personal Exposure and Outdoor Fixed Monitors: A Two-Year Colocation Study.*

Fifty Atmotube Pro personal monitors and five QuantAQ Modulair-PM outdoor units were co-located four times over two years with a GRIMM EDM180 before deployment in the SJEQ-Denver community study. Raw 1-h agreement was modest (Atmotube rho 0.770, RMSE 5.12 μg m⁻³; QuantAQ rho 0.720, RMSE 5.51), both under-read above 30 μg m⁻³, and unit-level response drifted between colocations; 5–11% of Atmotubes were not working at any comparison. An MLR correction on RH and dew point raised rho by 6.5–8.3% and cut RMSE by 42–50%, and reduced drift. The GRIMM is itself an optical FEM, so 'reference' carries its own humidity and composition bias, and a correction fitted to Denver's high-altitude semi-arid aerosol will not transfer. Relevance: direct evidence that periodic re-colocation, not one pre-deployment fit, is what holds a wearable fleet in calibration.

doi: 10.1021/acsenvironau.6c00020

Kim C-h et al. 2026 Atmosphere*Block-missing monitoring data, five imputation models, and an honest sensitivity test*

Eleven months of 20-minute measurements of 22 designated odour compounds at a livestock farm, a wastewater plant and an organic-waste plant in Eumseong, South Korea, with **multi-week to multi-month block gaps** — the failure mode that actually afflicts long-running low-cost networks, as opposed to the scattered single-point dropout most imputation papers assume. Compound-specific performance was assessed by **block holdout** against BiLSTM, BiGRU, TCN, Transformer and HistGradientBoosting. Odour activity values were computed *primarily from observed concentrations*, with the fully imputed series used only as a sensitivity comparison to quantify interpolation-induced bias, which is the correct discipline and is rarely followed. Ammonia was excluded outright for a low valid-observation rate. Odour contribution was dominated by volatile fatty acids and trimethylamine rather than hydrogen sulfide. Transferable to PM networks in method, not in analyte.

doi: 10.3390/atmos17090868

Krajnc Pogačnik et al. 2026 — metadata only Build Environ*Selective real-time monitoring of respiratory droplets for indoor air safety in educational environments*

Elsevier same-day deposit, abstract absent from Crossref, Europe PMC and PubMed on per-DOI retry. Retained because selective real-time discrimination of respiratory droplets against a background of ambient particles is the hard problem for any optical counter deployed indoors: a scattering sensor reports total number in a size bin and cannot by itself separate exhaled aerosol from resuspended dust, so any claimed selectivity mechanism is worth reading. Whether that selectivity is optical, chemical or inferential is not recoverable from the title, and no performance claim is asserted. On the retry list.

doi: [10.1016/j.buildenv.2026.115281](https://doi.org/10.1016/j.buildenv.2026.115281)**LCS field calibration correction 2026** J Environ Health Sci Eng*Correction: Long-term performance evaluation and seasonal field calibration of a low-cost particulate matter sensor*

METADATA ONLY. Correction notice to 10.1007/s40201-026-00986-0. The notice carries no content of its own; logged here only so the archive records that the underlying long-term low-cost-sensor calibration paper has been amended, and so a later reader of that record checks which figures or coefficients moved before reusing them. No finding is asserted and the record is tier C on that basis

doi: [10.1007/s40201-026-01003-0](https://doi.org/10.1007/s40201-026-01003-0)**LLM-compiled adaptive sampling policies 2026** Sensors*Agentic LLM Compilation of Adaptive Sampling Policies for Low-Power IoT Sensing*

Directly relevant to battery-limited PM nodes. A generator LLM proposes adaptive-sampling policies as a handful of thresholds and integer periods under 100 bytes, a deterministic evaluator scores each exactly on the recorded trace, and an optimiser LLM refines over eight rounds; no weights are fitted and the model sees only eight summary statistics, never the raw series. Benchmarked on the Beijing PM_{2.5} series against classical baselines, budget-matched random search, NSGA-II and ParEGO on the same grammar, the loop beat zero-shot generation and random search (pooled Wilcoxon $p < 1e-5$): at a 0.20 duty cycle mean RMSE 21.79 vs 25.09, at 0.30 15.98 vs 20.57. Gains narrow at the tightest budgets, bounded by signal curvature. Single dataset, offline compilation only.

doi: [10.3390/s26175519](https://doi.org/10.3390/s26175519)**Lawson et al. 2026** Aerosol Sci Technol*Humidity-dependent photoacoustic absorption biases and effective water mass accommodation coefficients for multicomponent aerosol particles*

METADATA ONLY. Metadata-only record: Taylor & Francis deposited title, authors and DOI on 24 Sep with no abstract; Crossref, Europe PMC, PubMed efetch and Semantic Scholar all returned nothing. The title states a laboratory characterisation of humidity-dependent bias in photoacoustic absorption measurement and effective water mass accommodation coefficients for multicomponent particles (Lawson, Langridge, Allan, Cotterell). Tiered B despite the missing abstract: photoacoustic absorption is a BC reference method, and an RH-dependent bias in the reference propagates into every sensor calibrated against it. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: [10.1080/02786826.2026.2733732](https://doi.org/10.1080/02786826.2026.2733732)**Le et al. 2026** Meas Sci Technol*Real-time monitoring of ultrafine aerosol size distributions using parallel non-bouncing impactors coupled with corona charger.*

A Parallel Electro-inertial Particle Sizer: unipolar corona charger feeding six parallel non-bouncing impactor/Faraday-cup electrometer channels with kernel inversion, 10-560 nm. Impactor cut-offs landed within 5% of design; against an SMPS, unimodal and bimodal distributions over $1e4-1e7$ /cm³ were reproduced with under 10% deviation in total PNC, NMAD and GSD. In a 16-day rooftop co-location, total PNC and UFP gave slope 1.023 and r 0.995, and nucleation, growth and plume events were resolved. Weak points are stated: sub-20 nm and complex multimodal structure deviate, and one site over 16 days says nothing about drift, humidity or long-term electrometer zero stability. Cost is claimed lower but not quantified. Directly relevant: a cheaper, maintenance-light sizer is what UFP network monitoring currently lacks.

doi: [10.1088/1361-6501/aeaafe](https://doi.org/10.1088/1361-6501/aeaafe)**Li et al. 2026** Atmos Meas Tech*A multi-angle and polarization-based retrieval algorithm for aerosol layer height of smoke and dust*

Optimal-estimation retrieval adding multi-angle polarimetry to conventional spectral aerosol-layer-height methods, exploiting polarisation sensitivity to the Rayleigh-versus-aerosol scattering contrast. Information-content analysis quantifies the gain as increased degrees of freedom for signal over radiance-only observation — measured, not asserted. Applied to HARP2 on PACE and validated against ATLID on EarthCARE, RMSE 1.03 km across all collocated samples. That is the number to hold: about a kilometre of vertical uncertainty against a 1–2 km boundary layer constrains plume altitude but does not resolve surface coupling. Validated against a single spaceborne lidar with limited collocation statistics and no ground profile network. **Why it matters here:** vertical distribution is the dominant error term in AOD-to-surface-PM_{2.5} conversion, and smoke and dust are exactly where it bites.

doi: [10.5194/amt-19-5843-2026](https://doi.org/10.5194/amt-19-5843-2026)**Liu et al. 2026b** Atmos Environ*Morphology Change of Diesel Exhaust Black Carbon Particles during Humidification and Evaporation.*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors (Liu, Du, Li, Zhang) and DOI on 23 Sep without an abstract. Morphology change of diesel black carbon under humidification and evaporation is the physical mechanism behind humidity-dependent bias in every optical instrument that infers mass from scattering, so the topic is directly load-bearing for low-cost sensor correction - and the deposit gives no technique, no humidity range, no restructuring metric and no numbers. Recorded on metadata with its subtopic taken from the title, flagged for retry, and explicitly not summarised: a morphology claim inferred from a title would be a fabricated result.

doi: [10.1016/j.atmosenv.2026.122360](https://doi.org/10.1016/j.atmosenv.2026.122360)**Low-cutoff cyclone method 2026** Aerosol Sci Technol*An improved method for measuring the penetration efficiency of low cutoff cyclones.*

METADATA ONLY. Metadata-only record: Taylor & Francis deposited no abstract, the landing page is bot-blocked, and neither Crossref nor a Consensus sweep of 2026 cyclone-characterisation work surfaced it. This is the most directly relevant instrumentation title of the window and the least legible. Low-cutoff cyclone penetration efficiency is the calibration that defines what a size-selective inlet actually admits, and an error there propagates into every downstream gravimetric or optical measurement behind that inlet - including the reference samplers that low-cost sensor networks are co-located against, where an inlet cut error becomes a systematic multiplicative bias in the correction factor rather than added noise. Top of the retry list; worth a manual full-text pull rather than waiting on the deposit.

doi: [10.1080/02786826.2026.2731025](https://doi.org/10.1080/02786826.2026.2731025)

Mathissen et al. 2026 Atmosphere*Development of the Worldwide Harmonized Heavy Duty Brake Cycle (WHBC) for Brake Wear Particle Measurement*

Builds a regulatory brake-wear test cycle for heavy-duty vehicles (Euro 7 covers HDV non-exhaust PM but GTR 24 has only a light-duty cycle). Brake events were extracted from a global activity dataset and matched on empirical distributions of key brake metrics: 246 events over 271 km and 5 h 39 min. CO₂-monitoring data imply ~73% of kinetic energy is dissipated by friction brakes rather than retarders and drag. Dynamometer runs show the cycle is feasible and repeatable; particle results are described only as first insights, so no emission factor is reported. Limitation: retarder and regenerative braking share will shift with fleet electrification. Relevance: moderate - harmonised non-exhaust emission factors are what source-apportionment kerbside sensor signals.

doi: 10.3390/atmos17100964

Medellin MABI eBC 2026 Atmos Pollut Res*Multi-wavelength spectral attribution of equivalent black carbon in Medellin, Colombia (2010-2023): Disentangling fossil fuel and biomass burning using the MABI instrument.*

METADATA ONLY. Metadata-only record; no abstract deposited or recoverable. A 14-year multi-wavelength absorption record with fossil-versus-biomass attribution would be an unusually long instrument series for a Latin American city, and the same absorption-Angstrom-exponent dependence the Sonipat paper in this issue flags as its principal uncertainty applies here, so the two would be worth reading against each other once the text is available. Nothing further can be said without it; retry list.

doi: 10.1016/j.j.apr.2026.103203

Mohd Nadzir et al. 2026 Atmosphere*Minute-scale trace gas and PM variability in the Antarctic Peninsula from a low-cost node*

A single low-cost multi-sensor package, May–June 2023, **58,652 valid observations** of NO₂, O₃, CO, SO₂, PM_{2.5} and PM₁₀ (TVOC in May only). Medians: PM_{2.5} **6.29** and PM₁₀ **7.29** $\mu\text{g m}^{-3}$, O₃ 38.17 ppb, NO₂ 11.05 ppb, SO₂ 0.01 ppb. The **PM_{2.5}/PM₁₀ ratio of 0.96** indicates an almost entirely fine-mode aerosol, consistent with combustion rather than sea salt or crustal dust. Percentile-based event detection found **142 single- and 30 multi-pollutant enhancements**. The TVOC channel was 98% zeros and was rightly demoted. What is missing is any co-location or drift assessment — at these concentrations a few $\mu\text{g m}^{-3}$ of baseline offset would change every conclusion, and an uncalibrated nephelometric PM channel in high humidity is exactly where that offset lives.

doi: 10.3390/atmos17090855

Morel et al. 2026 Indoor Air*Evaluation of Two Cyclonic Bioaerosol Samplers for Hospital Air Monitoring in a Calibrated Test Chamber Integrating Aerodynamics and Biological Endpoints*

Coriolis+ and Coriolis Compact cyclones were challenged in a calibrated 20 m³ chamber with infectious SARS-CoV-2, *S. aureus* and *A. fumigatus* at several concentrations, scored on genomic and viability endpoints, and paired with CFD and airflow measurements. Neither detected infectious virus at the lowest level; Coriolis+ recovered nearly all viral genomes and more viable bacteria and fungi, while the Compact recovered more genomes of the two non-viral organisms at moderate-high loads. Ranking flips with endpoint, so 'collection efficiency' is not one number. No quantitative recoveries in the abstract. Relevance: low-moderate - bioaerosol, not PM mass, but a reference protocol for any sampler co-location.

doi: 10.1155/ina/2773156

NaCl aerosol turbulent size redistribution 2026 J Aerosol Sci*Turbulence-Induced Size Redistribution of Submicron NaCl Aerosols in Ribbed Confined Flows Prior to Electrostatic Collection*

METADATA ONLY. Metadata only. Crossref deposited this DOI on 12 September with no abstract, and per-DOI retries against Crossref, Europe PMC and PubMed all returned nothing - the expected same-day Elsevier deposit gap. Carried on title alone because submicron size redistribution in confined turbulent flow directly governs what reaches an inlet, and no finding is asserted here. On the retry list for a later issue.

doi: 10.1016/j.jaerosci.2026.106899

Ni et al. 2026 Atmos Meas Tech*An AI-based algorithm for retrieving aerosol optical depth and single scattering albedo from All-Sky Imager observations*

Machine-learning retrieval trained on co-located sun-photometer AOD and SSA at two sites, Beijing_PKU in East China and SGP in the central US. Independent validation gave $r > 0.86$ for AOD at all wavelengths with RMSE below 0.25, and for SSA $r = 0.74$ (Beijing_PKU) and 0.84 (SGP) with RMSE below 0.09. Retrieving SSA from a consumer-grade sky camera is the striking part — absorption is normally the hardest aerosol property to retrieve and demands a calibrated multi-channel photometer. Two sites means two aerosol regimes and two instrument configurations, and an AOD RMSE of 0.25 is large wherever ambient AOD is low. As a densification strategy this is the same economics that produced PurpleAir — cheap hardware plus a trained model substituting for a reference instrument — with the same unanswered transferability question.

doi: 10.5194/amt-19-5871-2026

Open-pit mine dust stacking model 2026 Atmos Pollut Res*METADATA ONLY — multi-point cold-season dust prediction at an open-pit coal mine*

No abstract existed in Crossref queried per-DOI, in Europe PMC or in PubMed on the entry date. What the record establishes: a multi-point dust-concentration prediction task at an open-pit coal mine during the cold season has been addressed with a stacking ensemble and an interpretability analysis. What it does not establish: the number and siting of monitoring points, the prediction horizon, the base learners in the stack, the interpretability method, any error figure, or the mine's location — none of which is inferable from the title and none of which is asserted here. Flagged for retrieval; open-pit mining dust is one of the few settings where a dense low-cost network has an unambiguous operational customer.

doi: 10.1016/j.j.apr.2026.103196

Organic aerosol self-diffusion MD 2026 ACS Omega*Predicting the Self-Diffusion Coefficient of Atmospheric Organic Aerosol Components via Molecular Dynamics Simulations.*

Molecular dynamics estimates of self-diffusion coefficients for atmospherically relevant organics in the particle phase, with three tiers of method: direct simulation at 298 K for mobile species, high-temperature simulation with WLF or VFT extrapolation for less mobile ones (oxomalonic, tartronic and cis-pinonic acids, dihydroxyacetone), and 420 K only for species too immobile to converge. Mobility falls in the order -COOH > -OH > C=O for monofunctional compounds, and multiple or adjacent polar groups suppress it further through dense hydrogen-bond networks. Molecular weight and elemental ratios correlate with D only within chemical families. Relevant because particle viscosity governs gas-particle partitioning and water uptake, and so the hygroscopic growth that biases optical sensors. Pure simulation, no measured diffusivities for validation.

doi: 10.1021/acsomega.6c04326

Komiljon et al. 2026 Atmosphere*Non-cumulative, size-specific calibration of low-cost PM sensors under simulated drilling*

Controlled laboratory evaluation of PMS5003 and Sniffer4D under simulated concrete-slab drilling, with six correction models — linear, polynomial, Random Forest, XGBoost, ANN, Kalman — fitted independently per size fraction. Random Forest and ANN perform best, $R^2 > 0.89$ for PM_1 and > 0.87 for $PM_{2.5}$ at 50 s averaging. The premise is the useful part: reporting non-cumulative rather than cumulative bins, since cumulative output conflates fractions with different deposition behaviour and different regulatory meaning. Against that, a single drilling source in a chamber produces a mineral-dust size distribution that will not transfer to combustion aerosol, and PM_{10} performance is not reported alongside the finer fractions. *Connector-sourced; Crossref resolves only the preprint of record.*

doi: 10.20944/preprints202604.1260.v1

PPDRTL leaf-image AQI 2026 PLoS One*Predicting roadside air quality from image texture of pollutant deposits on tree leaves*

High-resolution images of roadside leaves at high, medium and low traffic-density sites in Haryana, segmented with PSO, DPSO and FODPSO, reduced to contrast, entropy and standard deviation features and regressed with ordinary linear regression onto Haryana State Pollution Control Board reference AQI. The FODPSO variant reaches R^2 up to **0.894** with **RMSE 8.76–12.34 $\mu g m^{-3}$** and reported site accuracies of 88.9–91.4%. The inference is weak for a reason worth naming precisely: the **pilot dataset is ten days**, no held-out test set is described, and leaf deposition is a cumulative integrator being regressed onto an *instantaneous* daily index, so the model is fitting a mismatched target. Mixing an R^2 against a unitless AQI with an RMSE in $\mu g m^{-3}$ compounds the confusion. Filed as a proxy-sensing curiosity, not a method.

doi: 10.1371/journal.pone.0326588

Papathanasiou et al. 2026 Atmos Pollut Res*Evaluating Personal Air Quality Monitors for $PM_{2.5}$ and CO_2 : Insights from a Year-Long Pilot Study.*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 25 Sep with no abstract; Crossref, PubMed efetech and the publisher landing page returned nothing, and Europe PMC was down (HTTP 503) for the whole run. The title indicates a year-long evaluation of personal $PM_{2.5}$ and CO_2 monitors. High priority for this watch: it pairs with the two-year Denver colocation in this issue and the retry should target instrument model, reference and drift. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. Nothing on the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.j.apr.2026.103210

Parks et al. 2026 Aerosol Sci Technol*Evaluating performance of real-time monitors for quantifying elemental carbon in diesel particulate matter.*

METADATA ONLY. Metadata-only record: Taylor and Francis deposited title, authors (Parks, Finley, Raj, Wilson) and DOI on 23 Sep with no abstract, and doi.org returned 403 to an automated fetch, so no landing-page recovery either. This is the single most relevant title of the day for the watch - a head-to-head evaluation of real-time elemental-carbon monitors against diesel PM is exactly the reference-grade comparison that underpins occupational black-carbon sensing - and none of it can be reported: no instruments named, no reference method, no bias or precision figures, no exposure range. It is listed, not summarised, and sits at the top of the retry list.

doi: 10.1080/02786826.2026.2733733

Vance et al. 2026 J Occup Environ Hyg*Evaluation of the Partector 2 Pro for ultrafine particles from 3D printing*

Wind-tunnel comparison of a portable diffusion-charging monitor against SMPS and CPC during controlled printing of PLA, PETG, TPU and nylon. Number-concentration agreement is strong across all four filaments — regression slopes **0.88–1.05**, Pearson $r > 0.96$, mean deviations **–9.6% to +6.2%**, inside the $\pm 30\%$ manufacturer specification. Sizing is systematically biased low (slopes 0.82–0.90) and worst at the fine end: **–27.0% to +7.5% at D10** and **–28.3% to –7.3% across D25–D75**. That pattern is the operational point for any distributed ultrafine network — the instrument is fit for concentration surveillance but its size distribution must not be used to infer deposition fraction. Wind tunnel only, with no humidity or co-pollutant interference testing.

doi: 10.1080/15459624.2026.2724020

Peng et al. 2026 Environ Pollut*Inhibition of condensable particulate matter by co-firing biomass and high-sulfur coal*

50 kW one-dimensional self-sustaining furnace, high-sulfur coal co-fired with corn stover or pine wood at 10% and 20% blend. Baseline condensable PM was **6.72 mg/MJ**, **83.3% inorganic**, falling to 5.09 and 3.85 mg/MJ for corn stover at 10% and 20% and to 5.36 and 4.55 for pine wood — **a 43% reduction at 20% corn stover**, traced to suppression of sulfate-associated inorganics and organic precursors. PAHs and sulfur-containing organics fell below detection at 20% blending. **Why it matters here:** condensable PM is invisible to hot-filter methods and to every optical low-cost sensor, yet can rival filterable PM in mass, so inventories built on filterable-only sampling are biased low. One bench furnace, one coal, two feedstocks, no atmospheric ageing — and condensable PM measurement is method-dependent enough that absolute values should not travel across sampling protocols.

doi: 10.1016/j.envpol.2026.129163

Gupta et al. 2026 Aerosol Air Qual Res*Evaluating and calibrating low-cost air quality sensors in contrasting aerosol regimes*

PurpleAir PA-II co-located with FEM monitors for 2–4 weeks in Raleigh, North Carolina and Delhi, India, deliberately spanning a clean and a heavily loaded regime. A model on PA $PM_{2.5}$, temperature and RH holds mean absolute bias within **12% hourly and 6% daily**. The headline is negative: pre-deployment calibration specific to location or region is a prerequisite for scientific use, i.e. no single transferable correction exists. Two to four weeks is short against the seasonal cycle in either city, so drift and seasonal composition change lie outside what this design can see. *Connector-sourced; outside the 14 September entry window.*

doi: 10.1007/s44408-026-00146-x

RI-URBANS urban AQ mapping 2026 Atmos Meas Tech*What a continent of low-cost-sensor and citizen-science air-quality pilots actually taught*

Synthesis of the RI-URBANS mapping and hotspot-identification pilots across European cities, covering stationary low-cost sensor networks, mobile monitoring platforms and citizen-collected data alongside reference networks and modelling. The conclusion the programme reaches is organisational rather than technical: the limiting factor on producing comparable urban maps was **harmonisation between methods**, not the accuracy of any one method, and harmonisation was one of the project's stated primary aims for that reason. Two operational findings are worth carrying into any deployment — campaign design and sustained, direct communication with citizen participants determined data quality as much as hardware did. As a programme-level synthesis it reports no single effect estimate and cannot be appraised like a primary study; its value is the negative result about where the effort goes.

doi: 10.5194/amt-19-5729-2026

Raju et al. 2026 J Environ Sci Health A

Investigations of black carbon aerosol sources and strengths over a rural-valley location in Haryana, India

Three years of seven-wavelength **AE33 aethalometry** (January 2020 – December 2022) at Panchgaon, a rural Aravali valley site at 285 m AMSL in northwest India. Annual means: eBC 4.06 ± 2.15 , fossil-fuel eBCff 3.10 ± 2.40 , biomass-burning eBCbb $1.42 \pm 1.19 \mu\text{g}/\text{m}^3$, with eBCff carrying 68% of absorption at 880 nm. Winter maxima (eBCff 4.19, eBCbb $2.51 \mu\text{g}/\text{m}^3$) and monsoon minima (1.90, 0.52) track boundary-layer height and wet scavenging. The aethalometer-model split rests on **assumed absorption Ångström exponents that are never reported**, and the eBCff/eBCbb ratio of 3.02 ± 3.80 has a standard deviation larger than its mean — the apportionment is unstable even where the record is not. A three-year reference-grade absorption series at a rural South Asian site is exactly the co-location anchor low-cost networks there lack.

doi: 10.1080/10934529.2026.2734478

Raso et al. 2026 Atmosphere

Temporal resolution and the characterisation of episodic pollution events

Two consecutive New Year's Eve firework campaigns with certified reference instrumentation co-deployed with calibrated low-cost sensors, comparing daily, hourly and minute aggregation of PM₁₀, NO₂, CO and O₃. Regulatory averaging detects the episode and misses its structure: minute PM₁₀ reached $215 \mu\text{g m}^{-3}$, a **peak amplification factor of 2.90–2.93** over the hourly value. The direct implication for distributed sensing is that the case for a dense network is not redundancy against reference-grade accuracy but access to a time scale the reference framework discards — and that acute-exposure metrics built on hourly data are systematically low for episodic sources. Two events at one site; the PAF is a description, not a transferable constant.

doi: 10.3390/atmos17090860

Rosas et al. 2026 Environ Sci Pollut Res Int

The PM_{2.5}/PM₁₀ ratio separates two aerosol regimes that mass alone cannot

Continuous PM_{2.5}, PM₁₀ and particle-bound PAH at a near-coastal urban site in Merida, Mexico, joined to FIRMS fire detections and HYSPLIT back-trajectories, contrasting the March–May regional biomass-burning season against the June–August African dust season. Concentrations are the wrong discriminator: **PM_{2.5} and PM₁₀ rise only modestly during dust**, but the **PM_{2.5}/PM₁₀ ratio falls**, marking the coarse mineral contribution, while biomass burning loads the fine mode and the combustion tracers. A network reporting PM_{2.5} mass alone would score the dust season as the milder of the two and miss the toxicologically distinct composition entirely. Part I of a two-part study, so the health framing in the title is anticipatory rather than delivered here. Single site, single urban domain.

doi: 10.1007/s11356-026-38193-x

Rosca & Stancu 2026 Indoor Air

Indirect Machine Learning Prediction of a Composite Indoor Air Quality Index in an Airport Passenger Transit Area

A composite index (PIAQI) built from indoor CO₂, PM_{2.5} and PM₁₀ at Brindisi Airport is predicted without those three inputs, from other indoor aerosol channels, weather and public outdoor pollution data. Seven regressors were trained on 1 million high-frequency observations; XGBoost was best. Validation is a random 70/30 split of an autocorrelated series - the design Sahin et al. showed in the 29 Sep issue inflates R² - and the 'alternative aerosol measurements' retained are themselves close proxies for the excluded PM channels, so the task is closer to interpolation than to indirect sensing. Relevance: low - an example of the leakage-prone validation practice the field needs to retire.

doi: 10.1155/ina/3119205

Rural wintertime SIA metal catalysis 2026 Atmos Res

Important role of acidity-regulated metal catalysis in rural wintertime secondary inorganic aerosol formation in the North China Plain

METADATA ONLY. Crossref deposit of 13 September with no abstract released, so only the claim in the title is verifiable here: that transition-metal-catalysed oxidation, modulated by aerosol acidity, is a significant route to secondary inorganic aerosol in the rural North China Plain in winter. If sustained, this bears on low-cost optical sensing, because the sulfate and nitrate produced by this route are strongly hygroscopic and drive the humidity-dependent bias that every nephelometric PM sensor inherits, and because rural residential coal burning supplies both the metals and the acidity. Placed on the abstract retry list; no design, sample or effect size can be assessed until the record deposits, and the record should be revisited rather than cited.

doi: 10.1016/j.atmosres.2026.109322

SAQIN 2026 Neural Netw

Long-short-term decoupled residual learning for high-resolution spatio-temporal air quality inference

Inference — not forecasting — of high-resolution concentration fields from sparse reference stations. The model splits a static branch (semantic segmentation, remote-sensing context, population density, elevation) from a dynamic branch (multi-head self-attention over geographic relationships and inter-pollutant coupling), then treats inference as residual correction anchored to multiple stations, fused by distance-weighted averaging across stations. That last step is the substantive contribution: it removes the Voronoi-partition discontinuities that single-anchor schemes produce, yielding a continuous field. Reported as superior on large-scale real-world data with zero-shot cross-domain generalisation, but the abstract names no dataset, no station count, no RMSE and no baseline, so none of it is checkable here. The architecture is directly applicable to dense low-cost networks, where the anchor stations are the few reference-grade nodes.

doi: 10.1016/j.neunet.2026.109575

SE India monsoon aerosol 2026 Atmosphere

Seasonal aerosol optical properties and radiative effect at a rural Indian receptor site

MICROTOPS-II sunphotometry from April 2021 to December 2023 at a rural site in Southeast India, with trajectory-based source attribution (PSCF, CWT) and OPAC–SBDART radiative transfer. Annual mean AOD₅₀₀ 0.56 ± 0.22 , peaking pre-monsoon (0.66 ± 0.19) and in winter (0.64 ± 0.23) and falling to 0.49 ± 0.21 in the monsoon. **Ångström exponent 1.30 ± 0.24 in the dry seasons against 0.83 ± 0.37 in the monsoon** marks the fine-to-coarse shift, corroborated by negative spectral curvature. Precipitable water rose 2.00 to 4.35 cm winter to monsoon, which confounds the optical size retrieval through hygroscopic growth — the paper says so. Column optical depth is not surface mass, and a single rural receptor cannot be generalised; its value is as a South Asian background record against which urban surface networks can be referenced.

doi: 10.3390/atmos17090876

SIBS inexpensive single-particle 2026 Aerosol Sci Technol

METADATA ONLY — feasibility of a low-cost single-particle SIBS instrument

No abstract existed in Crossref, Europe PMC, Semantic Scholar or on the publisher page on the entry date, and Taylor & Francis refused the fetch. What the record establishes: a feasibility study for an *inexpensive* single-particle spark-induced-breakdown-spectroscopy instrument has been published in the field's instrumentation journal of record. What it does not establish: any detection limit, element set, particle-size range, spark energy, hit rate or cost figure — none of which can be inferred from the title, and none of which is asserted here. This is the most directly relevant record of the day to low-cost elemental speciation and is flagged for retrieval once the abstract deposits.

doi: 10.1080/02786826.2026.2721653

Sahin et al. 2026 Atmosphere*Validation Design Governs the Apparent Performance of Machine Learning Calibration for Low-Cost Optical Particle Counters: A PM₁ Study*

Three low-cost OPCs were co-located with reference nephelometers over seven indoor runs of 20–33 h (Jun–Sep 2022; reference PM₁ 8.4–9.2 $\mu\text{g m}^{-3}$). Factory output under-read by 5.3–6.1 $\mu\text{g m}^{-3}$ (RMSE 6.5–7.6, 74–82% of the mean). Six calibrators (MLR, SVR, RF, GB, XGBoost, LightGBM) reached R² 0.86–0.89 under a random 80/20 split, but R² collapsed under day-blocked and run-blocked splits and sat near or below zero under forward chaining and chronological hold-out. Calibration still cut prospective RMSE 22–48%, yet the calibrated slope was 0.18–0.44, compressing real variation. Validation design explained 63–91% of the R² spread, algorithm choice 3.6–10.6%. Limits: three sensors, ~200 h, narrow indoor range. Relevance: direct - random-split R² in LCS calibration papers is leakage from autocorrelation; require temporally blocked validation.

doi: 10.3390/atmos17100947

Seong & Hoque 2026 Front Pediatr*Attachment fraction ranks metal > wood > glass, and carpet is two microenvironments*

Batch attachment and centrifugal detachment of *Corynebacterium* sp. on carpet, wood, metal and glass, with near-surface velocity fields resolved by particle image velocimetry and surfaces characterised for roughness, contact angle and porosity. Attachment fraction rose with exposure time on all materials, reaching ~0.59 on metal, ~0.48 on wood, ~0.36 on glass. The useful observation is morphological: **carpet generates two distinct microenvironments** — a higher-velocity outer-fibre region and a sheltered low-velocity inter-fibre region — so a single resuspension coefficient for “carpet” is a category error. Wood produced the most spatially heterogeneous velocity field, glass the most uniform. Relevance to PM is indirect but real: resuspension source terms in indoor exposure models are usually fitted per material with no flow-field justification. One organism, no PM measurement, replicate count not stated.

doi: 10.3389/fped.2026.1895305

Shafi & Scafetta 2026 Environ Sci Pollut Res Int*Reconstructing a decade of missing Delhi monitoring records*

Delhi went from **4 stations in 2014 to 45 in 2024**, leaving a fragmented record that defeats trend analysis. For each missing daily value across six pollutants, the workflow selects the four most-correlated stations as predictors, evaluates **35 regression models** in MATLAB's Regression Learner, takes the best, and iterates until the matrix is full. Tree ensembles and Gaussian/rational-quadratic kernels consistently beat multilinear regression; artificial-gap experiments validate across pollutants, stations and years, best for PM_{2.5} and other regionally-driven species. The structural problem is unaddressed and important: stations are missing non-randomly, and imputing from correlated neighbours smooths exactly the local excursions that a network is deployed to detect. Reconstructed series should not be used for exceedance statistics.

doi: 10.1007/s11356-026-38181-1

Shen et al. 2026 Atmos Chem Phys*Apportioning light absorption of ambient aerosols to black carbon, brown carbon, and lensing effect using a PAX-ISS hybrid method: insights into absorption enhancement.*

Couples photoacoustic spectroscopy (total, coated absorption) with an integrating sphere after solvent mediation (lensing-free absorption); the difference is the lensing term, and a dual-wavelength iteration splits the remainder into BC and BrC. Across seasonal Beijing samples in 2023, BC dominated absorption and BrC contributed about 10% annually. Lensing averaged 40% of total absorption but ranged 4.6–52.0% by season and contracted at short wavelengths, consistent with a BrC ‘blocking’ effect previously only simulated; annual absorption enhancement was 1.69 +/- 0.10. The assumption to test is that solvent mediation removes coatings without altering BC morphology or extracting chromophores unevenly - the method's validity rests on it. One city, one year. For low-cost absorption sensing, a seasonally varying 40% lensing term is the correction a filter photometer's mass-absorption cross-section silently absorbs.

doi: 10.5194/acp-26-13303-2026

Shenzhen tower vertical PM_{2.5} DL 2026 J Hazard Mater*Leveraging multi-source observations for vertical PM_{2.5} estimation: A deep learning fusion model*

An AM-CNN-BiLSTM fusion model estimating PM_{2.5} at 200 and 300 m on the Shenzhen Meteorological Gradient Tower, July 2022–June 2023, combining lidar aerosol extinction coefficient, microwave-radiometer thermodynamic profiles, meteorology and tower PM_{2.5}. Chronological test split with an expanding window gave R² 0.87 at 200 m and 0.83 at 300 m, beating baselines; SHAP, ablation and sensitivity analysis all identify extinction coefficient as the dominant constraint, with RH, temperature and wind complementary. The authors state plainly that the learned extinction-PM_{2.5} and humidity relations are site-specific and need multi-year cross-site validation - with one tower and one year that caveat is binding. Still the cleanest demonstration that lidar plus profiler can substitute for vertical in-situ nodes.

doi: 10.1016/j.jhazmat.2026.143445

Shi S et al. 2026 npj Clim Atmos Sci*Vessel speed reduction trades CO₂ against plume composition — and against the tracer that identifies the plume*

UAV interception of 14 fresh ship plumes with single-particle mass spectrometry and population-balance inversion to reconstruct ultrafine and coarse modes. Two marker gaps emerge. **Only 35.8% of the reconstructed particle-number population carried detectable vanadium** (29.5% at 4.9 knots vs. 39.7% at 13.4 knots in a matched-vessel pair; 0–70.8% across a speed-blind manifold), so the canonical ship tracer is operation-dependent rather than conserved. Separately, **~20% of analysed particles met biomass-burning criteria**, a false-positive channel in ship-influenced samples. Sulfate-related ion signals ran 2.3–10.7-fold higher in V/VO-bearing particles. The authors are careful that laser desorption/ionization matrix effects may contribute to that contrast, which is the right caveat and limits how far the V–S co-location can be pushed mechanistically. Single campaign, 14 plumes, one matched pair — the speed comparison rests on very few vessels.

doi: 10.1038/s41612-026-01532-3

Shin et al. 2026 Environ Sci Technol*A Continuous Virus Aerosol-into-Liquid Sampler (VAiLS) Integrating Twin-Vortex Condensation Growth and Wet-Cyclone Enrichment.*

A twin-vortex condensation growth unit (5 C preconditioner, wick-wetted chamber at 20–60 C) feeding a wet cyclone grows virus-sized aerosol into micrometre droplets and concentrates them continuously into liquid at 16 L/min. On KCl nanoparticles, number-weighted collection rose from 7.3% for the cyclone alone to 31.0, 64.4 and 70.4% at growth temperatures of 20, 40 and 60 C, independent of inlet humidity across 16.8–71.8%. For viruses the optimum was 40 C (MS2 93 +/- 0.6%, phi6 90 +/- 2.1%), giving infectious concentrations 3.9e3 and 1.2e3 times a BioSampler, with viability above 92% except phi6 at 60 C. Two surrogate viruses on a bench rig say nothing about field backgrounds or enveloped respiratory pathogens. The humidity independence is the result worth carrying: it is the failure mode of every optical bioaerosol channel.

doi: 10.1021/acs.est.6c03875

Song 2026 Atmos Environ

Relocation-associated discontinuities in Korean air-quality monitoring: a Bayesian structural time-series analysis of PM₁₀ and PM_{2.5}.

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 25 Sep with no abstract; Crossref, PubMed efetch and the publisher landing page returned nothing, and Europe PMC was down (HTTP 503) for the whole run. The title indicates Bayesian structural time-series detection of step changes in PM₁₀ and PM_{2.5} series when regulatory stations are relocated. Directly relevant to network metadata and to every long-term trend or epidemiology study that treats a station ID as a fixed exposure site. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.atmosenv.2026.122369

TROPOMI SO₂ EAKF assimilation 2026 Atmosphere

Assimilating TROPOMI SO₂ columns into a regional chemical transport model

An Ensemble Adjustment Kalman Filter implemented in the MINNI regional CTM through DART, assimilating Sentinel-5P TROPOMI SO₂-COBRA total columns over continental Europe for August 2023, with a 20-member ensemble perturbing emissions and boundary conditions. The filter constrained ensemble variance around power-plant and volcanic plumes; the monthly domain-mean column correction was $2 \times 10^{-5} \text{ mol m}^{-2}$ with localised maxima of 3.3×10^{-4} , and near-surface concentration corrections reached $2.6 \mu\text{g m}^{-3}$, validated against in-situ data. The stated limitations are real ones: small ensemble, static vertical localisation, and the usual temporal fading of initial-condition corrections. Relevance here is indirect but concrete — SO₂ is a sulfate precursor, and column-to-surface transfer skill is the same problem satellite-based PM_{2.5} products have to solve.

doi: 10.3390/atmos17090875

Thomas et al. 2026 J Test Eval

Procedural Challenges in the Analysis of Toxic Metals in Electronic Cannabinoid Delivery System Aerosols Produced for Vaping Cannabinoids.

No subjects; a white paper reviewing ten years of trapping approaches. The central problem is that metal constituents sit in the $\mu\text{g/g}$ range, so the trap itself and the sample-preparation consumables contribute more metal than the aerosol does unless materials are chosen against a metals blank - media validated for organic constituents are simply wrong here. The second constraint is mechanical: several otherwise-clean media impose enough airflow resistance that the device will not puff-activate, so the aerosol never forms. No quantitative inter-method comparison is offered, which is the paper's limitation - it maps the failure modes without ranking the survivors. The transferable point for distributed sensing is that sampling-substrate blank, not detector sensitivity, sets the floor for any trace-metal PM measurement, which is the same constraint that governs filter-based reference sampling behind a low-cost network. Note: this record was rejected from the 19 September issue for carrying no DOI; the DOI has since been registered.

doi: 10.1520/jte20250320

Tu et al. 2026 Indoor Environ

Assessing the Need for Site-Specific Calibration of ATP Bioluminescence Sensors for Airborne Bacteria Monitoring in Indoor Environments

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title asks whether ATP-based airborne-bacteria sensors need site-specific calibration. The same transferability question PurpleAir regional-calibration work asks for optical PM sensors. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.indenv.2026.100197

UAV single-particle CCSEM 2026 Environ Sci Technol

Size-resolved aerosol composition and mixing state through the boundary layer from a UAV

A microscopy-based particle sampler flown on a UAV at up to 30 m vertical and 1 min temporal resolution in a coastal boundary layer; over 100,000 particles classified by computer-controlled SEM with EDX. The observed transition runs from sea-breeze aerosol dominated by aged sea salt and dust (46–83%) to a continentally influenced population dominated by organic-sulfur particles (57–89%), and the mixing state index falls from 53% to 27% as those particles form — i.e. toward external mixing. Mixing state showed strong diurnal and size dependence but weak vertical structure. Offline microscopy is not a real-time measurement and the sampling is necessarily intermittent, so the “temporal evolution” is a sequence of snapshots; one coastal site, and no companion mass measurement is described. The transferable point for network design is that composition and mixing state vary on scales a fixed filter sampler averages away entirely.

doi: 10.1021/acs.est.6c03627

US precipitation washout DoubleML 2026 Research Square

Observation-derived washout coefficients, and how wildfire smoke breaks them

Preprint. Ten years of hourly EPA observations — 760 PM_{2.5} and 451 PM₁₀ monitors across the contiguous US — merged with NLDAS-2 meteorology and NOAA HMS smoke classifications, with a double machine-learning causal framework separating precipitation from confounding meteorology. Removal rises with rainfall intensity, particle size and initial concentration, and **falls as wildfire smoke loading rises** — smoke aerosol is less efficiently scavenged, which matters more each year. The authors fit power-law washout coefficients ($a = 0.0404$, $b = 0.5951$ for PM_{2.5}; $a = 0.0812$, $b = 0.4508$ for PM₁₀) intended to drop straight into chemical transport models. The inference threat is the usual one for DoubleML on observational data: unmeasured confounding is assumed away rather than eliminated, and hourly monitor-scale precipitation from a reanalysis is a coarse proxy for what fell on the sampler.

doi: 10.21203/rs.3.rs-10855101/v1

Urban OOM flux 2026 Atmos Chem Phys

Urban surface-atmosphere fluxes of selected pptv-level oxygenated organic molecules (OOMs) from eddy covariance observations.

First urban eddy covariance measurement of OOM surface-atmosphere exchange at pptv level, and the contribution is as much methodological as observational - the paper works through how mass spectral integration time and water vapour variation corrupt flux retrieval at low signal-to-noise, which is the reason these fluxes had not been measured. Sixteen OOMs resolved into deposition-dominated, emission-dominated and bidirectional classes. Campaign-mean deposition was $4.6 \mu\text{mol/m}^2/\text{d}$, 16.3% of HNO₃ deposition and 2.2x the OOM emission flux, so the urban surface is a net sink. Formic acid dominated both directions (66.2% of deposition, 84.9% of emission), which means the 16-species budget is really a formic acid budget plus minor terms. One site, one month, hottest season - the seasonal extreme most favourable to the photochemistry being measured, so these are upper bounds. Bears on sensing indirectly but concretely: SOA formed from these precursors is secondary mass a low-cost optical sensor sees only after it has condensed, with a lag and a hygroscopicity that no dry-mass calibration captures.

doi: 10.5194/acp-26-13213-2026

Veselovskii et al. 2026 Atmos Meas Tech*Five-channel fluorescence lidar retrieves smoke mass inside the boundary layer*

Smoke and urban aerosol have distinct fluorescence spectra, and a multi-channel fluorescence lidar exploits that to separate them and retrieve smoke mass concentration with an estimated detection threshold of $\sim 0.1 \mu\text{g}/\text{m}^3$. Over Moscow, smoke was detected in **59 of 67 sessions between April and October 2024**, mostly from long-range Atlantic transport; only 12 episodes traced to southern Russian fires. Within the PBL, transported North American wildfire smoke contributed $\sim 1 \mu\text{g}/\text{m}^3$ against **up to $50 \mu\text{g}/\text{m}^3$** for regional fires. This is the source-resolved column-to-surface capability that mass-only networks lack, and $0.1 \mu\text{g}/\text{m}^3$ is a genuinely low threshold. Single site, single instrument, and mass retrieval requires an assumed fluorescence-to-mass conversion whose uncertainty is the main open term.

doi: 10.5194/amt-19-5587-2026

Wathore et al. 2026 Aerosol Air Qual Res*Environmental and temporal performance of ML calibration for a low-cost $\text{PM}_{2.5}$ sensor*

Long-term co-location in Vishakhapatnam comparing Random Forest and XGBoost against linear regression, adding temperature, relative humidity, hour-of-day and month-of-year stepwise, plus a hybrid ensemble. The best single model cuts RMSE **58%** against the linear baseline and the hybrid reaches **63%**, meeting US EPA low-cost $\text{PM}_{2.5}$ performance criteria. The contribution is the evaluation design rather than the models: performance is reported binned across environmental and temporal strata and across AQI categories, which exposes where a calibration degrades instead of burying it in a pooled R^2 . No absolute RMSE in $\mu\text{g m}^{-3}$ is given, and single-site training leaves transferability untested — the same limitation already flagged for the Sofia calibration-transfer work in this corpus. *Connector-sourced; outside the 15 September entry window.*

doi: 10.1007/s44408-026-00094-6

W Sub-Saharan Africa PM monitoring 2026 Atmos Pollut Res*METADATA ONLY — $\text{PM}_{2.5}$ and PM_{10} monitoring capacity in Western Sub-Saharan Africa*

No abstract retrievable on the entry date. The record establishes that a challenges-and-perspectives treatment of $\text{PM}_{2.5}$ and PM_{10} monitoring and estimation across Western Sub-Saharan Africa has been published; whether it is a systematic review, a narrative synthesis or a data-driven assessment cannot be determined from the metadata, and no finding is asserted here. Western Sub-Saharan Africa has the sparsest reference network of any populated region and is the canonical justification for low-cost sensor deployment, so a current capacity assessment is worth retrieving — particularly alongside the Accra hotspot record in this same issue.

doi: 10.1016/j.apr.2026.103195

Wang S et al. 2026 Fire Saf J*Smoldering chaparral emits 10–70× the particulate of flaming, and fuel moisture sets the multiplier*

FTIR quantification of gas and particulate emissions from controlled flaming and smoldering combustion of four dominant California chaparral species at varying fuel moisture. **Smoldering produced 10–70-fold higher PM emission factors than flaming across all species**, with the amplification ratio itself varying by fuel type and moisture; higher moisture enhanced incomplete-combustion products, principally CO and hydrocarbons, with live eucalyptus highest at $37.18 \pm 0.65 \text{ g/kg}$ hydrocarbons under smoldering. Cross-ecosystem comparison found chaparral broadly similar to western shrubland and grassland fuels except in nitrogen-containing species. The limitation that matters is scale: these are bench-scale burns, and the authors note combustion scale and mixing conditions shift emission profiles, so transfer to landscape fire inventories is an assumption rather than a result. Directly useful for anyone weighting wildfire smoke exposure by combustion phase rather than by total burned area.

doi: 10.1016/j.firesaf.2026.104687

Wang et al. 2026 npj Clim Atmos Sci*Artificial intelligence for atmospheric chemistry: from observation to process understanding*

METADATA ONLY. Metadata-only record: Springer Nature deposited title, authors and DOI on 28 Sep with no abstract; PubMed efetch, Crossref, Europe PMC and the publisher landing page returned nothing. The title indicates a review or perspective on AI from atmospheric observation to process understanding. Of interest only if it covers sensor-network calibration or PM retrieval. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1038/s41612-026-01559-6

Wildfire CHON $\text{PM}_{2.5}$ 2026 Environ Res*FT-ICR MS with rule-based prediction and QSAR for nitrogen-containing organics in wildfire $\text{PM}_{2.5}$*

Molecular-level characterisation of $\text{PM}_{2.5}$ from a temperate coniferous forest wildfire in South Korea, wildfire versus post-wildfire periods. GC-MS gave nitrated polycyclic aromatic compounds **$3.4\times$ higher during the fire (0.122 vs $0.036 \text{ ng}/\text{m}^3$)**, with 2-nitrofluorene then 3-nitrofluoranthene dominant; FT-ICR MS showed a relative shift toward CHON formulae, and rule-based SOA prediction from 12 VOC precursors matched limonene-derived products most often. QSAR LC_{50} values of 1.0, 1.6 and 8.3 mg/L for β -pinene-, α -pinene- and isoprene-derived compounds are ranked hazard, not measured toxicity, and the authors say so — the assignments are formula-level, so isomers with very different toxicity are pooled. One fire, one forest type; the compositional shift is the result, the hazard ranking is a hypothesis.

doi: 10.1016/j.envres.2026.125576

Wu et al. 2026 Atmosphere*Spatiotemporal Prediction of Sandstorm Utilizing Multi-Source Data and ConvLSTM Model: A Case Study in Northwest China*

Dust forecasting for photovoltaic operations in Gansu: surface PM_{10} and Himawari-8/9 AOD are assimilated with an ensemble Kalman filter and fed to a multi-scale ConvLSTM with a high-value-focused loss. At 4 h lead the model reached RMSE $30.1 \mu\text{g m}^{-3}$ and MAE $16.1 \mu\text{g m}^{-3}$, captured the evolution of an April 2024 regional event, and ran in under 30 s per simulation. Validation rests on one event, errors are not stratified by concentration (where dust peaks run to thousands of $\mu\text{g m}^{-3}$), and no persistence baseline is reported, so skill over naive forecasting is unknown. Relevance: EnKF assimilation of surface PM is the natural entry point for low-cost sensor data into nowcasts - provided the sensors' known coarse-mode undercount under dust is corrected first.

doi: 10.3390/atmos17100925

Xing et al. 2026 Environ Sci Technol*DeepAQM — observation-constrained ML surrogate for the NOAA forecast system*

A ConvLSTM-ResNet fusion reconstructs spatially continuous surface $\text{PM}_{2.5}$ and O_3 from VIIRS AOD, TEMPO NO_2 and EPA AirNow, and supplies them as initial conditions for 72 h UFS-AQM forecasts over the continental US. It reproduces UFS-AQM spatial patterns ($R^2 > 0.9$ for O_3 , to 0.80 for $\text{PM}_{2.5}$, normalised mean bias within $\pm 10\%$). The result that matters: observation-updated initial conditions lift site-level R^2 in the first 10 forecast hours from **~ 0.10 – 0.15 to 0.40 – 0.50 for $\text{PM}_{2.5}$** , and observation-based fine-tuning carries overall R^2 to **0.41 for $\text{PM}_{2.5}$** across the full 72 h. Note what that number is: a fourfold gain at short lead, decaying to a forecast that still explains under half the variance. TEMPO makes this a current-generation result rather than a reanalysis exercise.

doi: 10.1021/acs.est.6c04600

Xu et al. 2026 Environ Sci Technol

Integrating Fragmented Mobile Monitoring Data into a High-Resolution Spatiotemporal Exposure Model for Black Carbon

Vehicle-borne eBC measured in three seasonal campaigns (2023–2024) across residential Beijing, aggregated hourly, then gap-filled in time with XGBoost on external temporal predictors (regulatory pollutants, meteorology) and interpreted with SHAP; site-day estimates feed a hierarchical geostatistical model applied to pregnancy-cohort addresses. XGBoost reached 10-fold CV R^2 0.69 against site-hour eBC; the spatial stage is reported only as internal interpolation at monitored points, so there is no external validation at unmonitored cohort homes - the step the cohort actually uses. Two stacked models also propagate error the paper does not quantify. Relevance: high - the same temporal-gap problem affects any campaign- or sparse-node design, and the two-stage template transfers directly to low-cost BC/PM networks.

doi: [10.1021/acs.est.6c08120](https://doi.org/10.1021/acs.est.6c08120)

Zeng et al. 2026d J Hazard Mater

Oxidative potential and light extinction decouple, and aerosol water is the switch

Paired oxidative-potential and optical measurements at two urban sites in southern Taiwan, arranged into a quadrant framework on *volume-normalised* OP against light extinction. Under coupling, mass-normalised OP and mass extinction efficiency share their drivers, principally **water-soluble organic carbon**. Under decoupling, **high aerosol liquid water content** pushes the aerosol into an optics-dominant state through hygroscopic growth, while **low ALWC** yields the highest OP-to-extinction ratios — the “invisible killer” quadrant. Secondary nitrate loads extinction specifically; aerosol pH is a weaker secondary term. The direct consequence for distributed sensing: every nephelometric low-cost sensor estimates mass from scattering, so its error against the redox-active fraction is not random but conditioned on humidity, in the same direction as its already-known RH bias. Weakness: OP is one assay on one aerosol population at two sites, sampling duration is not stated in the abstract, and no causal chain from OP to a clinical endpoint is tested here.

doi: [10.1016/j.jhazmat.2026.143459](https://doi.org/10.1016/j.jhazmat.2026.143459)

Zhang Z et al. 2026 Anal Chem

Microfluidic multispectral diffraction for black carbon quantification

A microfluidic chip does inertial separation, low-velocity sheath-flow regulation and high-voltage electrostatic capture to enrich submicron BC in the imaging region (**92% enrichment efficiency**); a lensless diffraction system then acquires hyperspectral fingerprints over 400–800 nm, which a graph model maps to concentration. Over **0.01–0.20 mg/m³**, $R^2=0.98$. The discriminating claim is more interesting than the calibration: the spectral fingerprint separates BC from similarly-sized microplastics and pathogenic spores, which absorption-based instruments cannot do. Bench work only — no ambient co-location, no interference testing against real coated or aged BC, and 0.01 mg/m³ (10 µg/m³) is a high floor for ambient BC, which typically runs 0.5–5 µg/m³.

doi: [10.1021/acs.analchem.6c03051](https://doi.org/10.1021/acs.analchem.6c03051)

Zhang et al. 2026 J Hazard Mater

Adaptive robotic sensing for exposure-oriented mapping of hazardous airborne particle dispersion in indoor environments.

CFD virtual sampling plus a full-scale mechanically ventilated aerosol release. The claim worth taking is the region-of-interest one: adaptive sampling cut normalised reconstruction error in the high-concentration ROI to 9.5% at 65 points against 18.3% for uniform scanning, and cut mean error at four fixed reference locations from 59.7% to 34.3% in the physical experiment. Note the honest counter-result the authors report - optimised uniform scanning beat adaptive sampling on final global NRMSE (2.5% vs 3.0%), so the gain is purely in plume localisation, not in field-wide accuracy. Single ventilation condition, single room, no repeat releases, so the 34.3% figure has no dispersion around it. Directly relevant to mobile-node deployment: it quantifies what a moving sensor buys over a fixed sparse array for exactly the transient indoor plume case fixed networks miss.

doi: [10.1016/j.jhazmat.2026.143612](https://doi.org/10.1016/j.jhazmat.2026.143612)

Zhang et al. 2026 Build Environ

Are expensive sensors necessary for assessing indoor environmental quality? A data-driven approach using minimal sensing.

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 25 Sep with no abstract; Crossref, PubMed efetch and the publisher landing page returned nothing, and Europe PMC was down (HTTP 503) for the whole run. The title indicates a data-driven test of whether a reduced sensor set can stand in for a full indoor-environmental-quality instrument suite. Whether PM is among the measured parameters is not stated; that decides relevance. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: [10.1016/j.buildenv.2026.115306](https://doi.org/10.1016/j.buildenv.2026.115306)

Zhao et al. 2026 PLoS One

Deep learning PM_{2.5} prediction integrating improved MSTL decomposition and the FATA optimizer

FATA-GSPMSTL-CNN-LSTM: RFECV plus Pearson feature screening, a grid-searched power-spectral variant of Multiple Seasonal-Trend decomposition using Loess, the Fata Morgana metaheuristic for hyperparameter search, and a CNN-LSTM head. Evaluated separately for heating and non-heating seasons in Guangzhou and Xianyang, and the seasonal split is the sensible part given how differently those two cities are sourced. Beyond that it follows a familiar pattern: four stacked components, **no ablation isolating which one carries the gain**, and improvement measured only against author-selected baselines. No prediction intervals, so the output is a point forecast with no stated uncertainty and cannot drive threshold alerting. Two cities, no external-city test.

doi: [10.1371/journal.pone.0358271](https://doi.org/10.1371/journal.pone.0358271)

Zhou et al. 2026 Aerosol Sci Technol

Effects of geometrical parameters on eccentric cylindrical differential mobility analyzer

METADATA ONLY. Metadata-only record: Taylor & Francis deposited title, authors and DOI on 28 Sep with no abstract; PubMed efetch, Crossref, Europe PMC and the publisher landing page returned nothing. The title indicates a study of how geometry affects the performance of an eccentric cylindrical differential mobility analyser. Relevant to compact mobility sizers - the reference against which low-cost ultrafine and size-resolving sensors are characterised. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: [10.1080/02786826.2026.2734523](https://doi.org/10.1080/02786826.2026.2734523)

cCPC/OPC/SPS30 field evaluation 2026 Aerosol Sci Technol

Field Evaluation of Community Condensation Particle Counters, Optical Particle Counters, and a Low-Cost Particle Sensor

Three community CPCs, three OPCs and one TSI BlueSky (Sensirion SPS30) co-located at an urban regulatory station for about four months, data recovery above 98%. Against a research-grade reference CPC the cCPCs gave R^2 above 0.99 with concentration ratios near unity and a 3% coefficient of variation across the three units - genuinely reference-comparable ultrafine number at community cost. OPC fine and coarse volumes correlated with reference PM_{2.5} and PM_{10-2.5} but on different regression slopes, so a single mass factor cannot serve both fractions. The SPS30 tracked PM_{2.5} and failed on coarse mass, which is the known optical limit restated with numbers. One season at one site, so no drift or humidity-regime generalisation; this is the most directly usable record of the day for network design.

doi: [10.1080/02786826.2025.2590103](https://doi.org/10.1080/02786826.2025.2590103)

Zhou et al. 2026 Atmos Meas Tech*Performance modeling of a small mixing-type condensation particle counter*

Validated CFD model of the sMCPC particle channel resolving turbulent flow, temperature and species transport, with Kelvin, non-continuum and latent-heat terms in the growth calculation. Raising the saturator-cooler temperature difference ΔT raises supersaturation and lowers the critical activation diameter $D_{p,50}$; raising total flow Q_g suppresses supersaturation and shifts its peak downstream; both ΔT and vapour fraction steepen the $D_{p,50}$ slope. Everything is numerical — counting efficiency is asserted through one validation case, not measured across the swept space. This is design-space guidance of the kind a portable ultrafine network node needs before it goes to the field.

doi: 10.5194/amt-19-5815-2026

Exposure assessment & modelling**148 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (148 OF 148 RECORDS IN THIS SUBTOPIC)

Kim et al. 2026 PLoS One*LASSO-based reduced-form CMAQ for ozone and PM_{2.5} response to emission change*

Surrogate for CMAQ fitted by LASSO over **118 emission scenarios** spanning seven sectors and 17 provincial regions, with an adaptive logit transform to respect concentration bounds. Trained on 100 simulations and tested on 18: mean RMSE **0.1 ppb** for ozone and **0.1 $\mu\text{g m}^{-3}$** for PM_{2.5} against the parent model. The sparse set of retained sector–region terms is the practical output, making the control-design problem interpretable rather than opaque. Error is quoted against CMAQ and not against observations, so every CMAQ bias passes through undiagnosed; 118 scenarios is thin for a 119-dimensional predictor space and no out-of-envelope extrapolation test is reported.

doi: 10.1371/journal.pone.0347073

Aji et al. 2026 Environ Monit Assess*Fine-mode combustion aerosol events near Indonesian coal plants from AE-filtered Himawari-8*

41,119 daily Himawari-8 observations around 13 Java coal plants, 2015–2025, with events defined as Ångström exponent > 1.2 and AOD > 0.4 (8.52% of days). Suralaya has the highest mean AOD at 1.01 with 51.4% very-high-pollution days; Pacitan the lowest at 0.46. Random forest and XGBoost return accuracy, precision, recall, F1 and AUC-ROC all **exactly 1.000** — on a target that is a deterministic threshold function of the two inputs, so the models relearn the labelling rule rather than predict anything, and the 2026–2060 Monte Carlo projection built on them is uninterpretable. Column AOD without surface validation cannot separate plant plume from boundary-layer aerosol.

doi: 10.1007/s10661-026-15804-1

Zhang et al. 2026 Environ Pollut*Surface-based temperature inversion climatology across China and its relation to PM_{2.5}*

Hourly microwave-radiometer profiles from **33 stations** over 2024–2025, validated against radiosondes at 25 sites, used to build a surface-based inversion climatology and relate it to PM_{2.5} at hotspot stations. Winter SBI occurrence **39.2%** against **12.1%** in summer, with a northwest-to-southeast gradient peaking in the Tarim Basin and Hexi Corridor. SHAP and Spearman analysis place boundary-layer variables in the top predictor tier monthly and daily, with effective mixing-layer height third among all hourly predictors in every season. Two years is short for a climatology, the network is sparse relative to the terrain it spans, and SHAP importance over correlated meteorological predictors does not establish the causal pathway it is read as establishing.

doi: 10.1016/j.envpol.2026.129172

Kim & Park 2026 Atmosphere*Loading-rate-based analysis of PM_{2.5} and precursor gases in Korean agricultural areas*

547 days of hourly data from eight agricultural sites, comparing concentration-based with loading-rate-based interpretation, where loading rate is concentration scaled by a wind-speed-derived air exchange rate. Concentration correlations among PM_{2.5}, NH₃, NO₂ and SO₂ are weak and inconsistent; loading-rate correlations are strong and uniform at $r \geq 0.816$, and the diurnal shape flips from bimodal to unimodal. The authors concede the improvement may reflect the shared wind-speed term — it almost certainly does, since dividing every series by one common quantity induces correlation by construction. Recorded as a method to treat with suspicion: any dense open-field network normalising by a common meteorological variable manufactures the same result.

doi: 10.3390/atmos17090898

Accra traffic intersections 2026 Environ Monit Assess*Meteorological drivers of PM_{2.5} at nine Accra traffic hotspots*

An Aeroqual Ranger portable monitor at nine traffic intersections in Accra across the harmattan and non-harmattan seasons, against Ghana Meteorological Agency station data. Median PM_{2.5} was **54.90 $\mu\text{g m}^{-3}$** at high-density sites, **34.26** at commercial and **16.75** at low-density — the cleanest category still exceeds the WHO 24 h guideline of 15. Temperature correlated positively ($r = 0.353$, $p < 0.01$); rainfall, wind speed and humidity negatively outside the harmattan, while in the harmattan wind speed flipped weakly positive ($r = 0.079$, $p = 0.450$, i.e. null) consistent with dust resuspension. The analysis is Pearson correlation only, with no multivariable adjustment and no traffic counts, so “the dominant factor remains localised vehicular emissions” is a residual argument rather than a measured one. A single portable instrument across nine sites also gives no co-location error term.

doi: 10.1007/s10661-026-15889-8

Ahmad et al. 2026 Preprints.org (preprint)*Characterizing Ambient Fine Particulate Matter Pollution in Jordan: National Monitoring Evidence of Spatial Inequality, PM_{2.5} Hotspots, and Temporal Trends (2018-2024)*

Preprint analysing annual PM_{2.5} across Jordan’s national monitoring network, 2018-2024, by site class. Industrial-related sites averaged 26.51 $\mu\text{g m}^{-3}$ versus 29.24 $\mu\text{g m}^{-3}$ at urban-related sites (Mann-Whitney $U=297$, $p=0.604$), and the Sen’s slope was $-1.67 \mu\text{g m}^{-3}/\text{yr}$ with significant governorate-level differences. The authors themselves flag that the network expanded over the period, so the declining trend is confounded by a changing site mix - a fixed-site subset analysis is the obvious missing check. Annual means only; no instrument type or completeness criteria are stated. Not peer reviewed. Relevance: a network-expansion artefact masquerading as a trend is the same failure mode that growing low-cost networks face.

doi: 10.20944/preprints202609.1911.v1

Arpornwichanop & Chuersuwan 2026 Environ Geochem Health*Life-stage-resolved inhalation exposure to PM_{2.5}-bound elements, Greater Bangkok*

Secondary-data screening at one fixed site (selected 24 h filter days, Dec 2020–Sep 2021), combining seasonally weighted outdoor elemental concentrations with indoor penetration, time-activity, respiratory-deposition and simulated-lung-fluid solubility assumptions across pediatric life stages. Concentration and deposited-intake profiles rank **Al, Fe, B, Zn** highest; the solubility- and toxicity-adjusted comparative priority quotient ranks **As, Cd, Sb**. Modelled lifetime cancer risk spans 1.11×10^{-6} to 2.78×10^{-5} across 0–100% Cr(VI) assumptions — an order of magnitude driven entirely by a speciation parameter nobody measured. The authors state explicitly that this is within-model prioritisation, not measured child exposure. That honesty is unusual and worth noting.

doi: 10.1007/s10653-026-03455-7

Arunachalam et al. 2026 Research Square (preprint)*Building-sector combustion mortality is three times prior estimates, and it is mostly NO₂*

CMAQ Decoupled Direct Method sensitivities of NO₂, O₃ and PM_{2.5} to commercial and residential building-sector precursor emissions for the contiguous US (2021), propagated through BenMAPR. Net burden **≈14,000 premature deaths per year (95% CI 7,900–19,000): 12,000 from NO₂, 3,100 from PM_{2.5}, and 1,000 deaths avoided** through O₃ reductions. Residential gas alone accounts for ≈7,900. The headline “three times higher than previously reported” is carried almost entirely by the NO₂ term, so the result is a statement about which NO₂ concentration–response function you accept, not about building emissions being larger than thought. Preprint, not peer reviewed. The negative O₃ term is the sign of a titration effect and is correctly reported rather than suppressed, which is a point in the paper’s favour.

doi: 10.21203/rs.3.rs-10758198/v1

Athens NPF 2026 Atmos Res*The influence of air mass origin and ammonia emissions on new particle formation at a Mediterranean urban site (Athens, Greece).*

METADATA ONLY. Metadata-only record; Elsevier deposit carried no abstract and none was recoverable. The topic is squarely in scope - ammonia-mediated new particle formation is the ultrafine source that governs urban particle *number*, and number is precisely the metric that mass-calibrated low-cost sensors do not measure and that regulatory PM_{2.5} mass does not capture. Whether this paper quantifies nucleation rates, growth rates or only classifies event days cannot be determined from the deposit. Retry list.

doi: 10.1016/j.atmosres.2026.109334

Atik et al. 2026 Environ Sci Atmos*Inter-annual trends of PM_{2.5}-bound heavy metals in Dhaka, Bangladesh: a comparative study for source apportionment and health risk assessment*

Only the graphical-abstract summary was deposited: elevated-altitude PM_{2.5} sampling in Dhaka found PM_{2.5}-bound heavy metals far above global guidelines, positive matrix factorisation resolved six emission sources, and Cr, Pb, Cd and Mn drove both non-cancer and cancer risk across all age groups. No concentrations, years, sample counts or factor identities are available from the deposit, so the inter-annual trend in the title cannot be evaluated, and deterministic risk quotients from filter metals typically assume total-metal bioavailability. Elevated-altitude sampling also under-represents street-level exposure. Relevance: Dhaka is a high-priority low-cost network setting where mass sensing needs exactly this kind of periodic compositional anchoring. Admitted on the summary; re-check when the full abstract resolves.

doi: 10.1039/d6ea00084c

Atmos Pollut Res 2026 Atmos Pollut Res*Canyons and Canopies: Effects of building configuration and green infrastructure on urban air quality.*

METADATA ONLY. Crossref deposit of 19 September with no abstract; Semantic Scholar has not indexed the DOI and the OpenAlex per-work route returns no inverted index, the same dead end logged on 18 September. Carried on title and venue alone. Nothing is asserted about the method, the domain or the direction of the green-infrastructure effect, because the record does not support it. Flagged for pickup: street-canyon geometry crossed with canopy cover is directly a siting question for a distributed network - it determines whether two nodes a block apart are measuring the same air - so this is worth re-fetching once the abstract appears.

doi: 10.1016/j.apr.2026.103205

Benin PM_{2.5}-water grid screening 2026 Environ Monit Assess*Grid-based screening of fine particulate matter and water-optical conditions in Southern Benin: the Cotonou, Lake Nokoue, and Porto-Novo Corridor*

A 1 km screening framework over 2,589 grid cells in the Cotonou-Lake Nokoue-Porto-Novo corridor, 2001–2022. Mean PM_{2.5} was $32.17 \mu\text{g m}^{-3}$ for 2001–2022 and $32.70 \mu\text{g m}^{-3}$ for 2018–2022, roughly six times the WHO annual guideline, with no corridor-wide trend surviving false-discovery-rate correction. Sentinel-2 NDTI tracked field turbidity (Spearman rho 0.488, 95% CI 0.369–0.590) and a red-edge ratio tracked chlorophyll-a (0.432, 0.295–0.544) against 19 stations. Six cells co-flagged at the 80th percentile, but the count swings from 16 at the 75th to zero at the 90th - the threshold, not the data, sets the answer. No ground PM monitor exists in the corridor, so the PM_{2.5} layer is wholly unvalidated: a concrete argument for low-cost reference-grade co-location in West Africa.

doi: 10.1007/s10661-026-15908-8

Bhattarai et al. 2026 Research Square (preprint)*Spatio-temporal assessment of PM_{2.5} and PM₁₀ pollution in core and peripheral municipalities of the Kathmandu valley using regulatory and low-cost sensor data*

Preprint integrating regulatory monitors with a dense low-cost sensor network across core and peripheral municipalities of the Kathmandu Valley, 2023–2024. Annual mean PM_{2.5} was $40.71 \mu\text{g m}^{-3}$ in the urban core versus $27.22 \mu\text{g m}^{-3}$ in the periphery; the core exceeded WHO guidelines on nearly all winter and pre-monsoon days, with a core amplification factor up to 1.60 in the dry season, contrasting size dominance, and evidence of long-range transport during extreme events. The abstract gives no sensor model, no calibration or co-location statistics and no sensor count, so the core-periphery gradient cannot yet be separated from sensor bias - critical in a valley with strong humidity and dust seasonality. Not peer reviewed. Relevance: exactly the hybrid-network use case, in a data-poor basin; the calibration section decides whether it is evidence.

doi: 10.21203/rs.3.rs-10981275/v1

Bi et al. 2026 J Environ Manage*Associations of street morphology and local wind environment with annual cumulative ambient PM_{2.5} exceedance: Evidence from Wuhan*

6,785 Wuhan streets in five canyon typologies; the outcome is annual cumulative PM_{2.5} exceedance. CatBoost-SHAP and PLS path models explained >70% of variance on average; geometry plus wind contributed >50%. Enclosed canyons were governed by Building View Index and Sky View Factor; in open canyons street-wind angle and speed contributed 25% on average. Wind-speed breakpoints: 0.33 m/s (95% CI 0.31–0.44) enclosed, 1.19 m/s (0.78–1.31) open. Visible greenery was positively associated with exceedance in enclosed canyons, canopy ratio negatively. The street-level PM_{2.5} field is modelled, not measured, so the typology effects inherit the exposure model’s structure. Relevance: moderate-canyon type and wind thresholds are a siting stratification for street sensors.

doi: 10.1016/j.jenvman.2026.130987

Blond et al. 2026 Preprints.org (preprint)

Boundary-Layer Height Outweighs Wind Speed in the Meteorological Control of Urban Air Pollution: Unique-Variance Attribution and a City Typology for Ten French Cities

Semi-partial R² separates the unique variance of boundary-layer height and wind speed for PM_{2.5}, PM₁₀, NO₂ and O₃ in France's ten largest cities (2018–2025), across pooled, deseasonalised and detrended frames, checked with GAMs. BLH and wind correlate at $r \sim 0.70$ (0.81 on anomalies); on pooled levels BLH carries four (NO₂) to nine (PM) times the unique variance of wind, a lead robust for PM. A three-type city typology (ventilation-, transitional, BLH-controlled) predicts which covariate a city needs. BLH is reanalysis-derived, not measured, and the preprint is not peer reviewed. Relevance: high for sensor-network trend normalisation and short-term epidemiology that adjust for wind alone - BLH should be a first-tier covariate for PM.

doi: 10.20944/preprints202609.2417.v1

Bloomberg et al. 2026 medRxiv (preprint)

Associations of individual exposure to particulate matter with next-day physical activity and sedentary behaviour in older adults: a micro-longitudinal study

Micro-longitudinal preprint: 67 UK adults aged 50–85 wore personal air-quality monitors (waking hours) and 24-h accelerometers for five days (304 person-days). Within-person mixed models: a higher-than-usual day (+1 IQR, i.e. +2.6 $\mu\text{g m}^{-3}$ PM_{2.5} above the person's mean) was followed by 8 min less light activity (95% CI -14 to -2) and 483 fewer steps (-855 to -82); for PM₁₀ (+2.8 $\mu\text{g m}^{-3}$), 8 min less LPA (-13 to -3), 8 min more sedentary time (0 to 15) and 412 fewer steps. No MVPA association, and none with area-level PM. Not peer reviewed; the monitor model and any calibration are not stated; a 2.6 $\mu\text{g m}^{-3}$ contrast is within typical optical-sensor error, so residual weather or illness confounding of day-to-day behaviour is the main threat. Relevance: high - the association exists only in personal-sensor exposure, not area estimates.

doi: 10.64898/2026.09.23.26363783

Boyle et al. 2026 Cancer Epidemiol Biomarkers Prev

Cancer-specific air-pollution exposome metrics for the continental US

Combines CACES land-use-regression criteria-pollutant surfaces with the RSEI industrial-release model across US census tracts, 2005–2020, weighting each pollutant by meta-analytic breast- and lung-cancer risk estimates to produce two composite metrics (APBC, APLC). Both are strongly right-skewed — some tracts an order of magnitude above the mean — peak in large cities and along transport corridors, differ from RSEI notably in Los Angeles and New York, vary by race, ethnicity and housing, and cluster spatially. No cancer outcome is analysed here: this is metric construction, and the meta-analytic weights import the heterogeneity of their source studies into a tract-level surface that then reads as precise. Useful as a template for building a risk-weighted composite from a network, with that caveat attached.

doi: 10.1158/1055-9965.EPI-26-0493

Brake wear SO₂ uptake 2026 Nat Commun

Unexpectedly rapid SO₂ heterogeneous reaction on the surface of vehicular brake wear particles

Bench heterogeneous kinetics on collected brake wear particles. SO₂ uptake coefficient up to 2.87e-5 under dark ambient conditions, orders of magnitude above mineral dust or soot. The mechanism proposed is a self-sustained alpha-Fe₂O₃ / carbonaceous synergy: oxygen vacancies activate O₂ and H₂O to ROS and Fe-OH, while organics and elemental carbon promote H₂O dissociation and enhance SO₂ adsorption at carbon defects. Limitation: uptake coefficients are measured on collected, laboratory-handled particles under dry dark conditions, so the extrapolation to ambient RH and photochemistry is an assumption, not a measurement. Relevance: non-exhaust PM is now the dominant vehicular primary source in regulated fleets, and a reactive brake-wear pathway means roadside sulfate is not purely advected

doi: 10.1038/s41467-026-76820-w

Bravo et al. 2026 J Expo Sci Environ Epidemiol

Chicago's 2023 wildfire smoke, by tract

Hourly PM_{2.5} on a 500 m grid aggregated to daily 24-hour means for all 792 Chicago census tracts in nine city regions, 1 May–31 July 2023, examined against demographic, socio-economic and health profiles. The title carries the finding: during the long-range wildfire-smoke episodes, the usual within-city exposure gradient compressed — smoke reached every region — while the pre-existing disparity in baseline PM_{2.5} and in underlying health persisted underneath it. That combination is the argument for source-resolved rather than total-mass exposure metrics in environmental justice work, since a total-mass summer average would show convergence and hide the differential burden. Modelled rather than measured exposure at tract level, one summer, one city, and no health outcome is linked here.

doi: 10.1038/s41370-026-00956-6

Brisbane Aitken-mode PNC 2026 Environ Int

METADATA ONLY — long-term Aitken-mode particle number trends in Brisbane

No abstract retrievable on the entry date despite *Environ Int* being a PubMed-indexed journal — the DOI indexed in Crossref before either the abstract or the PubMed record deposited. The title commits to a long-term change in Aitken-mode particle number concentration attributed across meteorology, traffic and biomass burning. Neither the record length, the trend sign or magnitude, the instrument, nor the attribution method is recoverable, and none is asserted here. Long records of sub-100 nm number concentration are rare and this one would bear directly on whether mass-based networks are tracking the health-relevant fraction; flagged for retrieval.

doi: 10.1016/j.envint.2026.110514

Broda et al. 2026 Atmosphere

Vertical and Zonal Characteristics of Transatlantic Biomass Burning Aerosol Transport During the Extreme 2023 Canadian Wildfire Season.

May–September 2023 smoke transport from Canada to Europe against a 2003–2022 baseline. Source-region smoke AOD was the highest on record in every month except August. At source the 0–2 km layer held 78–93% of smoke over western Canada; during transport most smoke rose above 2 km, the 2–5 km layer dominated over the Atlantic and the extinction-weighted mean altitude climbed from 1–2 km to 2.5–4.3 km. Monthly transport efficiency to Europe was low, and satellite AOD did not reproduce the reanalysis zonal pattern. Monthly means mask episodes, and a reanalysis is not an observation. The surface-exposure implication is the useful part: when smoke is elevated, column AOD overstates surface PM_{2.5}, so AOD-to-surface products need vertical-profile correction, and surface sensors are the check.

doi: 10.3390/atmos17100913

Chen et al. 2026 Atmos Environ

Chemical Characteristics, Source Apportionment and Toxic Potential of PM_{2.5} in Southern Taiwan: Insights from the Kao-Ping Experiment (KPEX).

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors (Chen, Wu, Hsu, Hsu) and DOI on 23 Sep without an abstract. The Kao-Ping Experiment is a named campaign in southern Taiwan, and the title promises speciation, receptor-model source apportionment and a toxic-potential measure in one dataset - the combination that lets a source be ranked by harm rather than by mass. None of the instruments, campaign dates, factor solution or toxicity assay can be stated from the deposit. Carried because this is the same size-composition-source axis as the stroke record in this issue and would be the natural external check on it. Retry list.

doi: 10.1016/j.atmosenv.2026.122373

Chen et al. 2026 Atmos Res*Formation mechanism of rapid daytime nitrate production in a typical semi-arid city near dust source regions*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 24 Sep with no abstract; Crossref, Europe PMC, PubMed efetch and Semantic Scholar all returned nothing. The title states a formation-mechanism study of rapid daytime nitrate production in a semi-arid city near dust sources. Relevant in that nitrate is semi-volatile and dust-coated nitrate shifts particle size - both known optical-sensor bias terms. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.atmosres.2026.109361

Chen et al. 2026 Environ Sci Technol Lett*Solid Waste Burning: An Overlooked Atmospheric Source of p-Phenylenediamine Antioxidants and Their Quinones*

Field sampling of uncontrolled open burning and controlled incineration of solid waste with suspect screening annotated 47 PPD-related features in PM_{2.5} (43 in open-burning, 27 in incineration; 20 confirmed). Open burning emitted parent PPDs (median 11.54 ug/g PM_{2.5}); incineration had relatively more quinones (median 2.50 ug/g). A screening-level national estimate attributes >99.7% of the 10 targeted PPD + PPD-Q emissions from waste burning in China to open burning. Emission factors rest on few events and the national scaling on activity data of uncertain quality. Relevance: moderate - 6PPD-quinone toxicity makes this a composition-specific hazard PM_{2.5} mass sensors cannot register; open burning is detectable as short plumes in dense networks.

doi: 10.1021/acs.estlett.6c00796

China NOx isotopes 2026 Environ Pollut*Driving factors of atmospheric nitrogen oxides emissions and pollution in China.*

Twenty-two years of nitrate isotope data put fossil-fuel combustion at ~53% of national NOx and non-fossil sources (biomass burning, microbial N cycling) at 47%. The paper is unusually candid about what it cannot resolve: regional source-fraction differences 'lack statistical significance', and the isotope-derived NOx trend is a non-significant 7% decline (p=0.19), which the authors say precludes robust inference of emission-deposition decoupling - even though a 35% post-2011 drop in surface NO₂ against only 19% less nitrate deposition is the headline. A NOy deposition saturation threshold of 15.0 kg N/ha/yr is proposed. Relevant to PM as nitrate is a major secondary component; the saturation claim rests on a fitted inflection with no uncertainty stated.

doi: 10.1016/j.envpol.2026.129167

China aerosol vertical 2026 Atmos Pollut Res*Spatiotemporal Dynamics and Vertical Distribution of Aerosols in China: Insights from Multi-Source Evidence from 2000 to 2022.*

METADATA ONLY. Metadata-only record: Elsevier deposited title, journal and DOI without an abstract, and no abstract was recoverable from Crossref, Europe PMC or PubMed. A 23-year national synthesis of aerosol vertical structure is potentially a tier-A input to satellite-to-surface PM_{2.5} reconstruction - vertical profile shape is the dominant structural uncertainty when converting AOD to surface concentration - but nothing about method, platform mix or findings can be stated from the deposit, and inferring a design from a title is how this watch would start manufacturing results. Carried as a placeholder and added to the retry list; if the abstract appears it should be re-screened on its merits rather than admitted on the strength of its title.

doi: 10.1016/j.apr.2026.103198

Construction equipment PM-bound EPFR 2026 Atmos Environ*METADATA ONLY — persistent free radicals on PM from non-road construction plant*

No abstract retrievable on the entry date from Crossref per-DOI, Europe PMC or PubMed. What the record establishes: PM-bound environmentally persistent free radicals and downstream reactive radical species have been measured from non-road construction equipment under *real-world* rather than dynamometer conditions, which is the harder and rarer measurement. What it does not establish: radical concentrations or lifetimes, the engine tiers and duty cycles sampled, the EPR or spin-trapping protocol, or the site — none is asserted here. Relevant because EPFR content is a redox-activity metric that mass concentration does not predict, and construction plant sits inside the urban domain that dense networks are deployed to characterise.

doi: 10.1016/j.atmosenv.2026.122353

Delhi N2O5 2026 Atmos Chem Phys*Enhanced nocturnal N2O5-CINO₂ chemistry in Delhi.*

Early-spring 2023 measurements in Delhi against a winter 2019 baseline. Nocturnal N2O5 and CINO₂ averaged 13 and 80 ppt, roughly double 2019, and the authors attribute this to nighttime NO falling from 124+/-25 to 44+/-9 ppb, which removes the main N2O5 sink. Particulate chloride at 4.7 µg m⁻³ is among the highest reported anywhere and drives efficient N2O5-to-CINO₂ conversion, raising NO₃ and Cl radical production and organic oxidation. The comparison is the weak joint - different seasons (spring vs winter) and different years are confounded, so the NO-decline explanation competes with a straightforward seasonal one. Matters for PM because this chemistry feeds nitrate and secondary organic mass, the fraction low-cost optical sensors systematically mishandle under high humidity.

doi: 10.5194/acp-26-13121-2026

Dou et al. 2026 — metadata only Environ Pollut*Volatile organic compounds in Chengdu: functional zone disparities and four-year dynamics in a basin city of southwest China*

Same-day Elsevier deposit; no abstract in Crossref, Europe PMC core, Europe PMC fullTextXML or the OpenAlex per-work record. The title indicates a **four-year** ambient VOC record for Chengdu resolved by urban functional zone — the same Sichuan Basin whose humidity-driven nitrate chemistry Sun et al. model in this issue. In scope as precursor monitoring: functional-zone contrasts are the spatial scale distributed networks are built to resolve, and a four-year baseline is long enough to separate trend from meteorological variability. Carried forward for pickup.

doi: 10.1016/j.envpol.2026.129164

Douala households 2026 Spat Spatiotemporal Epidemiol*Particulate concentration and household respiratory health in urban Cameroon*

On-site campaign in Douala with an OC 300 Dust Particle Laser — a low-cost portable optical sensor — paired with a survey of 1,721 individuals in 344 households sited near the measurement points; instrumental Logit to address endogeneity of exposure. Reported effects on respiratory disease were **0.231 for PM₁₀**, **0.122 for PM₅** and **0.018 for PM_{2.5}**, and unsealed roads and traffic density dominated the PM levels. The size ordering is the problem: effect falls monotonically as particles get smaller, the reverse of what deposition physics predicts, and the most economical explanation is an uncorrected nephelometric instrument distributing mass across bins it cannot separate — no co-location against a reference monitor is described, and no humidity correction. Included because it is a rare health study built on a low-cost node, and because it shows precisely what the missing co-location arm costs.

doi: 10.1016/j.sste.2026.100836

EROD-HR/SOILHD Iberia 2026 Atmos Chem Phys*High-resolution soil erodibility and its effect on WRF-Chem dust and PM₁₀*

Two erodibility datasets tested in WRF-Chem with GOCART: EROD-HR, which folds fine-resolution topography in at 0.0625° globally and 1 km over Iberia, and SOILHD, which adds high-resolution sand/silt/clay texture and removes misclassified bare-soil areas at 1 km globally. Five dust episodes spanning 2022–2025 cover both local emission and long-range North African intrusion, evaluated against **PM₁₀ from the SINCLAIR network in Murcia**. Both datasets improved the magnitude and timing of modelled PM₁₀ peaks at inland stations; **improvements were limited at coastal stations**, which the paper does not fully resolve and which is the most interesting part of the result — coastal sites mix dust with sea salt and are the ones regulatory exceedance attribution most often disputes. Five episodes is a small evaluation set, and network PM₁₀ cannot separate dust from co-occurring anthropogenic mass.

doi: 10.5194/acp-26-12641-2026

Engeroff et al. 2026 Research Square (preprint)*PM exposure while running: park, residential and roadside routes by time of day*

Mobile personal PM₁₀, PM_{2.5} and PM₁ over **60 instrumented 2 km runs** (20 each on a major road, a residential street and an urban park; 30 morning, 30 midday; pace 8:02–4:31 min/km) with local urban background as reference and linear mixed-effects models adjusting for meteorology. Lowest concentrations of every fraction: **midday in the park**. Highest PM₁₀ anywhere: **morning on that same park route**. Exposure across all routes was lower at midday, tracking urban background, and humidity, pressure, wind speed and temperature all carried signal. The conclusion — park routes and rush-hour avoidance are sufficient — is reasonable but rests on 20 runs per route in one city, and the fact that the park produced both the minimum and the maximum should temper any general route advice. Preprint, not peer reviewed.

doi: 10.21203/rs.3.rs-10016764/v1

Environ Pollut 2026 Environ Pollut*Underlying Surface Heterogeneity Modulates Meteorological Drivers of PM_{2.5} and O₃ Pollution in the Sichuan Basin.*

METADATA ONLY. Crossref deposit of 19 September, abstract absent from the deposit and not yet indexed anywhere queried. Carried on title and venue. The claim in the title - that land-surface heterogeneity modulates how meteorology drives PM_{2.5} and O₃ - is the kind of effect that breaks a spatially stationary land-use regression, and the Sichuan Basin is the standard stagnation test case, but no magnitude, period or model class can be stated from what was deposited. Held for pickup.

doi: 10.1016/j.envpol.2026.129201

Fireman et al. 2026 Int J Public Health*Biomonitoring of urban ultrafine particle exposure and associated health effects*

99 hospital staff — 50 in the industrial Haifa Bay area, 49 Tel Aviv controls — with spirometry, FeNO, exhaled breath condensate and saliva at baseline and one year. Latent class analysis found a cluster with higher BMI, raised EBC ultrafine particle concentration and elevated salivary LDH-calcium, predominantly Haifa residents, reporting more cough, sputum and atopic complaints. The concept is worth tracking — EBC particle burden as an internal-dose biomarker would close the gap between a fixed monitor and what a person actually inhaled — but the execution does not yet support it. *n* = 99 **with LCA on a handful of variables will find clusters whether or not they exist**; no ambient UFP measurements accompany the biomarker, so the external-to-internal dose relation is never tested; and BMI loading on the same cluster makes an exposure reading hard to separate from a metabolic one.

doi: 10.3389/ijph.2026.1609855

Firozabad size-segregated lung deposition 2026 Environ Monit Assess*Seasonal variation of particulate matter deposition in human lungs through different exposure pathways in an industrial city.*

Two-season sampling in the Firozabad glass-industry belt: mean PM_{1.0} 77.14, PM_{2.5} 105.40 and PM₁₀ 277.64 $\mu\text{g m}^{-3}$, winter about 1.2x summer. Modelled pulmonary-region deposition fractions were highest for infants and children (78–79% for PM_{1.0} and PM_{2.5}), and total regional deposition was near-identical for nasal and oral breathing (16.67–16.68%) - a suspiciously exact agreement that suggests the two pathways were run with the same default ventilation parameters rather than route-specific ones. Cr exceeded the 1e-4 carcinogenic benchmark in both seasons and both age groups; Ni and Co dominated non-carcinogenic risk. Deposition is modelled, not measured, and no personal or activity data were collected.

doi: 10.1007/s10661-026-15885-y

Frueh et al. 2026 J Expo Sci Environ Epidemiol*Source apportionment of volatile organic compounds in a Philadelphia fenceline community: results from THRIVEair*

Year-long community-designed VOC campaign in South Philadelphia run with Philly Thrive, collecting one-week integrated passive thermal-desorption samples of **38 VOCs at 20 sites**, July 2023 – June 2024, around a former oil refinery under redevelopment. Concentrations were higher in industrial than non-industrial and high- than low-traffic areas. **Benzene, the community's declared concern, was low overall** (median 1.21 $\mu\text{g}/\text{m}^3$, range 0.31–9.05), while (*m,p*)-xylenes emerged as the industrial-area pollutant of concern; toluene:benzene and xylene:ethylbenzene ratios were higher at industrial and high-traffic sites, indicating fresher emissions, and PMF apportioned sources. **One-week integration resolves none of the short plumes fenceline residents actually report**. The design is the template distributed PM networks should copy: community-set siting and a declared pollutant of concern rather than compliance checking.

doi: 10.1038/s41370-026-00976-2

GPS+LCS schoolchild exposure model 2026 Environ Pollut*Integrating GPS and Low-Cost Sensors to Enhance Personal PM_{2.5} Exposure Modeling for Schoolchildren*

918 person-days from northern Taiwan schoolchildren over one year with GPS-equipped portable PM_{2.5} sensors, pre-deployment calibrated against a Met One BAM-1020 (mean R² 0.996, slope 0.962). Personal exposure was dominated by within-subject variability (ICC 0.125); school was the highest-concentration microenvironment at 22.7 $\mu\text{g m}^{-3}$. The fully GPS-resolved microenvironment model reached R² 0.619 (74.0% accuracy) and a home-plus-school-only model 0.616 (71.7%) - so essentially all the gain comes from identifying two microenvironments, and full trajectory resolution adds nothing. That negative result is the useful one for network design: it sets a ceiling on what finer location data buys. Validation is internal against concurrent personal measurement; whether the models reduce exposure misclassification in health analysis is explicitly not tested.

doi: 10.1016/j.envpol.2026.129149

Gao J et al. 2026 Environ Sci Technol

Are inverse ozone–health associations an artefact of measuring the wrong oxidant?

A perspective, not a study. Population studies not infrequently find *lower* O₃ associated with *higher* disease risk. NO titration near traffic and differential measurement error are the standard explanations and do not fully account for it. The alternative advanced here: low measured O₃ can mark locations and times of active oxidative chemistry — O₃ being consumed, or parallel oxidant pathways running — which generate toxic secondary products that no monitoring network measures. If so, O₃ is functioning as a negatively-signed proxy for an unmeasured oxidant field. The same logic transfers directly to PM mass as a proxy for oxidative potential, which is why this is carried in a PM watch. No data and no test proposed beyond a call for expanded measurement, so it stands or falls on plausibility.

doi: 10.1021/acs.est.5c17322

Garcia-Vara et al. 2026 J Hazard Mater

Expanding the chemical space of urban air: Identification and semiquantification of polar emerging contaminants in PM_{2.5} through LC-HRMS non-target analysis.

PM_{2.5} filters from eight Barcelona monitoring stations spanning traffic, urban background and port sites were extracted by pressurised liquid extraction and screened by data-dependent LC-HRMS. Forty contaminants of emerging concern were annotated at confidence level 2A - industrial chemicals, personal-care products, pesticides, pharmaceuticals and drugs of abuse - 21 reported in urban air for the first time, at pg/m³ to ng/m³; dicyclohexylamine, tributyl phosphate and DEET were highest. Semiquantification without authentic standards carries large uncertainty, level-2A identity is probable not confirmed, and the abstract gives no sampling duration or sample count, so site contrasts cannot be weighed. Relevance: optical sensors report mass only; this is the composition layer that decides whether equal mass means equal hazard.

doi: 10.1016/j.jhazmat.2026.143569

Godri Pollitt 2026 — metadata only Environ Sci Technol

Beyond inhalation: air as an integrative matrix for exposome research

ACS deposited this DOI on 18 September without an abstract and no text was recoverable from Crossref, Europe PMC or PubMed on per-DOI retry. Carried because it is the **second air-as-exposome-matrix Perspective in this same issue**, alongside Sunyer-Caldú, which makes the pair an editorial signal rather than a coincidence: air sampling is being argued into the exposome programme as an integrative matrix rather than merely an inhalation route. No argument or claim is asserted beyond the title. On the retry list.

doi: 10.1021/acs.est.6c05073

Gonzalez-Mendez et al. 2026 Environ Pollut

Morpho-chemical fingerprinting and distance-dependent attenuation of airborne Particulate Matter from abandoned mine tailings in Northwestern Mexico.

Three dry-season campaigns (2022–2024) placed Sigma-2 passive samplers at nine sites from abandoned tailings through farmland to a town. Automated SEM-EDX on 3,017 particles from oxidised and non-oxidised tailings and efflorescent salts built source fingerprints, classified by a three-step ML plus compositional-screening scheme. Mine-derived particle concentrations fell steeply with distance but were still detected about 1.1 km out; power laws explained 85.8% (PM_{10-2.5}) and 82.8% (PM₈₀₋₁) of the variance. Passive Sigma-2 deposition is converted to concentration through an assumed deposition velocity, nine sites constrain a decay curve loosely, and PM_{2.5} - the respirable fraction - is not the reported metric. Relevance: single-particle fingerprints are the ground truth a low-cost optical network near a point source cannot provide.

doi: 10.1016/j.envpol.2026.129233

Guo & Sudo 2026 Atmosphere

Global online PAH simulation in the CHASER chemistry-climate model

First global online simulation of naphthalene, phenanthrene, pyrene and benzo[a]pyrene in CHASER V4.0, coupling soot adsorption $K_{SA}(T)$ with organic-matter absorption $K_{OA}(T)$ under an extended Dachs–Eisenreich dual-mode scheme. Evaluated at urban, background and Arctic stations: pooled spatial R 0.78 on a log₁₀ scale, **FAC2 80%** for BaP excluding sub-grid hotspots, seasonal R 0.62–0.93. The K_{SA} term fixes the long-standing under-prediction of particle-bound fraction at cold high latitudes, raising polar ϕ_{BaP} by up to ~0.6, and elevated elemental carbon extends BaP lifetime to 1.72 d against 0.57–0.72 d for lighter congeners, sustaining transoceanic outflow. The dependence on modelled EC is also the weak point: the partitioning result is only as good as the soot field underneath it.

doi: 10.3390/atmos17090861

Guo et al. 2026 Toxics

Seasonal Formation, Sources, and Long-Term Evolution of PM_{2.5}-Bound Water-Soluble Inorganic Ions in Nanchang, China: The Increasing Importance of Wintertime Nitrate.

Day and night PM_{2.5} samples in summer and winter in Nanchang were analysed for nine water-soluble inorganic ions with source-apportionment, trajectory and fire-hotspot analyses, and set against 2009–2026 records. WSIs averaged 8.01 +/- 3.58 $\mu\text{g m}^{-3}$ in summer and 23.97 +/- 9.42 in winter (57.0% and 50.3% of PM_{2.5}); SNA made up 82.0% and 76.4% of WSIs, sulfate leading in summer and nitrate in winter, with nitrate formation favoured at night. Since 2019 the control problem has moved from sulfate to wintertime nitrate. Two seasonal campaigns at one site; campaign length is not in the abstract. Relevance: a rising ammonium-nitrate fraction is the composition shift that most inflates humidity-driven optical-sensor overestimation - calibrations fitted in the sulfate era will drift.

doi: 10.3390/toxics14090828

Guo et al. 2026 Atmosphere

Atmospheric Environment of the Northern Tianshan Urban Agglomeration, Xinjiang: A Comprehensive Review of Pollutant Characteristics, Photochemical Processes, and Health Implications

Narrative synthesis of air-quality research in the Northern Tianshan agglomeration, grading evidence as local, regional Central Asian, or extrapolated. Aerosol shifts from dust-dominated coarse mode in spring to secondary inorganic and carbonaceous fine mode in winter; BC is local fossil combustion, BrC sources are speculative; ozone regime diagnoses are thin and there are no radical or oxidative-potential measurements. Local epidemiology is essentially absent, so the health section rests on extrapolation. No systematic search. Relevance: moderate - an arid, dust-coal mixture region where optical sensors need dust-aware correction and where a dense network would fill a documented gap.

doi: 10.3390/atmos17100961

Gürler Akyüz & Akyüz 2026 J Environ Sci Health A

Size-resolved elemental composition in a coastal urban-industrial region

Seasonal PM_{2.5} and PM_{2.5-10} elemental composition at Zonguldak, Türkiye, where coal mining, coal combustion, iron-steel with coke production, maritime fuel oil, traffic and marine/crustal sources overlap. Heating-period PM_{2.5} carries elevated **As, Sb, Cd, Pb, Se, V, Ni, Cr**; non-heating PM_{2.5-10} is crustal, marine and resuspended dust with a persistent industrial residue. Enrichment factors against Ti show strong anthropogenic enrichment for combustion and metallurgy and near-natural levels for crustal and marine elements; PCA separates coal combustion, iron-steel, fuel-oil/shipping, non-exhaust traffic, crustal dust, marine aerosol and coal fly ash, and discriminant analysis confirms the season/size separation. Conventional in method, valuable as a composition baseline for a source mixture that is rarely characterised together.

doi: 10.1080/10934529.2026.2724190

Han et al. 2026 China CDC Wkly

Environmental Mixture-Health Associations: Current Analytical Practice and Reporting Recommendations.

97 original mixture-exposure studies in four high-impact journals, 2021-25. Median sample size 729 (IQR 396-1,992) and median 9 exposures (IQR 6-15); BKMR in 80.4%, WQS 36.1%, quantile g-computation 30.9%, with 42.3% running at least two methods. 88.7% used internal (biomarker) exposures and only 67.4% of those stated a detection-rate threshold; feature reduction was reported in 25.8%. Post-hoc variable-importance, interaction and nonlinearity testing appeared in 80-90% of papers, which with median $n=729$ and 9 correlated exposures is a multiple-comparison problem the field is not reporting. The review is descriptive and the four-journal frame is not the literature, so prevalence figures are upper bounds on best practice. The checklist matters here because PM component mixtures are the archetypal case and, as the frailty and MASLD records in this same issue show, component-level claims are being made routinely without the preprocessing documentation this paper finds missing.

doi: 10.46234/ccdcw2026.179

He B. et al. 2026 Atmos Environ

Contrasting seasonal behavior and inhalation risks of atmospheric PAHs, PAEs, and OPEs in a heating-influenced urban atmosphere

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes seasonal behaviour and inhalation risk of three semivolatile organic classes in a heating-season city. Composition data of this kind are what mass-only networks cannot supply. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.atmosenv.2026.122394

He et al. 2026 J Environ Manage

Interpretable machine learning for spatially differentiated nonlinear relationships between PM_{2.5} and landscape patterns in an urban-rural integrated zone

Random forest with SHAP relates PM_{2.5} to grey-green landscape metrics across an urban-rural zone of Xuchang. Three regimes emerge: elevation dominates in mountains; in dense cores the positive association with mean building height strengthens above 35-40 m; in transition zones green-space porosity of 0.18-0.22 (plains) or 0.12-0.15 (hills) and blue-space connectivity go with lower PM_{2.5}. The PM_{2.5} surface source and resolution are not given in the abstract, model skill is not reported, and SHAP thresholds on a single city are associations, not design standards - the authors say as much. Relevance: low-moderate; building-height and porosity thresholds are testable with a dense street-level sensor network rather than a gridded product.

doi: 10.1016/j.jenvman.2026.130981

Hien et al. 2026 Atmos Environ

Long-term Observations of Fine Particulate Sulfate in Hanoi during 1999-2023: Impact of SO₂ Regional Emissions and Monsoon Circulations.

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors (Hien, Vuong, Bac, Ha) and DOI on 23 Sep without an abstract; no leg recovered one. A 24-year fine-sulfate record from Hanoi set against regional SO₂ emissions and monsoon circulation would be one of the longest speciated series in Southeast Asia, and the title asserts both an emissions and a meteorological attribution. Sampling method, filter protocol, completeness and the size of any trend are all unavailable, and the title is not evidence of what the paper measured. Carried as a placeholder because long speciated records are the only way to validate the secondary fraction that low-cost optical sensors systematically misread. On the retry list.

doi: 10.1016/j.atmosenv.2026.122372

Hong Kong GEMA PM_{2.5} mobility 2026 Environ Res

1.7 million personal PM_{2.5} measurements against momentary stress and activity

Two-day geographic ecological momentary assessment on 800 Hong Kong adults, integrating **1.7 million personal PM_{2.5} measurements**, 2.1 million accelerometer records, 4,800 momentary assessments and eye-level greenspace derived from 4.3 million street-view images, summarised over 5- to 600-minute windows and analysed across eight activity-travel contexts with a DAG-guided multi-relation double-ML framework. Higher PM_{2.5} tracked more light-intensity and less moderate-to-vigorous activity overall, and predicted worse *perceived* air quality; perceived pollution in turn predicted higher stress while measured PM_{2.5} did not. The result that should travel: **context-specific estimates ran in opposite directions** while cumulative associations with stress showed nothing sustained. Two days per participant is thin for a stress outcome and reverse causation between activity and exposure is unresolved, but the decomposition itself is the finding — a pooled null here is an artefact of aggregation.

doi: 10.1016/j.envres.2026.125659

Hu et al. 2026b Environ Pollut

Long-term spatiotemporal evolution, relative hotspot persistence, and predictive attribution of PM_{2.5} across China.

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors (Hu, Shi, Zhang) and DOI on 23 Sep with no abstract recoverable from any leg. The title claims three things - long-term spatiotemporal evolution, persistence of relative hotspots, and predictive attribution of PM_{2.5} across China - of which the middle is the interesting one, since relative hotspot persistence under a falling national mean is what decides whether monitoring networks can be thinned. Dataset, period, grid resolution, attribution method and results are all unavailable. Listed on metadata; re-screen when the abstract is deposited rather than reading the title as a finding.

doi: 10.1016/j.envpol.2026.129192

Hulme et al. 2026 Environ Res Lett

Home and school exposure, jointly, for 919 Accra schoolchildren

Land-use regression annual means for PM_{2.5}, NO₂ and BC plus L_{day}, L_{night} and L_{den} noise, linked to geocoded home and school points for **919 of 1,034** children aged 8-12 across 90 Accra schools, then combined into a time-location-weighted total. Median home PM_{2.5} **33.8 µg/m³**, NO₂ 55.5, BC $5.2 \times 10^{-5} \text{ m}^{-1}$; school values slightly lower (29.9 / 53.5 / 4.9). Time-weighted totals exceeded WHO guidance for most children, and noise exceeded WHO road-traffic thresholds at home (L_{den} 66.3) and Ghana's educational-area standard at school. The joint air-plus-noise, home-plus-school design is the contribution — both are under-measured in sub-Saharan Africa. Annual LUR means cannot resolve the school day, so the time-weighting is coarser than it looks.

doi: 10.1088/1748-9326/ae8d40

Hurynovich et al. 2026 Atmos Environ

Transferability of dust-related PM_{2.5} models under contrasting arid environmental regimes.

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 25 Sep with no abstract; Crossref, PubMed efetch and the publisher landing page returned nothing, and Europe PMC was down (HTTP 503) for the whole run. The title indicates a test of whether dust-PM_{2.5} models fitted in one arid regime hold in another. Transferability is the same failure mode that limits sensor calibration models; worth recovering. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.atmosenv.2026.122386

Ireland Census linkage 2026 Health Place*Ambient air pollution at home, work and school: Evidence from Ireland.*

Whole-population census microdata linked to Irish EPA pollutant surfaces at two addresses per person. Mean annual $PM_{2.5}$, PM_{10} and - most sharply - NO_2 are higher at the work, school or college address than at the residence, while O_3 runs the other way, the expected NO_x -titration signature of the same urban gradient. The socioeconomic result inverts the usual assumption: among workers the residential-to-workplace contrast is *larger* for more advantaged groups, because advantaged workers commute into dense urban cores from cleaner suburbs. Residence-only assignment therefore does not simply add noise, it adds noise correlated with socioeconomic position, which is the one thing an exposure proxy must not do in a study adjusting for it. Limits: both endpoints are modelled ambient surfaces at an address, so nothing here captures infiltration, indoor sources or time-activity weighting, and the paper measures the size of a bias without estimating its effect on any health coefficient. For network design this is the strongest argument yet that siting density should follow daytime population rather than residential population.

doi: [10.1016/j.healthplace.2026.103732](https://doi.org/10.1016/j.healthplace.2026.103732)**Kagawa et al. 2026** Atmos Environ*Seasonal variations of aerosol selenium concentrations and oxidation states in central Japan.*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors (Kagawa, Nagao, Kagami, Kida) and DOI on 23 Sep with no abstract and none recoverable elsewhere. Selenium oxidation state is a coal-combustion and long-range-transport tracer, and a seasonal speciation record from central Japan would bear on transboundary attribution in the East Asian outflow. The deposit says nothing about site, sampling duration, analytical method, detection limits or the size of the seasonal contrast, so nothing quantitative can be reported and the subtopic assignment rests on the title alone. Listed rather than summarised, on the retry list, and to be re-screened on its merits when an abstract appears.

doi: [10.1016/j.atmosenv.2026.122364](https://doi.org/10.1016/j.atmosenv.2026.122364)**Kaur et al. 2025** Atmos Pollut Res*Long-term $PM_{2.5}$ source apportionment at 12 sites in the western United States from 2000-2020.*

Twelve monitoring sites, twenty years, five to eight resolved factors per site. $PM_{2.5}$ fell at ten of twelve sites (-0.38 to $-0.037 \mu g m^{-3}$ per year; no decline at Boise or Commerce City), and the decline is carried by secondary ammonium nitrate, ammonium sulfate and the two EC-rich factors - exactly the sources NO_x , SO_2 and mobile-source rules targeted. EC-rich fell despite rising vehicle miles travelled, isolating per-vehicle emission control rather than reduced activity. Winter OC-rich, the biomass-burning proxy, showed the largest seasonal decline. Dust rose at most sites. The weakness is the one the authors set out to address and only partly do: CSN speciation methods changed within the record, so a factor's 20-year trajectory is partly an instrument trajectory, and reconciling PMF with CMB constrains but does not eliminate that. For distributed sensing the finding that matters is compositional: the shrinking part of western $PM_{2.5}$ is the secondary and combustion mass an optical sensor sees reasonably well, while the growing part is dust, which nephelometric low-cost sensors under-report because coarse mineral particles scatter poorly at their sampling cut. Network bias against dust therefore worsens over time.

doi: [10.1016/j.apr.2025.102779](https://doi.org/10.1016/j.apr.2025.102779)**Kim et al. 2026** Environ Health*Residential indoor air monitoring for long-term exposure assessment in susceptible populations*

130 older adults with idiopathic pulmonary fibrosis, Seoul Metropolitan Area, 2021–23. Indoor hourly $PM_{2.5}$ averaged **14.90 $\mu g/m^3$ (SD 36.12)** against **20.44 (61.63)** outdoors, higher in winter (20.17) than summer (11.19) and by day than night, with hourly patterns that contrast rather than track outdoors. Infiltration factors from median indoor/outdoor ratios of sulfur, nickel and iron were **51–70%**. Source contrasts were large: pets 24.09 vs 13.35, humidifiers 19.19 vs 13.16, smokers 16.76 vs 11.79, cooking 15.14 vs 10.96 $\mu g/m^3$. Every standard deviation exceeds its mean, so the distribution is episodic and a seasonal mean is a poor summary; five days per season is thin support for a long-term claim and no health estimate is reported yet. The transferable piece is the non-contact protocol — instruments redeployed by postal mail, which is how an unattended network is actually serviced.

doi: [10.1186/s12940-026-01334-3](https://doi.org/10.1186/s12940-026-01334-3)**Klebach et al. 2026** Atmos Chem Phys*MSA and sulfuric acid in tropical Indo-Pacific aerosol from aircraft*

CAFE-Pacific, HALO aircraft, January–February 2024, nitrate chemical-ionisation mass spectrometry of methanesulfonic and sulfuric acid. Marine boundary layer concentrations reach **$4 \times 10^7 cm^{-3}$ MSA and $6 \times 10^7 cm^{-3}$ SA**; the MSA/SA ratio increases with altitude in the lower free troposphere, consistent with temperature-dependent DMS oxidation. Two flights show marine deep convection lofting DMS to the upper troposphere with particle formation and growth after **10–20 h of OH exposure**, matching the kinetically modelled DMS lifetime. MSA frequently exceeds SA, which the current generation of new-particle-formation schemes does not anticipate. Aircraft inlets heat and evaporate acidic particles, so the gas/particle split at altitude is instrumental as much as atmospheric — the authors are explicit about this.

doi: [10.5194/acp-26-12355-2026](https://doi.org/10.5194/acp-26-12355-2026)**Kong J et al. 2026** J Environ Sci (China)*Triplet-state brown carbon as an endogenous oxidant*

Photochemistry of 1,4-naphthoquinone as a brown-carbon proxy in deoxygenated aqueous systems under 355 nm irradiation, resolving two long-lived $^3(n, \pi^*)$ triplets ($^3NQ^*$ and $^3OH-NQ^*$). $^3NQ^*$ abstracts hydrogen from organic substrates **without any external oxidant**, with a rate hierarchy set by bond dissociation energy and one-electron oxidation potential: alkanes $\sim 10^4$, oxygenated alkanes $\sim 10^5$, phenols $\sim 10^6 M^{-1} s^{-1}$. Self-photoconversion regenerates a second triplet, amplifying the oxidative cascade. The claim is that $^3BrC^*$ is the dominant phenol sink in BrC-rich haze under acidic, O_2 -limited conditions — which is exactly the condition inside a haze particle and exactly the condition current models do not represent. Proxy compound, deoxygenated system; the rate constants are the transferable output.

doi: [10.1016/j.jes.2025.10.037](https://doi.org/10.1016/j.jes.2025.10.037)**Konon et al. 2026 — metadata only** Atmos Environ*Seasonal variation in volatile organic compound emissions and secondary organic aerosol formation from unconventional oil and gas development*

Same-day Elsevier deposit; nothing in Crossref, Europe PMC core, Europe PMC fullTextXML or the OpenAlex per-work record. The title indicates a seasonally resolved measurement campaign of VOC emissions from unconventional oil and gas development and the secondary organic aerosol those emissions form. In scope as a precursor study: **SOA is the fraction of basin $PM_{2.5}$ that no primary-particle emission inventory captures**, and seasonality in the precursor flux propagates straight into any exposure model built on annual-average emission factors. Carried forward for pickup.

doi: [10.1016/j.atmosenv.2026.122368](https://doi.org/10.1016/j.atmosenv.2026.122368)

Kuntz et al. 2026 J Hazard Mater

Dynamics of tire-related chemicals in roadside aerosol over one year in Leipzig, Germany-influencing factors, potential source materials and comparison with rural aerosol

119 roadside PM₁₀ samples over 14 months were analysed for 40 tire-related chemicals. Medians: 6-PPD 0.61 pg/m³, 6-PPD-quinone 2.9 pg/m³, diphenylguanidine 90 pg/m³, benzothiazolesulfonic acid 41 pg/m³. Concentrations spanned about four orders of magnitude, with precipitation the dominant control, modulated by temperature, sunshine, wind and oxidants. Tread particles carried one to two orders of magnitude more TRC per mass than roadside aerosol, and rural levels were up to tenfold lower. Traffic-normalised concentrations are offered for extrapolation, but one roadside and one rural site cannot support transfer across fleets and climates. Relevance: moderate - a toxic tyre-wear fraction that optical sensors cannot distinguish within PM₁₀.

doi: 10.1016/j.jhazmat.2026.143749

Ahmad et al. 2026 Atmosphere

Carbonaceous composition and multi-wavelength optical properties of PM in Lahore

Six-week urban campaign in Lahore, 30 quartz filters collected, 25 usable for gravimetry and optics, **only the first 15 analysed for OC/EC**. PM_{2.5} ranged **40–417 μg m⁻³** with a mean of **137 ± 84**. Organic carbon was 76.6% of total carbon against 23.4% elemental. Mean absorption Ångström exponent **1.17 ± 0.30**, with 56% of samples in 1.0–1.5, 32% below 1.0 and 12% above 1.5. The authors are appropriately careful that AAE intervals describe spectral behaviour rather than attribute sources. Fifteen OC/EC filters in one pre-monsoon season cannot characterise Lahore's annual cycle, and the gravimetric series has no co-located reference for traceability. Directly relevant to optical low-cost sensors, whose response depends on exactly this absorption behaviour.

doi: 10.3390/atmos17090899

Lai et al. 2026 Environ Sci Technol

Two-thirds of ambient organosulfates have no assigned source

Organosulfates contribute up to **30% of aerosol mass**. Of **366** ambient OS molecular formulas compiled from field studies, only about **36%** map to a known precursor or formation mechanism, leaving **64% unclassified**. The proposal here is that some unclassified species are not freshly formed at all but higher-generation products of heterogeneous and aqueous multiphase transformation of existing OS: identified transformation products already account for ~17% of reported formulas and explain an additional 5% of the unclassified set. Modest arithmetic, but it reframes a source-apportionment problem as an aging problem. Compilation-based, dependent on the sampling and HRMS coverage of the underlying field studies, and 5% is a small demonstration of a mechanism claimed to be general.

doi: 10.1021/acs.est.6c08394

Lei et al. 2026 Atmos Environ

Source transition in PM_{2.5}-bound metals in a coastal megacity

Ships on metadata only — no abstract in the Crossref deposit and no other index carries the record on this run. The title indicates a multi-year, high-time-resolution observational record of PM_{2.5}-bound metals with a source-transition analysis and a health-implication component. Counted and listed; no numbers claimed. High retry priority: multi-year hourly metals is exactly the dataset class against which a low-cost network's composition proxies would have to be validated, and there are very few of them.

doi: 10.1016/j.atmosenv.2026.122336

Leung et al. 2026 Indoor Air

Abandoned wells, indoor VOCs, and a pilot honest enough to report its own limits

Passive indoor sampling for 13 VOCs in **17 households (October 2024)** and **15 households (February–March 2025)** in southwestern Ontario, with abandoned-well exposure estimated by inverse-distance-squared weighting over wells within 3 km and self-reported acute symptoms over the previous six months and during sampling. This is gas-phase, not particulate — carried here because the exposure-assignment design is the one a distributed indoor PM study would use and because the MAPSH project measures both. At $n < 20$ per round with Spearman correlations and no personal monitoring, the study is powered to describe indoor concentrations against Canadian guidelines and not to detect a well-proximity association; the six-month symptom recall window compounds that. Read as a feasibility and concentration-range report.

doi: 10.1155/ina/8041546

Li et al. 2026 Atmosphere

A theoretical model for predicting visibility during dust events

Physically based chain from surface wind erosion through dust emission and near-surface concentration to horizontal visibility, with an analytical concentration–visibility relation closed heuristically and a moment-balance entrainment model carrying median particle size and soil cohesion into the threshold friction velocity. Against three published datasets the relation explained $R^2 = 0.62$ and had the highest R^2 of the formulations compared under identical calibration. The appeal is economic: visibility is observed almost everywhere dust matters and PM₁₀ usually is not, so a defensible physical inversion turns an existing network into a concentration proxy. $R^2 = 0.62$ supports event detection and screening, not exposure assignment. The authors are explicit that **only the concentration–visibility link was evaluated** — the emission and entrainment stages are untested, and closing the chain needs synchronised observations that do not currently exist.

doi: 10.3390/atmos17090902

Liu S et al. 2026 J Environ Sci (China)

Terrain blocking and monsoon coupling in North China Plain PM_{2.5} and O₃

Quantifies mountain-blocking effects on pollutant dispersion in the semi-closed North China Plain now that anthropogenic emissions have fallen far enough for terrain and meteorology to dominate the residual pattern. Within 120 km buffers of the major ranges, concentrations rise with altitude and then fall sharply above a threshold; the **effective blocking altitude is 22–929 m for winter PM_{2.5} and 19–1060 m for summer O₃**. Concentrations rank plains > platforms > hills > mountains, and winter PM_{2.5} and summer O₃ high-value zones aggregate in *opposite* north–south directions. Descriptive rather than causal — emissions and terrain co-vary strongly in the NCP — but the altitude thresholds are a concrete siting constraint for any regional network.

doi: 10.1016/j.jes.2026.01.013

Liu S et al. 2026 Environ Pollut

Wind-shear regimes and the vertical PM_{2.5} profile in a basin

Six instrument types — surface network, wind profiler radar, microwave radiometer, aerosol lidar, radiosonde and UAV — plus WRF, over Taiyuan Basin. Unsupervised clustering gives three vertical-wind-shear regimes: strong directional near-surface shear with near-surface accumulation and the shallowest boundary layer; speed shear above 500 m with an *elevated* PM_{2.5} maximum near that height; and deep shear below 2000 m with stronger winds and low concentrations. UAV profiles over 0–500 m confirm the three shapes. The 17 January 2024 episode makes the operational point: strong shear did not enhance vertical exchange while the shear layer sat above a shallow stable layer, and only coupled once daytime heating deepened the boundary layer into it. Directly relevant to siting — a surface monitor cannot see regime 2 at all. One basin, one winter.

doi: 10.1016/j.envpol.2026.129103

Liu et al. 2026 Life (Basel)

Airborne Bacterial and Eukaryotic Communities in Temuco, Chile, a City Dominated by Residential Wood Combustion.

First characterisation of airborne eukaryotic communities alongside bacteria in Temuco, a city whose PM is dominated by residential wood burning. Mean PM_{2.5} and PM₁₀ were 75.0 and 87.7 $\mu\text{g m}^{-3}$, PM_{2.5} exceeding the Chilean 50 $\mu\text{g m}^{-3}$ standard; nitrate (5.6), sulfate (2.6) and K⁺ (2.6 $\mu\text{g m}^{-3}$) were the main ions. Clostridium, Hymenobacter and Sphingomonas led the bacteria; Corioliopsis (41.0%) and Hyphodontia (26.9%) the eukaryotes. Communities varied more by season than size fraction, and RDA loaded SO₄ (0.71) among the drivers. Sample numbers and averaging periods are not in the abstract, and the ion-community correlations are descriptive. Relevance: a PM_{2.5}/PM₁₀ ratio of ~0.86 confirms a fine-mode smoke aerosol - the regime where optical sensors track reference best if humidity is corrected.

doi: 10.3390/life16091433

Liu et al. 2026a Environ Health (Wash)

Effect of Skin Types and Microenvironment of Dermal Exposure to Polycyclic Aromatic Hydrocarbon Derivatives and Heavy Metals.

Polyethylene balloons coated with graded doses of glycerol trioleate stood in for oily, neutral and dry skin, then were deployed across four microenvironments to measure adsorption of nitrated and oxygenated PAHs and 11 metals. Adsorption followed first-order kinetics and ranked oily > neutral > dry; it was highest in dusty weather and lowest indoors on campus, with NPAHs exceeding OPAHs in campus air, and metal uptake ordered V > Zn > As > Ti, with Cu detected only during sand-blowing. A coated balloon is not skin - no sebum turnover, no perspiration, no penetration - so these are surrogate deposition rates, not doses. Useful anyway: it quantifies a route that PM mass monitoring ignores entirely, and dust episodes dominate it.

doi: 10.1021/envhealth.5c00843

Liupan tethered eBC 2026 Atmos Chem Phys

Vertical structure of carbonaceous aerosol over the Loess Plateau from tethered balloon

Coordinated tethered-balloon profiling of carbonaceous aerosol and meteorology at Pingliang, eastern foothills of the Liupan Mountains, during July 2023 and July 2024 campaigns. Near-surface means were 0.84 $\mu\text{g/m}^3$ eBC and 1.24 $\mu\text{g/m}^3$ UVPM, both decreasing with height. The substantive observation is that in the early morning and at night, when convection is weak, UVPM aloft exceeded eBC — attributed to nucleation from gaseous precursors rather than transport. Thermodynamic processes dominated vertical transport near the surface while enhanced dynamics aloft drove horizontal dispersion, and southerly air masses contributed disproportionately to UVPM. Two July campaigns at one site cannot separate seasonal from interannual variation, and eBC and UVPM are optical attributions from a filter-based absorption instrument, not chemical speciation — the nucleation interpretation is inference from the eBC/UVPM divergence, not a measured formation pathway.

doi: 10.5194/acp-26-12671-2026

Ma M. et al. 2026 Atmos Environ

Atmospheric Redistribution of Microplastics Reshapes Environmental Risk Across China

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes atmospheric transport of microplastics and its effect on risk across China. Airborne microplastics are part of the coarse fraction optical sensors mis-size. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.atmosenv.2026.122397

Mainka & Mainka 2026 Toxics

Modelled Physical Activity Benefits and PM_{2.5} Risks in Two Urban Cyclists.

Re-analysis of 162 previously published cycling trips by two commuters in Gliwice, each treated as a hypothetical five-day-a-week habit and compared with resting for the same time at the concurrent station concentration, using assumed ventilation and published dose-response functions. Median net RR was 0.840 (16.0% relative risk reduction; 95% UI 0.792-0.892 when the epidemiological coefficients were varied); the median break-even concentration was 196.7 $\mu\text{g m}^{-3}$, and a +/-10% personal-concentration error flipped the sign of the dose increment on 12 trips without pushing net RR above 1. Everything is modelled, n=2, and the personal PM_{2.5} came from uncalibrated optical monitors, as the authors state. Relevance: a worked example of how sensor error propagates into risk-benefit claims - here it does not change the conclusion.

doi: 10.3390/toxics14090836

Han et al. 2026 Environ Pollut

Source apportionment of PM_{2.5} oxidative potential in metropolitan Melbourne

Published PMF source contributions linked by multiple linear regression to oxidative potential measured on three complementary acellular assays (DTT, AA, DCFH); models explain 73–86% of OP variance. Combustion dominates OP_v-DTT at 52–53% with biomass burning above traffic; OP_v-AA is driven by metal-rich sources including diesel at 20–22%; secondary nitrate contributes 28–33% and secondary sulphate 10% at the second site. The assay-dependent ranking is the result — a single-assay OP metric would nominate a different mitigation target. Reusing an earlier PMF solution propagates its rotational ambiguity, and no uncertainty is quoted on the source-resolved shares. First Southern Hemisphere estimate of its kind.

doi: 10.1016/j.envpol.2026.129159

Melssen Pfahl et al. 2026 Environ Sci Technol

Renoxification via Reduction of Inorganic Nitrate by Photoacids in Aerosols.

Chamber study, no human subjects. Catechol and related phenols directly reduce particulate nitrate to HONO and NO under UV-visible light; nebulised gallic acid with nitrate produced 23 times more NO_y than nitrate photolysis alone. HONO formation persisted at pH 8, above the pK_a of HONO, implicating the phenols as photoacids rather than as bulk acidifiers. The limitation is the usual one for chamber aerosol chemistry - gallic acid at nebuliser concentrations is a proxy for a biomass-burning phenol pool that is far more heterogeneous, and no ambient measurement anchors the 23-fold enhancement. Relevance to distributed sensing is indirect but real: a smoke plume that renoxifies this efficiently generates ozone downwind, so a PM-only network mis-reports the health burden of the same plume hours later and tens of kilometres away.

doi: 10.1021/acs.est.6c04742

Milan pizza-oven dispersion 2026 Preprints.org

Assessment of Air Quality Impacts from Wood-Fired Pizza Ovens: Comparison with Major Urban Emission Sources

PREPRINT. Milan. Measured wood-fired-oven emission factors feed a dispersion model, with road traffic and domestic heating simulated on the same framework and all three compared against ARPA Lombardia monitors. Oven contributions concentrate in the city centre: 0.085-3.34 $\mu\text{g m}^{-3}$ of primary PM₁₀, only 0.0044-1.75 $\mu\text{g m}^{-3}$ of NO_x, but 19.70-34.28% of levoglucosan and 34.95-37.47% of benzo(a)pyrene. That split is the finding - a source that is minor by mass and major by tracer, which is exactly the pattern that breaks levoglucosan-based residential-wood-burning apportionment in a restaurant-dense centre. Limitations: preprint, single city, emission factors from a small appliance sample, and the ranges are model spreads rather than measured confidence intervals

doi: 10.20944/preprints202609.0792.v1

Momen et al. 2026 Research Square (preprint)

Explainable Machine Learning for Predicting Thermal Discomfort from Multipollutant Exposure on Dhaka Roadways

Preprint. Roadside PM_{2.5}, noise and microclimate were measured on Dhaka arterials in the monsoon alongside thermal sensation votes; mean PM_{2.5} was 58.35 $\mu\text{g m}^{-3}$ and noise 82.96 dB, and 99.46% of votes were hot, warm or slightly warm. ML classifiers reached AUC 0.96 with SHAP attributing discomfort across heat, PM and noise. The outcome is thermal sensation, not a health endpoint; sample size, instrument type and validation scheme are not in the abstract, and an AUC of 0.96 on a 99% prevalent outcome says little. Not peer reviewed. Relevance: low; co-located PM, noise and heat is the multi-stressor sensor package pedestrian-exposure networks increasingly deploy.

doi: 10.21203/rs.3.rs-10487427/v1

Lai et al. 2026 Atmosphere

Monte Carlo model for wind-driven PM emissions from industrial granular materials

First quantitative evaluation of a saltation-and-sandblasting emission model for wind erosion of granular stockpiles, with the Monte Carlo impact count fixed rather than run to convergence. Applied to lead and zinc sulphide concentrates at a Sardinian plant, cutting simulated impacts from 10,000 to 500 changed the median emission by a small relative difference at a twentyfold saving, and the revised model reproduced emission magnitude, sandblasting efficiency and their friction-velocity dependence. Validation is internal and physical-consistency based — no measured emission flux is compared against, which is the central weakness. Fugitive stockpile emission is exactly the source class regulatory networks miss and fenceline arrays are asked to capture.

doi: 10.3390/atmos17090891

Montreal smoke advisory mobility 2026 Health Place

Urban mobility responses to wildfire smoke-related air quality advisories in Montreal, Canada.

Time-series forecasting against four Montreal mobility streams - bike-share, metro, bus and recreational running - with counterfactuals controlling season, weather, holidays and events. Advisory days shifted travel from active to enclosed modes: bike-share fell, bus and metro rose, running did not move, and the effect varied by neighbourhood. This matters for exposure assessment because mode substitution changes both ventilation rate and microenvironment concentration in opposite directions, so an ambient-monitor exposure metric mis-specifies dose on exactly the days it matters most. The abstract reports no effect magnitudes or intervals, only significance, and advisory issuance is a coarse proxy for smoke concentration; running data are self-selected app users.

doi: 10.1016/j.healthplace.2026.103736

Multi-Tree 2026 J R Stat Soc Ser C Appl Stat

Multi-Tree Model for Precision Environmental Health with Longitudinally Assessed Mixture Exposure.

A Bayesian ensemble of tree triplets - two interacting binary tree types - that estimates individualised and subgroup distributed-lag effects for mixture exposures assessed repeatedly, with built-in modifier selection over a high-dimensional candidate set. Applied to Colorado birth registry data it recovers a PM_{2.5}-birth weight association concentrated in mothers under 25 with low income. The applied finding is the weaker half: effect-modification by maternal age and income is already well established, so it functions as a sanity check rather than a discovery, and no interval is reported for the subgroup estimate in the abstract. The methodological contribution is the part worth using - most distributed-lag mixture methods return one population-average curve, and this returns a per-individual one.

doi: 10.1093/jrsssc/q1ag047

Mutic et al. 2026 medRxiv (preprint)

Paired silicone wristbands across home and early-childhood-education settings

A protocol, not results. Prospective repeated-measures pilot targeting 44 preschool children and 8 staff across two sociodemographically distinct Atlanta early-childhood centres, with home- and centre-designated silicone wristbands exchanged between settings over three consecutive days and nights, one spot urine per child, and continuous indoor air-quality monitoring in two classrooms per site. Eight pre-specified feasibility indicators. The design premise is sound and underexploited — children spend >90% of time indoors and early-childhood centres are high-occupant-density microenvironments almost absent from the exposure literature — but wristbands integrate semi-volatile organics, not particle mass, so the PM component rests entirely on the fixed classroom monitors. Listed for the design; nothing to conclude from it yet.

doi: 10.64898/2026.08.27.26361470

N China indoor Finf 48 homes 2026 Atmosphere

Determinants of indoor PM_{2.5} and infiltration factors across 48 dwellings

Paired indoor/outdoor PM_{2.5} at 5 min in 48 residences across eight Northern Chinese cities, seven consecutive days in each of four seasons, 31,094 valid hourly observations from 32,256 scheduled. A linear mixed-effects model with a residence random intercept and AR(1) residuals gives marginal $R^2 = 0.71$, conditional $R^2 = 0.91$, ICC = 0.34: dwelling identity carries a third of the variance that the fixed effects — outdoor PM_{2.5} ($\beta = 0.872$), window-open fraction (0.548), cooking (0.046), purifier-on fraction (-0.902) — cannot reach. Continuous purifier operation associates with 59.4% lower indoor PM_{2.5}, which the authors correctly decline to call causal. Threat to inference: window and purifier use are self-reported and dominate the same model, and coal-heated Northern China is a specific regime. For distributed sensing this is the cleanest recent argument that per-dwelling measurement is not redundant with a good ambient field.

doi: 10.3390/atmos17090874

NZ tractor fleet PM inventory 2026 Sci Total Environ

An uncertainty-propagated PM_{2.5} inventory for an entirely unregulated source class

Non-road mobile machinery is under-regulated and rising in relative importance as on-road sources are controlled; New Zealand is an instructive case because it pairs ultra-low-sulphur diesel and tightening on-road standards with no NRMM emission regulation at all. The inventory joins row-level registration microdata for 43,367 tractors, a licensing table defining 31,253 active units, and 29,278 dealer sales records (2017–2025), under an IPCC Tier 2 fuel-anchored framework with EMEP/EEA factors, tier hours calibrated against a CARB monitoring study, and fuel anchored to a 204 Ml national consumption target. 10,000-iteration Monte Carlo gives 115 t PM_{2.5} (95% UI 73–190) and 2258 t NO_x. The distributional result is the useful one: Gini 0.51 for PM_{2.5}, with the dirtiest 30% of tractors producing 69% of it, and modernisation running at only +2.7 percentage points a year. Activity hours remain the weakest link — they are calibrated, not measured.

doi: 10.1016/j.scitotenv.2026.182302

Na et al. 2026 Atmos Environ

Satellite-based estimation of PM_{2.5} chemical composition under sparse observations in Seoul metropolitan area

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes satellite-based estimation of PM_{2.5} chemical components from sparse speciation observations. Relevant to satellite-surface fusion and to composition-specific health models. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.atmosenv.2026.122402

Niamien et al. 2026 Atmos Pollut Res*Spatio-temporal analysis of the physical properties of aerosols in urban and rural environments in Cote d'Ivoire*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes urban-rural aerosol physical properties in Cote d'Ivoire. West Africa is among the least monitored regions, where low-cost networks carry most of the observational load. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.apr.2026.103220

Ningombam et al. 2026 Atmos Environ*Assessment of global aerosol burden over land and oceans using ground-based, satellite, and reanalysis data*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 26 Sep with no abstract; Crossref, PubMed efetch, Europe PMC and the publisher landing page returned nothing. The title indicates a global land/ocean aerosol-burden comparison across ground-based photometry, satellite retrievals and reanalysis. Retry priority: the reanalysis-versus-photometer bias by region is what decides whether reanalysis AOD is a usable prior for satellite-to-surface PM_{2.5} fusion. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.atmosenv.2026.122370

ORACLES 4STAR 2026 Atmos Chem Phys*Subseasonal and spatial variability of biomass burning aerosol radiative properties observed over the Southeast Atlantic during ORACLES.*

All three ORACLES deployment months (2016-2018) of 4STAR sky-scans over the Southeast Atlantic smoke-over-stratocumulus regime. Single scattering albedo rises through the burning season - the plume brightens - and the absorption Angstrom exponent analysis gives no evidence of a brown carbon contribution, so the authors attribute it to composition change rather than a change of aerosol type. Comparison against a 30-year climatology from 31 southern African AERONET sites shows the same seasonal direction over land and sea. Sampling is the caveat the paper handles well: the September SSA dip is shown to be an artefact of 2016's more southerly flight tracks. Three campaign years cannot separate a seasonal cycle from interannual variability, which they concede.

doi: 10.5194/acp-26-13083-2026

Odu-Onikosi et al. 2026 Atmos Pollut Res*PM_{2.5}-bound toxic elements in Lagos*

Ships on metadata only, same Elsevier deposit gap as above: no abstract, not in PubMed or Europe PMC, ScienceDirect blocked. Sampling duration, element list, apportionment method and risk estimates are all unretrievable at digest time. Counted and listed; no numbers claimed, and flagged for the abstract retry list. Noted because the author list includes P. A. Solomon and P. K. Hopke, and because speciated PM_{2.5} from Lagos — a megacity of roughly 15 million with almost no published speciation record — is exactly the geographic gap the two African records above document.

doi: 10.1016/j.apr.2026.103193

Ilies et al. 2026 Front Public Health*Urban PM₁₀ and population exposure from satellite, meteorological and socio-spatial data*

MODIS-MAIAC AOD combined with four ground stations and meteorology in a regression for PM₁₀ over Oradea, Romania, then folded with population density, built form, NDVI and elevation into a composite exposure index. Cold-season maxima reach 41.8 $\mu\text{g m}^{-3}$ against an annual mean near 20.5 — above WHO, inside EU limits — and 20.1% of the metropolitan population falls in high-predisposition areas. The AOD-PM₁₀ relation alone is weak ($R = 0.28$) and meteorology carries the model, which is the authors' own admission and the standing reason satellite-only surfaces underperform at city scale. Four stations cannot validate an intra-urban field, and the composite index is unweighted and never checked against measured personal exposure.

doi: 10.3389/fpubh.2026.1940025

PRD limona ketone 2026 Environ Pollut*Ground-gridded observations of high concentrations of ambient limona ketone suggest overlooked limonene oxidation in subtropical megacities.*

Large-scale gridded plus intensive diurnal measurements of limona ketone across the Pearl River Delta found unusually high concentrations during spring and autumn ozone episodes. A parameterised inversion using the product as a constraint implies more limonene precursor than inventories carry, with a pronounced nighttime enhancement; modelled production is OH-dominated while the loss pathway shifts from OH by day to the nitrate radical at night. The inference is indirect by construction - precursor abundance is inverted from a product, so it inherits every error in the assumed yield and loss rates, and the authors only claim the nocturnal enhancement is robust 'across plausible parameter ranges' rather than quantifying it. Episode-specific sampling limits it further. Matters as biogenic SOA is an under-constrained part of urban PM mass.

doi: 10.1016/j.envpol.2026.129177

Pachés et al. 2026 Atmos Pollut Res*Regional background PM_{2.5} in the Valencian Community by clustering*

Ships on metadata only. The Elsevier deposit carries no abstract, the record is not yet in PubMed, Europe PMC or Semantic Scholar, and ScienceDirect refused retrieval from this sandbox, so the network size, the clustering method and the resulting background estimates are all unavailable. Counted and listed; no numbers claimed. Flagged for the abstract retry list. Background/local-increment separation is directly relevant to siting a low-cost network — the increment is what a dense network is deployed to resolve — so this is worth revisiting once the abstract deposits.

doi: 10.1016/j.apr.2026.103192

Paediatric exposome 2026 Pediatr Res*Assessing environmental exposures in children: toward a comprehensive exposome approach*

Narrative review of exposure-assessment tooling for paediatrics — questionnaires, biomarkers, wearable and fixed sensors, geospatial indicators and pooled databases — alongside longitudinal cohort design, multi-omics integration and methods for mixtures and critical windows. No systematic search protocol, no pooled estimate, and the closing argument is professional advocacy rather than evidence. Useful only as a citation anchor for the claim that single-pollutant assessment is the limiting factor in child environmental epidemiology, and for its framing of sensors as one instrument among several rather than the exposure itself.

doi: 10.1038/s41390-026-05460-z

Papetta et al. 2026 Atmos Chem Phys*Source-dependent optical and mineral signatures of Mediterranean dust outbreaks*

Four major 2021–2022 dust outbreaks from the Eastern, Western and Central Sahara and the Middle East, tracked across **24 AERONET stations** with IASI and MODIS MIDAS dust optical depth and HYSPLIT back-trajectories. Saharan events were coarse and scattering; the Middle East event carried **finer, more absorbing particles**, likely anthropogenically influenced. DOD-to-AOD ratio separates them — only one East-Central Saharan event stayed above 0.8 (near-pure dust), while the Middle East event had the lowest ratios, indicating mixing with anthropogenic or marine aerosol. A Cyprus in-situ case study with elemental and absorption measurements anchors the remote-sensing inference. For network work the transferable idea is the DOD/AOD purity screen: it is a cheap way to decide whether a coarse-mode excursion is genuinely dust before attributing it.

doi: 10.5194/acp-26-12395-2026

Paris Saharan dust heatwave 2026 Atmos Chem Phys*Dust aerosol cools Paris by up to 1 °C — and that does not mean comfort*

Three Meso-NH simulations of the 15–19 June 2022 Paris heatwave — no dust, CAMS reanalysis dust, and doubled dust — validated against the PANAME-Urban campaign, which the model reproduces for aerosol optical depth and incoming solar radiation, with improved air temperature and boundary-layer height once aerosol is included. The dust plume's radiative effect cuts surface shortwave by up to **75 W m⁻²** and air temperature by up to 1 °C. The result worth the page is the one that refuses the easy conclusion: thermal comfort improves by up to 1 °C in *sunlit* urban and suburban areas at 16:00 UTC, but in shaded suburban areas the increase in diffuse radiation and humidity plus the drop in wind speed cancels the temperature benefit entirely. A single five-day episode in one city, and comfort indices are not health outcomes — but the case for coupling aerosol forecasts to heat-health warning systems is made on the disaggregation, not on the mean.

doi: 10.5194/acp-26-12729-2026

Paygude et al. 2026 Environ Monit Assess*Heavy metal exposure and source profiling from leaf-deposited particulate matter in Pune*

27 roadside, industrial, peri-urban, highway and rural sites. Elemental order Ca > K > Fe > Na > Mg > Mn > Zn > Cu > Cr > Pb, consistent with crustal material, construction and resuspended road dust; enrichment factors flagged Cu, Zn and Cr at traffic- and industry-influenced sites; PCA separated construction/road-dust, vehicular, biomass-or-food and localised industrial factors. Hazard quotients ordered Cr > Pb > Cu > Zn with hazard indices below 1 for adults and children. The method is the contribution: leaf-deposited dust is a zero-power, zero-maintenance passive sampler that resolves metal-enriched microenvironments a sparse reference network cannot. Its limits are equally clear — deposition velocity varies with leaf morphology, species and washoff history so concentrations are not convertible to airborne mass, integration time is uncontrolled, and an **inhalation hazard quotient computed from deposited dust requires an airborne concentration that is never observed here**.

doi: 10.1007/s10661-026-15906-w

Petersen et al. 2026 Environ Sci Technol*Sulfate ions in aerosol water are their own source of radical oxidants*

Direct spectroscopic evidence that the sulfate anion absorbs surface-level sunlight and ejects an electron, forming **SO₄^{•-} on a sub-microsecond timescale** in a concentration-dependent manner; long-timescale production rates quantified by radical trapping with LC-HRMS. Compounds emitted from biomass fires photosensitise the pathway, raising radical formation **up to five-fold** over direct photolysis across pH 1.6–5.5, more strongly at low pH. Estimated SO₄^{•-} production is **comparable to measured hydroxyl-radical production in aerosols** — i.e. a second oxidant budget nobody was counting. The downstream claim is the striking one: secondary brown carbon from this chemistry **turns the direct radiative forcing of sulfate-containing aerosol from negative to positive**, and the organosulfates formed are cloud-active. Bulk solutions at controlled ionic strength; extrapolation to real aerosol water is the open step.

doi: 10.1021/acs.est.6c05704

Ajibade et al. 2026 Proc Natl Acad Sci U S A*Gentrification, displacement, disadvantage and exclusivity predict urban environmental and health inequities*

PCA on 2010 and 2020 ACS and Census data yields four contemporary segregation typologies for 342 Portland tracts, compared against historical HOLC zones across 104 environmental and health variables. HOLC grade predicted persistent exclusivity in the urban core but not injustice elsewhere; **persistently disadvantaged tracts, often outside HOLC boundaries entirely**, carried the highest pollution and hazard exposure and the worst health and disability outcomes. Exclusive neighbourhoods had more canopy and better preventive care *and* more air pollution — an awkward result the paper does not resolve. Wholly ecological and cross-sectional. For monitor siting the message is that a single historical redlining layer is an inadequate stratifier.

doi: 10.1073/pnas.2422760122

Qin et al. 2026 Microorganisms*Seasonal Variation of Airborne Bacteria and Potential Pathogens Adhere to Fine and Coarse Particles in a Coastal City.*

Size-resolved airborne bacterial communities on PM_{2.5} and PM₁₀ in Qingdao were profiled by 16S sequencing across seasons. Staphylococcus, Lactobacillus, Escherichia-Shigella and Klebsiella dominated everywhere; Staphylococcus was enriched in autumn (LEfSe LDA > 3.5). Summer and autumn communities diverged, but particle-size groups did not differ significantly; Chao and Shannon indices were higher on PM₁₀ than PM_{2.5}, significant only in autumn. The abstract gives no sample count or sampling duration, pathogen potential is predicted from 16S taxonomy (not viability or virulence), and no PM mass is reported. Relevance: marginal; optical sensors cannot distinguish biological particles, so this fraction sits unresolved inside every coastal PM_{2.5} reading.

doi: 10.3390/microorganisms14091924

Rana & Guzman 2026 ACS Earth Space Chem*Heterogeneous Nitrate Radical Oxidation of Catechol Produces Nitrocatechols, Oligomers, and Brown Carbon.*

Flagged backfill: Crossref created this DOI on 20 Aug; it reached this window via a 22 Sep Europe PMC deposit and had not been carried or rejected. Catechol thin films oxidised by NO₃ under dry conditions produced nitroaromatic IR bands near 1590 and 1350 cm⁻¹, a broad UV-vis absorption centred near 349 nm characteristic of brown carbon, and 4-nitrocatechol, dinitrocatechols, dimers and trimers; the proposed route runs through semiquinone radicals, then nitration and radical coupling. Thin films under dry conditions are far from real humid particles, and no yields or mass absorption efficiencies are given, so atmospheric significance is asserted rather than shown. Paired with the PAX-ISS record in this issue: nighttime BrC formation is one source of the seasonal BrC term that record apportionments.

doi: 10.1021/acsearthspacechem.6c00226

Rawat et al. 2026 Atmos Environ*Size-resolved aerosols from residential fuel combustion in the northwestern Himalayas: potential effects on light absorption and human health*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes size-resolved aerosol from household fuel combustion and its absorption and health implications. Relevant to household-air-pollution exposure, today's tuberculosis burden paper being the health side of the same source. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.atmosenv.2026.122399

Shukla et al. 2026 Atmos Chem Phys*Organic vapors from savannah and boreal fire emissions under photochemical and dark aging*

Controlled burns of understudied African savannah and European boreal fuels sampled by high-resolution PTR-MS, then aged photochemically and in the dark in a 29 m³ Teflon chamber. Total primary organic-vapour emission factors spanned **69–161 g/kg** by fuel type. Photochemical ageing depleted furanics, phenolics and oxygenated aromatics and enriched carbonyls and small O-containing species; dark low-NO_x ageing produced almost no compositional change, and clustering attributed **73% of group-level compositional variance to oxidation regime**. Chamber burns of prepared fuels do not reproduce field moisture, combustion phase or dilution, so these are upper-bound laboratory factors. Any smoke PM_{2.5} forecast assuming a fixed downwind secondary yield is contradicted here.

doi: 10.5194/acp-26-12977-2026

Sawan et al. 2026 Environ Sci Technol*Aviation drives ultrafine number, freight drives black carbon, in the same neighbourhood*

Hybrid mobile and fixed-site campaign in a marginalized northwest Toronto neighbourhood downwind of Toronto Pearson and a freight corridor. UFP correlated with flight activity ($r = 0.58$, $p < 0.05$ **at 3.5 km**) and was enriched **1.7–3.3-fold** downwind of the airport, measured as the ratio of interquartile ranges under downwind versus other wind conditions, reaching **3.2-fold at residential sites ~8 km away**. BC instead tracked roadway proximity and freight and correlated weakly with flight counts. BC–UFP correlation was strong under northerly winds (mean $r = 0.62$) but collapsed to $r = 0.1$ downwind of the airport. Random Forest models ($R^2 = 0.79$ UFP, 0.87 BC) identified industrial land use and a highway as shared drivers, airport proximity as UFP-specific and freight trucking as BC-specific. The inference is observational and wind-sector stratified, so airport attribution rests on a co-location argument rather than on any tracer unique to jet exhaust. The lesson for network design is that mass-based siting would have missed the aviation signal entirely.

doi: 10.1021/acs.est.6c06666

Scalabrini et al. 2026 Atmos Pollut Res*Chemical tracers for tire wear particles in atmospheric aerosol: analytical approaches and challenges*

Metadata only. Crossref registers the title, 11 authors and a 5 September creation date with no abstract, and the record has no Europe PMC counterpart. The title places it as a review of tracer chemistry for apportioning non-exhaust tire wear — a live problem, since tire wear is a growing share of traffic PM precisely as exhaust controls tighten, and the standard markers each fail differently. **No findings, recommended tracer or study count are asserted here.** A targeted Scholar Gateway full-text query returned only adjacent June 2026 reviews, not this paper. Added to the abstract retry list.

doi: 10.1016/j.apr.2026.103194

Schaefer et al. 2026 Atmos Chem Phys*Deep convection sorts ice-nucleating particles by activation temperature*

Immersion-mode INP filter samples from the HALO aircraft up to **14.5 km** during CIRRUS-HL (summer 2021), taken in the inflow and outflow of deep convective clouds and from cloud-particle residuals. Inflow spectra contain INPs active at both high ($T > -15^\circ\text{C}$) and low ($T < -20^\circ\text{C}$) temperatures; **outflow spectra contain only the low-temperature population**. The reading is precipitation scavenging in the updraft removing the warm-active INPs while the cold-active ones are carried efficiently into the free troposphere. Cloud microphysics rather than exposure relate, carried for the particle-transport result: convection is a size- and composition-selective vertical filter, which matters for any attempt to isolate column-integrated aerosol to surface composition. Offline analysis, filter sampling, one campaign.

doi: 10.5194/acp-26-12505-2026

Seibert et al. 2026 Environ Pollut*National-scale NMF source apportionment reveals component-specific west-to-east asymmetries in PM_{2.5} concentrations associated with transboundary transport.*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 25 Sep with no abstract; Crossref, PubMed efetch and the publisher landing page returned nothing, and Europe PMC was down (HTTP 503) for the whole run. The title indicates non-negative matrix factorisation of national PM_{2.5} data resolving transboundary components with a west-east gradient. Country, input species and factor count are not in the deposit. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.envpol.2026.129243

Sen et al. 2026 Environ Microbiol Rep*Genomic and phenotypic classification of Staphylococcus species isolated from Delhi air*

Outdoor culturable bioaerosol sampling at six Delhi sites in November 2019 — the peak pollution month, pre-pandemic — using a Merck MAS Eco-100. *S. gallinarum*, *S. haemolyticus* and *S. arlettae* were recovered and characterised by whole-genome sequencing, phenotypic microarray, antibiotic-resistance profiling and biofilm formation in the presence of black carbon. Comparative genomics found the urease cluster *ureABCDEFG* in *S. arlettae* Naraina and *S. haemolyticus* Yojna Vihar 1, unusual because urease is normally reported negative in both species and may confer a niche fitness advantage. The black-carbon biofilm assay is the link to PM and the part worth following, but **no quantitative result for it is reported**. Culture-based sampling recovers a small biased fraction of the airborne microbiome, six sites in one month give no seasonal or spatial contrast, and no particle mass was measured alongside the biology, so the association with the pollution season is contextual.

doi: 10.1111/1758-2229.70411

Shahadat et al. 2026 Environ Geochem Health

Spatiotemporal variability, trends, and health risk assessment of ambient PM_{2.5} and PM₁₀ in Bangladesh during 2018–2025.

Analysis of 2018–2025 PM_{2.5} and PM₁₀ from the 16 Department of Environment continuous stations in Bangladesh, with seasonal Mann-Kendall trends and ELCR/HQ risk indices. Every station exceeded national standards and WHO guidelines every year. Network annual mean PM_{2.5} fell from 91.54 to 77.64 $\mu\text{g m}^{-3}$ (-15.2%) and PM₁₀ from 155.7 to 132.3 (-15.0%), non-monotonically, rebounding after the 2020 minimum; winter/monsoon ratios were ~4.2 (PM_{2.5}) and 3.0 (PM₁₀). Most seasonal trends were not significant. The authors note kriging performed poorly and was used only for display - 16 stations cannot map a country of 170 million. ELCR applied to PM_{2.5} mass is a screening construct, not an epidemiological risk. Relevance: the station-data quality control is undescribed, and this network is exactly what low-cost sensor densification in Bangladesh would calibrate against.

doi: 10.1007/s10653-026-03506-z

Shen et al. 2026 Atmos Pollut Res

A Hybrid Deep Learning-Geostatistical Framework for PM_{2.5} Spatial Estimation with SHAP Interpretation

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 24 Sep with no abstract; Crossref, Europe PMC, PubMed efetch and Semantic Scholar all returned nothing. The title states a hybrid deep-learning and geostatistical PM_{2.5} spatial estimator with SHAP interpretation. Squarely in the exposure-modelling scope; the validation design (spatial vs random CV) is what would decide its value. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.apr.2026.103208

Liu et al. 2026 J Environ Manage

Decadal changes in PM_{2.5} sources and secondary aerosol drivers, Shenzhen 2014–2024

Eleven years of speciated filter sampling with PMF apportionment and machine-learning attribution of secondary formation drivers. Annual mean fell 37.7 to 17.3 $\mu\text{g m}^{-3}$. Secondary sulphate was historically dominant and declined fastest; vehicles became the largest single contributor after 2019; **secondary nitrate barely moved and is now the binding constraint**. NO_x and VOC cuts drove nitrate and SOA declines but rising atmospheric oxidation capacity offset them, while sulphate tracked oxidation and regional transport over local SO₂. Attributing a decadal trend to oxidation capacity from a correlational model at one site cannot exclude meteorological confounding, and the PMF factor series carry no uncertainty bounds.

doi: 10.1016/j.jenvman.2026.130913

Sichuan Basin 2026 Environ Pollut

Topographic Modulation of High-Temperature Ozone Suppression: Core-Periphery Threshold Structure in the Sichuan Basin.

High-temperature suppression - ozone flattening or falling under extreme heat rather than rising with the usual climate penalty - appeared in 16 of 18 analysed Sichuan Basin cities. The critical suppression temperature fell from 36.8 C in the basin interior to 33.8 C at the periphery, and the decline rate above it reached -9.65 $\mu\text{g m}^{-3}/\text{C}$ in the inner zone, about 2.1x the outer. SHAP attributions put boundary-layer ventilation ahead of cloud and circulation terms. Ozone, not PM, so it sits at the edge of scope; included because the same basin ventilation structure governs PM accumulation and because the threshold-by-topography result is a caution against fitting a single regional temperature response. Five years of data and a closed basin limit transfer elsewhere.

doi: 10.1016/j.envpol.2026.129161

Singh et al. 2026 Atmos Environ

Aerosol Absorption Coefficients Across Global Sites: Impacts of Chemical Composition, Emissions, and Meteorology

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 29 Sep with no abstract; PubMed efetch, Crossref, Europe PMC and the publisher landing page returned nothing. The title indicates a multi-site analysis of aerosol absorption coefficients against composition, emissions and meteorology. Relevant to absorption-based black-carbon sensors and their site-dependent mass-absorption cross-sections. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.atmosenv.2026.122391

Solaiman et al. 2026 Environ Sci Atmos

Nighttime NO₃ oxidation of myrtenal: SOA yields and HOMs

Chamber study of secondary organic aerosol from the gas-phase oxidation of myrtenal, a biogenic aldehyde, by NO₃ radicals, reporting SOA yields, chemical composition and highly oxygenated molecules. The Crossref deposit truncates the abstract mid-sentence and no full record was retrievable from Semantic Scholar on this run, so the yields and HOM distributions cannot be quoted. Counted and listed; the nighttime NO₃ channel for oxygenated biogenics is under-represented in this watch's corpus and the record is worth revisiting once the full abstract deposits.

doi: 10.1039/d6ea00092d

Sonipat ACSM campaign 2026 Atmos Chem Phys

Chemical characterization and source apportionment of carbonaceous aerosols during post-monsoon biomass burning and Diwali at an upwind site of Delhi.

25 October to 15 November 2023, spanning two severe haze episodes and Diwali. Composition-based PM_{2.5} exceeded 300 $\mu\text{g m}^{-3}$ in both episodes; organic aerosol was 65% of non-refractory PM_{2.5} with a daily mean near 140 $\mu\text{g m}^{-3}$, and oxygenated OA was 57–60% of total OA, reaching 42.8 $\mu\text{g m}^{-3}$ of more-oxidised OOA at peak haze - so the haze is substantially aged secondary mass, not fresh plume. Biomass- or solid-fuel eBC was ~78% of campaign-integrated eBC. The authors are honest that this split is sensitive to the assumed absorption Angstrom exponent pair, which is the paper's real limitation: the headline 78% is a function of a prior, and the sensitivity range is not propagated into the PMF factors. Single site, single season, three weeks. The result that matters upstream of a sensor network is that the mass is already loaded before the air reaches Delhi, so an urban-only network measures an urban increment on a large regional background and will misattribute regional transport to local sources unless it has an upwind anchor.

doi: 10.5194/acp-26-13227-2026

Groenen et al. 2026 Sci Total Environ

Characteristics and trend of wildfire activity in the Southern Hemisphere

Two decades of MODIS active-fire detections across Southern Africa, South America and Oceania. Seasonality is strongly heterogeneous: Southern Africa shows a stable 20-week May–October season with low inter-annual variability, South America a 14-week July–October season with high variability, and **Oceania no clear season at all**, which the authors read as co-existing fire regimes. Active-fire counts are a detection proxy, not emitted mass — insensitive to smouldering, biased by overpass timing and cloud — so no PM emission inventory follows directly. Note that the abstract's stated period (2004–2024) disagrees with the title's (2003–2022); check the paper before citing either.

doi: 10.1016/j.scitotenv.2026.182240

St Elias proglacial dust 2026 Atmos Chem Phys*Sub-kilometre modelling of mineral dust emission from proglacial valleys*

A sub-km WRF-Chem implementation built to resolve dust emission from the proglacial valleys of the St. Elias Mountains, Canada — sources absent from current emission modelling and unaccounted for in global inventories. Land-surface inputs were rebuilt with explicit treatment of glaciofluvial deposit heterogeneity and inundation state; 5–19 d simulations across 2019–2022 were evaluated against in situ meteorology, dust measurements, camera stations and surface Doppler LiDAR. Total emission is estimated at $1.0 \times 10^4 \text{ kg km}^{-2} \text{ d}^{-1}$ from erodible deposits over 47 simulated days, offered explicitly as a first-order figure. Skill in reproducing surface meteorology and in-valley vertical dispersion is **season-dependent**, and the emission scheme remains sensitive to soil texture, so the flux estimate inherits that sensitivity. High-latitude dust is a genuinely under-measured PM source and this is a careful first quantification, but 47 days at one mountain range does not support extrapolation.

doi: 10.5194/acp-26-12613-2026

Stergiou et al. 2026 Atmos Pollut Res*Spatiotemporal graph neural network bias correction for CMAQ over CONUS*

Ships on metadata only. Crossref carries no abstract for this DOI and the record is not in PubMed, Europe PMC or Semantic Scholar as of this run, so nothing beyond title, authors and venue can be attributed to it. Counted and listed; no numbers claimed. Flagged for the abstract retry list — graph-structured bias correction of a regulatory CTM against monitor networks is directly on this watch's topic and should be summarised once the abstract deposits.

doi: 10.1016/j.apr.2026.103191

Sun et al. 2026 Environ Sci Technol*Nonlinear nitrate formation governed by aerosol water and VOC-sensitive oxidation in a humid urban basin*

Wintertime field observations in a humid, topographically confined Chinese urban basin, combined with a box model and thermodynamic calculations. Nitrate became the dominant $\text{PM}_{2.5}$ component during pollution episodes, co-varying with NO_2 , O_x and aerosol liquid water content. Pathway-resolved modelling split formation almost evenly between daytime $\text{OH} + \text{NO}_2$ (52%) and nighttime heterogeneous N_2O_5 hydrolysis (46%), the latter rising sharply during episodes. Binning regressions located thresholds at RH 87% and ALWC $76.5 \mu\text{g}/\text{m}^3$ above which formation efficiency rose abruptly; VOCs exerted slightly stronger control than NO_x , with oxygenated VOCs and alkenes dominant. One site, one winter, one box model — the thresholds are not validated out of sample. The RH threshold is directly load-bearing for sensor correction: it is the same humidity range in which optical low-cost sensors diverge from reference monitors.

doi: 10.1021/acs.est.6c08621

Sunyer-Caldú 2026 Environ Sci Technol*Beyond blood and urine: decoding the human chemical exposome across matrices and time*

A Perspective arguing that high-resolution mass spectrometry can profile thousands of chemical features per sample while study design has not kept pace — no investigation has combined external source monitoring, multi-matrix internal biomonitoring and toxicokinetic normalisation in one longitudinal framework. Three elements are proposed: multi-matrix sampling extending beyond blood and urine to accumulation archives, developmental matrices and external media; a Unified Exposure Scale placing heterogeneous matrices on a common toxicokinetically normalised dosimetric axis; and a two-track hazard-prioritisation pipeline. No data, no validation, and the Unified Exposure Scale is circular where it matters — it needs toxicokinetic parameters that do not exist for the molecular dark matter it is meant to rank. Logged for how air monitoring is being positioned inside the Global Human Exposome Project.

doi: 10.1021/acs.est.6c06509

Surdu et al. 2026 Environ Sci Technol*Summertime Aerosol Composition in Southern Greenlandic Fjords: Impacts of Sources and New Particle Formation Events*

First aerosol composition measurements in the fjords near Narsaq, South Greenland, from two summer ship campaigns using nanoelectromechanical-systems FTIR. Median organic matter was $1.1 \mu\text{g m}^{-3}$, above High Arctic and mid-latitude coastal sites; PMF split the aerosol into primary marine (53%) and MSA-bearing secondary marine (47%) factors, amines exceeded open-ocean levels, and NPF events shifted composition toward secondary material. Campaign length and sample counts are not in the abstract, and two summers cannot separate interannual bloom timing from sampling differences. Relevance: marginal for health; NEMS-FTIR is a low-mass, high-time-resolution composition method that could pair with network optical sensors at supersites.

doi: 10.1021/acs.est.6c09104

Taiwan LRT bioaerosol 2026 Environ Res*Long-range transport effects on airborne fungal concentration and persistence*

Daily wintertime fungal monitoring in northern Taiwan, comparing periods before, during and after Asian dust storms and frontal pollution events. Long-range transport raised fungal concentrations by **57–193% during events** and **17–69% in the post-event window**, with elevation persisting **one to four days** after passage. Frontal pollution enhanced concentrations more broadly than dust events did, and the assemblages differed by regime: dust storms enriched Basidiospores and *Cladosporium*, frontal events enriched *Aspergillus/Penicillium* and Ascospores. The persistence result is the useful one, because event-triggered sampling designs that stop at event end will systematically under-count exposure. Weaknesses are the usual ones for this design: event periods are defined meteorologically and confounded with the temperature, humidity and mixing changes that accompany them, and no source apportionment separates transported spores from locally resuspended ones.

doi: 10.1016/j.envres.2026.125640

Taiwan $\text{PM}_{2.5}$ -bound metals 71 sites 2026 Res Sq (preprint) *$\text{PM}_{2.5}$ -Bound Metals in a High-Tech Industrial Area of Northern Taiwan: Spatial Distribution, Sources, and Health Risks*

71 ambient sites sampled across two campaigns (2019, 2020) in a northern Taiwan high-tech industrial region. Mean $\text{PM}_{2.5}$ was $19.33 \pm 8.97 \mu\text{g m}^{-3}$, higher in autumn (23.51) than summer (14.57), with hotspots near industrial parks; total metals averaged $736.85 \pm 751.20 \text{ ng}/\text{m}^3$, dominated by Zn, Ba, Cu, Pb and Mn. PMF gave fossil-fuel combustion 33.1%, soil and road dust 29.3%, metal industry 22.3% and tyre/brake wear 22.3%. Monte Carlo risk assessment returned a combined lifetime incremental cancer risk of 2.60×10^{-4} , driven by As and Cr, and a children's hazard index of 4.65 against 0.69 in adults, mostly via dust ingestion. A preprint, not peer-reviewed; source percentages sum past 100% as printed; two short campaigns cannot support a lifetime risk integral. The usable claim is that $\text{PM}_{2.5}$ mass alone misranks risk where composition varies - the case for composition-aware monitoring.

doi: 10.21203/rs.3.rs-10850297/v1

Ullah et al. 2026 Environ Pollut

Transboundary smoke moves almost nobody across China's standard and almost everybody across WHO's

Daily PM_{2.5} from **51 cities** in Yunnan, Guangxi and Guangdong for March–April 2015–2019, gridded onto WorldPop, with the Indo-China Peninsula fire contribution isolated by paired GEOS-Chem runs with and without the fires. The population-weighted share of days above the **WHO 2021 24 h guideline** (**15 µg/m³**) **reached 78–96% every year**; above **China's Grade II standard** (**75 µg/m³**) **it was 1–6%**, an average daily exposed population of about **6 million**, under 3% of the corridor's residents. The two benchmarks describe different worlds from identical data, which is the paper's real contribution to monitoring policy. Principal weakness: the attribution runs at 2° × 2.5°, roughly 200 km, so the fire contribution is a regional average imposed on city-scale observations and cannot support any within-province ranking.

doi: 10.1016/j.envpol.2026.129081

Vilnius grey-green PM_{2.5} SHAP 2026 J Environ Manage

Unraveling associations between grey-green spatial trade-offs and PM_{2.5} concentrations using explainable machine learning

Vilnius elderships 2016–2024 under a Grey-Green Space Coordination Index with LISA clustering and XGBoost-SHAP. GGSCI is negatively associated with PM_{2.5}; persistent high-high PM_{2.5} clusters sit on imbalanced grey-green structures. SHAP flips sign by range: NDVI above ~0.3 contributes negatively in balanced classes, while population density above ~2,000 km² and nighttime light above ~18 contribute positively in imbalanced ones. The awkward result, stated but not resolved, is that the most imbalanced GGSCI classes show the lower pollution - which inverts the paper's own framing and points to confounding by peri-urban land use. SHAP attributions are model explanations, not causal effects, and the PM_{2.5} field is itself modelled at a resolution the index outruns.

doi: 10.1016/j.jenvman.2026.130895

W. India BC-dust 2026 Environ Sci Atmos

Seasonal Variability and Interactions of Black Carbon and Mineral Dust over Semi-Arid Western India: Satellite and Ground-Based Observations.

Only the opening of the abstract was deposited, so the design is legible but the numbers are not: the study characterises seasonal co-variation of anthropogenic black carbon and natural mineral dust in a semi-arid region where both are large, using satellite retrievals alongside ground-based observations. No effect sizes, retrieval-validation statistics or station counts are recoverable from the deposit, and nothing is claimed here beyond scope. Flagged for retry rather than summarised further. The reason to carry it at all is the mixing problem it addresses: BC and dust are the two components that most degrade low-cost optical PM sensing in arid regions, BC because it absorbs rather than scatters and dust because its coarse mode falls outside a nephelometric sensor's effective cut, and a seasonal climatology of their interaction is the calibration prior a regional network would need.

doi: 10.1039/d6ea00109b

Wang J et al. 2026 J Environ Sci (China)

Water-soluble aerosol matter photosensitises guaiacol into aqueous SOA

Mechanism and kinetics of aqSOA formation from guaiacol mediated by photosensitive water-soluble inorganic and organic matter extracted from real aerosol. Photolysis of the extract generates both reactive oxygen species and triplet WSOM; **³WSOM* is the predominant initiator** of guaiacol transformation, with steady-state concentration **5.35 ± 0.12 × 10⁻¹³ mol/L**, generation rate **3.43 × 10⁻⁷ mol L⁻¹ s⁻¹**, quantum yield **4.86%** and a rate constant with guaiacol above **10⁹ L mol⁻¹ s⁻¹**. ROS then convert soluble intermediates into the insoluble ethers and low-volatility phenols that constitute the aqSOA. Bulk aqueous extracts are not cloud or particle water — ionic strength differs by orders of magnitude — which is the standing limitation of this whole method.

doi: 10.1016/j.jes.2025.11.050

Wang T et al. 2026 J Environ Sci (China)

IAQMS-Industry: 100 m PM_{2.5} inside industrial parks

Couples the regional NAQPMS to a city-scale chemical transport model with explicit point-source locations and Gaussian plume dispersion, resolving PM_{2.5} at **100 m** against a 3 km regional baseline, evaluated at Beijing Yizhuang and Tangshan. Normalised mean bias improves from **-16.9% - -7.7% to 3.1–6.2%**, and industrial point sources account for **22.9–26.4%** of park PM_{2.5} versus 1.6–13.7% in the regional model. The mechanism is straightforward — coarse grids smooth the concentration gradient and over-disperse — but the size of the attribution error is the useful number. This is the exact error a dense in-park sensor network would be deployed to measure, and the paper supplies the modelled expectation to test against.

doi: 10.1016/j.jes.2025.12.020

Wang X et al. 2026 Environ Monit Assess

Coupling coordination between bus-stop dust and soil metals

52 paired surface-dust and soil samples (104 total) at bus stops in Weiyang District, Xi'an, assayed for Cr, Mn, Cu, Zn, As and Pb and scored with Igeo and the Nemerow index. All but As and Mn exceeded Shaanxi soil background. Spatial patterns differ by medium — dust higher east, soil higher south — and sources resolve to traffic, natural and industrial. Hand-to-mouth ingestion dominated the exposure route and non-carcinogenic risk was higher for children than adults. The novel element is applying a coupling-coordination degree between the two media, which came out *moderate* and is read as partial interactive migration with shared sources. That index is a descriptive statistic borrowed from regional economics, not a transport model, so “interactive migration” is an interpretation rather than a demonstrated flux.

doi: 10.1007/s10661-026-15820-1

Wu Q et al. 2026 Environ Sci Technol

Nighttime new particle formation, and it is not always sulfuric acid

Roughly **1.5 years** of molecular-level measurement in coastal Dalian. Spring and winter nocturnal NPF is driven by H₂SO₄, whose dimer formation proceeds near the collision limit. In summer and warm autumn, H₂SO₄ and its dimers are scarce while oxygenated organic molecules are abundant, and **OOM dimers correlate with nascent particles at mean r=0.55**. Organic-driven nucleation has been reported for biogenic environments; showing it under anthropogenic influence at night is the novel claim. Implication for number-concentration monitoring: an ultrafine source term exists at night that photochemically-anchored parameterisations will miss. Correlation of 0.55 is suggestive rather than mechanistic, and one coastal site cannot establish how general the seasonal switch is.

doi: 10.1021/acs.est.6c04493

Xiamen chlorine NO₂ uptake 2026 Atmos Chem Phys*Chlorine-enhanced nocturnal NO₂ uptake and its nitrate yield in a coastal city*

Field observations in coastal Xiamen combined with a machine-learning attribution and a multiphase box model, contrasting sea-land-breeze nights against non-SLB nights. HONO and particulate nitrate both rose on SLB nights, with a mean NO₂ uptake rate constant of $k_{\text{NO}_2} = 9.70 \times 10^{-6} \text{ s}^{-1}$; chlorine ranked as the single most important driver of the enhancement. Feeding that constant back into the box model resolved the standing underestimation of both HONO and nitrate production, with nocturnal uptake accounting for **83.9% of HONO** and **46.4% of nitrate formation**. The machine-learning importance ranking is associational and cannot separate chlorine from the marine air mass that carries it, and one coastal city in one season limits transfer. Relevance is to secondary inorganic PM: nitrate is a large and poorly forecast fraction of coastal urban PM_{2.5}.

doi: 10.5194/acp-26-12715-2026

Xiao S et al. 2026 Environ Geochem Health*PAHs and oxygenated PAHs in Xi'an PM_{2.5} under the clean air action*

Winter PM_{2.5}-bound parent PAHs and OPAHs in Xi'an, a Fenwei Plain basin city, with PMF source apportionment and toxic-equivalent concentrations. Wintertime PAHs are lower than historical levels under the clean-air actions; biomass-burning-related mixed combustion is the largest PMF factor. During haze episodes both classes rise, but the increase is **statistically significant for OPAHs and not for parent PAHs, and OPAHs — BcdPQ in particular — dominate the toxic-equivalent burden**. Regional pollution is in-basin accumulation plus short-range transport from northern Shaanxi industry, modulated by synoptic pressure pattern. The finding that matters: a control programme that lowers parent PAHs need not lower the oxygenated derivatives that carry the toxicity, and routine monitoring measures the former.

doi: 10.1007/s10653-026-03449-5

Xu L et al. 2026 J Environ Sci (China)*Marine aging compacts soot and thickens its coating*

TEM single-particle analysis of soot collected over the Bohai Sea and the Northern and Southern Yellow Sea after outflow from East Asia. **Over 98% of soot-containing particles are coated**, mean mixing-state index **0.83**. Fractal dimension rises along the transport path — 1.84 ± 0.05 (N. Yellow Sea), 1.90 ± 0.08 (Bohai), 1.96 ± 0.07 (S. Yellow Sea) — indicating progressive structural compaction, while D_p/D_{core} of 3.9–5.3 exceeds the 3.54 typical of continental regional transport. Compact, uniformly coated, thickly shelled soot absorbs substantially more than the lacy fresh particle that optical models assume. Any absorption-based black-carbon instrument calibrated on fresh soot will misreport aged marine air, and this paper quantifies by how much the morphology has moved.

doi: 10.1016/j.jes.2026.01.047

Xu X et al. 2026 J Environ Sci (China)*Ammonia enhances isoprene SOA through gas-phase cluster formation*

Smog-chamber series on isoprene photooxidation by OH with and without NH₃, with nitrate-CIMS for the gas phase and HR-TOF-AMS for the particle phase. NH₃ increased SOA mass and accelerated isoprene oxidation. The mechanism is not the expected particle-phase acid-base reaction: CIMS shows NH₃ reacting homogeneously in the *gas* phase with organic acids and low-volatility oxygenated organic molecules to form extremely and ultra-low-volatility **NH₃-OOM clusters**, which then nucleate and condense. Quantum chemistry confirms spontaneous hydrogen-bonded association with RCOOH, R-OOH and R-OH groups. **Cluster formation accounts for 78% of the SOA enhancement**. Chamber conditions with elevated NH₃; the relevance is to agricultural and livestock regions where NH₃ and biogenic VOC co-occur.

doi: 10.1016/j.jes.2026.01.041

Yang & Li 2026 Environ Sci Technol*Modelling the whole indoor-radon distribution, not the area mean*

Over **6 million** short-term radon measurements across the contiguous US, 2000–2021, used to predict *percentiles* at ZIP Code Tabulation Area level and then to model the within-ZCTA distribution rather than summarising each area by a mean or median. The consequence is quantitative: roughly **half of US ZCTAs have at least one in five households** above policy-relevant thresholds — a figure an area-average surface cannot produce, because the exceeding fraction is a property of the spread. Same argument applies to any PM exposure surface used for threshold-crossing statistics: modelling the mean and modelling the tail are different tasks. Radon test data are self-selected (homeowners who chose to test, often at point of sale), which biases both the level and the spread in unknown directions.

doi: 10.1021/acs.est.6c01333

Yang et al. 2026c Atmos Chem Phys*Complexation strength between organic carbon and transition metal ions dominates the photochemical conversion of SO₂ to sulfates.*

Photooxidation of SO₂ on coal-combustion organic carbon with and without Fe³⁺, Cu²⁺ and Mn²⁺. Steady-state SO₂ uptake coefficients spanned (0.39–22.24)e-6 and sulfate masses (2.36–26.05)e-3 mg. Fe³⁺ was antagonistic while Cu²⁺ and Mn²⁺ enhanced both uptake and sulfate yield; absorbance loss and fluorescence quenching confirmed metal-chromophore complexation, strongest for Fe³⁺, and sulfate mass rose linearly with hydroxyl-radical intensity. The Fe³⁺ result inverts the usual expectation that iron catalyses sulfate formation, and the proposed reason - iron binds the chromophores that generate the radicals - is testable. Films under laboratory irradiation are not ambient particles and no atmospherically scaled yields are offered. Secondary sulfate is the fraction optical sensors mis-size worst under humidity, which is why formation pathways matter here.

doi: 10.5194/acp-26-13387-2026

Youn et al. 2026 Atmos Pollut Res*Seasonal distributions and source identification of water-soluble arsenic species of PM_{2.5} in the urban atmosphere of Northeast Asia*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 24 Sep with no abstract; Crossref, Europe PMC, PubMed fetch and Semantic Scholar all returned nothing. The title states seasonal distributions and source identification of water-soluble arsenic species in urban PM_{2.5}. Speciation matters because As(III)/As(V) toxicity differs; low relevance to sensing. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.apr.2026.103207

Yu H et al. 2026 Environ Sci Technol*Sea-land breeze recirculation of black carbon in Qingdao*

A full year of online BC (2021) split into fossil-fuel and biomass-burning fractions by the Aethalometer model, then attributed to meteorology by random forest and trajectory analysis. Annual mean **1.78 ± 1.29 μg/m³**; fossil fuel **88.2%** of mass, biomass burning rising to **19% in winter**. The two fractions respond differently — BC_{ff} to wind-driven ventilation and transport, BC_{bb} to low temperature — and BC enhancement under sea-land breeze peaks at **2.0–3.5 m/s**, with diurnal persistence and directional shifts consistent with return transport and reaccumulation. Coastal siting guidance follows directly: a single coastal monitor under SLB conditions is sampling a recirculated air mass, not a fresh one. Aethalometer-model source splitting depends on assumed absorption Ångström exponents, which is the dominant uncertainty here.

doi: 10.1021/acs.est.6c10060

Yu et al. 2026 Atmos Chem Phys

Continental pollutants modulate organic nitrogen and light absorption of marine organic aerosols

Shipboard sampling over the Yellow and Bohai Seas across three seasons. Aerosol organic nitrogen averaged $0.35 \pm 0.25 \mu\text{g N m}^{-3}$, 4–60% of total nitrogen, highest in autumn. Anthropogenic secondary pollutants — aged biomass burning and secondary nitrate — supplied **36–76% of ON**, 46–83% of the water-soluble and 39–89% of the water-insoluble fraction; in spring dust accounted for 55% of ON and above 80% during a dust episode, while summer carried a marine biogenic component. Organic-aerosol light absorption rose with both ON concentration and ON/OC ratio, so absorption capability per unit carbon increased with nitrogen enrichment rather than tracking mass. Three cruises give no annual coverage and apportionment is receptor modelling on a small campaign. The sensing consequence is direct: an optical instrument calibrated on one absorption regime is misapplied when composition shifts.

doi: 10.5194/acp-26-13175-2026

Zahid & Morris 2026 Preprints.org (preprint)

Demand or Congestion? Disentangling Event-Related Emissions Using a Directional Control on I-15.

One-minute freeway detector data from the LATTICE platform during the 2026 Electric Daisy Carnival. Northbound (arrival) lost 9.5 mph over a five-hour window; southbound lost nothing, yet CO₂ rose almost equally in both directions (+11.6% vs +11.8%), so the headline increase is weekend demand, not congestion. Using the uncongested direction as control in a difference-in-differences, only PM_{2.5} kept a substantial congestion residual (+8.9 percentage points), and residuals ranked by low-speed emission sensitivity (Spearman rho 1.00, n = 4). Emissions are modelled from speed and volume, not measured, so the PM_{2.5} result inherits the emission model's low-speed factors; n = 4 pollutants makes rho uninformative. Preprint. The control-corridor logic transfers directly to near-road sensor pairs.

doi: 10.20944/preprints202609.1747.v1

Zero-filling DLM bias 2026 Int J Epidemiol

Gestational misalignment with fixed exposure windows: the potential dangers of zero-filling in distributed lag models

Distributed lag models require equal-length exposure histories, and perinatal analyses routinely pad short gestations with zeroes to the maximum length. The authors show analytically that zero-filling manufactures an association between gestational age and late-pregnancy exposure, introducing confounding absent from the observed data. In simulation, zero-filling produced the largest bias, worst coverage and highest RMSE of all approaches tested; applied to a Colorado birth cohort it produced implausibly strong late-pregnancy effects that adjustment for gestational age attenuated. Carrying the last pre-birth value forward, using observed post-birth exposures, or truncating at 37 weeks eliminated the bias. This invalidates a common analysis choice across the critical-window literature, and the fix is trivial. Simulations are the authors' own generative model.

doi: 10.1093/ije/dyag162

Zhang 2026 medRxiv (preprint)

Within-Year PM_{2.5} Exposure Structure Provides Mortality-Predictive Information Beyond the Annual Mean.

Preprint. Daily modelled PM_{2.5} was linked to age-adjusted mortality for 27,289 county-years across 3,064 US counties, 2003–2011. PMRI is the largest k for which at least k days reached $k \mu\text{g m}^{-3}$ — a scale-free summary of how a year's concentrations are distributed across days. Under county-grouped 10-fold cross-validation, adding PMRI to the annual mean raised held-out R² from 0.0802 to 0.1391 (delta R² 0.0589, 95% CI 0.0499–0.0677) and reduced RMSE and MAE, holding across geographic, temporal and all six cause-specific splits. It is ecological, unadjusted for county composition, and an R² of 0.14 remains weak prediction. The relevance is direct: daily-resolution networks carry mortality-relevant structure that annual means discard.

doi: 10.64898/2026.09.20.26363510

Zhang N et al. 2026 J Environ Sci (China)

Aging, composition and light absorption of nitrogen-containing organics in PM₁

59 Shanghai PM₁ samples analysed by UHPLC-Orbitrap MS. Nitrogen-containing organics are **31% of detected species in ESI– and 64% in ESI+**. Aging reduces molecular diversity: CHON– compounds fall in both number and mass fraction with aging degree, and carboxylic-rich alicyclic molecules drop from **37.7% to 21.2%** of the CHON– contribution. Mass absorption efficiency at 365 nm declines with aging alongside the Bep/(Bep+Bap) ratio and relative humidity, pointing to aqueous-phase oxidation as the dominant mechanism, with CHON– and CHN+ identified as the principal chromophores. Semi-quantitative by design — ESI response factors are compound-specific and uncalibrated — so the fractions are ranks, not concentrations.

doi: 10.1016/j.jes.2025.12.003

Zhao C et al. 2026 Environ Pollut

Mass fell a third; toxicity per microgram did not move

Sixteen priority PM_{2.5}-bound PAHs in Kunming, compared between a higher- and a lower-PM_{2.5} stage. Raw declines: PM_{2.5} –**32.7%**, ΣPAHs –**27.4%**, BaP_{eq} –**31.4%**; covariate-adjusted, –**28.8%** and –**33.5%**. But the particle-normalised intensity BaP_{eq}/PM_{2.5} **changed by –7.6%** (*p*=0.274) and BaP_{eq}/ΣPAH by only –6.5% (*p*=0.026). Four-to-six-ring species held above **81%** of ΣPAH and BaP plus dibenzo[a,h]anthracene supplied over **82%** of BaP_{eq} in both stages; **40.3%** of simulated adult incremental lifetime cancer risks still exceeded 10^{–6}. PMF patterns are read conservatively as source-related rather than as resolved sources, which is the right caution. The inference is threatened by the two-stage design: “stage” bundles season, meteorology and policy, and no counterfactual separates them.

doi: 10.1016/j.envpol.2026.129094

Zhao H et al. 2026 Environ Monit Assess

Lead isotopes resolve dustfall sources across urban functional areas

Ten trace elements (V, Mn, Co, Cr, Ni, Cu, Zn, As, Cd, Pb) in dustfall from **16 sites** across residential, busy-road, park and industrial areas of Baoding, with enrichment factors, geo-accumulation index and Pb isotope fingerprinting. Cu, Pb, Zn and Cd reach considerable pollution levels; the spatial ranking is **residential > busy road > industrial > park**, which is not the ordering most source-apportionment intuition predicts and the paper does not fully explain. Total non-carcinogenic risk stayed below threshold but **total carcinogenic risk exceeded the acceptable threshold** in several areas, worst on busy roads and in residential zones. Pb isotopes attribute **over 50%** to a combined coal-combustion plus traffic-emission group in most areas — combined, i.e. the isotope system does not separate the two here, which is the honest limit of the method in a city that burns coal and drives cars.

doi: 10.1007/s10661-026-15859-0

Zhao et al. 2026 Environ Sci Technol*Unveiling the importance of organic hydroxy acids-related autoxidation pathway in isoprene secondary organic aerosol formation*

Year-long PM_{2.5} measurements at **nine Pearl River Delta sites** quantifying six isoprene-derived highly oxygenated molecules (iHOMs). Annual mean **2.69 ± 3.52 ng/m³**, higher in summer and autumn, dominated by methyltartaric acids (71%). Particle-phase tracer shares for the HOM, HMML and IEPOX pathways were 10%, 11% and 79%; box-modelled gas-phase intermediates gave 25%, 22% and 53%, so IEPOX dominates isoprene SOA in the polluted PRD with the other two non-negligible in winter. Pathway partitioning tracked the ISOPO₂ isomer split. The standard deviation exceeds the mean, so the annual average describes episodes rather than a typical day, and tracer-to-SOA scaling relies on chamber yields measured well above ambient. These are **nanogram-level organics** — the composition layer no optical network sees.

doi: 10.1021/acs.est.6c06525

Zhao et al. 2026 Atmosphere (Basel)*Spatial-Temporal Variability of PM_{2.5} and Ozone Pollution in the Huaihai Economic Zone of China.*

PM_{2.5} and O₃ concentrations (2000-2023) were set against MEIC inventory emissions (2000-2020) for the Huaihai Economic Zone. PM_{2.5} emissions and concentrations fell after about 2013 while O₃ rose. The raw monthly correlation between PM_{2.5} emissions and O₃ was strongly negative ($r = -0.773$) but collapsed to $r = 0.137$ after adjustment for meteorology, season and trend - the headline is that the unadjusted relationship is mostly shared seasonality. The abstract does not say whether pre-2013 PM_{2.5} is measured or reanalysis (China's national network began in 2013), and correlation on a regional inventory cannot identify chemical regime. Relevance: low; a caution that emission-concentration correlations without meteorological adjustment mislead.

doi: 10.3390/atmos17100929

Zhou et al. 2026 GeoHealth*Can Electric Vehicles Reduce Summertime Air Pollution Exposure Inequalities in Metro Atlanta? A High-Resolution Analysis*

1-km CMAQ simulations of 30% light- and heavy-duty EV adoption in metro Atlanta for summer. Domain-wide means fell 0.22 ppb NO₂, 0.94 ppb MDA8 O₃ and 0.06 μg m⁻³ PM_{2.5}, with local NO₂ and PM_{2.5} increases near power plants; the largest reductions fell in communities with higher Asian, Hispanic and Black shares, though disparities persisted. The PM_{2.5} change is well inside CMAQ's typical error and a single season. The most transferable result is methodological: 12-km runs underestimated minority exposure by up to 25% for NO₂ and 10% for PM_{2.5} relative to 1 km. Relevance: that resolution bias is a quantitative argument for dense sensor networks in environmental-justice assessment, where coarse grids systematically understate the disparity.

doi: 10.1029/2025gh001654

Zhu et al. 2026 Atmos Chem Phys*Stabilized Criegee intermediates and the sensitivity of sulfuric acid formation*

Criegee reactions in MCM v3.3.1 updated to current IUPAC recommendations, driven through a large AtChem box-model ensemble, with interpretable XGBoost surrogates used to attribute the sCI-mediated SO₂ loss rate and its fractional share to controlling factors. The sCI loss rate is governed chiefly by **relative humidity and isoprene**; the drivers of the sCI fraction split diurnally — alkene speciation, O₃ and alkene share of VOCs by day, alkene speciation, RH and NO_x by night. Under high-sCI-fraction regimes, the positive response of total SO₂ oxidation to O₃, VOCs and alkene fraction is *amplified*, which a summer episode at Wuhai supports. Policy consequence: where alkene ozonolysis is active, **cutting VOCs controls sulfuric acid formation more effectively than cutting SO₂ alone** — and sulfuric acid is the new-particle-formation precursor. Box model with a machine-learned surrogate; the surrogate inherits every mechanistic assumption underneath it.

doi: 10.5194/acp-26-12479-2026

da Silva et al. 2026 Toxics*Chemical Regimes and PMF-Resolved Sources Linked to Acute PM₁₀ Ecotoxicity in Luanda, Angola.*

116 daily PM₁₀ samples from Luanda (27 Jun-5 Nov 2023) were speciated, clustered by centred log-ratio composition and apportioned by PMF, with aqueous extracts tested in the Microtox Aliivibrio fischeri assay. TU15 ranged 0.84-8.54; only 3 samples were below 1. A metal-enriched regime (Zn, Pb, Cd) had the highest mean TU15 (4.22), matching a PMF non-ferrous metallurgy / scrap-burning factor that correlated most with toxicity (rho 0.63 for TU15, 0.78 for dTU), then a Ni-rich metallurgical factor (rho 0.51); marine aerosol was inversely associated. Bacterial bioluminescence of water extracts is not a human-relevant endpoint and misses insoluble fractions; correlations are univariate. Relevance: one of very few speciated PM₁₀ records from sub-Saharan Africa - baseline for any sensor network sited there.

doi: 10.3390/toxics14090773

Cardiovascular & metabolic**46 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (46 OF 46 RECORDS IN THIS SUBTOPIC)

Aysin et al. 2026 Metabolites*Effects of Traffic-Related Diesel Exhaust Exposure and Moderate Exercise on Obesity, Metabolic Dysfunction, and Inflammatory Responses in Rats.*

48 female Wistar rats randomised to six groups: diesel exhaust at a mean 300 μg m⁻³ PM_{2.5} for 2 or 4 h/day over eight weeks, with or without 30 min/day moderate treadmill exercise. Exhaust raised body weight, BMI, Lee index, visceral adiposity index, LDL, triglycerides, glucose, leptin, CRP, IL-1b, IL-6 and TNF-α and lowered HDL and insulin (all $p < 0.001$), most strongly at 4 h/day; exercise attenuated but did not abolish these changes. Only p-values are reported - no effect sizes - and whole exhaust cannot separate particles from NO₂ and CO. 300 μg m⁻³ is a high-traffic peak, not a typical mean. Relevance: the exposure-by-activity interaction argues for pairing wearable PM with accelerometry, since ventilation rate rises with exercise.

doi: 10.3390/metabo16090682

Bikdeli et al. 2026 Eur Heart J Cardiovasc Pharmacother*Coated or chewable Aspirin and a hybrid air pollution mitigation strategy in patients with atherosclerotic disease: Rationale and Design of the COATED-AIR Randomized Trial.*

Design paper for a 2x2 factorial randomised trial in patients with established ASCVD on low-dose aspirin: double-blind enteric-coated versus chewable aspirin 81 mg, and a hybrid air-pollution mitigation package versus usual care. The package combines an educational flashcard, SMS air-quality alerts, and on high-pollution days advice to limit outdoor activity or wear a KN95 mask and to raise citrus intake. No results. This is among the first outcome-powered trials of individual-level pollution mitigation, which is why it is tier B despite having no data. The weakness is built in: a bundled behavioural intervention cannot attribute any benefit to exposure reduction unless adherence and personal exposure are measured, which the abstract does not describe. The alert signal presumably comes from a monitoring network, so alert latency and spatial resolution are unreported design parameters.

doi: 10.1093/ehjcvp/pvag071

CHARLS coexposure 2026 Int Arch Occup Environ Health*Indoor solid fuels, outdoor PM and heatwave vs. cardiometabolic multimorbidity*

Five CHARLS waves 2011–2020, four disease cohorts (CMM $n = 10,273$; diabetes 9,893; heart disease 9,536; stroke 10,466), time-varying Cox plus a cumulative risk index and quantile g-computation. Single-exposure: solid fuel HR **1.118 (1.016–1.231)** for heart disease and **1.149 (1.007–1.310)** for stroke; PM₁, PM_{2.5} and PM₁₀ were the most consistent risk factors ($P < 0.001$). Mixture ORs: CMM **1.650 (1.375–1.979)**, diabetes **1.675 (1.475–1.903)**, with solid fuel carrying the largest weight. Fuel use is self-reported and correlates with income, education and rurality, so the weight assigned to it is the estimate most exposed to residual confounding. Outdoor PM is modelled, not measured, at residence.

doi: 10.1007/s00420-026-02232-4

COPDB Beijing 2026 J Hazard Mater*Associations between air pollution and cardiorespiratory function and metabolic indicators: New insights into mixture exposures.*

112 older adults in the Beijing COPDB study across 410 follow-up visits. Single-pollutant BKMR flagged SO₂ and UFPs for fasting blood glucose, PM_{2.5} Ca²⁺ for heart rate, and O₃ and UFPs for reduced peak expiratory flow. The novel step is removing those pollutants and re-fitting: the residual mixture associations got stronger, with the gaseous group over a 14-day window tied to a 19.1% FBG increase (10.8–27.4%) and a 16.0% PEF decrease (8.7–23.3%). That is a real warning about single-pollutant screening, but the design invites it - dropping the strongest predictors and finding the rest matter more is partly a collinearity artefact, and $n=112$ with 410 visits is thin for pairwise interactions across a dozen-plus exposures. Component-resolved measurement is the strength.

doi: 10.1016/j.jhazmat.2026.143550

California PM_{2.5} diabetes mortality 2026 Front Public Health*Associations of PM_{2.5} exposure with diabetes-related mortality for California residents*

60,824 diabetes-related deaths (ICD-10 E11, primary or contributory) and 119,053 matched controls from California vital records, 2010–2021. Per 2.65 $\mu\text{g m}^{-3}$ IQR increase in the one-year rolling mean PM_{2.5}, conditional logistic regression gave OR 1.18 (95% CI 1.15–1.22), rising to 1.21 (1.17–1.25) after NO₂ adjustment - the association strengthening on co-pollutant adjustment is itself a signal that NO₂ is acting as a negative confounder here. The design's weakness is structural: controls are drawn from the same mortality file and are people who had not yet died, so both arms are decedents and the contrast is timing of death, not death versus survival. The \$31.2 million per 100,000 avoided-loss figure rests on that design.

doi: 10.3389/fpubh.2026.1903823

Campagna et al. 2026 Front Public Health*Tracking biochemical indicators of air pollution exposure in acute cardiorespiratory emergencies: a scoping review.*

A PCC-framework scoping review (PubMed, Scopus, Web of Science, Embase, CINAHL; 2000–2025) of studies measuring biochemical markers in emergency-department presentations after short-term outdoor air pollution. Twelve studies qualified; PM_{2.5}, PM₁₀ and NO₂ were most often linked to CRP, IL-6, D-dimer and troponins within hours to days. Twelve heterogeneous studies with no quality appraisal or pooling cannot support any causal or quantitative claim, and ED biomarkers are dominated by the presenting illness. Relevance: low; the stated need for standardised exposure windows is where hourly sensor data would add over daily station means.

doi: 10.3389/fpubh.2026.1892038

Campailla et al. 2026 Int J Cardiol Cardiovasc Risk Prev*Ambient particulate matter exposure and acute type A aortic dissection: A retrospective time-stratified case-crossover study in northern Italy.*

Time-stratified case-crossover of 40 Stanford type A acute aortic dissections at a single northern-Italian centre, 2018–2022, referent days matched on weekday within month and year, temperature-adjusted. A doubling of same-day PM_{2.5} gave adjusted OR 4.78 (95% CI 1.61–14.29); 30-day PM_{2.5} and both PM₁₀ windows pointed the same way. Forty cases is the whole story - the interval spans nearly an order of magnitude, the authors label the estimates exploratory, and a single urban-background monitor stands in for everyone in the catchment. Reported here because it is a rare European estimate for a high-lethality outcome, not because the magnitude is credible. This is exactly the low-count, high-misclassification setting where a dense network changes the variance more than the point estimate.

doi: 10.1016/j.ijcrp.2026.200708

Chen N et al. 2026 Eur J Prev Cardiol*Elemental carbon leads the PM_{2.5} components on venous thromboembolism, barely*

483,610 UK Biobank participants (mean age 56.5, 54.3% female), median 11.83 years, 9,988 incident VTE from hospital and death records. Per-IQR hazard ratios: elemental carbon **1.052 (1.005–1.101)**, sulfate **1.064 (1.001–1.130)**, ammonium **1.090 (1.010–1.176)**; organic matter and nitrate not significant. Residential greenness **0.944 (0.916–0.972)**, partly mediated by elemental carbon and ammonium. Heatwave hazard ratios span **1.149 (1.137–1.161)** to **1.407 (1.396–1.417)** across definitions. Two cautions. The three positive component intervals all have lower bounds within 0.01 of unity at 3×3 km resolution, so component ordering is not established. And the heatwave intervals are roughly three times narrower on the same events because that exposure is daily rather than annual — a real difference in information, not a real difference in certainty about air pollution.

doi: 10.1093/eurjpc/zwag465

Chiang et al. 2026 J Am Heart Assoc*Ambient carbon monoxide as a delayed trigger of acute aortic dissection: a time-stratified case-crossover study in a tropical industrial metropolis*

Time-stratified case-crossover of 798 emergency-department-confirmed acute aortic dissections, 2018–2021, exposures assigned from 24 monitoring stations, conditional logistic regression across lags 0–3 adjusted for meteorology. **No pollutant was associated on the event day.** At 2-day lag each 1 ppm CO increment carried a 2.67-fold risk (95% CI 1.207–5.900) and each IQR increment of 0.19 ppm raised risk **113% (95% CI 1.075–4.239)**; PM_{2.5}, PM₁₀, NO₂, O₃ and SO₂ showed nothing at any lag. Two problems: a 1 ppm contrast is roughly five IQRs, so the headline number extrapolates past the observed range; and with six pollutants across four lags the one surviving association is not multiplicity-corrected. The PM nulls are the relevant part here — station-assigned exposure at this network density returns nulls that a denser deployment might resolve.

doi: 10.1161/jaha.126.049936

China WBC homeostasis 2026 Environ Pollut*Long-term exposure to ambient air pollution and circulating white blood cell homeostasis among adults in China: Time-window-specific associations and the mediating role of fasting blood glucose*

29,603 adults with routine health examinations at Liangshan General Hospital, 2016–2024, pollutants assigned to geocoded residences. Associations were strongest at the six-month window and were negative for every particulate pollutant: per IQR, WBC fell 0.503 $\times 10^9/\text{L}$ (95% CI 0.454–0.552) for PM_{2.5} and 0.349 (0.307–0.391) for PM₁₀, while O₃ was positively associated. Quantile g-computation put PM in the dominant negative position at six months and SO₂ at two years; fasting glucose mediated 4.97–10.98%. The direction is the problem: a fall in WBC with rising PM runs against the inflammatory-recruitment model the authors invoke, and a single-hospital examinee population selected on health-seeking is a plausible source of that inversion. Not included in the forest plot - the estimates are absolute counts, not ratios

doi: 10.1016/j.envpol.2026.129140

Constituent multiomics cardiopulmonary 2026 Environ Sci Technol*PM_{2.5} Constituents and Cardiopulmonary Dysfunctions in Older Adults: A Multiomics Approach Unveiling Molecular Pathways*

Longitudinal panel, 313 repeated measurements in older adults, personal rather than ambient PM_{2.5} with constituent speciation, screened by a linear mixed model plus three machine-learning methods. Chloride, nickel, barium and selenium emerged as the key constituents. Per IQR, chloride was associated with a 3.48% (95% CI 1.50-5.46) rise in diastolic blood pressure and nickel with a 0.09% (0.05-0.12) fall in FEV1/FVC. Mediation implicated 212 CpG sites acting on both cardiovascular and pulmonary endpoints and 7 genes spanning more than two constituent-function pairs, with 22 CpGs linked by eQTM to 14 coding genes including CERS6 and LRCH3. Limitations: 313 measurements is a small panel for a screen over this many constituents and CpGs, the constituent set is collinear so the four winners are correlates rather than isolated causes, and the mediation is cross-sectional within visit. Personal monitoring is the strength - it removes the spatial misalignment that defeats constituent studies built on central-site speciation

doi: 10.1021/acs.est.6c04206

Cui et al. 2026 Environ Sci Technol*Differential Risks of Incident Heart Failure Subtypes Associated with PM_{2.5} Chemical Constituents in China between 2006 and 2021.*

Kailuan prospective cohort of 101,521 adults followed a median 14.83 years (2006-2021) for incident heart failure by subtype. Every PM_{2.5} constituent was associated with higher risk across subtypes, more strongly for early- than late-onset HF. In mixture weighting, BC contributed 40.2-48.7% of the effect for overall, late-onset, HFrEF and HFmrEF; NH₄⁺ (28.2%) and BC (28.1%) led early-onset HF; OM (34.8%) led HFpEF. The abstract reports no per-constituent HRs, so magnitudes cannot be entered in the forest plot. Kailuan is a single-city cohort, so contrast comes from within-city variation, and constituents are highly collinear; mixture weights are unstable under that. Relevance: BC is measurable by low-cost aethalometers, so a BC-led HF signal favours adding absorption to network nodes.

doi: 10.1021/acs.est.6c02927

De Potter et al. 2026 Eur Heart J Acute Cardiovasc Care*Socioeconomic and environmental determinants of STEMI incidence and in-hospital mortality*

Nine years of Belgian national hospital discharge data, **51,544 STEMI admissions** (51.4 per 100,000 per year), with residential PM_{2.5}, NO₂, O₃, temperature, noise and greenspace from high-resolution models. A time-stratified case-crossover handled the acute question and multivariable logistic regression the mortality question. Per IQR, same-day: PM_{2.5} **OR 1.022 (1.008–1.035)**, NO₂ **OR 1.044 (1.025–1.063)**; O₃ peaked over lag 0–2 at **1.063 (1.034–1.090)**. In-hospital mortality was 11.8% and tracked *socioeconomic* variables — lower individual and area income, retirement status — rather than any pollutant. Two readings matter: NO₂ outperforming PM_{2.5} in single-pollutant models is the standard traffic-proximity confounding, and the incidence/mortality split says exposure and outcome severity are governed by different mechanisms.

doi: 10.1093/ehjacc/zuag118

Dzhambov et al. 2026 Arh Hig Rada Toksikol*Residential exposures associated with adiposity and proinflammatory markers in obese adults with metabolic syndrome: a cross-sectional study*

90 obese adults with metabolic syndrome in Bulgaria (2023-2024) had serum IL-6, TNF- α , CRP and adipokines, and adiposity measures, linked to residential NO₂, PM_{2.5}, PM₁₀, Lden noise, domestic solid-fuel burning and greenness. Per 10 $\mu\text{g m}^{-3}$ NO₂, IL-6 was 41.63% higher (95% CI 8.16-85.46), CRP 37.80% (5.89-79.32) and visceral fat 7.52% (1.30-14.12); time in nature lowered TNF- α and raised adiponectin. The PM_{2.5} and PM₁₀ associations were 'largely null, inconsistent, or occasionally protective' and are not quantified. n=90 with many exposures and outcomes and no multiplicity control; cross-sectional. Relevance: a PM null in a small, narrow-contrast sample - uninformative either way.

doi: 10.2478/aiht-2026-77-4124

Fang et al. 2026 Eur J Prev Cardiol*Lifestyle matters more where the air is worse, and most for early-onset heart failure*

233,341 UK Biobank participants, 2006–2010 baseline to 2023, with an air pollution score over NO₂, NO_x, PM₁₀, PM_{2.5} and PM_{2.5–10} crossed against a six-component lifestyle score. Additive interaction is present and larger for early onset: RERI **0.44 (0.14–0.74)** for all heart failure, **1.33 (0.03–2.62)** early-onset, **0.39 (0.08–0.70)** late-onset, with the early-versus-late contrast at $p=0.08$ — i.e. the headline age difference is not itself statistically resolved, and the early-onset interval very nearly touches zero. Read the other direction instead, where the contrast is firmer: lifestyle differences map to **3.35-fold (1.76–9.79)** larger early-onset incidence differences under high pollution against **1.52-fold (1.33–1.74)** under low. A composite pollution score cannot attribute anything to particles specifically.

doi: 10.1093/eurjpc/zwag456

Fang et al. 2026 J Thromb Haemost*Venous Thrombosis and Air Pollution: A Systematic Review and Meta-Analysis of 1.4 Million Adult Patients.*

23 studies, >1.4 million venous thrombotic events, searched to 1 July 2025. Long-term particle exposure was associated with VTE - RR 1.0804 (95% CI 1.0015-1.1656) per 10 $\mu\text{g m}^{-3}$ PM_{2.5} and 1.0369 (1.0260-1.0479) per 10 $\mu\text{g m}^{-3}$ PM₁₀ - while short-term particle exposure and every gaseous metric were null. The PM_{2.5} lower bound sits at 1.0015, so the headline association is barely separated from the null and rests on a handful of long-term cohorts; PM₁₀, with the far tighter interval, is the more defensible estimate despite being the cruder metric. The projected 98,233 events averted by meeting the 5 $\mu\text{g m}^{-3}$ WHO guideline is a linear no-threshold extrapolation well below the exposure range of the contributing studies and should be read as illustrative. The long-short contrast is the useful signal: chronic annual mean, not the daily excursion a dense sensor network resolves best, is what carries the thrombotic risk here.

doi: 10.1016/j.jtha.2026.09.022

Fleming & Fleming 2026 — metadata only Am J Med*Tire wear pollution and cardiovascular risk*

Elsevier deposited title and authorship on 18 September with no abstract in Crossref, Europe PMC core, Europe PMC fullTextXML or the OpenAlex per-work record. The title places it as a commentary linking tire-wear particle pollution to cardiovascular risk; it asserts no finding here. Logged because the particle class matters for measurement: **non-exhaust emissions now dominate traffic PM in fleets with modern aftertreatment**, and tire wear differs from combustion PM in both size distribution and composition — a combination that mass-only optical sensors and exhaust-derived emission factors each mis-handle. Carried forward for pickup when an abstract appears.

doi: 10.1016/j.amjmed.2026.09.003

Hangzhou AAD PM components 2026 J Am Heart Assoc*Which PM_{2.5} component triggers acute aortic dissection, and by how much*

Time-stratified case-crossover on Hangzhou acute aortic dissections, 2015–2023, with conditional logistic regression for single components and weighted quantile sum for the mixture. At **lag 0–1 days**, interquartile-range increases in black carbon, organic matter and sulfate were associated with **42% (OR 1.42, 1.15–1.75)**, 35% (1.35, 1.10–1.64) and 36% (1.36, 1.09–1.69) higher odds of Stanford *type A* dissection; type B associations were positive but not significant. In the joint model **black carbon carried more than half the weight**. Age, smoking, alcohol use and hypertension modified the associations. The case-crossover design self-controls for time-invariant confounding, which is its strength; the weaknesses are that component concentrations are strongly collinear (WQS mitigates but does not resolve this), exposure is assigned from ambient monitors rather than personally, and a single city's aerosol composition determines which component appears to dominate.

doi: 10.1161/jaha.126.049278

Hua et al. 2026 Environ Health (Wash)*Heterogeneous Risks of Chemical Components and Sources in Size-Resolved Particulate Matter on Ischemic and Hemorrhagic Stroke.*

Time-series analysis of 58,763 ischaemic and 2,588 haemorrhagic stroke hospitalisations against 41 PM components and six source factors resolved across nine size fractions below 10 μm - the finest joint size-composition-source contrast this watch has carried. Ischaemic stroke tracked components below 1.1 μm and above 3.3 μm ; haemorrhagic stroke tracked As, Cu, Fe, Mn, Ni, Pb and PAHs. Coal combustion gave RR 1.05 (95% CI 1.01–1.10) per IQR for ischaemic stroke and industrial emissions RR 1.18 (1.00–1.41) for haemorrhagic, whose lower bound sits on the null with 2,588 events. Source factors from receptor modelling are collinear by construction, so component-specific attribution is softer than the size resolution implies. Size-resolved sensing, not PM_{2.5} mass, is what this design needs at scale.

doi: 10.1021/envhealth.6c00019

Huang et al. 2026 Zhonghua Liu Xing Bing Xue Za Zhi*Association between air pollution exposure and metabolic dysfunction-associated steatotic liver disease in adults in China.*

23,806 adults, China Kadoorie Biobank third resurvey, mean age 65.3 (SD 9.1); 13,428 with MASLD of whom 2,457 had significant fibrosis. Per 1 SD: PM_{2.5} OR 1.27 (95% CI 1.05–1.53), PM₁₀ 1.77 (1.36–2.31), NO₂ 1.20 (1.08–1.33), warm-season O₃ 1.53 (1.20–1.95). One unit of the mixture score gave 40% higher MASLD odds (18–65%) and 42% higher fibrosis odds (1–100%) - the fibrosis interval touching unity makes that second estimate uninformative. PM_{2.5} was non-linear (P=0.007), flat at low concentration and steepening above it. The design is cross-sectional, so reverse causation from residential sorting by health status is unaddressed, and the reported PAFs of 25.4–66.4% are causal-language attached to a single-timepoint contrast and should not be quoted as avoidable burden.

doi: 10.3760/cma.j.cn112338-20251103-00783

Hyperuricemia low-concentration Korea 2026 Clin Rheumatol*Long-term exposure to ambient air pollution and an increased risk of hyperuricemia.*

Health-screening series of 20,117 Korean adults (2015–2019), 15.3% hyperuricaemic. All five pollutants were associated with hyperuricaemia per IQR: PM_{2.5} OR 1.06 (95% CI 1.01–1.10), PM₁₀ 1.15 (1.08–1.21), NO₂ 1.07 (1.00–1.14), SO₂ 1.13 (1.07–1.19), CO 1.06 (1.01–1.11), with adjustment for eGFR so the signal is not simply renal clearance. PM₁₀ outperforming PM_{2.5} is unusual and points at a coarse-fraction or resuspended-dust component rather than combustion aerosol. Cross-sectional design with a single serum urate measurement cannot establish temporality, and CMAQ assignment at address gives no indoor or commuting contribution - the exposure error that street-scale sensing exists to reduce.

doi: 10.1007/s10067-026-08398-z

Jeong et al. 2026 Environ Int*Joint effects of air pollution and temperature on glycemia: An exposome analysis in European cohorts.*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors (Jeong, Edstorp, Lovison, Pickford and colleagues) and DOI on 23 Sep with no abstract, and Crossref, Europe PMC and PubMed all returned nothing. The title states a joint air-pollution-temperature exposome analysis of glycaemia across European cohorts, which is a design this watch tracks - temperature is the covariate most likely to be doing the work in short-term PM cardiometabolic associations. Nothing about the pollutants, exposure model, cohorts, sample size or estimates can be stated from the deposit, and inferring a method from a title is not something this watch does. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.envint.2026.110538

Jiang Z et al. 2026 Atmos Environ*Multipollutant exposure and cardio-renal-metabolic multimorbidity trajectories — metadata only*

No abstract was deposited to Crossref, Europe PMC or Semantic Scholar at index time. Design, cohort, exposure model, sample size and setting are all unknown and are recorded as such rather than inferred from the title. It is the **32nd Atmospheric Environment** record to enter this corpus abstract-free since 1 August, which makes this a chronic publisher deposit gap rather than an incident; the DOI is queued for the weekly re-check, and the record occupies a *Cardiovascular & metabolic* slot on the title's face only.

doi: 10.1016/j.atmosenv.2026.122340

Jin et al. 2026 Neuroepidemiology*Ambient air pollution and stroke in two population-based cohorts: roles of systemic inflammation and polygenic risk.*

Two separate analyses. CHARLS 2015 cross-section (n=21,095): prevalent self-reported stroke OR 1.26 (95% CI 1.04–1.53) for PM_{2.5} and 1.31 (1.08–1.59) for PM₁₀, increments not in the abstract, with positive multiplicative interaction with hs-CRP. UK Biobank (n=368,644): incident ischaemic stroke HR 1.29 (1.15–1.45) per 5 $\mu\text{g m}^{-3}$ PM_{2.5}; high PRS plus high PM_{2.5} HR 1.44 (1.32–1.57), without a formal interaction test. The UKB estimate is large for a 5 $\mu\text{g m}^{-3}$ contrast in a low-exposure cohort and conflicts with Wang et al. today (null for stroke per IQR), which points to specification sensitivity. Relevance: moderate - inflammation-stratified susceptibility argues for better exposure resolution in high-CRP groups.

doi: 10.1159/med/adwag036

Kriti et al. 2026 Front Public Health*Ecological associations between ambient air pollution, gut microbiome composition, and metabolic markers in urban populations of central India.*

Cross-sectional ecological study of 95 adults from three Bhopal localities: station PM_{2.5}, PM₁₀, NO_x and SO₂ assigned by locality, T2D markers, and full-length 16S rRNA gut profiles. NO_x correlated positively with glycaemic indices; PM_{2.5} and PM₁₀ showed inconsistent, non-directional associations; community structure was broadly similar and most genus differences did not survive multiple-testing correction. With three exposure levels shared by all participants in a locality, the design cannot separate pollution from anything else that differs between neighbourhoods - the authors call it hypothesis-generating. Relevance: low; a textbook case where personal or dense sensor exposure would be the minimum to make the question answerable.

doi: 10.3389/fpubh.2026.1834285

Li et al. 2026 Zhonghua Liu Xing Bing Xue Za Zhi

Joint effects of short term exposure to heat and ozone on circulatory disease mortality in Beijing-Tianjin-Hebei region and surrounding areas.

59 counties, warm seasons 2013–18, median 4 circulatory deaths per county-day. Independently, each $10 \mu\text{g m}^{-3}$ of O_3 raised circulatory mortality 0.16% (95% CI 0.02–0.31%) and a P75-to-P99 temperature rise raised it 21.43% (16.32–26.76%). The interaction is the finding: heat-associated mortality rose from 9.93% (2.52–17.89%) at $65 \mu\text{g m}^{-3}$ O_3 to 27.17% (20.48–34.23%) at $214 \mu\text{g m}^{-3}$, and the O_3 coefficient only became significant above roughly the 89th temperature percentile, peaking at 0.92% circulatory and 1.26% cerebrovascular at P99. Ozone is not PM, but the structure transfers directly: a pollutant coefficient estimated as a main effect is an average over a temperature distribution and will be wrong at both tails. Four deaths per county-day is thin, and the counties are not independent - regional synoptic episodes drive heat and O_3 together, so the product term partly encodes shared meteorology rather than biological synergy.

doi: 10.3760/cma.j.cn112338-20260118-00051

Lin et al. 2026 Research Square (preprint)

Joint association of $\text{PM}_{2.5}$ and AIP with advanced Cardiovascular-Kidney-Metabolic syndrome in middle-aged and older Chinese adults: a national cohort study.

CHARLS 2011–2015 participants classified by median $\text{PM}_{2.5}$ and median atherogenic index of plasma, logistic regression for advanced (stage 3–4) cardiovascular-kidney-metabolic syndrome. High AIP was strongly associated; $\text{PM}_{2.5}$ alone showed no positive association; joint high- $\text{PM}_{2.5}$ and high-AIP exposure raised risk, with no additive or multiplicative interaction. Effect sizes and n are not in the abstract. The analysis is a median split of both exposures, so the 'joint' effect is almost entirely the AIP effect restated, and absence of interaction contradicts the framing. Preprint, not peer-reviewed. Carried for the null $\text{PM}_{2.5}$ main effect, which is informative in a cohort with a wide exposure range, and as a caution against dichotomising a continuous exposure that sensor and satellite products now resolve finely.

doi: 10.21203/rs.3.rs-10585989/v1

Lin et al. 2026 Front Public Health

Short-term lagged exposure to ambient air pollutants and hospitalization costs among AMI patients: health-economic evidence from a subtropical city

Single-centre retrospective cohort of 3,650 PCI-treated AMI admissions (2018–2024) in Jiangmen, where concentrations met China Grade II. Gamma GLMs with FDR correction: SO_2 , $\text{PM}_{2.5}$, PM_{10} , NO_2 and CO were associated with higher hospitalisation cost; SO_2 largest (2.40%, ~729 CNY per log unit); $\text{PM}_{2.5}$ and PM_{10} peaked at lag 4–5; O_3 raised costs at lags 3 and 5 in 1,590 never-smokers. PM-specific percent changes and CIs are not in the abstract. Cost under DRG/DIP is a weak severity proxy, confounded by coding and length of stay, and city-wide monitors give no spatial contrast. Relevance: low; illustrates why lag structures need higher-resolution exposure.

doi: 10.3389/fpubh.2026.1939605

Pan SG et al. 2026 Front Public Health

Ozone, not PM, survives two-pollutant adjustment against cardiac arrest — in one city-year

7,996 out-of-hospital cardiac arrests from the Tianjin prehospital emergency system, July 2023–June 2024 (mean 21.9/day), with a quasi-Poisson DLNM over lag 0–14 for temperature and lag0–1 single- and two-pollutant models for $\text{PM}_{2.5}$, PM_{10} , NO_2 , SO_2 , CO and O_3 . **Ozone was the only pollutant consistently associated:** per-IQR lag0–1, +15.3% overall, men **RR 1.22 (1.08–1.39)**, women **1.22 (1.04–1.43)**. The temperature result is reported with unusual honesty — a monotonic cumulative risk curve, minimum-mortality temperature -1.9°C , apparent heat peak at lag 1 (RR ≈ 1.57 at the 95th percentile) whose **confidence intervals include unity**, and the authors say so in the abstract. The binding limit is exposure length: **one city, twelve months**, so season and pollutant are almost fully confounded and the null PM result is uninformative rather than negative.

doi: 10.3389/fpubh.2026.1871041

Pellegrino et al. 2026 Sci Total Environ

A possible mediating role of SASP in the association between environmental air pollution and incident atherosclerotic cardiovascular disease

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 24 Sep with no abstract; Crossref, Europe PMC, PubMed efetch and Semantic Scholar all returned nothing. The title states a mediation analysis of senescence-associated secretory phenotype markers between air pollution and incident atherosclerotic cardiovascular disease. Mechanistically interesting - ageing biology as a pathway - if the exposure is resolved below county level. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.scitotenv.2026.182315

Pollutant mixtures atherosclerosis 2026 Int J Mol Sci

A Perfect Storm of Pollutants: Environmental Mixtures in the Pathogenesis of Atherosclerosis

Structured narrative review of 37 studies on how co-exposure to particulate matter, metals, pesticides, PAHs and endocrine-disrupting chemicals converges on atherogenesis. The mechanistic through-line is oxidative stress, endothelial dysfunction, chronic inflammation and lipid dysregulation feeding monocyte recruitment, foam-cell formation, smooth-muscle remodelling and plaque instability, with epigenetic changes in DNA methylation and non-coding RNA proposed as the long-term susceptibility carrier. The authors note that additive, synergistic and antagonistic interactions cannot be predicted from single-agent data and argue for mixture-based risk assessment. As a narrative review with no protocol, search strategy or quality appraisal, the 37 studies are a selected rather than systematic sample, and the human evidence cited is predominantly associative.

doi: 10.3390/ijms27177624

Shahriary et al. 2026 Toxicol Lett

Cardiovascular and pregnancy-related health effects of ultrafine particulate matter

Mechanistic synthesis of experimental, controlled-human-exposure and epidemiological evidence linking ultrafine PM to cardiovascular injury, with emphasis on hypertensive disorders of pregnancy. Pathways covered: pulmonary and systemic inflammation, oxidative stress, mitochondrial dysfunction, autonomic imbalance, endothelial injury and disturbed angiogenic balance, extended to maternal cardiovascular adaptation and placental vascular development. The review's honest core is its attribution problem — it states plainly that **UFPM-specific findings are hard to separate from $\text{PM}_{2.5}$, diesel exhaust particles, engineered nanoparticles and traffic-related pollution generally**, and that direct developmental evidence is largely animal. Uncertainties named: particle translocation, exposure metric, critical gestational windows. No pooled estimates. The exposure-metric point is the operational one — UFP is a number-concentration phenomenon and mass-based networks, including nearly all low-cost optical nodes, do not measure it.

doi: 10.1016/j.toxlet.2026.113203

Sichuan Basin CAA 2026 Int Arch Occup Environ Health*Temporal changes in short-term PM_{2.5}-cardiorespiratory mortality associations under the Clean Air Action*

Multi-county time-series, Sichuan Basin 2014–2022, extended two-stage design with county-specific estimates pooled and longitudinal meta-regression on annual excess risk. ER per 10 µg/m³ traced 0.7% (0.6–0.9) → 0.4% (0.3–0.6) → 1.4% (1.2–1.6) across the three policy periods, $P < 0.001$ for the pattern; scenario decomposition gives 52,831 deaths averted by concentration decline against 10,041 added by the risk change, and the late rise held in both sexes and was larger at ages 65+. The threat to the inference is that the final period is 2020–2022: pandemic-era changes in care-seeking, mortality coding and time-activity are not separable from a true slope change in this design. Still the most consequential result here — a falling mean does not imply a falling per-unit risk, and networks must resolve smaller contrasts as policy succeeds.

doi: 10.1007/s00420-026-02237-z

Sun D et al. 2026 Lancet Planet Health*China's Clean Air Act and individual-level cardiovascular risk*

China Kadoorie Biobank, $n=34,862$ CVD-free at baseline, mean age 51.3, split into intervention ($n=25,497$) and control ($n=9,365$) by the local government's PM reduction target under the 2013 Clean Air Act, with 10-year predicted CVD risk as the outcome and a difference-in-differences estimator. The intervention group's predicted risk rose 3.95% (3.18–4.72) less than control, with larger differences under stricter enforcement. Per 10 µg m⁻³, 2013–2021: PM_{2.5} 1.80 (1.34–2.27) percentage points, PM₁₀ 1.24 (0.84–1.63), O₃ 0.58 (0.33–0.83); overall pollutant change corresponded to a 6.6 percentage-point reduction in predicted risk. Two real limits: the outcome is a validated prediction, not observed incidence, and treatment assignment is a municipal policy target, so residual regional confounding is structural rather than adjustable. O₃ rose in both arms.

doi: 10.1016/j.lanplh.2026.101513

Sun et al. 2026 Front Public Health*When air gets heavy: joint PM(2.5) and ambient humidity exposure, frailty, and stroke.*

CHARLS ($n=17,197$) plus a single-centre hospital sample ($n=782$), with annual PM_{2.5} and relative humidity dichotomised at database medians. High PM_{2.5} gave HR 1.184 (95% CI 1.083–1.295) for incident stroke in CHARLS and OR 1.846 (1.247–2.733) in the hospital sample; high humidity HR 1.112; joint high-high exposure HR 1.361 (1.171–1.582). Frailty mediated 18.1% of the PM_{2.5} association. Median dichotomisation discards the exposure-response shape, the hospital arm is cross-sectional, and humidity at annual resolution is largely a climate-zone proxy confounded with region. Relevance: humidity is also the dominant bias term in optical PM sensors - an epidemiologic humidity effect estimated from sensor-derived PM would be partly an artefact, so correction must precede the health model.

doi: 10.3389/fpubh.2026.1883147

Sun et al. 2026 Indoor Air*Solid Fuel Use and Risks of All-Cause and Cardiovascular Disease Mortality in Urban Northern China: A 12-Year Cohort Study*

Retrospective cohort of 39,054 urban residents in four northern Chinese cities followed 1998–2009: 1,435 non-accidental and 677 cardiovascular deaths. Each additional year of solid-fuel heating was associated with HR 1.15 (95% CI 1.14–1.17) for non-accidental, 1.15 (1.13–1.18) for cardiovascular and 1.16 (1.12–1.20) for cerebrovascular mortality; cooking HRs were slightly higher; adults ≥ 60 were most sensitive. A 15% hazard per year of use is implausibly steep for a cumulative-exposure metric and points to confounding by socioeconomic position, housing age and co-exposure; fuel-years is self-reported and no indoor PM was measured. Relevance: moderate - the exposure contrast is real, but quantifying it needs the indoor sensors this study lacks.

doi: 10.1155/ina/9645092

UK PH metabolomic mediation 2026 Ecotoxicol Environ Saf*Air pollution, pulmonary hypertension, and how much of it runs through metabolism*

Prospective cohort of 444,346 participants free of pulmonary hypertension at baseline with NMR metabolomics, 2,328 incident cases over a 13.58-year median follow-up. PM_{2.5}, PM₁₀, NO₂ and NO_x entered individually and as a joint air pollution score. The score gave HR 1.12 per SD (95% CI 1.08–1.17); individual pollutants ran 1.06–1.93 per 10 µg/m³. Elastic-net regression produced a 105-metabolite signature dominated by lipoprotein measures, fatty acids and amino acids, itself associated with incident PH (HRs 1.12–1.24 per SD), mediating 11.74% (7.91–18.95%) of the score–PH association. The mediation estimate is the part to hold loosely: metabolites were measured once at baseline, so the mediator is cross-sectional with respect to a 13-year outcome, and the causal ordering of exposure, metabolome and disease cannot be established from that design. Residential exposure models add the usual non-differential error, which biases the 11.74% downward.

doi: 10.1016/j.ecoenv.2026.120778

Wang et al. 2026 Front Public Health*Long-term exposure to fine particulate matter constituents and incident stroke risk among middle-aged and older adults in China: a cohort study.*

CHARLS 2013–2018: 8,724 adults, 610 incident strokes over a median 5.0 years, with PM_{2.5} and five constituents from the TAP reanalysis. Per IQR, HRs were 1.55 (95% CI 1.35–1.79) for PM_{2.5}, 1.55 for sulfate, 1.37 for nitrate, 1.42 for ammonium, 1.59 for organic matter and 1.77 (1.54–2.03) for black carbon; quantile g-computation gave BC the largest mixture weight (28.6%). Susceptibility was higher above 65 and in southern China. Stroke is self-reported physician diagnosis, constituents are strongly collinear model outputs, and IQRs differ by constituent, so per-IQR HRs are not directly comparable. Still the clearest signal this week that carbonaceous particles carry the cerebrovascular risk. Relevance: argues for adding absorption (BC) channels to low-cost nodes, not just scattering mass.

doi: 10.3389/fpubh.2026.1909901

Wang et al. 2026 Bioengineering (Basel)*Long-Term Air Pollution Exposure, Cardiovascular Disease, and Cardiometabolic Risk Factors in Middle-Aged and Older Chinese Adults: A Prospective Cohort Study.*

CHARLS adults aged ≥ 45 : cross-sectional analysis of 7,420 at 2011 baseline and a prospective analysis of 8,792 followed to 2020, with CHAP-derived exposures and a GBD burden component. Incident CVD HRs were 1.09 (95% CI 1.03–1.16) for PM_{2.5}, 1.07 (1.04–1.10) for PM₁₀, 1.12 (1.00–1.24) for NO₂ and 1.08 (0.93–1.27) for O₃; exposure was also associated with hypertension, diabetes, dyslipidaemia, obesity and TyG indices. The abstract does not state the exposure increment, CVD is self-reported physician diagnosis in CHARLS, and a random forest with AUC 0.656 adds nothing. This is at least the fourth CHARLS PM-CVD paper in this watch in two months, so overlap of sample and outcome definition should be assumed. Relevance: low; single-pollutant model on 1-km reanalysis.

doi: 10.3390/bioengineering13090958

Wang et al. 2026 Front Public Health*Long-term PM_{2.5} exposure and incident heart failure in the UK Biobank: relative and absolute risk estimates.*

149,410 UK Biobank participants, median 17.6 y follow-up, 4,964 incident heart-failure and 3,795 ischaemic-stroke events. PM_{2.5} per IQR: HR 1.045 (95% CI 1.005–1.086) for heart failure; top vs bottom quartile HR 1.133 (1.037–1.238), approximately linear. No effect modification by diabetes, CKD, obesity, multimorbidity or CRP, and no pollutant was associated with ischaemic stroke. The 20-year absolute risk difference between extreme quartiles was ~ 0.2 per 100. Exposure is a single-year LUR surface over a narrow UK range, and the null stroke result contrasts with Jin et al. today in the same cohort. Relevance: moderate - reporting absolute risk alongside HR is the right norm.

doi: 10.3389/fpubh.2026.1864868

Xu et al. 2026 Front Public Health*Ambient fine particulate matter and haemorrhagic stroke in Yancheng, China*

Individual-level time-stratified case-crossover on 2,312 haemorrhagic stroke admissions across four Yancheng hospitals, 2022–2024, conditional logistic regression. Strongest at lag0: each $10 \mu\text{g m}^{-3}$ of $\text{PM}_{2.5}$ associated with a **3.24% increase (95% CI 0.55–6.00)**, with lag0 2.50% (0.11–4.95) and lag1 2.76% (0.32–5.27). The association survives adjustment for O_3 but **loses significance once SO_2 or NO_2 enters** — which is the finding that matters, because it means the $\text{PM}_{2.5}$ coefficient is not separable from co-emitted combustion products at this station density. City-average exposure assigned to residence leaves within-city gradients unmeasured, and a four-hospital catchment is not the full case population.

doi: 10.3389/fpubh.2026.1882030

Yang et al. 2026 Environ Res*Mitochondrial DNA methylation as a bridge: Linking indoor air purification to alleviated cardiovascular impairments in young adults.*

A randomised double-blind crossover of true versus sham air purification in 38 healthy young adults. Purification cut indoor $\text{PM}_{2.5}$ by 47.45%. Each $10 \mu\text{g m}^{-3}$ rise in personal $\text{PM}_{2.5}$ at lag 24–48 h was associated with mtATP6 hypomethylation, and mtDNA hypomethylation mediated 14.87% ($p=0.04$) of the $\text{PM}_{2.5}$ association with D-dimer at lag 0 h, among 16 cardiovascular indicators tested. The mediation analyses run across many CpG sites, lags and outcomes with no stated multiplicity correction, and the headline $p=0.04$ would not survive one. Short crossover periods in young adults cannot show clinical benefit. Relevance: personal $\text{PM}_{2.5}$ was the analysed exposure, so the intervention's value rests on personal-monitor accuracy - the device and its calibration are not reported in the abstract.

doi: 10.1016/j.envres.2026.125744

Yu et al. 2026 Public Health*Effect modification of greenness on the associations between air pollutants and out-of-hospital cardiac arrest: Evidence from a multi-city study*

Individual-level time-stratified case-crossover of 62,915 OHCA cases in Chinese cities, 2020–2023, with exposure at onset location. Pollutants were associated with OHCA and greenness attenuated the association: per 0.1 NDVI(250 m), the percent change in OHCA odds per $10 \mu\text{g m}^{-3}$ fell by 1.62% (95% CI 0.39–2.84) for $\text{PM}_{2.5}$ and 1.13% (0.24–2.01) for PM_{10} ; results held at 500 and 1000 m and across lags. The main-effect estimates are not in the abstract, exposure is area-level at onset rather than personal, and greenness is confounded with SES and baseline pollution. Relevance: moderate - greenness-dependent slopes argue that sensor-derived exposure contrasts should be stratified by vegetation context.

doi: 10.1016/j.puhe.2026.106515

Zhang et al. 2026 Nat Commun*Cardiopulmonary hospitalisation risks from wildfire-specific and non-wildfire $\text{PM}_{2.5}$ in 20 US states*

Self-controlled design over 20 states, 2006–19, contrasting two-year average wildfire-specific against non-wildfire $\text{PM}_{2.5}$. Per $1 \mu\text{g m}^{-3}$, wildfire $\text{PM}_{2.5}$ carried **10% (heart failure) to 16% (asthma)** higher hospitalisation risk; non-wildfire $\text{PM}_{2.5}$ **4.7% (COPD) to 8.5% (hypertension)**. Associations were stronger among minorities, metropolitan residents, the less educated and more deprived communities. The toxicity contrast at equal mass is the contribution, and it undercuts the mass-only metric every regulatory network and low-cost sensor reports. Two limits: no confidence interval is given for any of the eight headline risks, so precision cannot be judged; and wildfire-specific $\text{PM}_{2.5}$ is itself modelled, so the toxicity difference is conditional on a smoke-attribution step whose error is not propagated.

doi: 10.1038/s41467-026-76974-7

Zhang et al. 2026 Environ Health (Wash)*Air Pollution Exposure Induces Vascular Endothelial Cell Dysfunction via Elevated Circulating F2: Multiomics Insights from an Ex Vivo Biosensor Assay.*

Travel-panel design: eight college students moved from Qingdao to the substantially more polluted Shijiazhuang, with serial serum sampling, proteomics, metabolomics and an ex vivo endothelial biosensor. Prothrombin (F2) emerged as the hub node correlated with exposure and was nominated again by weighted gene coexpression analysis; metabolomics showed bile-acid and arginine-proline perturbation, and endothelial transcriptomes showed activation. $n = 8$ with no counter-travel arm is the ceiling on all of it - city is confounded with diet, water, sleep and stress, and the omics search space dwarfs the sample. Hypothesis-generating only. The design itself is cheap to strengthen: a personal-sensor-instrumented reciprocal travel panel would give an exposure axis instead of a city label.

doi: 10.1021/envhealth.5c00833

Zhongshan AMI 2026 Front Public Health*Daily changes in $\text{PM}_{2.5}$ and acute myocardial infarction hospitalisation*

Individual-level time-stratified case-crossover, 1,795 AMI admissions to one tertiary hospital in Zhongshan 2020–2024, 5,877 control days matched on weekday within calendar month; residential $\text{PM}_{2.5}$ from the CHAP 1 km product; conditional logistic regression adjusted for temperature, humidity and holidays. A $10 \mu\text{g m}^{-3}$ lag0–1 increment gave **+7.89% (0.86–15.40)** odds, with 22.58% in women, 22.16% at ages >65, 10.85% in the cold season and 17.90% for STEMI; age modified the effect ($P_{int} < 0.05$). Robust to co-pollutants except NO_2 , which is the honest weakness — traffic co-exposure is not excluded. Single-hospital catchment limits generalisability, and the 1 km modelled field is assigned to address, not measured. The transferable idea for sensing is the first-difference metric: it cancels the additive bias that dominates low-cost PM error.

doi: 10.3389/fpubh.2026.1911027

Neuro / mental health**30 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (30 OF 30 RECORDS IN THIS SUBTOPIC)

Pearson et al. 2026 J Epidemiol Community Health*Environmental exposure and risk of suicide, attempted suicide and ideation*

Forty-five studies screened across air, noise, light and water pollution, 29 pooled. Short-term $\text{PM}_{2.5}$ **OR 1.0139 (95% CI 1.0001–1.0279)** over 13 studies and PM_{10} **1.0100 (1.0007–1.0195)** over 7; long-term $\text{PM}_{2.5}$ **1.5254 (1.1351–2.0481)** over 6 and NO_2 **1.3173 (1.0641–1.6308)**. The short-term $\text{PM}_{2.5}$ lower bound sits at 1.0001, so significance is nominal. More seriously, the **38-fold gap between short- and long-term $\text{PM}_{2.5}$ point estimates is an artefact of pooling per-increment against high-versus-low categorical contrasts under one label**; the two are not on a common scale and must not be read as a duration–response. Newcastle-Ottawa assessment only, with no weighting for exposure misclassification.

doi: 10.1136/jech-2025-225491

4HAIE hair cortisol 2026 Stress

Intensity-dependent associations of physical activity and cardiorespiratory fitness with hair cortisol concentrations in the context of lifetime air pollution exposure

133 adults (87% women) from the 4HAIE industrial-environment study. Hair cortisol over ~3 months by LC-MS/MS against Fitbit and questionnaire activity, VO_{2peak}, and lifetime PM_{2.5}/PM₁₀/NO₂ reconstructed from residential histories. Vigorous activity was positively associated with hair cortisol (LTEQ B = 0.010, p = 0.042; Fitbit B = 0.006, p = 0.044); VO_{2peak} attenuated to null once vigorous activity entered. Lifetime air pollution showed no association with hair cortisol. A clean null, but an underpowered one: n = 133 against a lifetime exposure contrast within one industrial region leaves little variance to detect, so this constrains nothing about the PM-cortisol question

doi: 10.1080/10253890.2026.2721526

ABCD amygdala 2026 (preprint) bioRxiv (preprint)

Air pollution, early-life adversity and amygdala volume in preadolescents

3,601 ABCD Study children aged 9–10, annual residential PM_{2.5} total mass plus 15 components, linear mixed-effects adjusted for sociodemographics, co-pollutants and imaging covariates. Adversity predicted general, internalising and externalising symptoms; **PM_{2.5} moderated the adversity-externalising association**, with the highest symptoms in children high on both, and smaller basolateral paralaminar volumes tracked symptoms within the high-adversity group. Cross-sectional, so the mediation reading the framing invites is not supported; annual average exposure at residence cannot speak to the timing the developmental argument needs; and with 16 exposure terms the component results want a correction the abstract does not describe. Not peer reviewed. The component panel is the part worth watching — it is the kind of speciated contrast that mass-only networks cannot support.

doi: 10.64898/2026.08.31.748438

Barbalat et al. 2026 Environ Sci Technol

Heat, cold and ambient pollution in the first 1000 days and autism risk indicators

ELFE national birth cohort, n=12,139 two-year-olds, M-CHAT-R screening score against weekly temperature and PM_{2.5}/PM₁₀/NO₂ from 1 km daily predictions (200 m for NO₂ and temperature in large urban areas), modelled with confounder-adjusted distributed lag non-linear models. **Prenatal air pollution was not associated with the screening score**; night-time heat in the early third trimester was, and survived adjustment for PM_{2.5} and PM₁₀. Postnatally the pattern is sex-specific: higher PM_{2.5} and PM₁₀ at months 7–11.5 raised scores in males relative to females. The outcome is a screening instrument, not a diagnosis, and the sex-stratified postnatal windows are the kind of finding a DLNM will produce by construction unless pre-registered. Useful as a negative result on the prenatal PM window.

doi: 10.1021/acs.est.5c18062

CHARLS neurodegenerative composite 2026 Environ Health

The association of long-term air pollution with a composite neurodegenerative risk score among Chinese elderly: evidence from the CHARLS.

CHARLS middle-aged and older adults, outcome a composite of grip strength, CES-D-10, MMSE and IADL at 2015 follow-up adjusted for the 2011 baseline score. Only PM_{2.5} tracked the composite: adjusted beta 0.010 (95% CI 0.008-0.016) and ordered-logit OR 1.52 (95% CI 1.01-2.31) for a higher risk grade, confined to the 60-74 stratum. PM₁₀, NO₂ and O₃ were null, and neither hs-CRP nor WBC modified the effect, which undercuts the systemic-inflammation pathway the authors set out to test. The composite is not a validated neurodegeneration instrument and mixes motor, affective and cognitive domains, so the estimate is not interpretable as dementia risk; a three-year exposure contrast is also short for a neurodegenerative endpoint.

doi: 10.1186/s12940-026-01335-2

Corrigendum 2026 Brain Behav Immun Health

Corrigendum to 'Prenatal and childhood air pollution exposure, cellular immune biomarkers, and brain connectivity in early adolescence'.

METADATA ONLY. Corrigendum notice only; the record carries no abstract beyond the pointer to the 2024 article it corrects, and the deposited text does not state which results changed. Logged rather than summarised because the corrected paper sits squarely on a line this digest tracks - modelled early-life PM_{2.5} against immune biomarkers and adolescent brain connectivity - and because a correction to that specific inference chain is worth flagging to anyone citing the 2024 original. No effect estimate can be extracted from the notice, and the direction and magnitude of the correction remain unknown until the corrected version is read directly. Treat any citation of the 2024 estimates as provisional until then.

doi: 10.1016/j.bbimh.2026.101317

Curzio et al. 2026 Toxics

The Exposome-Brain Axis: A Scoping Review of Biological Biomarkers in Environmental Neurotoxicity, Neuroinflammation, and Neurodegeneration.

PRISMA-ScR review (PubMed, Scopus, WoS, Embase, no date limit) of 52 human observational studies linking environmental exposures to objective neurobiological markers. PM_{2.5}, metals, agrochemicals and POPs were associated with changes in NfL, GFAP, sTREM2, tau and amyloid-beta, with vulnerability windows from early-life DNA methylation onward. Scoping design: no appraisal, no pooling, and PM is one exposure class among many, so the PM-specific evidence base is not separable from the abstract. Relevance: low; flags plasma NfL/GFAP as candidate short-latency outcomes for sensor-exposure panels.

doi: 10.3390/toxics14090817

ENVIRONAGE eye-track 2026 EBioMedicine

Early-life air pollution and oculomotor markers of cognitive workload

196 children aged 9–11 from the ENVIRONAGE birth cohort completed eye-tracking paradigms 2021–2023; residential BC, NO₂ and PM_{2.5} from a high-resolution spatiotemporal model combining fixed-site monitors, land use and dispersion; multiple linear regression then BKMR distributed-lag models across windows. After multiple-testing correction the surviving associations are *recent* exposure with saccadic outcomes — per IQR of last-month BC **−10.97°/s** (−18.67, −3.26) and NO₂ **−10.81** (−18.13, −3.49) saccade velocity — plus first-trimester and first-1000-days windows on the same outcome. Pupil diameter, fixation count and attention-bias outcomes did not survive. n = 196 against many windows and pollutants means the surviving hits are the ones to trust and the null ones are underpowered rather than negative. Notable for sensing: the effect attaches to last-month exposure, a window a personal or neighbourhood monitor can actually measure.

doi: 10.1016/j.ebiom.2026.106469

Falkenstein et al. 2026 BMC Neurosci

Investigating effects of day-to-day variations in environmental exposure on the human brain: study protocol for the Day2Day Environment project.

Protocol and dataset description for a dense longitudinal study: 30 participants, 25 testing sessions each, with wearable devices, GPS-based geospatial tracking and geographic ecological momentary assessment capturing air quality, noise, light, activity, stress, cognition and affect, paired with repeated MRI (T1/T2*, quantitative multi-parameter mapping, hippocampal proton density, DTI). No results. The abstract does not name the air-quality sensor, its calibration or which pollutants are logged, which is the component this watch would need to judge. n=30 is small but the within-person repeated design is where short-term neuroplasticity signals are detectable. Relevance: one of few designs coupling personal low-cost PM tracks to imaging; worth following when the sensor methods paper appears.

doi: 10.1186/s12868-026-01052-z

Hong Kong momentary stress 2026 Environ Sci Technol

An Explainable Multi-Modal Deep Learning Framework for Investigating the Influences of Dynamic Environmental Exposure on Momentary Stress.

4,342 momentary stress reports against 125,440 minute-level environmental exposure records; the model reaches 78.26% accuracy and 0.828 AUC-ROC. Two results are worth extracting from the architecture. First, exposure in the preceding 25 minutes dominates prediction, which is a direct empirical argument for minute-resolution mobile monitoring over daily or residential-annual assignment for acute endpoints. Second, the explainability analysis recovers thresholds rather than slopes - $25 \mu\text{g m}^{-3}$ for $\text{PM}_{2.5}$, 55 dBA noise, 32 C, 40-70% humidity, 25% greenery visibility - consistent with nonlinearity that a linear exposure-response term would average away. The weaknesses are structural and serious: stress is self-reported into the same phone that carries the sensors, so reporting and exposure share a device and a context; the design is correlational with no within-person randomisation of exposure; the thresholds are read off a gradient-based attribution of a black-box model, not estimated with uncertainty, so they should be treated as hypotheses; and $\text{PM}_{2.5}$ co-varies with noise and temperature in ways a deep model can blend invisibly. Still the clearest current demonstration of what wearable-grade PM sensing buys over address-based assignment.

doi: 10.1021/acs.est.6c09466

Huang et al. 2026 Int J Mol Sci

Environmental Pollution, Biomarkers, and Dementia: An Evidence Review.

Searched PubMed and Embase for 2020-2026 epidemiology on air pollution, environmental chemicals and dementia, with mechanistic evidence for plausibility. Long-term $\text{PM}_{2.5}$, NO_2 and black carbon were associated with higher dementia risk; PM_{10} and ozone were inconclusive; phthalates, bisphenols, Pb, Cd and PFOS were linked to poorer cognition. No pooled estimate, no risk-of-bias assessment and no count of included studies in the abstract, so it adds nothing to the existing meta-analyses. Relevance: low; the BC signal is the one piece that points at a metric low-cost absorption sensors can supply.

doi: 10.3390/ijms27188224

Hubei Memory & Aging 2026 Environ Pollut

Associations between compound heat exposure and cognitive impairment and the potential mediating role of plasma sphingolipid metabolism.

3,617 non-dementia adults aged 65+ in the Hubei Memory and Aging Cohort, repeatedly assessed 2016-2025. Excess cumulative temperature in heatwaves was linearly related to cognitive impairment for daytime-only, nighttime-only and compound heatwaves; temperature-humidity indices gave near-U-shaped curves with Humidexmean best fitting (OR 2.18, 95% CI 1.31-3.64). Among 77 plasma sphingolipid measures, only the ceramide C22:0 / sphingomyelin C26:0 ratio mediated, and at -14.3% - a suppression effect, not an explanatory pathway, which the abstract soft-pedals. $\text{PM}_{2.5}$ enters only as a covariate, so this is a heat paper that adjusts for particles rather than a PM paper. Testing 77 metabolites for a single mediation signal invites a multiplicity problem that is not addressed.

doi: 10.1016/j.envpo1.2026.129182

India-Aldana et al. 2026 Environ Pollut

The chemical exposure during peri-pregnancy, maternal sleep outcomes post-pregnancy, and the potential mediating role of body mass index.

About 393 PROGRESS mothers with 52 exposure markers (estimated air pollutants, metals, phthalates, pesticides, PFAS) in pregnancy or one month postpartum, and actigraphy plus NIH-Toolbox sleep outcomes eight years later. BWQS regression: a quartile rise in the air-pollutant mixture associated with about 8 minutes shorter sleep (beta -0.14 h, 95% CrI -0.29 to 0.00); non-essential metals lowered sleep efficiency (-0.95%, -1.81 to -0.17). No BMI mediation. The air-pollution result sits on the null boundary, the mixture's $\text{PM}_{2.5}$ share is not separable in the abstract, and a single pregnancy exposure window is asked to explain sleep eight years later, over which residential and occupational exposure will have moved. Hypothesis-generating. The long lag is where repeated personal-sensor exposure histories would add the most.

doi: 10.1016/j.envpo1.2026.129220

Januel et al. 2026 J Neurol Neurosurg Psychiatry

Air pollution and the risk of multiple sclerosis relapses in the Paris region, the CONFISEP study

1,152 adults with relapsing MS (898 relapses, 2018-2021) from the French MS registry; weekly Airparif street-level exposure at lags 0-15 weeks, analysed by self-controlled case series and cohort models adjusted for time-varying DMT, infections, temperature, sunlight and lockdown. PM_{10} raised relapse risk at the 2-week (IRR 1.28, 95% CI 1.05-1.57) and 11-week lags (1.26, 1.02-1.54); $\text{PM}_{2.5}$ at 2 weeks (1.20, 1.03-1.39); NO_2 at 2, 3 and 12 weeks. The exposure increment is not stated in the abstract, and with 16 lags x 4 pollutants the isolated 11-12-week hits are likely multiplicity, as the authors concede. DMTs dominated. Relevance: moderate - week-scale personal or neighbourhood sensing would sharpen exactly this lag structure.

doi: 10.1136/jnnp-2026-339534

Kuznetsov et al. 2026 Environ Res

Particulate matter and life satisfaction, across and within twin pairs

TwinLife, $n=4,504$ young-adult twins (2,249 MZ / 2,255 DZ), mean age 23.0, up to six waves 2014-2022, decomposing the PM-wellbeing association into between- and within-pair components. The effect runs **between families** ($b=-0.05$, SE 0.01, $p<.001$) and strengthens with age and calendar year for PM_{10} . Within pairs, after removing shared family variance, the association is null in DZ twins and only nominally significant in MZ twins for PM_{10} ($b=-0.09$, SE 0.04, $p=0.04$) and not at all for $\text{PM}_{2.5}$ ($p=.15$). That is the honest reading: most of the apparent PM effect on subjective wellbeing is confounded by family SES and neighbourhood, and the design is what makes that visible. A rare case of a study whose null is the contribution.

doi: 10.1016/j.envres.2026.125565

Kwon 2026 Front Public Health

Fine particulate matter ($\text{PM}_{2.5}$) and suicide in South Korea: an instrumental variable analysis using thermal inversion.

Instrumental-variable panel analysis of South Korean suicide mortality 2015-2019 using thermal inversions as the instrument for $\text{PM}_{2.5}$. Each $1 \mu\text{g m}^{-3}$ raised suicide deaths by 0.66 per million (95% CI 0.10-1.22), about 2.94% of the rate; effects were larger in males, older adults, the less educated and non-metropolitan residents. The IV design addresses confounding by economic activity, but inversions also trap NO_2 and other pollutants and co-vary with cold, stagnant weather that may affect mood, so the exclusion restriction is questionable; temporal and spatial units are not stated in the abstract. Relevance: moderate - quasi-experimental PM designs need the exposure resolution that inversion-sensitive vertical sensing could supply.

doi: 10.3389/fpubh.2026.1909365

Morrel et al. 2026 Biol Psychiatry Glob Open Sci

Outdoor air pollution, perivascular space morphology, and cognition in preadolescence

Cross-sectional analysis of **6,949 ABCD participants aged 9–10**, with annual-average PM_{2.5}, O₃, NO₂ and **15 PM_{2.5} components** assigned to residential addresses by spatiotemporal models, perivascular-space (PVS) count and volume from T1- and T2-weighted MRI, and cognition from the NIH Toolbox. **Zinc, ammonium and bromine** were positively associated with regional PVS count; higher count in five regions tracked poorer cognition; **calcium, Zn and NH₄⁺** were associated with poorer cognition (FDR-corrected $p < 0.01$); frontal-lobe PVS count **mediated** the Zn–total-cognition association ($p < 0.01$); and weighted-quantile-sum PM_{2.5} mixtures predicted higher temporal and cingulate PVS with poorer working memory and crystallised intelligence. The design cannot order exposure, PVS and cognition, and the abstract reports significance without a single effect size. The component-specific pattern is the measurement claim: speciation, not mass, carries the signal.

doi: 10.1016/j.bpsgos.2026.100790

Oh et al. 2026 Mov Disord

Association Between Long-Term Exposure to Air Pollution and Parkinson's Disease: Administrative Cohort of Older Adults in South Korea.

5.3 million adults aged 65 and over in the Korean National Health Insurance database, 2010–2019, with 143,557 incident Parkinson's disease cases ascertained from claims requiring hospitalisation or repeated outpatient diagnoses. Per interquartile increase, PM_{2.5} ($4.3 \mu\text{g m}^{-3}$) gave HR 1.018 (95% CI 1.008–1.028) and O₃ (7.3 ppb) HR 1.019 (1.006–1.033); NO₂ was inverse after full adjustment, and no demographic or clinical effect modification reached significance. The inverse NO₂ is the tell: exposure is CMAQ output assigned by administrative district, so within-district contrast is zero and urban socioeconomic structure is absorbed into the pollutant. Claims also date diagnosis, not onset. District-resolution assignment at this cohort size is precisely what a dense sensor network would replace.

doi: 10.1002/mds.70518

Older-adult cognition review 2026 Rev Environ Health

Air pollution and cognitive outcomes in older adults: a systematic review

PubMed searched to February 2025 for observational studies from 2000 onward on ambient air pollution and cognition in adults 50+; 36 studies included. PM_{2.5} is the most-studied pollutant, and most studies report worse global or domain-specific cognition, faster decline, or higher impairment and dementia risk with higher PM_{2.5}. Other pollutants trend adverse; ozone is the least consistent. Heterogeneity is attributed to exposure-assessment method, cognitive instrument, design and confounding control. Two limitations bound what this can support: a single-database search with no quantitative pooling and no formal risk-of-bias tool, so the conclusion is a vote count, not a weighted synthesis, and publication bias in a literature this size is unaddressed. The exposure-assessment heterogeneity it names is the actionable part

doi: 10.1515/reveh-2026-0022

Passive smoking acute stress 2026 Front Public Health

Self-reported secondhand smoke and acute stress in 4,460 rescue personnel

Cross-sectional survey of 4,460 grassroots rescue officers and soldiers, screening with the Acute Stress Disorder scale and the PCL-5. Univariate odds were **2.509 (1.621–3.883)** for acute stress disorder and 3.298 (1.459–7.451) for PTSD; on full adjustment the ASD association survived and **the PTSD association did not** (2.513, 0.900–5.151). Among never-smokers, exclusive passive exposure still carried OR 1.910 (1.071–3.408). Three problems limit what can be drawn from this. Exposure is self-reported with *no particle measurement of any kind*, so the dose is unknown and likely differentially misreported by people who are already distressed. The design cannot order exposure and outcome. And the reported multivariate ASD interval — 1.959–3.144 around a point estimate of 1.990 — is internally inconsistent, so the univariate estimate is the one plotted here. Filed as a signal that SHS exposure belongs in trauma screening, not as a dose-response result.

doi: 10.3389/fpubh.2026.1790527

Paulino et al. 2026 Int J Neurosci

Exposome burden and cognitive recovery after mild TBI

44 adults with mild TBI (GCS 13–15, 50% female) prospectively enrolled at an academic clinic; 15 a priori exposome variables reduced by forward selection to 13 and summarised by PCA into a cumulative burden score, against residualised 6-month MoCA dichotomised into worse- and not-worse-than-expected recovery. The latent axis explains **16.0%** of variance and is loaded by education, area deprivation, insurance coverage, PM_{2.5} and adverse childhood events; burden was higher in the worse-than-expected group (median PC1 1.34 vs –0.76, $p=0.010$, holding under permutation). Carried for the PM_{2.5} loading only. At $n=44$ with 13 variables and an outcome-informed selection step, this is hypothesis-generating; the loading structure is mostly socioeconomic and PM_{2.5} may be riding on it.

doi: 10.1080/00207454.2026.2728613

Ren X et al. 2026 Front Public Health

Black carbon dominates a PM_{2.5} mixture on adolescent self-injury — in cross-section

18,398 adolescents in grades 7–12 across 15 schools in four Chinese provinces, with non-suicidal self-injury by ANSAQ and multidimensional biological rhythm by SBRDA. PM_{2.5} and its constituents were positively associated with NSSI risk, black carbon most strongly: **OR 1.76 (1.60–1.94)**, and dominant in the mixture models (**WQS weight 0.810**, Qgcomp effect 0.671). This is the largest effect on today's forest panel and rests on the weakest design there. A single cross-sectional survey round cannot order exposure and outcome in time; exposure is assigned by school or residence, so the black-carbon signal is also a traffic-proximity and urbanicity signal; and rhythm disorder is plausibly a mediator, a confounder or a consequence, which stratification cannot distinguish. The mixture weight is the informative part — it says the components are not interchangeable — and the odds ratio is not the part to quote.

doi: 10.3389/fpubh.2026.1887542

Rodulfo-Cardenas et al. 2026 Neurotoxicology

APOE genotype shapes persistent behavioral and molecular alterations following early postnatal particulate matter exposure in humanized apoE3 and apoE4 mice.

Humanized apoE3 and apoE4 mice of both sexes inhaled NIST fine PM (SRM 2787) or saline for 1 h/day across postnatal days 4–14, then underwent behavioural testing to PND96 and hippocampal/striatal expression profiling. Genotype, not PM, was the main determinant: apoE4 animals were more anxiety-like with hippocampal reorganisation of synaptic, glutamatergic, glial and redox pathways. PM produced only subtle social-behaviour effects and suggestive sex- and region-dependent marker changes. The authors state the chamber concentration was a nominal theoretical maximum of $3789 \mu\text{g m}^{-3}$ and was never measured - an unquantified dose, which makes the null-leaning PM result uninterpretable in dose-response terms. Relevance: a clean example of why exposure chambers need the same real-time mass verification as field sensors.

doi: 10.1016/j.neuro.2026.103580

Sex modification dementia meta 2026 Environ Pollut*126 million participants: sex does not modify the PM_{2.5}–Alzheimer's association*

Systematic review and meta-analysis of 24 longitudinal cohorts, **over 126 million participants**, searched to September 2025, using the female-to-male ratio of risk ratios as the modification metric — the correct estimand, and one that many sex-difference papers substitute a pair of stratified estimates for. For Alzheimer's disease the pooled PM_{2.5} **RRR is 0.963 (95% CI 0.919–1.009)** and for vascular dementia **1.003 (0.954–1.054)**: null, and precisely so. The one positive is all-cause dementia (1.046, 1.024–1.069), a 4.6% relative excess in women that the authors decline to build policy on. Heterogeneity was negligible to low except PM₁₀ in Alzheimer's ($I^2 = 61\%$), no publication bias was detected, and GRADE certainty is moderate. The stated limitation is the one that matters: methodological heterogeneity in how the constituent cohorts defined and adjusted sex could mask a true difference, so this is evidence against current *claims*, not proof of no effect.

doi: 10.1016/j.envpol.2026.129138

Shupler et al. 2026 Environ Sci Technol*Preconception, Prenatal and Postnatal PM_{2.5} Exposure and Neurodevelopmental Disorder Risk among Medicaid Recipients*

1,547,244 Medicaid mother-child pairs (2001-2013), ZIP-code PM_{2.5} with distributed-lag models over preconception, pregnancy and the first three years. Per 10 $\mu\text{g m}^{-3}$ cumulative prenatal PM_{2.5}, ASD HR 1.17 (95% CI 1.00-1.37), with a critical window at gestational weeks 23-31 (HR 1.09, 1.02-1.17). Prenatal exposure raised intellectual-disability risk in Hispanic children only (HR 1.57, 1.02-2.43). Postnatal ADHD (1.17, 0.94-1.46) and DCD (1.72, 0.97-3.07) estimates include the null; no other NDD was associated. ZIP-code exposure, claims-based diagnoses and residual area SES confounding limit the inference; the ASD lower bound sits at 1.00. Relevance: moderate - a third-trimester window argues for time-resolved residential exposure rather than annual means.

doi: 10.1021/acs.est.6c03766

Temporao et al. 2026 medRxiv (preprint)*Daily associations between outdoor air pollution and sleep quality*

Budapest Sleep, Traits and Experiences Study — seven days per participant of daily activity and sleep-quality reports plus home EEG giving sleep efficiency, %REM, onset latency and total sleep time. Linear mixed models with participant random intercepts, adjusted for age, sex, BMI, education, weekday, season, temperature, physical activity, screen exposure and travel. **Day-1 ozone predicted lower day-2 sleep efficiency and poorer perceived quality**, surviving multiple-comparison correction, multi-pollutant models and sensitivity analyses for depression, insomnia and chronotype; **no robust association for PM_{2.5} or NO₂**. The null is confounded with exposure-model quality — city-aggregated station data give near-zero within-city contrast, and misclassification is worse for PM_{2.5} than for the more spatially homogeneous ozone. Sample size is not stated in the abstract.

doi: 10.64898/2026.09.15.26363130

WTC responder exposure mixture 2026 Nat Commun*Urban exposures, frailty, and mental illness in World Trade Center Health Program responders*

18,734 WTC Health Program general responders, later-life residential exposures. A decile increment in the joint PM_{2.5}/temperature/greenness mixture raises frailty index by 0.56% (95% CI 0.14-0.98) and PTSD score by 0.31% (0.07-0.55); effects are larger in responders with higher original WTC exposure, read as acquired biological vulnerability. The estimates are percentage differences on continuous scores, not ratio metrics, so they are small in absolute terms and the mixture decile has no physical unit - a decile shift is not a concentration. Residential assignment at annual resolution cannot separate PM_{2.5} from the temperature and greenness terms it co-varies with. For distributed sensing the relevant point is that the mixture framing needs co-located PM, thermal and greenness data at a resolution annual products cannot give.

doi: 10.1038/s41467-026-77401-7

Wang et al. 2026 Environ Int*Environmental exposures and mental health: longitudinal results from the German National Cohort (NAKO)*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 27 Sep with no abstract; Crossref, Europe PMC and the publisher landing page returned nothing. The title indicates longitudinal analyses of environmental exposures and mental health in the German National Cohort; which exposures (air pollution, noise, greenness, heat) are included cannot be stated. NAKO's size (~200,000) makes it a priority retry. Nothing about exposure metric, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title.

doi: 10.1016/j.envint.2026.110539

Zhang et al. 2026 Front Public Health*Household solid fuel use, kitchen ventilation and depressive symptoms among older adults in China*

11,697 CLHLS participants aged 65+, 2018 wave. Depressive symptoms (CES-D-10) in **26.87%**, solid fuel use in 29.67%. Solid fuel carried **OR 1.25 (95% CI 1.09–1.44)** against clean fuel and **B = 0.57 (0.33–0.82)** on the continuous score — consistent but small, roughly a fifth of a scale point per unit. The charcoal subgroup showed a larger estimate on $n = 39$ exposed and the authors correctly flag it as unreliable. **No interaction was found between fuel type and kitchen ventilation**, which undercuts the paper's own framing that ventilation modifies the exposure. Exposure is entirely self-reported category with no PM_{2.5} measurement, so dose is unknown and misclassification is certain to be substantial; and the cross-sectional design cannot exclude reverse causation, since depressed older adults may differ in both household energy choice and reporting.

doi: 10.3389/fpubh.2026.1920233

Zou et al. 2026 Environ Res*Association of preconception to early childhood residential greenness with child neurodevelopmental outcomes and the mediating role of air pollution: a prospective cohort study.*

5,284 mother-child pairs in the Jiangsu Birth Cohort, with satellite NDVI from preconception to early childhood and Bayley-III screening at age three. Per 0.1 NDVI, preconception and third-trimester greenness lowered risk of the emerging/at-risk cognition category (RR 0.85, 95% CI 0.74-0.98; 0.82, 0.71-0.95) and gross motor (0.87, 0.77-0.98; 0.77, 0.68-0.88), surviving FDR at 0.10. Single-pollutant mediation assigned 68.0% (PM₁₀) and 63.5% (NO₂) of the preconception-cognition association, and 59.1% via PM_{2.5} for third-trimester gross motor. Separate single-pollutant mediators summing well past 100% show the pollutants are proxies for one another and for urbanicity; mediation of that size in a greenness model needs sequential or joint mediation to be credible. Relevance: greenness-PM trade-offs are measurable only with sub-kilometre PM surfaces.

doi: 10.1016/j.envres.2026.125746

Respiratory & allergic**44 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (44 OF 44 RECORDS IN THIS SUBTOPIC)

Amazon PM_{2.5} x heat 2026 Research Square*Interaction Effects of PM_{2.5} and High Temperatures on the Respiratory Health of Vulnerable Age Groups in the Brazilian Amazon*

PREPRINT. Monthly municipal data 2009-2019 for the Legal Amazon, hierarchical Bayesian models with 0-2 month distributed lags. Per 10 $\mu\text{g m}^{-3}$ PM_{2.5} under high-temperature conditions, respiratory mortality in adults 60+ rose 2.5% and hospitalisations in those aged 1-19 rose 2.7%, both at lag 0 with no reported persistence at longer lags. Limitations: ecological design at monthly resolution, so the interaction is estimated well below the timescale on which wildfire smoke and heat actually covary, and no confidence intervals are given in the abstract. Relevance: a monthly aggregate cannot resolve smoke episodes, which is exactly the gap a dense low-cost network in the burning arc would fill

doi: 10.21203/rs.3.rs-10973377/v1

Bengaluru COPD 2026 Indian J Community Med*Prevalence and risk factors of COPD in an urban poor locality*

Community-based cross-sectional survey, adults over 30 in an urban poor locality of Bengaluru, PPS sampling to a calculated 1,560, spirometry-confirmed diagnosis. Prevalence **10.38%** overall, 14.4% in men and 7.1% in women, with 74.06% at moderate-to-severe stage; multivariate logistic regression retained age, smoked and smokeless tobacco, fuel type, biomass exposure, environmental tobacco smoke and indoor air pollution. No air measurement of any kind — every exposure is a questionnaire item, so the “indoor air pollution” association is a self-report contrast and the effect sizes are not reported in the abstract. Value here is the spirometry-confirmed prevalence in a population that ambient monitoring networks systematically under-serve.

doi: 10.4103/ijcm.ijcm_145_25

COPD anaemia metals 2026 Int J Chron Obstruct Pulmon Dis*Air-pollution-associated intracellular metals, immune gene expression and erythrocyte indices*

61 COPD patients and 10 controls; annual PM₁₀, PM_{2.5}, NO₂ and NO_x by land use regression; metals by ICP-MS and single-cell ICP-MS in PBMCs and red cells; PBMC RNA-seq. PM_{2.5} was associated with higher RBC count ($\beta = 0.0314 \times 10^6 / \mu\text{L}$, 0.0066–0.0561), red cell distribution width, and with intracellular chromium, zinc and cadmium; those metals in turn tracked *ALPL*, *CXCR1* and *PI3* expression, and anaemic patients showed a distinct signature. Ten controls against 61 cases and a cross-sectional design make every link correlational, and LUR annual means cannot support a claim about cellular metal kinetics. Read it as an assay-development result: single-cell ICP-MS as an internal-dose readout is the transferable piece, and it is the kind of endpoint a speciating monitor could eventually be validated against.

doi: 10.2147/COPD.S626348

Cardiorespiratory review 2026 Med Gas Res*Narrative review of air pollution and cardiorespiratory disease*

Orientation-level review of the mechanisms linking ambient air pollution to cardiometabolic risk and cardiovascular death, with sections on heart failure, COPD and asthma exacerbation, on vulnerable groups (pregnancy, neonates, children, pre-existing cardiorespiratory disease), on the COVID-19 lockdown natural experiment, on international divergence in air quality standards, and on mitigation from policy through technology. There is no systematic search protocol, no PROSPERO registration, no pooled estimate and no risk-of-bias assessment, so it cannot be cited as evidence for any magnitude — and it does not claim to be. Listed because the lockdown-period material and the standards comparison are a useful map of where the regulatory targets currently sit, and because naming it here prevents it being mistaken for a systematic review later.

doi: 10.4103/mgr.MEDGASRES-D-25-00201

Carroll et al. 2026 Ann Am Thorac Soc*Global Tuberculosis Burden Attributable to Household and Outdoor Air Pollution*

140 studies (to Aug 2024) pooled by random effects and Burden of Proof meta-regression. Household solid fuel: tuberculosis incidence RR 1.82 (95% CI 1.52-2.17), 2.25 in children and 2.62 in women. Short-term outdoor PM_{2.5}: RR 1.03 (1.00-1.05) per 10 $\mu\text{g m}^{-3}$. Attributed to PM_{2.5} in 2023: 4.15 million cases, 23.6 million DALYs and ~535,000 deaths - 49.9% of global TB. The attributable fraction is driven by the household-fuel RR, which comes from fuel-type contrasts confounded by poverty, crowding and undernutrition, and is far larger than the outdoor estimate whose lower bound touches 1. Treat the 50% as an upper bound. Relevance: moderate - household PM measurement is the missing exposure link.

doi: 10.1093/annalsats/aaog299

Chen et al. 2026 Int J Chron Obstruct Pulmon Dis*Residential green and blue space, genetic susceptibility and incident COPD*

385,502 UK Biobank participants, Cox models, greenness at 300 m and 1000 m buffers against a COPD polygenic risk score. At 1000 m: **green space HR 0.87 (95% CI 0.83–0.92), natural environment HR 0.83 (0.78–0.87)**; blue space weak and inconsistent. No multiplicative or additive interaction with genetic risk, and the natural-environment association held in every genetic stratum — so the protective signal is not running through a susceptible subgroup. The inferential problem is what is absent: **no air-pollution term is reported**. Residential greenness is inversely correlated with traffic PM_{2.5} and NO₂ nearly everywhere, so an unadjusted greenness–COPD association may simply restate a pollution–COPD association.

doi: 10.2147/copd.s627458

Cui L et al. 2026 Clin Rev Allergy Immunol*Epidemic thunderstorm asthma in China as an Artemisia phenotype*

54 primary studies (43 international, 11 Chinese — Yulin, Hohhot, Chongqing, Lanzhou, Shenzhen) from 436 records, risk of bias by Newcastle–Ottawa and Murad, certainty by GRADE. Three patterns recur: Chinese events are associated with **mugwort (*Artemisia*)** rather than ryegrass sensitisation; **40–60% of Chinese cases and up to 78% of Australian and UK cases had no prior asthma diagnosis**; and individual Chinese localities report recurrence. No meta-analysis, and GRADE certainty for direct pollen rupture, pollution-mediated severity and pre-emptive treatment is **low to very low**. The measurement gap is explicit and is the actionable part: Chinese reports identify sensitisation only to genus level and **do not speciate the airborne pollen**, so no aerobiological trigger threshold can be set. The authors’ first research priority is field aerosol sampling.

doi: 10.1007/s12016-026-09194-y

Dinh et al. 2026 Front Med (Lausanne)*Association of air pollution with chronic obstructive pulmonary disease: anemia-related outcomes and exploratory genetic susceptibility analyses.*

Taiwan Biobank case-control of 7,566 COPD cases and 22,698 age- and sex-matched controls. Each 10 $\mu\text{g m}^{-3}$ of PM_{2.5} carried OR 1.252 (95% CI 1.222-1.283) at a one-month averaging window and 1.642 (1.588-1.699) at twelve months; the within-COPD anaemia analysis showed no consistent association, and a haemoglobin polygenic score interacted with SO₂ only nominally. Those confidence intervals are implausibly narrow for 30,000 participants under correlated multi-pollutant exposure, which usually means unmodelled spatial correlation rather than precision, and COPD is defined partly by self-reported emphysema or chronic bronchitis. The monotone rise with averaging window is as consistent with confounding by place as with cumulative dose - a distinction only within-person exposure variation can settle.

doi: 10.3389/fmed.2026.1925502

Disaster respiratory review 2026 Thorac Res Pract*Respiratory emergencies in disasters*

Narrative review spanning earthquake silica and asbestos dust, wildfire fine particulate, flood-related mould, volcanic ash and acidic gases, chemical release, blast barotrauma and pandemic threats, synthesised from PubMed/MEDLINE, Web of Science and Scopus with EM-DAT, WHO and UNDRR reports. It reports that acute respiratory infection is a leading cause of disaster-related morbidity and mortality across age groups, that chronic respiratory patients face elevated exacerbation risk, and that ARDS management remains hardest in resource-limited settings. There is **no stated search protocol, no inclusion criteria and no risk-of-bias appraisal**, so it indexes the literature rather than weighing it, and no quantitative estimate here can be traced to a defined evidence base. Carried as a citation anchor for the disaster-PM framing and for its explicit treatment of climate change as a driver of rising burden — not as evidence.

doi: 10.4274/ThoracResPract.2026.2026-6-13

Dou et al. 2026 Front Immunol*Risk factors for allergic rhinitis in children: an umbrella review of systematic review and meta-analysis.*

PROSPERO-registered umbrella review of meta-analyses to May 2025, AMSTAR and GRADE appraisal. Of 75 candidate risk factors, 22 raised paediatric allergic-rhinitis risk, 7 were protective and 46 were indeterminate; air pollution sits among the risk-raising factors with antibiotics, caesarean delivery and prenatal smoking. Evidence quality was low or very low overall, and only early-life food sensitisation reached moderate certainty - so air pollution is on the list at low certainty. Pollutant, metric and pooled effect size are not given in the abstract. Umbrella reviews inherit the heterogeneity of their inputs, and ambient-pollution meta-analyses in children pool incomparable exposure assignments. A calibration of how weak the paediatric AR evidence is, not a new estimate.

doi: 10.3389/fimmu.2026.1849271

Gómez-Olarte et al. 2026 Environ Pollut*Early-life air pollution, respiratory outcomes and stimulated cytokine response*

German birth cohort, $n=606$ one-year-olds for bronchitis and wheezing, cytokine subcohort $n=508$ (LPS-stimulated TNF- α , MCP-1, IL-8, IL-6, IL-10), residential PM_{2.5}/PM₁₀/NO₂ per IQR of 2.0/2.8/9.1 $\mu\text{g m}^{-3}$, with BKMR for the mixture and stratification by sex and temperature. In boys, PM_{2.5} gave **aOR 1.50 (1.02–2.23)** for bronchitis and **+23.8% (11.2–37.9)** IL-8; in girls, **+31.9% (18.6–46.7)** IL-8, **+12.8% (3.9–22.6)** TNF- α and **+35.7% (14.4–61.1)** IL-10. The mixture association with IL-8 and TNF- α was non-linear and appeared only above the 60th percentile of all pollutants. Small cohort, and the sex split doubles the tests; the consistent direction across five cytokines is what carries it. PM_{2.5}, not NO₂, is the primary predictor.

doi: 10.1016/j.envpol.2026.129079

HEPA RCT synthesis 2026 Expert Rev Respir Med*HEPA air filtration for asthma and COPD: lessons from randomized controlled trials.*

The consistent finding across trials is the one that should worry anyone who equates exposure reduction with clinical benefit: portable HEPA reliably and substantially lowers indoor PM_{2.5}, but clinical effects are heterogeneous. Several asthma trials improved symptoms, peak flow or quality of life - concentrated in highly symptomatic or higher-exposure subgroups - while most spirometric endpoints moved little or not at all. COPD evidence is thin and only suggestive. The authors attribute the gap to trial design rather than to absent efficacy, and list the specific failures: short follow-up, almost no adult studies, device specifications unreported, adherence unmeasured, no economic evaluation, and near-total absence of low-resource settings. That list is itself the contribution. Being a narrative rather than systematic synthesis with no pooled estimate and no formal risk-of-bias assessment, it cannot bound the effect, and the subgroup pattern it highlights is exactly the pattern selective reporting produces. For a sensing programme the actionable point is that unmeasured adherence and unreported device spec are measurement problems a logging sensor inside the intervention arm would largely solve.

doi: 10.1080/17476348.2026.2735502

Kim et al. 2026 Int J Environ Health Res*Heat exposure and COPD morbidity and mortality: systematic review and meta-analysis*

Thirty-six observational studies pooled in a three-level random-effects model that handles non-independent estimates. Heat raised overall COPD risk (**RR 1.15, 95% CI 1.09–1.21**), with mortality (**1.26, 1.18–1.34**) well above morbidity (**1.05, 1.01–1.09**). Heat-wave and threshold definitions both gave RR 1.21 against 1.05 for continuous temperature metrics, and the continuous metrics were also the least heterogeneous — so definitional choice, not biology, drives much of the spread. The finding that earns this record its place here is negative: **adjusting for air pollution did not substantially alter the associations**, which argues against PM confounding of heat–COPD associations while saying nothing about interaction. No publication-bias or exposure-misclassification assessment is reported in the abstract.

doi: 10.1080/09603123.2026.2734444

Huang et al. 2026 Tob Induc Dis*Global, regional, and 204 countries burden of disease attributed to secondhand smoke-related COPD, 1990-2021: A secondary data analysis of the Global Burden of Disease Study 2021.*

GBD 2021 secondary analysis, 1990-2021, with an ARIMA projection to 2031. Age-standardised mortality for secondhand-smoke-attributable COPD roughly halved, 11.36 (95% UI 4.52-18.55) to 5.87 (2.29-9.72) per 100,000, and ASDR fell 234.01 (93.58-380.47) to 120.94 (47.2-197.93) - yet absolute deaths rose, the familiar ageing-and-population-growth divergence that makes rate-only reporting misleading. China carries the largest absolute mortality (117,388; 47,776-195,915), Papua New Guinea the highest rates. The SDI relationship is non-linear with an inflection at SDI 0.42. The binding limitation is inherited, not introduced: GBD secondhand-smoke exposure is modelled from survey-reported household and workplace smoking, not measured particulate or cotinine, so the uncertainty intervals understate exposure misclassification and every country contrast partly reflects differing survey quality. An ARIMA extrapolation on a modelled series compounds that. Relevant here because secondhand smoke is the one fine-particulate exposure with an intervention that works, and it anchors the indoor end of the PM dose-response.

doi: 10.18332/tid/230940

IPF acute PM_{2.5} hospitalisation 2026 Ann Am Thorac Soc*Short-term air pollution exposure and hospitalizations in idiopathic pulmonary fibrosis: data from the IPF-PRO Registry.*

Case-crossover on respiratory hospitalisations among IPF-PRO Registry patients at 46 US sites. Each 5 $\mu\text{g m}^{-3}$ higher PM_{2.5} on lag day 0 raised the odds of respiratory hospitalisation 22% (95% CI 14% to 47%); lags 1-6 were null, and there was no effect modification by age, sex, Social Vulnerability Index, MUC5B genotype, telomere length or antifibrotic use. The same-day-only signal is biologically plausible for a fibrotic lung with no reserve, but the reported interval is not symmetric about the point estimate on the log scale (0.07 below, 0.19 above), which suggests a transcription or rounding error somewhere in the reporting chain and should be checked against the published table. Exposure is monitor-based, not personal, so Berkson error will attenuate. Nearest-monitor assignment is exactly the resolution a dense low-cost network would replace.

doi: 10.1093/annalsats/aaog285

Johansson et al. 2026 Indoor Air*Associations Between Indoor Air Pollution and Asthma, Allergic Rhinitis and Eczema in a Cohort of Children in Durban, South Africa*

Cross-sectional analysis of 300 children from the MACE birth cohort in Durban with indoor PM_{2.5} measurements, ISAAC interviews and a household walkthrough. ETS exposure was associated with doctor-diagnosed asthma (aOR 3.51, 95% CI 1.20-10.30) and allergic rhinitis (aOR 21.58, 2.80-166.38); biomass fuel use with rhinitis score (aOR 4.97, 1.13-21.95) and eczema score (aOR 6.61, 1.48-29.60). The abstract reports no estimate for measured indoor or prenatal PM_{2.5}, which suggests it was null. CIs spanning two orders of magnitude show very few cases; exposure categories are self-reported. Relevance: the measured-PM result is what this watch needs, and it is absent - the categorical proxies carried the association.

doi: [10.1155/ina/8341339](https://doi.org/10.1155/ina/8341339)**Kang et al. 2026** PLoS One*Asthma Intervention with Residential Ventilation and Air Cleaner (AIRVAC) Study: A 4-arm parallel-group randomized controlled trial protocol.*

Protocol for a single-blind, sham-controlled, four-arm RCT (active or sham energy recovery ventilator; active or sham portable air cleaner) in >80 Dallas-Fort Worth households with a physician-diagnosed asthmatic aged 5-17 or adult, with up to one year pre-intervention and one year post. Primary outcome asthma control; secondary pulmonary function, asthma QoL, stress and sleep. Exposure by low-cost plus research-grade PM, NO₂, CO and VOC sensors, and mold sampling. Registered as NCT07196436, where spirometry and ACT are both listed as primary - a protocol/registry mismatch to watch. ~20 households per arm is small for ACT differences, and no PM exposure endpoint is pre-specified. Relevance: a sham-controlled trial whose exposure arm is a low-cost sensor network - the calibration protocol will determine what the IAQ results mean.

doi: [10.1371/journal.pone.0358221](https://doi.org/10.1371/journal.pone.0358221)**Korea 8.25M asthma cohort 2026** Allergy*Cumulative Vulnerability to Particulate Matter and Incident Asthma: Joint Effects of Urbanicity, Poverty, and Age in a Nationwide Cohort*

8,252,498 adults in the Korean National Health Insurance Service database linked to 1-km long-term PM estimates, Cox models for incident asthma. Per 10 $\mu\text{g m}^{-3}$, PM_{2.5} gave HR 1.33 (95% CI 1.32-1.34) and PM₁₀ HR 1.38 (1.37-1.38). Stratification by age, Medical Aid status and urbanicity produced a monotone gradient from HR 1.16 (1.14-1.18) in younger rural adults without Medical Aid to 1.80 (1.66-1.94) in older urban Medical Aid recipients, p for interaction < 0.001. Limitation: incident asthma in claims data is diagnosis-coded, and urban Medical Aid recipients have higher healthcare contact, so part of the 1.16-to-1.80 gradient is plausibly ascertainment rather than susceptibility - the confidence intervals are narrow enough to make that distinction matter. Exposure contrast at 1 km is coarse relative to the urbanicity modifier being tested, which is the resolution argument for dense low-cost networks

doi: [10.1111/all.70502](https://doi.org/10.1111/all.70502)**Korea paediatric severity IV 2026** medRxiv*Air pollution drives concentrated clinical severity of respiratory illness in early childhood*

PREPRINT. Nationwide claims for 3,210,381 children, 99,261,945 child-months, 2015-2019, with thermal inversions used as a quasi-random instrument for monthly PM_{2.5}. A 1-SD rise (7.23 $\mu\text{g m}^{-3}$) adds 0.29 respiratory visits per child-month at age 1, significant through age 6; expenditure rises 46.7%, 43.0% and 35.4% at ages 0, 1 and 2 and is indistinguishable from zero past age 4, giving a 32.2% increase in annual respiratory spending under age 5. The severity gradient, not the visit count, is the contribution. Limitation: the inversion instrument is plausibly exogenous to care-seeking but not to temperature, and no temperature control is described in the abstract; that is the exclusion restriction the peer review will test

doi: [10.64898/2026.09.03.26362213](https://doi.org/10.64898/2026.09.03.26362213)**Korean J Intern Med 2026** Korean J Intern Med*Health effects of air pollution on chronic respiratory diseases.*

Narrative synthesis of the epidemiological evidence tying air pollution to incidence, exacerbation and mortality across asthma, COPD, interstitial lung disease and lung cancer, then a section on non-pharmacological exposure-reduction measures at individual and societal level. No search strategy, no inclusion criteria, no quantitative pooling, so it cannot be read as evidence in its own right - the value is the intervention section, which is the part least covered by the recent primary literature. Included as a reference point for what the field currently treats as settled rather than for any new estimate. The interventions discussed (purifiers, mask use, activity rescheduling) all presuppose exposure information at a spatial resolution regulatory networks do not provide.

doi: [10.3904/kjim.2025.423](https://doi.org/10.3904/kjim.2025.423)**Kulkarni et al. 2026** JAMA Netw Open*Short-Term Exposure to PM_{2.5} and Emergency Department Visits for Asthma and Chronic Obstructive Pulmonary Disease.*

METADATA ONLY. Metadata-only record: PubMed carries only the one-line plain-language summary - a cohort study of short-term PM_{2.5} and asthma- and COPD-related emergency department visits in Singapore (Duke-NUS / SGH authors). No estimate, lag, exposure source or sample size is deposited; Crossref and Europe PMC returned nothing and Scholar Gateway failed with an identity error. This watch does not infer results from a title. Priority retry: Singapore's transboundary-haze PM_{2.5} is a clean natural experiment, and a tropical-humidity city is where sensor-derived exposure would be most biased.

doi: [10.1001/jamanetworkopen.2026.36306](https://doi.org/10.1001/jamanetworkopen.2026.36306)**Kuwait tobacco smoke morbidity 2026** BMC Health Serv Res*Exposure to tobacco smoke, concurrent short-term symptoms of respiratory, allergic, cardiovascular morbidities and health-services use among young adults in Kuwait*

Convenience sample of 1,323 Kuwait University undergraduates, October 2022, electronic questionnaire. First-hand smoke exposure 13.4%, second-hand at home 50.8%, any exposure 55.9%. Self-reported symptoms were common: shortness of breath 59.1%, cough 46.3%, chest pain 26.9%. Limitations are severe and the authors name most of them: convenience sampling in one university, 75.8% female, all exposure and outcome self-reported, cross-sectional so no temporality. The prevalence figures are usable as a local exposure baseline; the symptom associations are not evidence of effect

doi: [10.1186/s12913-026-15530-x](https://doi.org/10.1186/s12913-026-15530-x)

Li H et al. 2026 ERJ Open Res

Multi-omics Mendelian randomisation of PM_{2.5} and childhood asthma

Two-sample MR on large-scale GWAS summary data gives **OR 1.897 (1.063–3.388)**, $p=0.030$ for genetically instrumented PM_{2.5} on childhood asthma. A TWAS identified **70 genes** co-expressed with both, enriched in lysosome- and leukocyte-mediated immunity, with **MEAF6** validated as protective and **RNF40** as a risk gene. Two-step MR put **FUT10 as a positive mediator carrying 19.3%** of the effect, with CD200 and MANBA negative mediators; a phenome-wide MR flagged associations of these targets with other diseases, which the authors correctly present as a caution against treating them as drug targets. The limits are the usual MR limits and they are severe here: the PM_{2.5} instrument is a genetic proxy for a residentially correlated environmental exposure, so horizontal pleiotropy through socioeconomic position is not excluded, and the interval spans a factor of three.

doi: 10.1183/23120541.01444-2025

Maunakea et al. 2026 Proc Natl Acad Sci U S A

Climate-amplified wildland-urban interface disasters create persistent internal metal mixtures associated with pulmonary impairment.

1,400 adults enrolled 6–18 months after the August 2023 Maui wildland-urban interface fires had 24 urinary metals and spirometry at the same visit. Burdens exceeded US reference levels - antimony 40-fold, manganese 3.8, barium 2.3, arsenic 2.2 - highest in the most affected areas. In WQS models each quantile of the positive mixture index (driven by As, Cd, Cu) raised the odds of FVC, FEV₁, FEV₁/FVC and FEF₂₅₋₇₅ below LLN (OR 1.71, 1.53, 1.83, 1.84; CIs not in the abstract). Cross-sectional, no pre-fire baseline, urinary metals partly dietary (As) and smoking-related (Cd), and no air measurement links the burden to smoke or ash. Relevance: air PM mass would not have captured this; WUI smoke needs metal speciation, not just optical PM_{2.5}.

doi: 10.1073/pnas.2613208123

Meng et al. 2026 Front Public Health

Exposure-response thresholds for ambient PM_{2.5} and SO₂ vs COPD and small airway dysfunction, Yunnan

11,095 residents aged 20+, multistage random cluster sampling 2021–22, with spirometry in questionnaire-screened high-risk individuals. COPD prevalence **27.2%**, small airway dysfunction **53.6%**. Restricted cubic splines gave significant nonlinearity for both pollutants with thresholds at **21.33 µg/m³ PM_{2.5}** and **8.67 µg/m³ SO₂**; above the PM_{2.5} threshold, COPD and SAD prevalence were 33.4% and 58.4%. Random-forest importance ranked air pollution above smoking and biomass fuel — a strong claim from a cross-sectional design that should be read as variable importance, not effect size. No temporality, exposure assigned without residential history, and the enriched high-risk sampling frame means these prevalences do not transfer to a general population. A 21 µg/m³ threshold sits well above the WHO 5 µg/m³ guideline and close to the regional mean, so it is plausibly a data-support artefact rather than a biological inflection.

doi: 10.3389/fpubh.2026.1874156

MobiliSense ambulatory spirometry 2026 Research Square

Sub-hourly personal black carbon and a measurable drop in FEV₁

Preprint. 199 adults in metropolitan Paris, 2018–2020, each carrying two portable pollutant monitors, a GPS receiver and an accelerometer through ordinary daily mobility, with **2,504 spirometry tests** taken morning and evening over three days. Mixed-effects models tested lags from 15 minutes to 6 hours. In multipollutant models, a 1 µg/m³ increase in BC one hour before testing was associated with FEV₁ lower by **0.016 L (95% CI –0.024 to –0.008)**, and at two hours by **0.021 L (–0.034 to –0.007)**; PM_{2.5} at 15, 30 and 60 minutes reduced FEV₁/FVC by 0.60, 0.70 and 0.50 percentage points. O₃ ran positive, which is the expected artefact of O₃ anticorrelating with traffic. The threats are real: 199 participants, three days, no personal-monitor validation reported against reference instruments, and spirometry quality in the field is hard. But this is the design that resolves an exposure window a fixed monitor cannot see at all.

doi: 10.21203/rs.3.rs-10386483/v1

Naka et al. 2026 Pediatr Allergy Immunol

Impact of particulate matter 2.5 exposure on severity of pediatric chronic rhinitis: A prospective cohort study.

146 children with chronic rhinitis in the Bangkok Metropolitan Region followed weekly for a median 52 weeks (Jul 2024–Aug 2025), with symptom severity on a visual analogue scale and the preceding week's PM_{2.5} interpolated by inverse-distance weighting. Per 10 µg m^{–3}, VAS rose by 0.260 (95% CI 0.215–0.306); the association held in both allergic and non-allergic rhinitis, with wet season, infections and medication also associated. The effect is small on a 10-point scale, the VAS is self- or parent-reported and unblinded to haze days, and IDW over sparse monitors ignores indoor exposure. Relevance: repeated weekly outcomes over a year are an ideal fit for home or school low-cost sensors, which would replace IDW with measured exposure.

doi: 10.1111/pai.70493

Zhu et al. 2026 J Asthma Allergy

Extreme cold events and paediatric asthma visits, modified by ambient CO and SO₂

62,167 paediatric asthma outpatient visits at one tertiary hospital in Nanjing, 2019–2024, quasi-Poisson DLNM over lag 0–14. Extreme cold events show **no overall association**, but cumulative RR reaches **2.60 (95% CI 1.13–5.95)** in the high-CO stratum and **3.41 (1.15–10.11)** in the high-SO₂ stratum. The intervals are very wide and the strata are subsets of an already null main effect, so this is hypothesis-generating; notably PM_{2.5} and PM₁₀ did not emerge as modifiers. Single-hospital catchment with city-average exposure. Relevant here mainly as a reminder that pollutant-by-meteorology interaction terms demand far denser exposure data than one municipal station provides.

doi: 10.2147/jaa.s483101

Martinez & Nadeau 2026 Curr Allergy Asthma Rep

Early-life exposure to air pollutants and childhood asthma

Narrative synthesis of recent cohort and mechanistic work on the prenatal and early postnatal window. Reports epigenetic modification in airway epithelium, a convergence point shared between tobacco smoke exposure and genetic asthma predisposition, new serum biomarkers of exposure, and a murine result in which asthma-predisposing information passes from exposed dam to offspring **via NK cells**. No pooled estimate, no effect size, and no stated search strategy or inclusion criteria, so study selection is unauditably and the piece cannot be weighted against the systematic reviews in this window. Useful as a pointer to the NK-cell transmission mechanism and the biomarker candidates, and for nothing quantitative.

doi: 10.1007/s11882-026-01293-2

PM_{2.5} lung cancer never-smokers review 2026 J Thorac Dis

Breathing carcinogens: PM_{2.5} air pollution and lung cancer in never-smokers, a narrative review

Narrative review of PM_{2.5} and lung cancer in never-smokers, PubMed/MEDLINE 2000 to May 2026 plus reference-list snowballing. The central claim is mechanistic rather than quantitative: PM_{2.5} acts predominantly as a tumour promoter rather than a direct mutagen, operating through inflammatory signalling, metabolic reprogramming and epigenetic change, and is associated with more advanced stage at diagnosis and worse survival after resection. Lung cancer in never-smokers is described as molecularly and immunologically distinct, with screening and precision-oncology implications. No pooled estimate, no risk-of-bias assessment and no search-yield count are reported, so it carries no quantitative weight and the selection is author-curated by design. Useful as a mechanism map, not as evidence.

doi: 10.21037/jtd-2026-1264

Samdarshi et al. 2026 Thorax*Systematic misclassification of lung function in Indian adolescents using GLI-2012 reference equations: evidence from the multicity APEAL cohort.*

Baseline of the Air Pollution Exposure on Adolescents' Lungs cohort: 4,175 healthy non-smoking 11-14-year-olds with ATS/ERS spirometry in four cities. Against GLI-2012 (Asian category), mean z-scores were -2.44 for FEV1, -1.34 for FVC and -0.78 for FEV1/FVC; 74.6% fell below the LLN for FEV1 against an expected 5%. The shift was uniform across sex, city, age, height and BMI. No PM analysis yet. It is carried because it bears directly on the outcome metrology of every South Asian PM-lung study: using GLI-2012 would label most exposed and unexposed adolescents abnormal, compressing contrasts and inflating 'impairment' prevalence. The counter-risk is that part of the deficit is itself pollution-driven, so a local reference set derived from these same cities may absorb the very effect later analyses seek.

doi: 10.1136/thorax-2026-225309

San Juan-Reyes et al. 2026 Environ Sci Pollut Res Int*Short-term ambient air pollution and respiratory hospitalisation in children under five*

266 children under five admitted to one regional IMSS hospital in the Toluca Valley Metropolitan Zone during 2023; matched case-crossover with conditional logistic regression, primary window lag 1-7, adjusted for temperature and humidity. PM_{2.5} was associated with admission for viral pneumonia, bacterial pneumonia and asthma, CO with bronchiolitis, with heterogeneous municipality-level associations. **The abstract reports direction only — not one odds ratio, interval or unit appears** — so the effect size is unverifiable from the published summary. 266 events split across four diagnoses, five pollutants, several lag windows and three municipalities is thin, and exposure is entirely fixed-station with no personal or indoor component in a setting where indoor sources plausibly dominate a young child's dose.

doi: 10.1007/s11356-026-38237-2

Shi et al. 2026 Indoor Air*Hidden Hazards at Home: Indoor Air Pollution and Adolescent Asthma - A Systematic Review and Meta-Analysis*

Meta-analysis of 77 observational studies of 11 indoor exposures and adolescent asthma. Dust mite OR 2.03 (95% CI 1.41-2.65; moderate certainty), pollen 1.70, mould/dampness 1.13 and pets 1.08. Cooking/heating was borderline (OR 1.09, 1.00-1.17; very low certainty) and open-fire sources raised risk (OR 1.26, 1.07-1.44). No association for climate-control appliances, VOCs or plants. Almost none of the combustion studies measured PM, so fuel type stands in for exposure, and GRADE certainty is mostly low. Symmetric CIs on OR scale suggest pooling on the linear scale. Relevance: moderate - the combustion signal is the one indoor PM sensors can quantify, and the evidence base lacks measurement.

doi: 10.1155/ina/7971301

Shijiazhuang AECOPD 2026 COPD*Seasonal and short-term environmental associations with COPD exacerbation admissions*

City-wide retrospective time-series of **22,468 AECOPD admissions**, 2020-2024, from the municipal medical insurance database covering all tertiary, secondary and community hospitals — unusually complete ascertainment, though ED discharges without admission are excluded. Mean age 71.7 (SD 9.1), male-to-female 3.7:1, winter peak 6,009 against summer 5,051. On the 7-day mean of lags 0-6, each 10 µg/m³ of winter PM₁₀ gave **IRR 1.069 (1.052-1.085)** and NO₂ **1.089 (1.050-1.129)**; ozone was positive in summer **1.035 (1.015-1.056)** and autumn **1.022 (1.005-1.039)**; a 1°C wider diurnal range gave **1.055 (1.034-1.077)**. The authors are explicit that the many lag × pollutant × season tests are exploratory and that the prediction models are not a deployable early-warning system — a welcome discipline, and the reason the apparent protective multipollutant IRRs should not be read as findings. Collinear pollutants in a single city cannot be separated by adjustment.

doi: 10.1080/15412555.2026.2723102

Song et al. 2026 Atmos Environ*Air Pollution Effects on Influenza-like Illness Modified by Temperature and Humidity in Central China*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 24 Sep with no abstract; Crossref, Europe PMC, PubMed efetch and Semantic Scholar all returned nothing. The title states effect modification of air-pollution associations with influenza-like illness by temperature and humidity in central China. Relevant to the humidity-confounding thread running through this issue. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.atmosenv.2026.122374

Kremer et al. 2026 Nat Microbiol*Pneumococcal population structure modifies the effect of air pollution on invasive disease*

Nineteen years of South African national surveillance — **59,017** invasive pneumococcal disease cases with **4,350** sequenced isolates — in Bayesian spatiotemporal models linking environment to pathogen diversity. PM_{2.5} is positively associated with IPD, but **the lag structure varies by serotype**: only serotypes 4, 8 and 23F confer immediate risk, and high prevalence of lineage GPSC21 (serotype 19F) shifts the population-level response to immediate. Cumulative 8-week effects for moderate humidity and cold minimum temperature are +5% and +4%. No confidence interval accompanies the headline pollution association and the serotype-stratified lag models present a large multiple-comparison surface. Still the first demonstration that pathogen genotype modifies an air-quality exposure-response, and a direct argument for pairing dense PM monitoring with pathogen surveillance.

doi: 10.1038/s41564-026-02458-5

Tang et al. 2026 Zhonghua Liu Xing Bing Xue Za Zhi*Analysis on epidemiology trends of chronic obstructive pulmonary disease in residents aged 40 years and above in Hubei Province, 2014-2024.*

China COPD surveillance, Hubei, waves 2014-15, 2019-20 and 2024, complex weighted sampling. Prevalence in adults 40+ fell 12.5% (95% CI 10.1-14.8) to 10.7% (7.8-13.6) to 8.1% (5.8-10.5), but the trend test was null (t=-2.06, P=0.079) - the point estimates drop by a third and the study still cannot reject a flat line, which is what three waves and intervals this wide buy you. Occupational dust exposure, household coal heating and respiratory symptom rates all fell significantly over the same decade while protective-equipment use rose, so the particulate exposure story is consistent even where the outcome trend is not. Adjusted risks: rural 1.38 (1.06-1.81), male 3.45 (1.80-6.61), age 70+ 8.43 (4.51-15.73), current smoker 2.48 (1.75-3.52). The protective BMI gradient (obese OR 0.07) is almost certainly reverse causation from advanced disease, not a benefit of adiposity.

doi: 10.3760/cma.j.cn112338-20251218-00906

Thoracic tissue microplastics malignancy 2026 J Clin Med*Microplastics in Human Thoracic Tissues: Increased Burden in Patients with Lung Malignancy*

Fifty adults undergoing thoracic surgery for suspected malignancy; 62 particles were recovered from 50 specimens, with microplastics detected in 32 patients (64%). Fragments made up 97%, median burden one particle per specimen. Detection was more frequent in primary lung malignancy than in metastatic or benign disease (81.8% vs 50.0%, p = 0.020), and the association held after adjusting for age and smoking, OR 4.09 (95% CI 1.06-15.79). Particles in mediastinal nodes were smaller, consistent with size-dependent translocation. The interval spans an order of magnitude on 50 patients and the lower bound is barely above unity; reverse causation is unaddressed, since tumour tissue architecture may retain particles rather than particles causing tumours. Contamination control is described but blank counts are not given.

doi: 10.3390/jcm15176503

Umea wood smoke chamber 2026 Environ Res*Controlled wood smoke exposure, sensory irritation and small airway mechanics*

Randomised, double-blind crossover chamber study: 14 healthy non-smokers (11 M, 3 F, mean age 26) exposed for 2 h to wood smoke at PM_{10} 453 $\pm$ 13 $\mu\text{g}/\text{m}^3$ and to filtered air, three-week washout, with forced oscillometry (tremoFlo) alongside spirometry and Borg symptom scoring. Wood smoke produced significant **nasal irritation** ($p = 0.003$) with trends for cough and odour, but **no exposure-specific change** in any spirometric or oscillometric resistance parameter ($R5$, $R5-R19$, $R5-R20$). Reactance and resonant frequency moved with time irrespective of exposure, which is the control the design exists to provide. With $n = 14$ healthy young adults this is badly underpowered for small-airway effects and the null should be read as such, not as evidence of no effect — the authors say as much. The methodological point stands independently: irritative symptoms preceded any mechanical signal, so spirometry-only designs will under-detect early response.

doi: 10.1016/j.envres.2026.125629

Vance et al. 2026 Scand J Public Health*Greenspace, air pollution and respiratory health outcomes: a systematic review of cohort studies*

14 cohort studies to May 2026, all rated good on the Newcastle-Ottawa Scale, predominantly European. Greenspace was generally protective against air-pollution-associated respiratory mortality, and the methodological finding is the transferable one: **effects appeared more consistently through mediation than moderation**, implying greenspace acts by improving local air quality rather than by raising individual resilience. That has a direct modelling consequence — if the pathway is mediation, greenspace and $\text{PM}_{2.5}$ should not be entered as independent covariates without care. No pooled estimate is reported, so the synthesis is narrative and magnitude is unavailable; greenspace and pollution are both strongly confounded by socioeconomic position; and near-complete European coverage limits transfer.

doi: 10.1177/14034948261481328

Wangupadcha et al. 2026 J Glob Health*A null result, reported as null, at 42 $\mu\text{g}/\text{m}^3$*

Time-series of 924 respiratory visits (ICD-10 J00–J99, mean 6.37/day) among residents aged ≥ 60 living within 5 km of a newly established industrial area in central Thailand, November 2023–March 2024, against daily $\text{PM}_{2.5}$ from the nearest government monitor, with negative binomial single-lag and moving-average models adjusted for temperature, humidity, wind speed, day of week and trend. Mean $\text{PM}_{2.5}$ was 42.43 $\mu\text{g}/\text{m}^3$ (SD 15.12), far above WHO guidance. The largest estimate was at lag 4: **IRR 1.055 (0.995–1.119)** per 10 $\mu\text{g}/\text{m}^3$ — positive, not significant, and described that way by the authors rather than spun. Subgroup estimates were numerically larger in women and at 60–69 with non-significant interaction tests. The study is underpowered by construction: **five months, 924 events, one monitor** for a 5 km radius. Worth carrying as a well-behaved null and as a monitoring-density argument.

doi: 10.7189/jogh.16.04269

Wright et al. 2026 Curr Environ Health Rep*Air pollution and respiratory outcomes in Africa — a review that mostly documents absence*

PRISMA search of PubMed, ScienceDirect and Elicit for African case-control studies 2015–2025 returned **13 studies from 10 countries**. Reported concentrations are the finding: indoor $\text{PM}_{2.5}$ 96–177 $\mu\text{g}/\text{m}^3$ in biomass-using homes and ambient $\text{PM}_{2.5}$ to 259 $\mu\text{g}/\text{m}^3$, an order of magnitude above WHO guidance. Nitrogen oxides tracked reduced lung function in children; household air pollution tracked under-five mortality and low birthweight; associations with acute respiratory infection were inconsistent across settings. **Only three studies measured temperature and none examined heat-pollution interaction**, which is the stated motivation for the review and which it therefore cannot deliver. Restricting to case-control designs also excludes the cohort and panel literature, so this is a floor on the African evidence base, not a census.

doi: 10.1007/s40572-026-00561-7

Xia et al. 2026 Commun Health*Short-term associations between air pollution and nocturnal cough frequency.*

Nightly cough counts inferred by an AI cough detector from Sleep Cycle audio, aggregated to city-day across 32 cities in 12 countries over ~500 days. Each 10 $\mu\text{g}/\text{m}^3$ $\text{PM}_{2.5}$ was associated with a 2.4% rise in nocturnal cough frequency (RR 1.024, 95% CI 1.013–1.035), with a non-linear exposure-response showing increases at low concentrations and a visible excursion during a wildfire episode. Two threats dominate: the cough detector's accuracy is never validated against a clinical reference in this paper, so outcome misclassification is unquantified, and Sleep Cycle users are a self-selected smartphone-owning population whose composition varies by city. Relevant as proof that a consumer sensor stream can act as a population respiratory endpoint alongside a PM network.

doi: 10.1038/s44528-026-00030-5

Zhang et al. 2026 Respir Med*Associations between household particulate matter and inflammatory cytokines in cystic fibrosis.*

Pilot in adults with cystic fibrosis: 4–11 days of in-home $\text{PM}_{2.5}$ monitoring with real-time sensors, then multiplex serum cytokines. Two metrics were compared - mean $\text{PM}_{2.5}$ and percentage of monitoring time above 35 $\mu\text{g}/\text{m}^3$. The mean was unrelated to any cytokine; percentage of time above 35 correlated moderately and significantly with PDGF-AA/BB, with non-significant positive trends for IL-7 and IL-8. Sample size, sensor model and any calibration are not in the abstract; Spearman correlations across a cytokine panel with no multiplicity correction; one blood draw per person. Relevance: high in kind - it is the peak-exceedance metric, only obtainable from continuous sensors, that carried the signal, not the mean a filter would give.

doi: 10.1016/j.rmed.2026.109180

Reproductive & developmental**29 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (29 OF 29 RECORDS IN THIS SUBTOPIC)

Ammons et al. 2026 Environ Res*Environmental chemical exposures and ovarian and endometrial cancers: a systematic review*

145 studies screened — 54% ovarian-cancer only, 53% North American, 40% cohort — with outdoor air pollution the single most studied exposure class (27 studies, 18%). The pooled $\text{PM}_{2.5}$ result is a clean null for ovarian cancer: **HR 1.02 (95% CI 0.91–1.13)** per 5 $\mu\text{g}/\text{m}^3$ across four studies, and **0.94 (0.81–1.09)** restricted to the three with co-pollutant adjustment. Endometrial cancer was weakly positive and implausibly precise — 1.02 (0.99–1.05) overall but **1.01 (1.00–1.01)** with co-pollutant adjustment and 1.01 (1.01–1.02) for incident cases only. **A confidence interval that narrow from five studies signals near-zero between-study heterogeneity or shared cohorts rather than genuine precision**, and no I^2 is reported. Worth recording as a negative: gynaecologic cancer is one endpoint where the $\text{PM}_{2.5}$ signal does not appear, against a literature that increasingly reports positive associations wherever it looks.

doi: 10.1016/j.envres.2026.125661

Bae et al. 2026 J Plast Reconstr Aesthet Surg

Exposure to ambient air pollutants during early gestation and phenotypic variation in non-syndromic microtia.

Single-centre case-only study of 589 patients (625 ears) with non-syndromic microtia, relating spatiotemporally modelled exposure in gestational weeks 5-9 to bilateral involvement and anatomical severity. O₃ was nominally associated with bilateral involvement; PM₁₀, SO₂ and CO showed nominal inverse associations; after FDR only an inverse SO₂ - left-ear severity association in males survived, plus an SO₂ ear-side interaction. No PM association survives. Address at clinical registration (often years after gestation) misclassifies embryonic exposure, and without unaffected controls nothing can be said about occurrence. Relevance: low; a multiplicity-driven null for PM.

doi: 10.1016/j.bjps.2026.09.003

Chen Y et al. 2026 Sci Rep

PM_{2.5} components in early pregnancy and congenital heart disease

2,601,300 live births and stillbirths, Guangdong 2014–2017, with satellite-derived component exposures resolved to the week of gestation and quantile g-computation for mixture weights. Effects peak in **pregnancy week 1**: sulfate **OR 2.23 (2.19–2.27)**, black carbon **1.95 (1.91–1.98)**, joint weights 86.8% and 13.2%. Read the intervals before the point estimates. A near-doubling of CHD odds with a 95% CI of width 0.08 implies a precision no exposure model at this spatial scale supports, and a week-1 peak in a satellite product with month-scale effective resolution is more consistent with seasonal and spatial confounding than with a window-specific teratogenic mechanism. The scale of the register is real; the causal contrast is not established.

doi: 10.1038/s41598-026-59940-7

ECHO prenatal wildfire PM_{2.5} 2026 Environ Epidemiol

Prenatal exposure to wildfire fine particulate matter and fetal growth outcomes in the United States: Findings from the ECHO Cohort

20,034 births (2006–2020) across 30 ECHO sites with daily census-tract wildfire PM_{2.5} and smoke-day counts. Mean pregnancy wildfire PM_{2.5} and smoke days were not associated with term birth weight, birth-weight-for-gestational-age z-score or small-for-gestational-age, and distributed-lag and trimester models showed no consistent sensitive window. The single positive signal is large-for-gestational-age, OR 1.015 per smoke day (95% CI 1.001–1.029), which is borderline and biologically unexplained. This is a well-powered null and should be read as one rather than as a failure; the binding limitation is tract-level assignment of a plume-scale exposure, which attenuates any true effect toward the null. Exactly the misclassification that dense low-cost networks exist to reduce.

doi: 10.1097/ee9.0000000000000526

Eick et al. 2026 J Expo Sci Environ Epidemiol

Prenatal exposure to aromatic amines and associations with infant cognition in the ECHO.CA.IL cohorts.

N=250 across two pooled prospective birth cohorts (IKIDS and Chemicals in Our Bodies). Six aromatic amines were detected in over half of second-trimester urine samples; infant visual recognition memory was measured by infrared eye tracking at 7.5 months, with log-transformed individual analytes and a molar sum entered in linear regressions and child sex examined as a modifier. The relevance to a PM watch is indirect and worth stating plainly: aromatic amines are combustion and tobacco-smoke products, cotinine is measured alongside them, so urinary AAs partly index particulate-bound combustion exposure rather than only consumer-product contact - but the study makes no air measurement and cannot separate the two routes. At N=250 with six correlated analytes and a sex-interaction analysis, this is hypothesis-generating; any single-analyte association at this sample size is underpowered for the multiplicity involved. Carried at tier C for the exposure biomarker panel, not for the cognitive estimate.

doi: 10.1038/s41370-026-00905-3

France SUDI case-crossover 2026 Paediatr Perinat Epidemiol

Short-term air pollution and sudden unexpected death in infancy, national registry

1,078 SUDI cases from the French national registry, 2015–2022, with modelled daily PM₁₀, PM_{2.5}, NO₂ and O₃ assigned at each infant's residence. A time-stratified case-crossover design with conditional logistic regression, adjusted for public holidays, influenza-like illness and daily mean temperature, examined lags to six days. **PM exposure was consistently associated with higher odds of SUDI at lags 3–5 and across the last week of life**, strongest in cumulative and multi-pollutant models; NO₂ showed a trend only, and effects were larger in spring. The case-crossover design self-matches on fixed confounders, which is its strength, but the abstract reports no point estimates or intervals — so the association's magnitude cannot be judged and this record contributes nothing to the forest plot. Residence-level modelled exposure also ignores where infants actually sleep and travel. SUDI remains an aetiologically open outcome; a short-term environmental trigger is a plausible and previously little-tested component.

doi: 10.1111/ppe.70196

Gao et al. 2026 Environ Sci Technol

Urinary biomarkers of airborne organics in pregnancy and small-for-gestational-age risk

TIMFEM prospective cohort, n=1,425 mother–infant pairs, 20 urinary biomarkers of airborne organic pollutants by LC-MS/MS (p-phenylenediamines, nitro-PAHs, phenylguanidines, benzothiazoles, cotinine), 18 detected in >40% of participants, with leave-one-out in vitro screening in human trophoblast cells at exposure-guided doses. Each ln-unit increase in **6PPD was associated with 27% higher SGA risk and 6PPD-Q with 23%**; mixture models added TPG, cotinine and 1-aminopyrene. The mechanism the cell work supports is metabolic rather than cytotoxic — reduced pyruvate entry into the TCA cycle and impaired isocitrate-to- α -ketoglutarate conversion at doses that barely touched proliferation. No confidence intervals are given for the epidemiological estimates in the abstract, so they are not plotted here. Relevance: these are tyre-and-road and traffic-associated organics that no PM mass channel resolves.

doi: 10.1021/acs.est.6c04451

Gielen et al. 2026 Neonatology

The association of neighborhood socioeconomic status with birth weight is not confounded by ambient air pollution in twin pregnancies

Population-based twin cohort stratified by placentation. In dichorionic twins, each EUR 3,000 higher neighbourhood median income raised birth weight by 46.83 g (21.92 g after adjusting for gestational age) and gestation by ~26 h; adding PM₁₀ and NO₂ did not change these estimates. PM effects themselves are not reported in the abstract, n is not stated, and kriged exposure over 1.55 km² sectors has limited contrast in Flanders. The finding concerns confounding of SES by PM, not a PM effect. Relevance: low; supports treating SES and PM as separable in birth-outcome models.

doi: 10.1159/neo/adsag016

Hu et al. 2026 Reprod Toxicol

Real-world PM_{2.5} exposure triggers early testicular epigenetic responses in the absence of significant changes in measured reproductive endpoints.

Male mice breathed real-world ambient PM_{2.5} in Hangzhou through a whole-body inhalation system for 4, 8 or 12 weeks - roughly one to three spermatogenic cycles. Sperm quality, reproductive hormones, histopathology and oxidative DNA damage were all unchanged, while global testicular 5mC fell at 4 weeks and rose at 12 weeks relative to control ($P < 0.05$), and 5hmC rose significantly at 12 weeks. The claim is that epigenetic remodelling precedes measurable injury. Two limits: the record gives no exposure concentration, so dose is unquantified, and global 5mC/5hmC has no locus resolution, so a bidirectional time course is as compatible with adaptation as with damage. Real-ambient inhalation studies live or die on continuous characterisation of what the animals actually breathe - an instrumentation problem.

doi: 10.1016/j.reprotox.2026.109366

Korean birth-cohort growth mixture 2026 Environ Pollut

Prenatal bisphenols and ambient air pollutants are differentially associated with early childhood length and weight from birth to 36 months: a multi-domain mixture analysis in a Korean birth cohort

METADATA ONLY. No abstract was available on the entry date from Crossref per-DOI, Europe PMC, PubMed efetch or Semantic Scholar. Title establishes a Korean birth cohort with a multi-domain mixture analysis placing prenatal bisphenols alongside ambient air pollutants against length and weight trajectories to 36 months, and states that the two exposure domains behave differently - but not in which direction, for which pollutant, or at what magnitude. Nothing is asserted here and the record is tier C. Flag for re-reading: multi-domain mixture methods that carry chemical and air exposures in one model are the methodological question this archive should track

doi: 10.1016/j.envpol.2026.129145

Li et al. 2026a BMJ Open

Cohort profile: the Shenzhen Birth Cohort (SZBC) study.

Cohort profile rather than a result: 4,208 women recruited at 19 weeks' gestation or earlier from a single Shenzhen maternity hospital since March 2018, 3,962 with complete baseline data, 3,150 deliveries, up to 20 follow-up points through age seven, with questionnaires, examinations and blood, urine and faecal samples. Findings cited to date include prenatal greenness associated with better motor development and postnatal PM_{2.5} - nitrate in particular - with delayed motor and problem-solving development. No effect sizes, intervals, exposure model or adjustment set are given, and single-hospital recruitment in a migrant-dense city limits transportability. Carried as infrastructure: a component-resolved prenatal and postnatal cohort with biosamples is the design that could eventually separate nitrate from mass.

doi: 10.1136/bmjopen-2026-121066

Sun et al. 2026 Environ Res

Air pollutant mixtures and endometrial receptivity in the peri-implantation window

1,349 women across 14 cities in Liaoning, with quantile g-computation and BKMR on a six-pollutant mixture against sonographic endometrial thickness and volume. A simultaneous one-quartile increase in the mixture is associated with 0.5 mm less thickness and 0.28 ml less volume, with CO and O₃ carrying the largest weights and PM_{2.5} and PM₁₀ not dominating. **No confidence intervals are reported for the mixture effects**, which for a BKMR result is a substantive omission. A 0.5 mm shift is at or below the reproducibility of transvaginal endometrial measurement, so the clinical meaning is unclear even if the association holds. Monthly gridded exposure with no personal or indoor component, against an implantation window measured in days.

doi: 10.1016/j.envres.2026.125682

Mbishi et al. 2026 Environ Pollut

Perinatal PM and infant head circumference: unambiguous, and small

119,110 Flemish children with BMI-, length- and head-circumference-for-age z-scores from routine well-baby clinic records, perinatal PM₁₀ and PM_{2.5} from the RIO detrended-kriging model, and mixed-effects models carrying a 5-df natural cubic spline in child age with exposure-by-age interaction and child-level random intercepts and slopes. At 12 months, per IQR: head circumference -0.0499 (95% CI -0.0606, -0.0393) z for PM₁₀ and -0.0645 (-0.0771, -0.0518) for PM_{2.5}, with inverse associations sustained between 6 and 24 months. BMI- and length-for-age associations change sign across age, and green-space associations are inconsistent across measures. The authors state that estimates are small and that most nominally significant associations are of limited magnitude - roughly a fifteenth of a standard deviation. Exposure is municipality-scale interpolation, which at this n makes precision a poor guide to accuracy.

doi: 10.1016/j.envpol.2026.129064

Nelin et al. 2026 Ann Epidemiol

Associations of short-term changes in air pollution with spontaneous preterm birth: A time-stratified case-crossover analysis.

A well-powered null: 109,735 spontaneous preterm births in Michigan (2003-2020) and 5,321 in metropolitan Philadelphia (2008-2020), analysed with nested daily lags, averaged lags and DLNMs, all temperature-adjusted, with likelihood ratio tests comparing linear-lag and DLNM fits. No significant association emerged in either cohort under any specification, with point estimates near null and narrow intervals - the narrowness is what makes this informative rather than merely underpowered. The exposure is modelled, not measured, so non-differential error at the residential address would bias exactly toward this result, and the authors do not quantify that error. Case-crossover also removes only time-invariant confounding. Worth pairing with the positive short-term literature rather than reading alone.

doi: 10.1016/j.annepidem.2026.110291

Preeclampsia nationwide 2018-23 2026 Eur J Intern Med

METADATA ONLY — *nationwide air pollution and preeclampsia, 2018–2023*

The PubMed record carries no abstract, and neither Crossref nor Europe PMC held one on the entry date. What the record establishes: a nationwide study covering 2018–2023 has examined the association between ambient air pollution and preeclampsia. What it does not establish: the country, the pollutant set, whether PM was resolved at all, the case count, the exposure-assignment method, the trimester windows examined, or the direction and magnitude of any association - none of which follows from the title, and none of which is asserted here. Flagged for retrieval. Hypertensive disorders of pregnancy are one of the better-powered PM endpoints and a nationwide six-year series would be a substantial addition if the exposure model holds up.

doi: 10.1016/j.ejim.2026.107175

Prenatal PM childhood NHL Spain 2026 Cancers

Exposure to Particulate Matter (PM) During Pregnancy and Non-Hodgkin Lymphomas in Children

Population-based case-control study across Spain (2009-2016) with 289 incident non-Hodgkin lymphoma cases from the Spanish Childhood Tumor Registry against a base population of 3,682,038 children under 15. Continuous prenatal PM_{2.5} was associated with higher NHL odds and PM₁₀ more weakly; tertile analyses put the PM_{2.5} signal in the third trimester, and PM₁₀ in the first and second trimesters among children diagnosed before age five. The abstract reports no odds ratios or intervals at all, only directions, so effect size cannot be judged from it. With 289 cases split across three tertiles and three trimesters the trimester-specific cells are small, and random-forest exposure surfaces at residential level carry error that is itself spatially structured.

doi: 10.3390/cancers18172746

Prenatal greenspace ASD window 2026 Environ Pollut*A mid-to-late pregnancy window for green space — and PM_{2.5} is not the mechanism*

1:3 age- and sex-matched case-control from a tertiary maternal and child health hospital in central China: 207 children with autism spectrum disorder, 621 controls. Monthly residential NDVI and PM_{2.5} were estimated from 10 months before to 3 months after birth and entered a distributed lag nonlinear model, with product terms testing NDVI $\times$ PM_{2.5} interaction, bootstrapped mediation, and Bonferroni correction. Cases had lower NDVI in all 14 months; the protective window localises to mid-to-late pregnancy, roughly 6 to 1 month before term birth. The negative result is the one that earns the entry: **PM_{2.5} neither mediated nor modified the green-space effect** after correction, which closes the most obvious alternative explanation rather than leaving it hanging. Caveats are the standard ones for this design — residential address as the exposure proxy, hospital-based case ascertainment, 207 cases against a 14-month lag structure, and green space as a proxy for an unmeasured bundle (noise, activity, socioeconomic position).

doi: 10.1016/j.envpol.2026.129118

SIP prenatal Bayley 2026 Child Neuropsychol*In utero PM_{2.5} and NO₂ and cognitive-behavioural development at 1–3 years*

Stress in Pregnancy cohort, $n = 210$, Queens, New York; Bayley-III and ABAS-3 at 18, 24 and 36 months; prenatal PM_{2.5} and NO₂ taken from public monitoring databases. The reported pattern is **sex-specific and pollutant-specific**: PM_{2.5} associated with poorer communication, social skills and gross motor function in **boys**, NO₂ with poorer communication, gross motor and social-emotional skills in **girls**. No effect estimates or intervals appear in the abstract, which is why nothing from this paper enters the forest plot. With $n = 210$ split by sex across three timepoints and multiple Bayley and ABAS subdomains, the multiplicity burden is severe and the crossed sex-by-pollutant pattern is exactly the shape that subgroup analysis produces by chance. Exposure from public monitors rather than a spatial model adds classical misclassification. Hypothesis-generating, and the authors largely frame it that way.

doi: 10.1080/09297049.2026.2723475

Shriyan et al. 2026 Wellcome Open Res*Ambient and Indoor Air Pollution in Pregnancy and the risk of Low birth weight and Ensuing Effects in Infants (APPLE): A cohort study in Bangalore, South India.*

516 pregnant women recruited before 18 weeks in Bangalore urban slums, infants followed to age one. This is a protocol, not a result - no effect estimate is reported, and it is a fourth-version revision of a design first posted years ago, so its appearance in this window is a deposit event rather than new evidence. Carried because the measurement plan is the relevant part: paired indoor and ambient PM₁₀, PM_{2.5} and CO in solid-fuel slum households is exactly the setting where a fixed regulatory monitor is furthest from personal exposure, and where a low-cost indoor node earns its keep. The planned mediation analysis of LBW on infant adiposity is ambitious for $n=516$ and will be underpowered for anything but a large effect.

doi: 10.12688/wellcomeopenres.14830.4

Tamire et al. 2026 J Expo Sci Environ Epidemiol*Prenatal fine particulate matter (PM_{2.5}) exposure differentially impairs fetal skeletal and soft-tissue growth: a longitudinal study in rural Butajira, Ethiopia.*

Pregnant women recruited at or before 24 weeks, followed over three visits, March-June 2025. An IQR rise of 27.0 $\mu\text{g m}^{-3}$ in personal PM_{2.5} was associated with a 0.43 SD fall in femur length (beta -0.43, 95% CI -0.51 to -0.35) and a 0.17 SD fall in abdominal circumference (-0.17, -0.21 to -0.13); head circumference and biparietal diameter showed no significant association under either concurrent or lagged exposure. The dissociation is the finding - skeletal and soft-tissue growth respond while cranial growth is spared, which is the brain-sparing pattern and argues against a purely nutritional confounder. Three weaknesses bite. The exposure contrast is enormous by design (an IQR of 27 $\mu\text{g m}^{-3}$ implies a biomass-cooking population far outside the range of the cohorts these coefficients will be compared against), the sample is small and the paper does not report n in the abstract, and three 24-h samples cannot represent a 16-week gestational window - classical measurement error that, being non-differential, biases toward null and so does not explain the femur result. Directly relevant to distributed sensing: the exposure instrument is a consumer-grade optical wearable, and effect sizes this large resting on three wearable-days per pregnancy is exactly the design low-cost sensing makes newly affordable in LMIC settings - and exactly where sensor humidity and biomass-smoke calibration error, unreported here, would propagate straight into beta.

doi: 10.1038/s41370-026-00966-4

Thirumalazhagan et al. 2026 Reprod Toxicol*A Comprehensive Meta-Analysis on the Impact of airborne Particulate Matter on Trophoblast Cell Function In Vitro.*

Twelve in vitro studies from 1,506 records to September 2025. PM_{2.5} reduced trophoblast viability by a pooled 20.7%, disrupted steroidogenic enzymes (CYP11A1, 3 β HSD1, FDXR), uncoupled hCG from progesterone synthesis, induced oxidative DNA damage with ATR-mediated p53-independent apoptosis, suppressed VEGF-VEGFR2-MEK-ERK-mTOR signalling and impaired invasion via TIMP, RhoA and collagen I. The central problem is that pooling relative risks of gene expression across studies with different particle sources, extraction methods, doses and cell lines produces a number with no exposure meaning; the dose axis is absent from the abstract, and cell-line trophoblasts are not placenta. Google Scholar as a search source compromises reproducibility. Useful as a pathway catalogue for placental mechanisms behind birth-weight associations, not as quantitative evidence.

doi: 10.1016/j.reprotox.2026.109364

Wang et al. 2026 Biomolecules*Associations of ASIC2, IKZF3, and GSDMB Variants with Gestational Diabetes Mellitus and Their Interactions with Environmental Exposures Among Pregnant Women.*

Case-control study of 1,564 pregnant women genotyped for ASIC2 rs12450465, IKZF3 rs2060941 and GSDMB rs2305479, with gene-environment interaction tests for PM_{2.5}, O₃ and urinary MIBP. The T-T-T haplotype was associated with GDM (OR 1.741, 95% CI 1.208-2.509; FDR p 0.0133) and, in the 289 with MIBP, OR 2.980 (1.378-6.446). Interaction signals surviving FDR mainly involved O₃, not PM_{2.5}; primary analyses were covariate-unadjusted. Candidate-gene interaction testing in one sample without replication is weak evidence, and exposure assignment is not described. Relevance: low; no PM_{2.5} main effect or robust interaction is reported.

doi: 10.3390/biom16091373

Wang et al. 2026 J Heart Lung Transplant*Long-term Exposure to PM_{2.5} and Risk of Adverse Pregnancy Events in Women with Pulmonary Hypertension: A Nationwide Multicenter Study*

751 pregnant women with pulmonary hypertension at 24 Chinese tertiary hospitals (2016-2022); 26.4% had maternal death or heart failure and 59.8% adverse fetal or neonatal events. Median 3-year PM_{2.5} 32.8 $\mu\text{g m}^{-3}$. Per 10 $\mu\text{g m}^{-3}$, OR 1.169 (95% CI 1.014-1.349) for maternal and 1.169 (1.018-1.348) for fetal/neonatal events, robust in two-pollutant models, strongest in late pregnancy. Retrospective and hospital-based, so referral patterns track region and pollution; the outcome is common, so ORs overstate relative risk; PH subtype and severity adjustment are not described. Relevance: moderate - a small, high-risk group where targeted indoor filtration with sensor feedback is a plausible intervention.

doi: 10.1016/j.healun.2026.09.021

Wanta et al. 2026 Cells Tissues Organs

Ultrastructural Changes in Human Placental Tissues and Evidence of Particulate Matter Transport into the Umbilical Vein Following PM_{2.5} Exposure in Chiang Rai.

Placentas from 10 women with third-trimester exposure during Chiang Rai's February-April burning season and 10 without were examined by SEM-EDS and TEM. Exposed villi were flattened and irregular; metal-containing particles were adherent to the placental surface and inside the umbilical vein, with an elemental profile (Al, Ca, Cr, Cu, Fe, K, Mg, Mn, Na, Ni, P, Si, Ti) resembling filter-collected PM_{2.5}; syncytiotrophoblast abnormalities and barrier thickening were seen. Exposure is season of delivery, not a measured dose; n=10 per group; SEM-EDS cannot exclude contamination during processing, and these elements are also endogenous. Relevance: supports tissue translocation, but a personal-monitor exposure axis would be needed to make it dose-response.

doi: 10.1159/cto/acbag006

Wei Q et al. 2026 Int J Epidemiol

Weekly component windows: black carbon at 12–15 weeks, nitrate at 5–12 and 22–24

2,680 mother–newborn pairs and 9,339 growth ultrasounds (abdominal circumference, head circumference, femur length, estimated fetal weight), Guangzhou 2017–2020, with *weekly* black carbon, organic matter, nitrate, sulfate and ammonium from the TAP model entered into distributed lag models with GEE, plus machine-learning mixture analysis. Femur-length reduction is most strongly associated with **black carbon at gestational weeks 12–15**; head-circumference deficits with **nitrate at weeks 5–12 and 22–24**; estimated fetal weight with **nitrate at weeks 6–8**. Black carbon and nitrate dominate the mixture. This is the exposure resolution the Flemish cohort above lacks, at 45× fewer children. The threat to inference is that five components drawn from one gridded product co-vary strongly, so “which component” is partly a question about the exposure model rather than about biology; the abstract reports no effect sizes for the identified windows.

doi: 10.1093/ije/dyag185

Xu ZL et al. 2026 Biomed Environ Sci

Component-specific PM_{2.5} windows and oocyte yield in assisted reproduction

51,122 ART cycles with individual PM_{2.5} and its sulfate, nitrate, ammonium, organic-matter and black-carbon fractions assigned over recent (0–3 months), distal (4–12 months) and cumulative windows; negative binomial regression on total, mature and normally fertilised oocyte counts, DLNM for window identification, mixture models for joint effect. Higher exposure tracked poorer oocyte outcomes throughout, with **stronger effects for the distal window** and two sensitive periods — 1 month and 6–11 months before retrieval. Sulfate and ammonium dominated the mixture. The 6–11 month window is biologically coherent with primordial-to-antral follicle transit, which is the paper's strongest argument. Its weakness is that ART patients are a selected population and the exposure model is residential, so a distal window is also the window over which residential misclassification is largest.

doi: 10.3967/bes2026.072

Zanini et al. 2026 Environ Pollut

Maternal air-pollution exposure and fetoplacental Doppler markers

Ships on metadata only. The Crossref deposit carries no abstract, the record is not in PubMed or Europe PMC, and Semantic Scholar returns 404 for the DOI, so cohort size, exposure model and Doppler effect sizes are all unavailable. Counted and listed; no numbers claimed. Flagged for the abstract retry list — fetoplacental haemodynamics is a mechanistically informative intermediate between maternal PM exposure and birth outcome and is thinly represented in this watch's corpus.

doi: 10.1016/j.envpol.2026.129046

Zhang et al. 2026 Front Public Health

Association between ambient PM(2.5) exposure during pregnancy and risk of preterm birth: a retrospective birth cohort study in Chengdu, Southwest China.

35,136 singleton births at one Chengdu hospital, 2015–2024; 2,622 (7.46%) preterm. Mean whole-pregnancy PM_{2.5} was 49.1 $\mu\text{g m}^{-3}$ (satellite) and 51.9 $\mu\text{g m}^{-3}$ (monitors). Per 10 $\mu\text{g m}^{-3}$, whole-pregnancy ORs were 1.089 (95% CI 1.010–1.174) satellite and 1.091 (0.996–1.195) monitor, with first- and third-trimester signals and landmark ORs 1.06–1.10 for weeks 32–35; associations were confined to late, not earlier, preterm birth. Single-hospital sampling, district-level fixed effects and residence at delivery limit the exposure contrast. The methodological result is the useful one: the two exposure products gave near-identical point estimates, with satellite giving complete coverage and narrower intervals. Relevance: a direct, if coarse, benchmark for when a sparse monitor network is adequate - the question dense sensor networks exist to answer at finer scale.

doi: 10.3389/fpubh.2026.1936547

Zhang et al. 2026 Environ Pollut

Second-trimester PM_{2.5} and PM₁₀ exposure is associated with neonatal metabolome alterations and elevated risk of pregnancy loss

Propensity-matched cohort of 30,654 singletons (106 losses). Second-trimester PM_{2.5} OR 2.32 (95% CI 1.12–4.78) and PM₁₀ OR 1.86 (1.18–2.95) for pregnancy loss; increments not in the abstract. A five-metabolite risk score tracked second-trimester PM_{2.5} in single-window models ($b=0.052$) but not after mutual adjustment across windows ($b=0.003$, $p=0.29$); glycine, proline, valine and tyrosine were lower. With 106 events the CIs are wide, and the authors note single-window and sensitivity analyses were inconsistent. Neonatal metabolites cannot measure a lost pregnancy, so the mechanistic link is indirect. Relevance: low-moderate; window-specific estimates need residential exposure series, not annual surfaces.

doi: 10.1016/j.envpol.2026.129218

Mechanistic toxicology

55 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (55 OF 55 RECORDS IN THIS SUBTOPIC)

6PPD-Q AC16 p53 2026 Ecotoxicol Environ Saf

Cardiotoxic mechanism of the tire-derived transformation product 6PPD-quinone

Network toxicology and docking followed by wet-lab confirmation in human AC16 cardiomyocytes. **144 putative targets** with TP53 the top hub; IC₅₀ 13.92 μM at 24 h. 6PPD-Q drove ROS accumulation and **p53 Ser15 phosphorylation without raising total p53 protein or TP53 mRNA** — i.e. post-translational stress activation — alongside TRAF6 upregulation, MAPK1 phosphorylation, IL6/TNF induction, mitochondrial apoptosis (membrane potential loss, Bax/Bcl-2 rise, cleaved caspase-9 and -3) and blocked late autophagic flux (LC3-II accumulation with p62 depletion). Pifithrin- α attenuated all of it, which is the load-bearing experiment. The exposure gap is total: a micromolar dose to an immortalised cardiomyocyte line has no established relationship to any airborne 6PPD-Q concentration, and no inhalation route is demonstrated. Relevant here only because 6PPD-Q is a measurable constituent of road dust and traffic-derived PM.

doi: 10.1016/j.ecoenv.2026.120754

AS-IV vs PM vascular senescence 2026 Cardiovasc Toxicol

Replicative and Air Pollution-Induced Senescence: Telomere Dysfunction, Mitochondrial Stress, and Modulatory Effects of Astragaloside IV in Endothelial and Vascular Smooth Muscle Cells

HUVECs and human coronary smooth muscle cells under replicative senescence (passages 6-7, 10-11, 17-20) and fine-dust-induced premature senescence (48 h, 50 and 100 ug/mL certified urban PM₁₀-like reference material), then Astragaloside IV at 50 uM for 24 h. Both cell types showed SA-beta-gal positivity, telomere shortening, p16INK4a/p21Cip1 activation and mitochondrial dysfunction; fine dust drove premature senescence, telomere shortening, TERRA upregulation and acute mitochondrial impairment mainly in endothelial cells, with smooth-muscle cells relatively resistant. AS-IV partially rescued endothelial but not smooth-muscle damage. The dose is the problem: 50-100 ug/mL in culture is orders of magnitude above any inhaled ambient mass dose, so the mechanism is plausible and the magnitude is not transferable.

doi: 10.1007/s12012-026-10173-0

Ameli et al. 2026 Environ Pollut

Acute co-exposure to urban particulate matter and aircraft noise elicits lung-centered immune-redox proteomic remodeling in mice.

DIA LC-MS/MS proteomics of lung, brain and heart from male C57BL/6J mice given three days of NIST SRM 1648a urban PM, aircraft noise, both, or control - the exposure model previously characterised by Kuntic et al. The lung dominated: PM+noise gave the broadest induction of acute-phase, phagosome, interferon and macrophage proteins, with NADPH oxidase, iron handling, barrier and endothelial/ECM changes that agree with the parent study's NOX-2, HO-1, ROS, blood-pressure and vascular endpoints. Most stress proteins scaled proportionally rather than synergistically; brain and heart responses were smaller. Dose is a standard reference material, not a measured airborne concentration in the abstract, and only males were used. Relevance: noise and PM co-vary along the same roads, so co-located acoustic and PM sensing is the matching exposure design.

doi: 10.1016/j.envpol.2026.129165

Arora et al. 2026 J Environ Sci (China)

Oxidative potential of fresh versus ozone-aged PM_{2.5} across Chinese sources

A commentary, not primary data — it summarises Ma et al. (10.1016/j.jes.2024.04.023), which measured DTT-assay oxidative potential on fresh and O₃-aged PM_{2.5} from multiple source types. **Biomass-burning particles carried up to 35× the OP of suburban PM_{2.5}**, driven by water-soluble organics and transition metals, and O₃ aging generally *lowered* OP while producing complex chemical transformation. Counted and listed here because the source-dependence claim is the one this watch tracks, but no number in it is new and the aging direction runs opposite to the usual assumption that atmospheric processing increases toxicity. Treat as a pointer to the 2024 primary paper.

doi: 10.1016/j.jes.2026.05.042

Berberine PM_{2.5} breast cancer 2026 Apoptosis

Berberine shows potential in mitigating PM_{2.5}-induced breast cancer progression by inducing DNA damage and inhibiting error-prone DNA repair pathways

Water-soluble Delhi PM_{2.5} applied to MCF7 breast cancer cells increased proliferation, migration and ROS and produced multinucleated giant cells, i.e. genomic instability. Transcriptomic profiling gave a PM_{2.5} carcinogenic signature enriched in MAPK, ROS, metabolic, lysosomal and ribosomal pathways, with BIRC5, WSB1 and RCC1 dysregulated in patient tumours and reversed by berberine. The inference is weak on two counts: only the water-soluble fraction was used, which discards the insoluble carbonaceous and metal-bound mass that carries much of PM_{2.5} oxidative potential, and no dose-response or in vivo confirmation is reported. Relevant to sensing only as a reminder that mass concentration is a poor proxy for the bioactive fraction.

doi: 10.1007/s10495-026-02423-x

CSE HASMC isoforms 2026 PLoS One

A comprehensive full-length transcriptome landscape of cigarette smoke-exposed HASMCs reveals extensive remodeling of mRNA isoforms and regulatory networks.

Oxford Nanopore long-read sequencing of human aortic smooth muscle cells at 0.6% cigarette smoke extract, the midpoint of a biphasic viability curve (0.2% mildly proliferative, 0.4-0.8% progressively cytotoxic). 11,222 novel transcripts; 1,866 transcripts up in ferroptosis and cell division pathways, 493 down in oxidative phosphorylation and glutathione metabolism, 1,753 alternative polyadenylation events remodelling 3'UTRs, and 269 alternative splicing events hitting cell-cycle regulators and transcription factors including CREM, FOSL1 and HBP1. The claim is that high-dose cytotoxicity is executed post-transcriptionally rather than through differential gene expression alone, and the isoform resolution is what makes that visible - a short-read study would have reported the pathway shifts and missed the 3'UTR remodelling entirely. Admitted at tier C with an explicit caveat: cigarette smoke extract is not ambient PM, the cell line is not a vessel, a single 0.6% dose is not a dose-response for the mechanism claimed, and no particle mass or size metric is reported, so nothing here is transferable to an ambient exposure-response coefficient. Carried for the ferroptosis and glutathione-depletion axis, which recurs in ambient PM vascular toxicology and is worth tracking.

doi: 10.1371/journal.pone.0354604

Cai et al. 2026 Ecotoxicol Environ Saf

Prenatal PM_{2.5} exposure disrupts cortical excitatory/inhibitory balance in male offspring by suppressing Lhx8-regulated cholinergic homeostasis and cell adhesion

Male offspring of PM_{2.5}-exposed dams showed a persistent cortical shift toward inhibition (gephyrin up at PND 1, 7 and 21). RNA-seq and microarray identified 14 differentially expressed transcription factors; Lhx8 was consistently suppressed, with downstream ChAT loss, compensatory GABRA1 up-regulation and repression of adhesion genes (Cadm2, Cdh10, Pgrmc1, Cd36). Dose, route, PM source and litter-level analysis are not given in the abstract, males only, and no behavioural phenotype or Lhx8 rescue is reported, so causality within the proposed pathway is not tested. Relevance: low for sensing; supports the prenatal neurodevelopmental window seen in cohort data.

doi: 10.1016/j.ecoenv.2026.120862

Cao et al. 2026 Environ Sci Atmos

Black carbon interfacial chemistry and multiscale aging

A review that puts BC *particle state* — surface reactivity, coating thickness, morphology, aging timescale — rather than BC mass at the centre, and follows it through to optics, cloud activation, removal, lifetime and toxicity. The framing matters for instrumentation because every one of those downstream properties is a function of a variable that filter-based absorption and single-particle soot photometry constrain only partially, and that no low-cost optical sensor constrains at all. No primary data and no quantitative synthesis; the Crossref deposit carries only a graphical-abstract blurb, so the parameterisation recommendations were not verifiable at digest time. Listed as a reference framework rather than as evidence.

doi: 10.1039/d6ea00093b

Chen et al. 2026 Environ Sci Technol

A Quasi-Michaelis-Menten Mechanism Explains the Nonlinear Relationship between Transition Metals and PM_{2.5} Oxidative Potential Measured by the Dithiothreitol (DTT) Assay.

62 ambient PM_{2.5} samples, Beijing, one year, plus laboratory standards. The authors derive a quasi-Michaelis-Menten scheme for DTT, transition metals and oxygen, and the resulting Cu and Mn equations reproduce both the saturating single-metal response and the antagonism seen in Cu-Mn mixtures. A three-term model on water-soluble Cu, Mn and WSOC - WSOC entered linearly - accounts for 98.6% of measured OP-DTT. The weakness is that 98.6% is an in-sample fit on 62 filters from one city with no held-out set and no cross-site test, so the coefficients are calibration, not validation. Still, this is the most directly useful result of the day for sensing: it says OP-DTT is a saturating, not additive, function of composition, so any attempt to infer toxicity-weighted PM from a low-cost mass or optical channel must carry a nonlinear composition term or it will systematically overstate hazard in metal-rich air.

doi: 10.1021/acs.est.6c09293

Cherriere et al. 2026 Part Fibre Toxicol

Influence of lung overload and exposure scenarios on recovery following inhalation of an alumina nanoparticle and HCl gas mixture.

Rats inhaled alumina nanoparticles at 1-20 mg/m³ with 5 ppm HCl under two temporal patterns - four consecutive days, or once weekly for eight weeks - with assessment at 1 and 28 days post-exposure. The short scenario produced dose-dependent neutrophilic inflammation that resolved by day 28 with partial clearance recovery; the intermittent long scenario left significantly higher retained lung burdens, reduced clearance and persistent neutrophilia at day 28, with the top dose exceeding accepted overload thresholds but no remodelling or genotoxicity. The abstract gives no burden values or dispersion, so the overload claim cannot be checked against the usual volumetric criterion. The transferable point for exposure science: equal cumulative dose delivered intermittently is not equivalent, which time-averaged occupational metrics assume.

doi: 10.1186/s12989-026-00703-7

Driencourt et al. 2026 Rev Mal Respir

The impact of post-exposure Streptococcus pneumoniae infections on pulmonary immunity: The role of air pollution and IL-20 cytokines in respiratory infections.

French-language short review framing the experimental programme on how PM_{2.5} alters pulmonary immunity to Streptococcus pneumoniae, with IL-20-family cytokines as the candidate mediator. The abstract restates burden figures (>9 million deaths globally, 48,000 in France) and oxidative-stress/inflammation mechanisms, and reports no data. Relevance: low; flags IL-20 cytokines as a mechanistic target for PM-infection susceptibility studies.

doi: 10.1016/j.rmr.2026.07.009

Dulong et al. 2026 Biochimie

Effects of fine particulate matter on neuroblastoma cell line: metabolic and lipid deregulations and their consequences on the onset of neurological diseases

Narrative review built around the **SH-SY5Y** human neuroblastoma line as an in vitro neurodegeneration model. The proposed cascade runs mitochondrial damage → ER stress → oxidative stress → calcium dyshomeostasis → lipid dysregulation, converging on apoptosis and autophagy-dependent **ferroptosis**, with lipid peroxidation advanced as the emerging driver. No pooled estimate, no exposure-response and no dose metric: the review does not harmonise PM_{2.5} mass, source or exposure duration across the studies it assembles, and the in vitro doses it draws on generally sit far above ambient, so the chain is qualitative. It concedes its own in-vitro-to-human translation gap. Every mechanism it privileges is **composition-dependent** — redox-active metals, organic carbon — which a mass-only network cannot resolve.

doi: 10.1016/j.biochi.2026.09.011

Fetal macrophage FCER1G 2026 Nat Commun

Prenatal PM_{2.5}, fetal macrophage epigenetic reprogramming and atopic dermatitis

Birth-cohort association between **first-trimester** PM_{2.5} and early-childhood atopic dermatitis, carried into mechanism by integrative multi-omics including single-cell profiling of placenta and fetal immune cells. The central finding is **hypomethylation of FCER1G in fetal macrophages** driving sustained overexpression that persists across developmental stages and re-emerges in M2 macrophages of AD skin and peripheral blood; functional work links FCER1G networks to NADPH-oxidase-mediated ROS and Th2 inflammatory signalling. This is a rare case of a durable, cell-type-resolved epigenetic mediator proposed for a prenatal PM_{2.5} association rather than asserted. The exposure remains a modelled residential estimate assigned to a trimester, which is the weakest link in an otherwise strong chain — first-trimester windows are precisely where exposure misclassification is largest. The persistence claim rests on cross-sectional comparisons at different stages, not on following the same cells.

doi: 10.1038/s41467-026-76298-6

Gao et al. 2026 Front Public Health

CCL22-associated immune signatures in air pollution-related atopic dermatitis

Shared pollutant-atopic dermatitis targets enriched in cytokine and chemokine signalling, pattern-recognition receptor activity, IL-17 and Toll-like receptor pathways. A framework over 15 algorithms and 175 combinations selected plsRglm + AdaBoost, with AUC 0.963 in training and 1.000, 0.909 and 0.966 across three validation cohorts, giving a ten-gene signature (CCL22, CCL5, CSF2, F2RL1, HRH4, ICAM1, IFNG, IL10, IL17A, IL18). CCL22 was upregulated in lesional samples, localised to dendritic cells and macrophages, and confirmed in DNFB-induced murine dermatitis. **The exposure link is the weak joint: no particulate matter was measured or administered anywhere**, so air pollution enters only as a database annotation. An AUC of exactly 1.000 indicates leakage or overfitting rather than performance, and DNFB is a contact hapten, not a particle.

doi: 10.3389/fpubh.2026.1906298

Ghio et al. 2026 Pathogens

Iron, Tuberculosis, and Particle-Related Exposures Including Smoking.

Mechanistic review from the EPA group proposing that particles - cigarette smoke, air pollution, biomass smoke, silica - complex host iron and induce a functional iron deficiency in the respiratory tract, which favours M. tuberculosis because of its many iron-acquisition pathways, explaining the epidemiological association of particle exposures with TB. A hypothesis-driven synthesis without systematic search or new data. It pairs, with opposite sign, with the Shen et al. finding in this issue that exogenous Fe relieves organic-induced iron deficiency in macrophages. Relevance: low; conceptual.

doi: 10.3390/pathogens15090982

Gradinaru et al. 2026 J Xenobiot

Atmospheric Particulate Matter as a Carrier of Pb, Cd, and Ni: From Environmental Transfer and Bioaccessibility to Molecular Toxicity and Predictive Modeling.

A narrative review tracing PM-bound lead, cadmium and nickel from emission through transport, deposition, food-chain transfer, bioaccessibility and toxicokinetics to molecular toxicity and predictive modelling. The central argument is that total PM mass and total metal concentration misrepresent biologically effective dose, which also depends on respiratory deposition, dissolution kinetics, gastrointestinal release and co-associated constituents; the three metals share oxidative, inflammatory, mitochondrial, genomic and epigenetic effects but differ in mobility and target organs. No search strategy or quality appraisal is described, so coverage cannot be judged, and no quantitative synthesis is offered. Relevance: a reminder that optical mass sensing is blind to exactly the metal fraction that drives toxicity; composition needs periodic filter sampling alongside.

doi: [10.3390/jox16050167](https://doi.org/10.3390/jox16050167)

Gulyaev et al. 2026 Biomed Rep

Polyphenols and inhalation-induced oxidative stress in chronic lung diseases: a systematic review

Systematic review to January 2026 across PubMed, Cochrane and Web of Science, with RoB 2 and SYRCL risk-of-bias assessment. Preclinical evidence was consistent — attenuated oxidative damage, suppressed inflammatory signalling and enhanced antioxidant defence, largely via Nrf2 activation and NF- κ B modulation, with histopathological improvement across several exposure models. **Clinical evidence was not:** some trials improved oxidative and inflammatory markers or symptom scores, but spirometric effects were inconsistent. That preclinical-to-clinical gap is the familiar signature of antioxidant intervention research, and the authors say so. Heterogeneity precluded meta-analysis, so there is no pooled estimate; trials were small and short. Recorded as the mitigation-side counterpart to the exposure literature, not as evidence that dietary intervention substitutes for reducing exposure.

doi: [10.3892/br.2026.2194](https://doi.org/10.3892/br.2026.2194)

Gulzar et al. 2026 Atmos Pollut Res

Mechanistic Insights into PM_{2.5}-Induced Cardiopulmonary Dysfunction: Bridging Pathways to Therapeutic Interventions

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 24 Sep with no abstract; Crossref, Europe PMC, PubMed fetch and Semantic Scholar all returned nothing. The title indicates a review of PM_{2.5} cardiopulmonary mechanisms oriented to therapeutic intervention. Low priority for this watch unless it carries a structured synthesis. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: [10.1016/j.apr.2026.103204](https://doi.org/10.1016/j.apr.2026.103204)

Han G et al. 2026 (preprint) Research Square (preprint)

Picropodophyllotoxin attenuates PM_{2.5} injury in pulmonary endothelial cells via mTOR-dependent autophagy

Human pulmonary artery endothelial cells exposed to PM_{2.5} at 25–100 μ g/mL for 24–48 h with picropodophyllotoxin (2–50 μ M) or dexamethasone (50 μ M). PM_{2.5} reduced viability dose- and time-dependently with increased LDH leakage and ROS and suppressed SOD/CAT and SGK1; PPT restored viability, SGK1 and mTOR phosphorylation and attenuated TLR4, MyD88, LC3II and Beclin 1 upregulation. **Not peer reviewed, single cell line, no animal or human data, and no dose-response statistics reported in the abstract.** The exposure concentrations are far above realistic alveolar deposition, which is the standard weakness of this design and limits the finding to pathway plausibility rather than therapeutic inference. Included for completeness of the mechanistic map, not as evidence of effect.

doi: [10.21203/rs.3.rs-10446885/v1](https://doi.org/10.21203/rs.3.rs-10446885/v1)

Heo et al. 2026 Tuberc Respir Dis (Seoul)

Three weeks of PM₁₀ at the air-liquid interface: barrier down, differentiation lost, Notch up

Differentiated human small airway epithelial ALI cultures exposed to PM₁₀ for 7, 14 and 21 days. Transepithelial electrical resistance **fell progressively**; club-cell and ciliated-cell markers fell while basal-cell markers were preserved, i.e. a differentiation block rather than uniform cytotoxicity. Surfactant **SP-A and SP-B mRNA fell, yet secreted SP-A rose while SP-B secretion fell** — a dissociation the paper reports without over-explaining. IL-1 β , IL-6, IL-8 and TNF- α rose; cleaved Notch1 and Notch3, phospho-NF- κ B, I κ B α , HES1 and HEY1 all increased. The chronic, repeated-dose ALI design is a genuine advance on single-dose submerged culture. Limits are the usual ones and are not minor: **PM₁₀ dose, source and donor number are not stated in the abstract**, and no pathway inhibition experiment establishes that Notch is causal rather than concurrent.

doi: [10.4046/trd.2026.0072](https://doi.org/10.4046/trd.2026.0072)

IL23A / NF-kB lung tumour 2026 Ecotoxicol Environ Saf

IL23A acts as a pivotal mediator of PM_{2.5}-induced lung tumorigenesis through the NF-kB pathway activation

PM_{2.5} raised proinflammatory cytokines including IL23A in lung epithelial and NSCLC lines; induction was blocked by p65 siRNA knockdown or chemical NF-kB inhibition, placing IL23A downstream of canonical NF-kB. IL23A is overexpressed in LUAD tissue and inversely associated with prognosis in non-smokers. Anti-IL23A neutralising antibody suppressed tumour formation and growth in orthotopic PM_{2.5}-treated mice. An atypical IL12B-independent secreted IL23A form was identified in lung epithelium. Limitation: PM_{2.5} dose and composition are not stated in the abstract, and instilled or extract-based dosing routinely exceeds inhaled ambient dose by orders of magnitude, so the axis is established and the exposure-response is not

doi: [10.1016/j.ecoenv.2026.120742](https://doi.org/10.1016/j.ecoenv.2026.120742)

Jiang et al. 2026 Environ Sci Technol

Nitration of Urban PM_{2.5}: A Hidden Driver of Enhanced Human Inflammation.

Longitudinal population exposomics in Beijing resolved roughly 1,000 nitrated aromatic compounds (NACs) on personal PM_{2.5}, at median concentrations above the traditional polycyclic aromatics measured alongside them. Secondary NACs formed by aqueous-phase nitration, represented by 3- and 5-nitrosalicylic acid, carried over 70% of summer NAC exposure, and their associations with lung and systemic IL-6 were 2.1 to 232 times stronger than those of PAHs or gaseous NOx. The upper end of that ratio range is doing a lot of work and no CIs are given; associations are cross-sectional within a panel and NAC exposure covaries with everything else in summer photochemistry. Still the strongest argument this quarter that NOx health effects are partly particle-phase - which means mass-only sensing misses the mechanism entirely.

doi: [10.1021/acs.est.6c06494](https://doi.org/10.1021/acs.est.6c06494)

Jiang et al. 2026 Redox Biol

Identification of shared and divergent molecular pathways linking ambient air pollutants to mortality and major chronic diseases through proteomic profiling

Olink plasma proteomics in 48,645 UK Biobank participants: proteins shared across PM_{2.5}, PM₁₀, NO₂ and NO_x were enriched in immune regulation; PM_{2.5}-specific proteins in vascular dysfunction, PM₁₀ in DNA damage, NO₂ in apoptosis, NO_x in matrix remodelling. Mediation and MR flag proteins causally linked to outcomes (vascular remodelling for CVD, oxidative stress for CKD, cell adhesion for dementia, AGER for chronic respiratory disease and lung cancer). No effect sizes or mediated proportions in the abstract. MR establishes protein-to-disease causality, not pollutant-to-protein, and the four pollutants come from the same 2010-era LUR surfaces, so 'pollutant-specific' pathways may be collinearity artefacts. Relevance: low-moderate for sensing.

doi: 10.1016/j.redox.2026.104411

Kamyabi et al. 2026 J Dairy Sci

A novel controlled wood smoke exposure model to evaluate physiological responses to smoke-derived air pollution in dairy calves.

24 pre-weaned Holstein bull calves were randomised to 6 h in purpose-built enclosures with wood smoke (PM_{2.5} 238.44 ± 62.97 µg m⁻³) or clean air (9.94 ± 4.50), with CO₂, CO, temperature and RH similar. Smoke-exposed calves had lower respiratory rate (17.50 vs 21.52 breaths/min) and higher tidal volume (0.90 vs 0.75 L), unchanged minute ventilation; BALF showed more granulocytes, CD68+ macrophages and NK cells, and circulating neutrophils fell. A single 6-h exposure with necropsy immediately after cannot address recovery or repeated smoke days; fresh wood smoke differs from aged wildfire plumes. Relevance: the enclosure PM_{2.5} record gives a controlled dose, rare in agricultural wildfire work, and the design transfers to sensor-guided barn ventilation trials.

doi: 10.3168/jds.2026-29100

Kou et al. 2026 Front Toxicol

Pollutants regulate changes in pathological markers of neurodegenerative diseases: a new perspective in environmental toxicology

Review organised around core neurodegenerative pathological markers — amyloid-β, tau, α-synuclein, TDP-43 — rather than around individual pollutants, detailing how air pollutants, metals and persistent organic pollutants modulate the generation, aggregation, propagation and toxicity of each, under a proposed "pollutants–pathogenic links–neural network failure" framework. No quantitative synthesis, no effect estimates, no exposure metric and no study-quality appraisal; it reorganises existing mechanistic claims rather than adding evidence, and **air pollutants are treated as a single undifferentiated class** alongside metals and POPs. The one useful implication for measurement is that marker-level convergence makes pollutant class, not PM mass, the exposure variable worth resolving.

doi: 10.3389/ftox.2026.1814795

Lang et al. 2026 Environ Toxicol Pharmacol

Particulate matter selectively promotes macropinocytosis and malignant transformation-associated phenotypes in p53-impaired epithelial cells.

Murine mammary and hamster respiratory epithelial cells with intact or impaired p53 were exposed acutely and repeatedly to a European certified reference PM_{2.5}. PM_{2.5} activated oxidative- and cellular-stress pathways in all models, but p53-impaired cells adapted with enhanced macropinocytosis and better growth under nutrient limitation, and repeated exposure increased anchorage-independent growth in a transformed mammary line. Rodent lines, a single reference material, and doses not stated in the abstract; soft-agar growth in an already transformed line is a promotion, not initiation, readout. Relevance: low; mechanistic support for PM as a tumour promoter in p53-deficient tissue.

doi: 10.1016/j.etap.2026.105177

Lin et al. 2026 Arterioscler Thromb Vasc Biol

Air Pollution Triggers Metabolomic Signature of Mitochondrial Dysfunction in Mice and Humans: Implications for Cardiovascular Disease Development.

A cross-species design: ApoE^{-/-} mice given two weeks of diesel exhaust, and 26 healthy adults who moved from Los Angeles (PM_{2.5} 14.4 µg m⁻³) to Beijing (67.6 µg m⁻³) for ten weeks. Multi-platform metabolomics and lipidomics converged on long-chain dicarboxylic acids and medium-to-long-chain acylcarnitines rising in both species, consistent with impaired fatty-acid oxidation; Seahorse respirometry confirmed hepatic mitochondrial dysfunction in the mice, and in humans the shifts tracked 12-/15-LOX lipid peroxidation products and PAH exposure biomarkers. Tryptophan, pyrimidine and lysine signatures did not replicate across species. n=26 with a city-level exposure contrast that also changes diet, time zone and stress - no personal monitoring is reported. Relevance: a 4.7-fold ambient contrast is exactly where calibrated personal sensors would sharpen the dose axis.

doi: 10.1161/atvbaha.125.324277

Ma et al. 2026 Ecotoxicol Environ Saf

Combined PM_{2.5} and microplastics exposure exacerbates allergic asthma via OPA1/SLC7A11-mediated ferroptosis

Ovalbumin-induced murine asthma plus OVA-stimulated MRE cells exposed to PM_{2.5} and/or microplastics. Co-exposure amplified inflammatory infiltration, goblet-cell proportion and lung-function impairment beyond either agent alone, with raised ROS and MDA and reduced T-AOC, CAT and SOD. Ferroptosis markers moved coherently — Fe²⁺ accumulation, lipid peroxidation, NCOA4/FTTH1/ACSL4 up, SLC7A11/GPX4/GCLC/GSS down — alongside mitochondrial depolarisation, ATP depletion and reduced respiratory complex activity. **OPA1 knockdown and ferrostatin-1 both attenuated the injury**, two independent interventions converging on one pathway, which is the strongest part of the design. What it cannot support: no dose-response against realistic ambient concentrations, and instillation or high-dose culture routinely exceeds inhaled doses by orders of magnitude. PM_{2.5} composition is unspecified, so this is one sample, not a class.

doi: 10.1016/j.ecoenv.2026.120808

Meganathan et al. 2026 Respir Res

Aryl hydrocarbon receptor (AhR) regulates organic dust and organic dust-associated bacterial extracellular vesicles-induced airway inflammation via ROS-NFκB signaling.

Beas2B bronchial epithelial cells were exposed to poultry-farm organic dust extract or dust-associated bacterial extracellular vesicles, with siRNA AhR knockdown, and AhR-knockout mice were challenged with the EVs. Both exposures increased AhR expression and nuclear translocation; knockdown reduced ROS and NFκB activation and attenuated IL-1b, ICAM-1, IL-6 and IL-8, though STAT3 responded in opposite directions to extract versus EVs. Dust extract is a soluble fraction, not the inhaled particle, and doses are not related to barn airborne concentrations in the abstract. Relevance: low for sensing; agricultural dust is coarse-mode, the size range where low-cost optical sensors are least reliable.

doi: 10.1186/s12931-026-03908-8

Meldrum & Leonard 2026 Front Immunol

Diesel particle priming reprograms the airway response to a subsequent allergen

Murine model with deliberate temporal separation between diesel exhaust particle exposure and house dust mite challenge — so neither co-exposure nor prior sensitisation — read out with compartment-resolved bulk transcriptomics (lung tissue vs airway lumen) plus targeted single-cell profiling. DEP priming amplified leukocyte recruitment on subsequent HDM challenge and did so **neutrophilically, with no eosinophil response**. DEP alone drove NF-κB programmes; HDM drove NF-κB and interferon; the primed response was qualitatively different from either, with elevated ISG expression and expanded neutrophil subpopulations concentrated in the luminal compartment. The design is the contribution: it isolates a memory-like effect of a particle exposure that has already cleared. Murine, single particle type, no dose-response.

doi: 10.3389/fimmu.2026.1869644

Messina et al. 2026 Front Toxicol*Organ-specific apoptotic pathways induced by micro- and nanoplastics: a systematic review of in vitro toxicological evidence.*

Eighty in vitro studies, PECOS framing, PRISMA reporting, search run 6 September 2024. Apoptosis recurs across gastrointestinal, hepatic, pulmonary, renal, reproductive, placental, neural, cardiovascular, dermal and haematological models, and the mechanisms converge: oxidative stress, mitochondrial depolarisation, ER stress, calcium dysregulation, MAPK and PI3K/AKT signalling, Bcl-2 family shifts and caspase activation, with ferroptotic and autophagic crosstalk. Particle size and surface chemistry modulate potency. Two limitations bound the inference hard. First, the search predates a fast-moving literature by two years. Second - and this is the one that matters for inhalation - the included work is overwhelmingly polystyrene spheres at mass doses far above any modelled human airway deposition, so the review establishes that these pathways can be engaged, not that they are engaged at ambient airborne microplastic levels. No dose-response anchor to a measured air concentration appears anywhere in the synthesis, which is the gap the monitoring side would have to fill.

doi: 10.3389/ftox.2026.1908508

Monge et al. 2026 Cells*Heart Function Changes During Environmentally Relevant Combined Exposure to Heat, Ultrafine Carbon Black, and Ozone: Role of NOD-like Receptor X1.*

Female Nlr1+/+ and Nlr1-/- mice received 3 h of filtered air, heat (36 °C), ultrafine carbon black plus ozone (2.5 mg/m³ + 1 ppm), or all three, with echocardiography, BAL and cardiac mitochondrial assays at 2 and 24 h. UfCB+O₃ alone cut stroke volume, cardiac output and complex III-V activity in both genotypes; the triple exposure produced larger falls in ejection fraction and fractional shortening, more pulmonary neutrophilia and dyssynchronous strain in the knockouts, and heat alone impaired mitochondrial function only in Nlr1-/- . Doses are two to three orders above ambient and the particle and ozone arms are never separated, so particle-specific attribution is not possible. Relevance: supports co-logging temperature with PM at every sensor node, since heat modifies the cardiac response.

doi: 10.3390/cells15181699

Na HW et al. 2026 J Toxicol*Airborne metals reconstructed from worker sebum, then dosed onto skin*

ICP-MS on sebum from 62 indoor and 40 outdoor workers in Shanghai across 24 metals, with large outdoor–indoor contrasts (cobalt 925×, arsenic 135×, barium 29×, manganese 17×). Sebum-mimicking mixtures were then formulated at the measured profiles and applied to keratinocytes, sebocytes and ex vivo scalp skin: the outdoor mixture reduced viability, raised lipid peroxidation, induced γH2AX foci, upregulated IL1A, IL1B, TNFA, IL6 and PTGS2, and activated NF-κB and MAPK. The design is the interesting part — a biomonitoring matrix used to specify the in vitro dose rather than an arbitrary slurry concentration. The weakness is that sebum metal load is a cumulative deposition proxy of unknown relation to inhaled or dermally absorbed dose, and indoor/outdoor is a crude exposure contrast with occupation confounded into it.

doi: 10.1155/jt/7925949

Nasal PM_{2.5} testicular pyroptosis 2026 Res Sq (preprint)*Nasal Exposure to Environmental PM_{2.5} Drives Testicular Pyroptosis and Tissue Damage: Evidence from Inhibitor Intervention and Multi-Source Data*

24 male Sprague-Dawley rats in four arms of six: saline, Caspase-1 inhibitor Z-YVAD-FMK, intranasal PM_{2.5} 10 mg/kg, and both, for four weeks. PM_{2.5} reduced sperm count and motility and raised abnormality rate, with elevated ASC, Caspase-1 and gasdermin D in testis, all reversed by the inhibitor - a coherent NLRP3/Caspase-1/GSDMD pyroptosis chain. Reanalysis of GEO GSE189187 (GC-2spd cells, 3 vs 3) gave 217 DEGs enriched in interferon/antiviral processes, with Hmox1 up and Stat1 down. n=6 per arm and a 3-vs-3 public dataset cannot support hub-gene claims, and intranasal instillation at 10 mg/kg bears no quantitative relation to ambient inhalation. Preprint, no peer review.

doi: 10.21203/rs.3.rs-8701799/v1

PM diabetic kidney MR transcriptomics 2026 Int J Mol Sci*Particulate Matter Exposure and Diabetic Kidney Dysfunction: Insights from Integrated Transcriptomic and Bioinformatics Analyses*

Bidirectional Mendelian randomisation put PM_{2.5}-10 causally upstream of lower eGFR, more strongly in diabetes, with no reverse causality. Transcriptomic integration yielded 168 genes shared between PM response and diabetic kidney disease, enriched for AGE-RAGE, IL-17, TNF and PI3K-Akt signalling; seven feature genes (AVPI1, DUSP1, FOSB, JUNB, PDK2, TPPP3, VIM) supported diagnostic models across training and external cohorts, with VIM localised to interstitial and fibrotic regions and TPPP3 to podocytes. The MR instrument set for coarse PM is weak and geographically confounded in UK Biobank-derived GWAS, which is the standard objection to air-pollution MR. Everything after the MR step is in silico pattern matching with no experimental validation, and the sanguinarine docking result is speculative.

doi: 10.3390/ijms27177615

PM skin AhR-Nrf2 framework 2026 Int J Mol Sci*Particulate Matter-Induced Skin Injury: A Dual-Pathway AhR-Nrf2 Framework for Epidermal Homeostasis and Therapeutic Targeting*

Review proposing that PM-induced skin barrier damage is best read as coordinated AhR and Nrf2 signalling rather than two separate pathways. Transition metals generate ROS through Fenton chemistry while PAHs are metabolised via AhR in keratinocytes, amplifying oxidative stress; Nrf2 then induces HO-1 and other antioxidant enzymes, and sustained PM exposure is argued to exhaust that compensation, degrading barrier protein expression and raising inflammatory cytokines. The dual-pathway framing is a reasonable organising device and identifies AhR-Nrf2 crosstalk as a therapeutic target, but it is a hypothesis assembled from single-pathway studies rather than a test of coordination, and the review does not distinguish which effects are PM-specific from those common to any oxidant or PAH source.

doi: 10.3390/ijms27177573

Pei et al. 2026 Environ Sci Technol*Deciphering Allergenic Components Responsible for Differential Sensitization Potentials of Source-Specific Atmospheric Particulate Matter*

h-CLAT and keratinocyte-dendritic cell co-culture tests of three reference materials: urban PM 1648a, coal fly ash 2693 and diesel particles 2975. 1648a raised CD54 in THP-1 monoculture via TNF-α and IL-1β; KC co-culture amplified DC activation for all three (CD86, CD54) through IL-1β and extracellular ATP. AhR signalling in HaCaT was required; amorphous silica drove 1648a's CD54 response, while soot and magnetite, with silica, raised CD86; PAHs acted synergistically. SRMs are aged bulk dusts, not size-fractionated ambient PM_{2.5}, and doses are not in the abstract. Relevance: low for sensing; identifies silica and soot as the sensitising components, both separable only by composition, not mass.

doi: 10.1021/acs.est.6c08887

Ren Y et al. 2026 Research (Wash D C)*Lung-on-chip for PM, appraised model by model*

Critical review of in vitro platforms for PM–lung interaction, from 2D monolayers through 3D bioprinted constructs and organoids to lung-on-chip devices, with each class assessed for what it can and cannot reproduce. The central argument is that breath-mimicking cyclic mechanical strain is not an incidental refinement but a determinant of how particles interact with the epithelial interface — deposition, clearance and mechanotransduction all change under motion, so static models systematically misrepresent the dose actually delivered. Useful as an orientation to which platform answers which question. It is a narrative review with no systematic search, no pooled comparison and a closing pitch for the authors' own "highly integrated and intelligent" platform, so the appraisal is not disinterested.

doi: 10.34133/research.1414

Retinal CRM 2026 Biochem Biophys Rep*Certified particulate reference materials in primary mouse retinal cells*

Two certified environmental particulate reference materials applied to primary mouse retinal cells, plus repeated ocular-surface exposure in mice. Effects on DCFDA fluorescence and ATP-dependent viability were condition-dependent and concentrated in glial conditions; neither α -tocopherol nor N-acetylcysteine gave a significant rescue, and glial coculture did not improve retinal ganglion cell survival. In postnatal mice topical particulate was associated with inner retinal thinning; in adult Thy1-CFP mice the within-animal comparison was null and only a *post hoc* between-animal pooling reached significance (-14.4 cells, -28.1 to -0.7 , $P = 0.040$). The authors state plainly that this establishes neither particle transport to the retina nor a causal oxidative mechanism nor any ambient risk — an unusually disciplined set of negative caveats, and the reason this is tier C rather than a finding. Ocular-surface dosing is not inhalation, and certified reference material is not ambient PM.

doi: 10.1016/j.bbrep.2026.102758

Vahdatbin et al. 2026 iScience*Is recycling sustainable: the promise and risks of scrap tire valorization*

Roughly **1.5 billion end-of-life tires** accumulate annually, and this review argues that devulcanization, pyrolysis and mechanical grinding each release hazardous species including VOCs, PAHs, heavy metals, microplastics and 6PPD-quinone. It is qualitative: no emission factors, exposure concentrations or dose–response relationships are given, so the risk claim is directional and cannot rank the three routes. Its value here is the explicit call for high-fidelity monitoring of recycling facilities — a fenceline low-cost-network application with essentially no published deployment literature. Tire wear is already a leading non-exhaust PM source; tire *reprocessing* as a second, point-source pathway is comparatively unmeasured.

doi: 10.1016/j.isci.2026.117462

Shen et al. 2026 Environ Sci Technol*A Brake on Toxicity: Iron Suppresses Organic-Induced Superoxide Generation in Macrophages*

Macrophages were exposed to brake-wear particles (BWPs), highway PM_{2.5}, and Fe alone or mixed with isoprene SOA or 9,10-phenanthrenequinone. BWPs slightly induced cellular superoxide below 10 $\mu\text{g/mL}$ and suppressed it dose-dependently above; highway PM_{2.5} did neither. Fe²⁺ and Fe³⁺ suppressed baseline and organic-induced superoxide, which the authors attribute to replenishment of the labile iron pool (relieving organic-induced iron deficiency and NADPH oxidase activity) plus direct chemical quenching. In vitro doses are not related to deposited lung dose, and superoxide is one ROS endpoint - Fe drives hydroxyl radical via Fenton chemistry, which this does not address. Relevance: undermines simple ‘more Fe, more toxicity’ weighting in composition-based exposure metrics for non-exhaust PM.

doi: 10.1021/acs.est.6c06554

Song et al. 2026 Toxicol Lett*Lung Toxicity Induced by Co-Exposure to Particulate Matter and Indoor Microorganisms in Mice.*

C57BL/6 mice received PM combined with dominant indoor airborne microorganisms by intratracheal instillation for two weeks. PM with the fungus *Bjerkandera adusta* increased lung eosinophil infiltration beyond either agent alone, with synergistic upregulation of C3 and IL-3; PM with *Bifidobacterium infantis* produced histopathology suggestive of fibrogenesis and synergistic collagen-gene induction. The PM type, dose and group sizes are not in the abstract, and instillation delivers a bolus unlike inhalation. Relevance: low for sensing; argues that indoor PM mass alone under-describes hazard where bioaerosol load is high.

doi: 10.1016/j.toxlet.2026.113207

Garner et al. 2026 Reprod Toxicol*Gestational nanoparticle inhalation disrupts placental zone structure in rats*

Pregnant Sprague Dawley rats exposed by whole-body inhalation to nano-TiO₂ at $9.23 \pm 0.39 \text{ mg m}^{-3}$ from gestational day 5 to 19, placentas harvested GD20 for histology. Placental weight and placental efficiency were **unchanged**, but decidua and labyrinth zone sizes fell, with compensatory increases in blood space number and maternal blood space area. The compensation is the interesting part: structural remodelling without a mass-level phenotype means placental weight is a poor screening endpoint for ultrafine exposure. The exposure is roughly three orders of magnitude above ambient ultrafine mass and engineered TiO₂ is a poor surrogate for combustion ultrafines in surface chemistry, so this bounds mechanism and not risk. No litter-level statistics are reported.

doi: 10.1016/j.reprotox.2026.109361

Tire wear particle phytotoxicity 2026 Curr Biol*Tire wear particles stunt a roadside grass and raise root oxidative stress*

Controlled tire-wear-particle exposure of *Bromus catharticus*, a common roadside weed in Japan, testing an endpoint the TWP literature has largely skipped — previous work concentrated on aquatic systems and on crops, not on the roadside and urban-park vegetation that sits metres from the source. Both **shoot and root growth were significantly inhibited**, and root malondialdehyde — a lipid-peroxidation marker — rose significantly, indicating oxidative stress as at least one operative mechanism. The abstract does not give the applied dose, the particle size distribution, the exposure route (soil versus foliar) or the exposure duration, and all four determine whether the treatment is environmentally realistic; a single annual grass in a pot cannot stand for roadside vegetation generally. Carried here because TWP is the fastest-growing non-exhaust PM source, is poorly sized by optical monitors, and now has a documented terrestrial effect at the point of emission.

doi: 10.1016/j.cub.2026.06.087

Tzagkaroulaki et al. 2026 Toxics*An A549 Cell-Based Approach Using Repeated Fluorescence Readouts for Assessing Reactive Oxygen Species Activity of Atmospheric Particulate Matter.*

Method-performance characterisation of an A549/DCFH-DA ROS assay read repeatedly from 15 min to 6 h, using zymosan and NIST SRM 2584 controls, CV and signal-to-noise. At 60 min the SRM response rose across 0.02, 0.05 and 0.10 mg/mL (descriptive R² 0.90); earlier readouts were non-monotonic. Eight matched PTFE-quartz filter pairs gave zero-intercept slopes of 0.90 (mass-normalised) and 0.99 (volume-normalised), R² ~0.99. Field samples showed DCFH-DA fluorescence did not track PM mass. Three concentrations and eight pairs are thin, and DCFH-DA is prone to auto-oxidation artefacts. Relevance: the filter-substrate equivalence is useful for networks archiving PTFE filters from reference co-location sites for oxidative-potential follow-up.

doi: 10.3390/toxics14090789

UPM instillation reproduction null 2026 Cells*Repeated Short-Term Intratracheal Exposure to Urban Particulate Matter Has No Apparent Adverse Effects on Female Reproductive Outcomes in Mice*

A clean null worth recording. Repeated intratracheal urban PM in female mice produced robust dose-dependent pulmonary inflammation - CD45+, F4/80+ and Ly6G+ infiltration, Il-6, Il-1b, Tnfa, Col1a1 and Tgfb1 induction - with no detectable effect on oocyte recovery, fertilisation competence, gestation length, litter size or postnatal pup development, and no measurable inflammatory transcript in ovary or uterus. Direct exposure of two-cell embryos in vitro did not impair blastocyst formation, though benzanthracene alone slightly reduced it. The result argues against a systemic-inflammation route from lung to ovary at these doses. Intratracheal bolus at 40-200 mg/kg is far above any ambient-equivalent deposited dose and bypasses the nasal and upper-airway deposition that governs real inhalation.

doi: 10.3390/cells15171522

Vancouver DE blood-lung concordance 2026 Environ Res*Concordant transcriptomics analysis of the lungs and blood after exposure to diesel exhaust: a randomized, controlled human exposure study*

Double-blind crossover controlled human exposure, 32 participants, two hours of diesel exhaust and of filtered air on separate occasions, with paired-end RNA-seq on blood and bronchial brushings. Airway DEGs at FDR < 10% include upregulated GPX2 and NQO1, i.e. the NRF2 antioxidant arm. Hallmark pathway mapping gives IFN- α , IFN- γ , inflammatory response and ROS activity in both compartments, and a concordant analysis finds correlated blood-airway expression that neither compartment showed alone. The authors are explicit that this is exploratory. $n = 32$ at a single two-hour dose, no exposure-response, and the concordance is correlational - but it is the right design question: whether a blood transcriptomic readout can stand in for bronchoscopy in panel studies of traffic-related particles

doi: 10.1016/j.envres.2026.125625

Vera-Colón et al. 2026 — metadata only Part Fibre Toxicol*Airborne fine particulate matter and human embryonic stem cell osteogenic differentiation*

BMC deposited this DOI on 18 September with no abstract; Crossref, Europe PMC core, Europe PMC fullTextXML and the publisher landing page all returned nothing on per-DOI retry — **the first time this gap has appeared on a BMC rather than an Elsevier deposit**. Carried on title because skeletal development is an endpoint the PM literature has barely touched, and a human-embryonic-stem-cell osteogenic model would be a new mechanistic line rather than another airway or endothelial assay. No finding is asserted: neither particle source, dose, duration nor the direction of the reported disruption is verifiable from metadata. On the retry list.

doi: 10.1186/s12989-026-00701-9

Wang Y et al. 2026 Environ Sci Technol Lett *α -Dicarbonyl aqueous products modulate cellular ROS in opposite directions*

Glyoxal or methylglyoxal reacted with ammonium sulfate or glycine in the aqueous phase; HRMS shows ammonium routes yield imidazole-type nitrogen heterocycles and glycine routes yield more pyrrole-related species. Alone, neither product set induces much ROS in A549 cells. Co-exposed with 1,4-naphthoquinone, Cu^{2+} or both, **imidazole-dominated mixtures enhance ROS and pyrrole-rich mixtures suppress it**, with acellular assays attributing the split to promotion versus suppression of quinone redox cycling and metal-associated pathways. This is a direct challenge to any oxidative-potential assay read as a scalar: the same secondary organic aerosol aging pathway can raise or lower measured ROS depending on the nitrogen source. Cell-line work at unstated atmospheric relevance of concentration, so treat as mechanism, not as an exposure-response.

doi: 10.1021/acs.estlett.6c00722

Wang et al. 2026 Environ Pollut*Ozone and $\text{PM}_{2.5}$ co-exposure exacerbates allergic rhinitis and is associated with PPAR/TRPV1-related barrier damage: Insights from nasal transcriptomic and microbial analyses.*

C57BL/6J allergic-rhinitis mice in five groups (control, AR, AR+ O_3 , AR+ $\text{PM}_{2.5}$, AR+ $\text{PM}_{2.5}$ + O_3). Co-exposure markedly aggravated sneezing, disrupted nasal mucosa and lowered ZO-1 and occludin; nasal transcriptomics implicated PPAR/TRPV1 signalling and 16S sequencing showed microbial dysbiosis correlated with inflammation and barrier injury. The authors state the doses were acute and high and not designed to model environmental exposure, which is the limitation - interaction at high dose need not persist at ambient levels, and group sizes and PM dose are not reported in the abstract. Associations between omics layers are correlational. Carried for the co-exposure design: O_3 and $\text{PM}_{2.5}$ peak in different seasons in most airsheds, so their joint effect is a question sensor networks measuring both can actually address.

doi: 10.1016/j.envpol.2026.129207

Wu et al. 2026 Ecotoxicol Environ Saf *β -carotene alleviates $\text{PM}_{2.5}$ -induced pulmonary fibrosis via suppression of the Frizzled5-mediated $\text{Wnt5a}/\text{Ca}^{2+}$ pathway*

C57BL/6 mice given **intratracheal $\text{PM}_{2.5}$** , paired with BEAS-2B human bronchial epithelial cells carrying Fzd5 overexpression. Instillation produced fibrosis, oxidative stress and calcium dyshomeostasis — more mitochondria-associated ER membranes, altered IP_3R , STIM1, ORA1 and SERCA2 — with activation of a **$\text{Wnt5a}/\text{Fzd5}/\text{Ca}^{2+}$** axis. Dietary β -carotene reversed these changes, and Fzd5 overexpression abolished the protection, identifying Fzd5 as the mediating receptor. **No instilled dose, particle source, treatment concentration, effect size or CI** is reported for any endpoint, so potency cannot be placed against ambient exposure. Intratracheal instillation also bypasses the deposition physics that a sensor-derived exposure metric is meant to approximate.

doi: 10.1016/j.ecoenv.2026.120816

Yang et al. 2026a Environ Health (Wash)*The Human Microbiome in Airborne Particulate Matter-Associated Aging-Related Chronic Diseases: An Evidence-Stratified Review.*

Evidence-stratified review proposing a PM exposure-microbiome-aging framework across respiratory, gut and skin compartments, with the gut microbiota argued to be the most mechanistically developed interface. The stratification is the contribution: the authors grade epidemiological, animal and multiomics evidence separately rather than pooling them into a single causal story. What it cannot do is quantify anything - no pooled estimates, no dose-response, and the reviewed studies use incompatible PM metrics, so the framework is a research agenda rather than a finding. Carried because microbiome mediation keeps being invoked to explain PM associations that survive adjustment, and because a graded map of where the evidence actually is saves the field repeated litigation of the same weak studies.

doi: 10.1021/envhealth.5c00852

Yang et al. 2026b Environ Health (Wash)*Molecular Mechanisms of Indoor Environmental Pollution tt-DDE-Induced Blood Pressure Elevation and the Protective Role of Human-Specific Gene CHRFAM7A.*

Cooking-oil fumes are stated here as roughly 90% of indoor particulate emissions, and tt-DDE is their dominant aldehyde. Inhaled tt-DDE raised blood pressure in mice through endothelial damage; CHRFAM7A-transgenic mice resisted the systolic rise and the endothelial dysfunction, with protection traced to STAT3/HO-1 activation, and in vitro tt-DDE drove KLF5 inflammatory signalling via METTL3-dependent m6A modification, reversible by CHRFAM7A overexpression and abolished by HO-1 inhibition. No exposure concentration, duration or blood-pressure magnitude appears in the record, and CHRFAM7A is human-specific, so the transgenic model has no wild-type counterpart. Relevance: kitchen PM sensing conflates mass with a gas-phase aldehyde that may carry the cardiovascular signal.

doi: 10.1021/envhealth.5c00703

Ye et al. 2026 Clin Exp Pharmacol Physiol*Mechanistic Study of SRSF1-Mediated Inhibition of Treg Cell Function via the mTORC1/ULK1 Signalling Axis Through Mitophagy in Asthma.*

SRSF1 was elevated in BALF cells from asthma patients and correlated negatively with Treg FOXP3 and FEV1 %pred. In OVA-asthma mice, SRSF1 upregulation in Tregs increased mTOR and ULK1-Ser757 phosphorylation and impaired mitophagy and metabolic reprogramming; SRSF1 knockdown or rapamycin restored Treg function and reduced airway inflammation, while an mTORC1 activator reversed the rescue. A PM_{2.5} arm exaggerated these changes in an SRSF1-dependent way. PM is secondary to the paper's question: dose, particle source and exposure route are absent from the abstract, and patient numbers are not given. The value is mechanistic - a Treg mitophagy axis through which PM could worsen established asthma - and it adds nothing to exposure-response. Tier C.

doi: 10.1111/1440-1681.70157

e-waste children mucosal immunity 2026 Front Immunol*iNOS/Arg-1 balance and sIgA levels are associated with intestinal mucosal immune defense in children chronically exposed to air pollutants and lead.*

238 children aged 3-6 from the Guiyu e-waste area and the Haojiang reference area. Guiyu children carried higher blood Pb and higher estimated daily doses of all five pollutants, and showed lower iNOS, higher Arg-1 and IL-4, and a lower iNOS/Arg-1 ratio - an M2-skewed macrophage profile. BKMR linked the mixture inversely to the iNOS/Arg-1 ratio, but serum sIgA did not differ between areas and was not associated with the mixture; only a non-linear iNOS-sIgA relation emerged. The headline endpoint is therefore null and the abstract leads with the secondary. Pb and combustion aerosol co-vary almost perfectly across these two sites, so the design cannot attribute anything to particles specifically; network toxicology adds no empirical weight.

doi: 10.3389/fimmu.2026.1903043

Occupational & indoor**72 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (71 OF 72 RECORDS IN THIS SUBTOPIC)

Albert et al. 2026 Innov Aging*Assessing home environmental hazards for older adults receiving in-home supportive services: the healthy homes assessment tool.*

Development and pilot validation of the Healthy Home Assessment Tool, a care-coordinator checklist for indoor air quality and moisture/mould embedded in home- and community-based services, in 41 low-income older-adult households (median age 76). Against consumer-grade IAQ monitors, 61.0% of homes exceeded 9 $\mu\text{g m}^{-3}$ PM_{2.5} and 21.9% exceeded the 4 pCi/L radon action level; PM_{2.5} was higher where smoking was reported and fungal burden higher with visible mould. The comparison of indoor snapshots with an annual ambient standard overstates exceedance, and the consumer monitors are uncalibrated in the abstract. Relevance: a concrete pathway for low-cost PM sensors into social-care workflows - but the 61% figure depends entirely on a sensor whose bias is not stated.

doi: 10.1093/geroni/igag078

Assan et al. 2026 Health Sci Rep*Ambient particulate matter, urinary arsenic and resident health in two Ghanaian mining communities*

65 residents each from an artisanal and an incorporated mining community, all resident at least five years; PM by Modular Air Quality System, urinary arsenic by AAS, kidney function by albumin-creatinine ratio. More artisanal residents fell in the at-risk band for kidney disease ($p = 0.006$); the incorporated community showed poorer general health status. Descriptive only: $n = 130$, chi-square on dichotomised uACR with no confounder adjustment and no exposure-response, and **neither PM concentrations nor arsenic levels are numerically reported** despite being the two measured exposures. The 35 and 70 $\mu\text{g/m}^3$ figures quoted as WHO standards are the 2005-era interim targets, not the 2021 guideline values of 15 and 45 $\mu\text{g/m}^3$ for 24-hour PM_{2.5} and PM₁₀ — so the exceedance framing understates the gap.

doi: 10.1002/hsr2.73246

Auditorium IAQ + 16S bioaerosol 2026 Sci Total Environ*Indoor Air Quality (IAQ) analysis and 16S rRNA gene sequencing of indoor air pollutants in a 3000-capacity religious auditorium*

METADATA ONLY. Metadata only; abstract absent from Crossref, Europe PMC and PubMed on per-DOI retry. Retained because a 3,000-capacity assembly space is a high-occupancy indoor microenvironment where occupant-generated coarse particles and bioaerosol dominate, and 16S sequencing of airborne material is particulate characterisation. PM scope is inferred from the title, no finding asserted, and the study location is not recoverable from metadata. On the retry list.

doi: 10.1016/j.scitotenv.2026.182237

Bi/Bi2Te3 workplace ICP-OES 2026 Zhonghua Lao Dong Wei Sheng Zhi Ye Bing Za Zhi*ICP-OES determination of bismuth and bismuth telluride in workplace air*

Method development for an occupational aerosol with no established Chinese workplace standard: mixed cellulose ester filter sampling, ultra microwave digestion at 130°C for 10 minutes, then simultaneous Bi and Te by ICP-OES with Bi₂Te₃ back-calculated from either element using on-site speciation information. Linear over 0.04–30 mg/L ($r > 0.9990$); on a 75 L sample diluted to 50 mL the minimum detectable Bi₂Te₃ is 0.006 mg/m³ via Bi and 0.008 via Te. Digestion efficiency 99.1–100.0% (Bi) and 101.2–103.3% (Te), intra-batch precision under 0.8%, 7-day room temperature stability, and no interference from Ag, Al or As up to 10 mg/L. Validated on simulated samples rather than a field inter-laboratory comparison, and thermoelectric workplaces are the only stated application. Included as the kind of low-visibility analytical groundwork on which occupational exposure limits eventually rest.

doi: 10.3760/cma.j.cn121094-20250319-00107

Build Environ 2026 (cleanroom) Build Environ*Multi-objective ventilation optimization in non-unidirectional cleanrooms: an integrated RSM-NSGA-II-TOPSIS approach.*

METADATA ONLY. The title names a response-surface plus genetic-algorithm plus TOPSIS pipeline applied to non-unidirectional cleanroom ventilation, which is a simulation-optimisation study, not a measurement one; the objectives in this literature are almost always particle count against fan energy. Relevance to this watch is narrow but real - cleanroom control is the upper bound on what mechanical ventilation can do to particle number indoors, and the trade-off surface is the same one that governs filtration interventions in ordinary buildings at far lower air-change rates. Carried at tier C on the strength of the venue and the topic, with no finding asserted.

doi: 10.1016/j.buildenv.2026.115290

Build Environ 2026 (subway) Build Environ*Spatiotemporal distribution characteristics of multiple air pollutants within subway stations.*

METADATA ONLY. The Crossref deposit carries no abstract and neither Crossref nor Europe PMC had text at the time of screening, so nothing beyond the title is asserted here. Subway microenvironments are the highest-concentration routine PM exposure most urban commuters encounter - platform PM_{2.5} commonly runs several-fold above street level and is Fe-dominated from rail-wheel-brake wear - so a spatiotemporal characterisation across a station is directly relevant to fixed-versus-personal monitor siting. Held for a deferred re-query once the publisher releases the abstract; the record is carried now rather than dropped so the DOI is on the ledger.

doi: 10.1016/j.buildenv.2026.115292

CFRP inhalation risk review 2026 Regul Toxicol Pharmacol*Carbon-fibre composite dust: fibre-like fragments, ultrafines, and thin evidence*

Life-cycle review of inhalation risk from carbon fibre-reinforced plastics across manufacturing, machining, recycling and end-of-life processing. Mechanical processing generates respirable high-aspect-ratio fragments; thermal processing generates ultrafine particles together with hazardous gases. The review's central judgement is a refusal: available evidence **does not demonstrate a magnitude or pattern of inhalation toxicity comparable to asbestos or to pathogenic high-aspect-ratio carbon nanotubes**, but it is too limited and heterogeneous — particularly for chronic exposure — to support any quantitative relative-toxicity comparison in either direction. It identifies the actual gaps: real-world occupational exposure data, harmonised emission scenarios, internal dose and biopersistence. As CFRP volumes rise with wind-blade and aerospace retirement, this is a source class whose workplace aerosol nobody is routinely monitoring.

doi: 10.1016/j.yrtph.2026.106243

Campiotti et al. 2026 Build Environ*Beyond leaf traits: context dependent particulate matter accumulation across outdoor, indoor and active green walls*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 29 Sep with no abstract; PubMed efetch, Crossref, Europe PMC and the publisher landing page returned nothing. The title indicates a comparison of leaf PM accumulation across outdoor, indoor and actively ventilated green walls. Relevant to vegetation-based mitigation claims, which need concentration measurements, not leaf loading, to show an exposure benefit. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.buildenv.2026.115347

Chen Y et al. 2026 Atmos Environ*Solid-fuel PAH exposure, phenanthrene metabolites, DNA damage and inflammation*

Ships on metadata only — no abstract in the Crossref deposit and no PubMed, Europe PMC or Semantic Scholar record on this run. Title and authorship indicate urinary phenanthrene metabolites as the exposure index in a rural northern Chinese solid-fuel population, with DNA-damage and systemic-inflammation biomarkers as outcomes. Counted and listed; no numbers claimed. High priority for the retry list: household solid-fuel combustion is the single largest PM exposure contrast available in observational work and this watch carries very little of it.

doi: 10.1016/j.atmosenv.2026.122339

Chen et al. 2026 Indoor Air*Indoor air protection under extreme air pollution risk: evidence from a sandstorm in China*

Difference-in-differences on Baidu search intensity for air purifiers and indoor air-quality information around China's most intense sandstorm in a decade (early 2021). Affected regions show roughly **18%** higher search intensity than unaffected ones, rising to about **33%** in historically cleaner areas and about **200%** where baseline purifier sales were lowest. The gradient is the useful result: protective response scales inversely with prior exposure experience and prior equipment stock. Search volume proxies intent, not purchase, use, or any realised reduction in indoor PM_{2.5} — no concentration data enter the study at all — and regional treatment assignment with no individual exposure makes the effect ecological throughout.

doi: 10.1155/ina/2274208

China gym co-exposure 2026 Environ Pollut*Association of indoor multiple airborne pollutants co-exposure and typical respiratory symptoms among gym employees: A nationwide observational study in China*

280 gyms across 32 Chinese provinces, 2022-2023. Median PM_{2.5} 35 $\mu\text{g m}^{-3}$, PM₁₀ 59 $\mu\text{g m}^{-3}$, fungal count 170 CFU m⁻³, plate count 226 CFU m⁻³; 22.13% of employees reported respiratory symptoms. PM_{2.5} OR 2.31, PM₁₀ 2.19, fungi 1.17, bacteria 1.24, and a quantile g-computation mixture OR 1.70 with PM_{2.5} and fungal count as the dominant components. Natural ventilation (OR 0.36) and high-efficiency filters (0.42) were protective; absent equipment disinfection (3.30) and air-return air conditioning (2.85) raised risk. Limitations: cross-sectional, self-reported symptoms, and no confidence intervals reported for any OR in the abstract, which is why the forest plot carries them as point estimates only. The ventilation and filtration coefficients are the operationally useful part

doi: 10.1016/j.envpol.2026.129141

Dennis & Ball 2026 AORN J*Strategies to Improve Compliance With Smoke Safety.*

Nursing practice review on surgical-smoke evacuation compliance: smoke from energy-based devices carries toxic gases, particulate matter and potentially viable cells; perioperative nurses report twice the general-population incidence of respiratory problems; barriers are education, equipment and culture; recommended strategies are gap analysis, training, leadership support and legislation. No exposure measurement or new data. Relevance: low; OR smoke is a high-concentration ultrafine source for which real-time particle counters would make evacuation compliance auditable.

doi: 10.1002/aorn.70159

Derwein et al. 2026 Build Environ*Masks, filtration and ventilation on exhaled-air transmission — metadata only*

No abstract was deposited to Crossref, Europe PMC or Semantic Scholar at index time, and ScienceDirect refused retrieval from this environment, so nothing beyond title, authorship and venue can be asserted. Recorded so the DOI enters `seen.json` and the record is not re-screened; queued for the weekly abstract re-check. Listed under *Occupational & indoor* on the title's face, which is a provisional assignment.

doi: 10.1016/j.buildenv.2026.115206

Dutch school air cleaner RCT 2026 Front Public Health*Protocol: 180 classrooms, HEPA versus ionisation versus nothing*

Protocol only, no results (ClinicalTrials.gov NCT07479420). 180 classrooms in 29 Dutch primary schools, clustered within schools and randomised to HEPA-filter portable air cleaners, ionisation/plasma cleaners, or none. Units were laboratory pre-tested, safety screened and operated at comparable clean air delivery rates — which is the design choice that makes the comparison mean something, since CADR is the confounder that sinks most field air-cleaner studies. The schedule is a three-week baseline or two-week post-intervention control plus three repeated three-week intervention periods, to 14 weeks per school. Airborne dust is collected passively on electrostatic dustfall collectors and assayed for bacterial and viral markers, with active sampling in 12 classrooms to validate the passive method; PM₁₀, PM₄, PM_{2.5}, PM₁, CO₂, temperature, humidity and VOCs are logged continuously. Absenteeism and parent-reported symptoms are retrospective, which is the weakest element. Worth tracking to results — ionisation devices rarely get a fair, CADR-matched test.

doi: 10.3389/fpubh.2026.1892902

Eghomwanre & Akharamé 2026 Environ Sci Pollut Res Int*Spatio-temporal assessment of classroom air quality and thermal comfort as determinants of respiratory health risks in urban pre-tertiary institutions.*

Ten weeks of classroom monitoring (PM_{2.5}, PM₁₀, CO, CO₂, thermal indices) in six primary and secondary schools in Benin City, with modified BMRC respiratory questionnaires. PM_{2.5} ranged 11.2–81.4 and PM₁₀ 20.1–138.6 $\mu\text{g m}^{-3}$; CO₂ peaked at 729.3 ppm; THI 28.3–30.1. HQs were all <1, yet cough (58.2%), phlegm (32.5%) and chest pain (26.7%) were common; higher PM was associated with cough ($p < 0.001$) and chest pain (crude OR 2.74), while poor sanitation predicted phlegm (aOR 5.54, 95% CI 1.35–22.80). Instrument type, calibration and student n are not in the abstract, and the PM ORs are crude. Relevance: West African classroom data are scarce; this is the setting where a school sensor network would be the only monitoring.

doi: 10.1007/s11356-026-38254-1

Environ Int 2026 Environ Int*Portable HEPA air filtration effects on particulate matter, radioactivity, and chemical constituents in occupied buildings.*

METADATA ONLY. Crossref deposit of 19 September, abstract not deposited and not recoverable from any queried index. Carried on title and venue. The title's three-endpoint framing - particulate mass, radioactivity and chemical constituents from the same intervention - is unusual and would speak to whether HEPA removal efficiency is constituent-dependent, which is the question a mass-only sensor cannot answer when validating a filtration intervention. No reduction fraction, building count or study design is asserted here. Held for pickup.

doi: 10.1016/j.envint.2026.110528

Environ Int 2026 (button mfg) Environ Int*Plastic button manufacturing as an under-recognized source of airborne multi-pollutant mixtures: process-resolved characterization.*

METADATA ONLY. No abstract in the Crossref deposit or Europe PMC at screening. The title claims a process-resolved characterisation, which in occupational aerosol work normally means emissions resolved to individual unit operations rather than a shift-average - the design that actually supports a control recommendation. Small-scale plastics fabrication is a plausible unrecognised source of mixed particulate and VOC exposure precisely because it sits below the size threshold at which most jurisdictions require monitoring. No effect size is claimed here because none is available. On the deferred re-query list.

doi: 10.1016/j.envint.2026.110529

Ethiopia dormitory ventilation 2026 Indoor Environments*METADATA ONLY — occupant-driven ventilation in naturally ventilated dormitories*

No abstract retrievable on the entry date. The title commits to a field study of occupant-driven ventilation and indoor exposure dynamics in naturally ventilated Ethiopian university dormitories. The pollutants measured, the ventilation-rate method (tracer gas, CO₂ decay or modelled), the number of rooms and the season are all unrecoverable and none is asserted here. Occupant-controlled natural ventilation is the largest uncontrolled term in indoor exposure models for warm low-resource settings, and measured air-change data from Sub-Saharan Africa are scarce enough that the study is worth retrieving on those grounds alone.

doi: 10.1016/j.indenv.2026.100192

Europe indoor PAH children 2026 Environ Res*Indoor-air PAH levels and child cancer risk across six European cities*

Twelve EPA PAHs in air and deposited dust from schools, homes and sports halls in Athens, Barcelona, Colchester, Copenhagen, Helsinki and Lisbon — four of the 16 target compounds were dropped for analytical limitations, which is stated. Spatial variability is large: medians above 50,000 pg/m^3 where intensive residential wood burning and poor ventilation coincide, against under 10,000 pg/m^3 in south-western Europe. High-molecular-weight PAHs ran higher in schools than homes, consistent with occupancy and accumulation of outdoor-derived particles, while low-molecular-weight species tracked household practices. Estimating air concentrations from deposited dust **underestimated HMW-PAHs**, which is a directly useful negative result for anyone tempted to substitute dust for active sampling. Excess cancer risk over a five-year primary-school exposure period spanned 10^{-8} to 10^{-5} , below the 10^{-4} reference. Cross-sectional sampling cannot capture seasonal wood-burning cycles.

doi: 10.1016/j.envres.2026.125643

Feng R et al. 2026 Environ Sci Technol Lett*Clean heating renovation cuts nitro-PAHs but not nitrated phenols*

Size-resolved *personal* exposure to particulate nitrated phenols (NPs) and nitro-PAHs among rural users of clean coal, raw coal chunk and biomass, paired with A549 cytotoxicity and NF- κ B reporter assays. Biomass users carried up to 11.5 $\times$ the PM_{2.5}-bound n-PAH exposure of clean-coal users, while NPs varied ≤ 1.3 -fold across fuels (clean coal 64.0 $\pm$ 12.1 vs raw coal 57.4 $\pm$ 25.8 ng/m^3 , $p=0.545$). Toxicologically the two classes diverge as well: n-PAHs were more cytotoxic by IC₅₀, NPs activated NF- κ B more strongly via p-I κ B- α and COX-2. The policy reading is sharp — a fuel switch evaluated on PM mass or on PAHs alone will score a benefit that the NP fraction does not deliver. Group sizes are not stated in the abstract and the in vitro work uses representative compounds, not the collected particles.

doi: 10.1021/acs.estlett.6c00629

Figueiredo et al. 2026 Environ Toxicol Pharmacol*Chemical and toxicological assessment of particulate matter in a charcoal barbecue restaurant.*

Indoor and outdoor PM₁₀ sampled over six working days plus a closure period at one charcoal-grill restaurant. Indoor PM₁₀ exceeded outdoor on every working day, carbonaceous matter reached half the mass, and K⁺, sulphate, nitrate and chloride dominated the ions; saccharides and cholesterol were the main organic tracers. Indoor extracts had higher DTT oxidative potential, lowered A549 viability and raised intracellular ROS, and ROS correlated with indoor phenolics and carboxylic acids. The design is a single site with a handful of filter days, so the correlations are descriptive, and PM₁₀ rather than PM_{2.5} blurs the combustion fraction. No worker personal sampling. The chemical profile is the useful output for sensing: biomass-like K⁺ and cholesterol from grilling is exactly the mixed smoke-and-cooking aerosol that optical sensors misread without source-specific correction.

doi: 10.1016/j.etap.2026.105173

Bellacov et al. 2026 J Occup Environ Hyg*Diesel-related pollutant migration in fire stations measured with low-cost direct-reading monitors*

Seven consecutive days of minute-level monitoring at fixed engine-bay and adjacent occupied locations in three fire stations, aggregated to daily means and fitted by hierarchical mixed-effects models. Coarse PM, NO₂, VOCs and CO₂ were significantly associated with engine activity, source distance and station identity, but the *direction* varied by pollutant — and **PM_{2.5} and PM₁ were not significantly associated with any of them**. Measurable concentrations beyond the bays indicate migration or shared indoor sources; this design cannot separate them. Daily aggregation of minute data discards precisely the transient peaks that dominate diesel exposure, so the null on the fine fractions reads as a resolution artefact rather than an absence.

doi: [10.1080/15459624.2026.2722264](https://doi.org/10.1080/15459624.2026.2722264)

Gaillard et al. 2026 Preprints.org (preprint)*Potential Link Between Indoor Air Quality and Coronary Artery Disease in Amsterdam, a Field Study*

Preprint: several weeks of in-home PM_{2.5}, PM₁₀ and CO₂ monitoring in 49 patients referred for coronary CT angiography in two socioeconomically contrasting Amsterdam neighbourhoods (Zuid n=24, Nieuw-West n=25), compared with CAD-RADS, calcium scores, vessel involvement and obstructive disease. No systematic neighbourhood difference in indoor air was found, but within-neighbourhood variability was large, and indoor PM showed no significant association with CAD or obstructive CAD. A weeks-long snapshot cannot represent the decades of exposure behind coronary atherosclerosis, and n=49 is badly underpowered for a null. Not peer reviewed. Relevance: the within-neighbourhood spread is the durable finding - household factors dominate indoor PM, which area-level outdoor sensing cannot capture.

doi: [10.20944/preprints202609.1981.v1](https://doi.org/10.20944/preprints202609.1981.v1)

Gulzar et al. 2026 Atmos Environ*PAHs Exposure, Health Risk Evaluation, and Renal Associations by Household Fuel Type in Rural Residents of Northwestern China*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 28 Sep with no abstract; PubMed efetch, Crossref, Europe PMC and the publisher landing page returned nothing. The title indicates PAH exposure, risk assessment and renal associations by household fuel type among rural residents of northwestern China. Pairs with the 25 Sep household solid-fuel records; retry for the exposure matrix (air, urine) and renal endpoint. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: [10.1016/j.atmosenv.2026.122388](https://doi.org/10.1016/j.atmosenv.2026.122388)

Fermoso et al. 2026 Atmosphere*Integrated hospital IAQ monitoring: continuous sensing with targeted chemical characterisation*

Thirty months of continuous low-cost sensor monitoring across hospital microenvironments (June 2023 – December 2025) yielding **1.79 million raw records, 1.42 million retained indoor measurements and 264,311 hourly aggregates**, paired with targeted VOC and aldehyde sampling. Distinct recurrent signatures emerged per area: CO₂ tracked occupancy and ventilation demand, while formaldehyde response was persistent and seasonal in a pattern consistent with indoor sources plus ventilation heterogeneity. The formaldehyde inference rests on a low-cost channel with known cross-sensitivity, and the campaign sampling is too sparse to validate it continuously. The **79% retained fraction** is the number to take away when sizing a long-horizon deployment.

doi: [10.3390/atmos17090895](https://doi.org/10.3390/atmos17090895)

Hu et al. 2026 Front Public Health*Analysis of carbon fiber dust exposure status and related occupational health risk factors among workers during the molding process of carbon fiber products.*

955 molding-line workers across 30 carbon-fibre firms in Shenzhen. Dust exposure was identified for 8.38% of employees and varied by post; ICMN tiering returned 3 tolerable, 36 potential, 37 high, 2 very high and 2 intolerable workstation risks. Elevated risk was associated with years of prior frontline work (OR 5.018, 95% CI 2.728-9.231) and grinding-shop employment (3.111, 1.123-8.624); protective were high prevention awareness (0.739, 0.595-0.919) and tertiary education (0.131, 0.024-0.699). The outcome is a risk score, not disease, so every OR describes exposure classification rather than health. The education coefficient spans an order of magnitude on 955 workers and should be treated as noise. Carbon fibre is a rising industrial aerosol with fibre morphology that mass-based instruments handle badly - worth tracking as a sensing gap.

doi: [10.3389/fpubh.2026.1926373](https://doi.org/10.3389/fpubh.2026.1926373)

Ibrahim 2026b Qeios (preprint)*The case for routine indoor PM monitoring, argued without data*

A commentary arguing that Malaysia's ICOP IAQ 2010 framework, though it contains preventive elements, is operated as a complaint-triggered investigation regime, and that **particulate matter and outdoor-to-indoor pollutant transport** should join CO₂ as routinely monitored parameters in building management. The specific climatic argument — that hot, humid conditions couple ventilation, condensation and biological growth in ways a temperate-derived code handles badly — is sound and underexploited in the LCS literature. Everything else is assertion: single author, non-peer-reviewed platform, no primary data, and the Malaysian office, hospital and school studies it cites are described rather than synthesised. Included as an argument about monitoring design, not as evidence.

doi: [10.32388/b506md](https://doi.org/10.32388/b506md)

Ikeda et al. 2026 — metadata only Build Environ*Characterizing layout-dependent breathing-zone contaminant concentration in office spaces with mixing ventilation using an occupant-to-exhaust distance metric*

Same-day Elsevier deposit; nothing in Crossref, Europe PMC core, Europe PMC fullTextXML or the OpenAlex per-work record. The title indicates a modelling study deriving breathing-zone contaminant concentration in mixing-ventilated offices from an **occupant-to-exhaust distance metric**. In scope for sensor siting specifically: it is a claim that indoor concentration is a predictable function of geometry, which is the assumption every single-point indoor monitor makes implicitly and almost never tests. Carried forward for pickup.

doi: [10.1016/j.buildenv.2026.115282](https://doi.org/10.1016/j.buildenv.2026.115282)

Yang et al. 2026 Environ Sci Technol*Organic contaminants of emerging concern in indoor air: global distribution and hazard burden*

Seven OCEC classes spanning over 130 compounds synthesised across residences, offices, schools and commercial spaces. Microenvironments separate cleanly — phthalate alternatives dominate residential air to 2.08×10^4 ng/m³, PFAS schools to 230 ng/m³, organophosphate esters offices to 159 ng/m³ — and the US, Germany, Sweden and China rank highest on inhalation hazard, which the authors say plainly may reflect monitoring intensity rather than contamination. **Microplastics were excluded from the hazard index because units and toxicity metrics are irreconcilable**: a candid admission that the particulate fraction of indoor emerging contaminants cannot currently be placed on a common scale with the rest.

doi: [10.1021/acs.est.6c06332](https://doi.org/10.1021/acs.est.6c06332)

JECS indoor PM Kawasaki 2026 Int J Hyg Environ Health*Household-specific indoor PM_{2.5} and PM_{10-2.5} concentrations and Kawasaki disease onset by age six: the Japan Environment and Children's Study*

3,846 mother-child pairs in the JECS sub-cohort with two one-week household PM measurements. Cumulative Kawasaki disease incidence by age six was 15.34 per 1,000. Higher indoor PM_{10-2.5} was associated with consistently lower odds of Kawasaki disease across exposure categories, with a weaker inverse trend for PM_{2.5} - an inverse association the authors report without over-reading. Intraclass correlations of 0.52-0.55 between the two measurement occasions mean roughly half the variance in a household's exposure is not captured by these snapshots, and with ~59 expected cases the categorical ORs are thinly powered. Two one-week integrated samples as a six-year exposure surrogate is precisely the gap continuous indoor sensing would close.

doi: 10.1016/j.ijheh.2026.114893

JP laboratory exposure 2026 Ann Work Expo Health*Assessment of personal exposure and spatial concentration variability in a small-scale chemical laboratory.*

Personal exposure concentrations ran consistently higher than co-located area measurements, mean values roughly an order of magnitude greater, reproduced across two independent measurement years. CFD then explains the gap mechanically: concentration varies strongly with height, so a fixed sampler placed away from the breathing zone of a person leaning over their own source cannot recover personal exposure, whatever its precision. The timing is regulatory - Japan makes personal exposure measurement mandatory from October 2026 - and the paper is effectively a demonstration that the outgoing area-based system was biased low by roughly 10x in this setting. Narrow and honest about it: one small teaching laboratory, student experiments rather than production work, passive samplers for vapour-phase analytes rather than PM. The 10x should not be transported to other geometries. It generalises to distributed PM sensing as a siting argument with a number attached - a fixed node samples the room, a person samples their own plume, and near a source the two differ by an order of magnitude, not a percentage.

doi: 10.1093/annweh/wxag082

Jaiswal et al. 2026 Research Square (preprint)*Nature-based indoor filtration in an EDM gear-manufacturing shop*

Single-site five-month before/after intervention in Gurugram, India. Baseline was severe: **PM_{2.5} 330 µg m⁻³ and CO 88 ppm**. Plant-based filter units plus a redesigned vent/exhaust layout, sited using airflow visualisation, cut **CO by 90% and PM_{2.5} by 42% on average** — leaving PM_{2.5} around 190 µg m⁻³, which the authors say plainly is still not a healthy range. No control period, no controlled comparison against a conventional HEPA unit, and the CO and PM reductions differ by a factor the paper does not explain mechanistically. Preprint, not peer reviewed. Listed for the occupational baseline number, which is the part that is hard to obtain.

doi: 10.21203/rs.3.rs-10511730/v1

Jamali et al. 2026 Int J Biol Macromol*All-cellulose HEPA nanofilter in a wooden frame*

Fog-sprayed, freeze-dried bacterial-cellulose nanofibres on a wood-derived cellulose-microfibre tissue substrate, in a modular wooden frame with a starch-adhesive gasket. Removal efficiency rises with BCNF grammage to **2 g/m²** and falls above it — consistent with pore closure raising velocity through the remaining paths. Optimum gives **99.998% PM_{0.3} at 150 Pa**, inside the EN 1822 H13-H14 band, and PM_{0.3} capture *peaks at high RH* under cyclic humidity, the opposite of the usual electret degradation. The claim being made is biodegradability at H13 performance, and the tests reported are single-pass efficiency and pressure drop; there is no loading lifetime, no microbial growth assessment on a cellulose substrate held at high RH, and no mechanical durability data. Promising material, incomplete filter.

doi: 10.1016/j.ijbiomac.2026.154178

Khan et al. 2026 Front Public Health*Cement industry emissions and respiratory health: global systematic review and meta-analysis, 2000–2025*

Eighteen observational studies, PROSPERO CRD420251268913. Random-effects pooling gave standardised mean differences of **−0.71 for FEV₁, −0.56 for FVC and −0.57 for FEV₁/FVC** in exposed groups; chronic cough was significantly more common while other symptoms were elevated but not significant. Occupational exposure produced greater FEV₁ impairment than residential — the expected dose ordering and the most reassuring internal check available. Reporting is the weakness: **no confidence interval, no I² and no prediction interval accompanies any pooled estimate**, so neither precision nor heterogeneity can be assessed, and SMDs pooled across studies using different spirometry reference equations with no common dust metric cannot be read as a dose-response.

doi: 10.3389/fpubh.2026.1887654

Kusi et al. 2026 Research Square (preprint)*Waste-tyre combustion toxicology: a systematic search that could not pool*

PRISMA-style search of PubMed, Scopus, ScienceDirect and Google Scholar for 2016–2025 human and experimental studies of waste-tyre combustion emissions — particulate matter, PAHs, VOCs and heavy metals — reporting biochemical, molecular, biomonitoring or toxicological outcomes, appraised with SYRCLE, JBI and NIH instruments as appropriate to design. **Heterogeneity in exposure conditions, designs and reported outcomes precluded quantitative synthesis**, which the authors state plainly. The value is in mapping organ-specific mechanistic pathways for an exposure that is ubiquitous in informal waste economies and almost absent from regulated monitoring networks. Preprint, not peer reviewed; a narrative synthesis of a heterogeneous literature inherits every selection effect in that literature.

doi: 10.21203/rs.3.rs-10530000/v1

Lee et al. 2026 J Hazard Mater*Comparative analysis of metal concentrations in respirable particles and nanoparticulate matter (NPM) from mild steel, aluminum, and stainless steel welding fumes.*

Breathing-zone respirable and nanoparticle respiratory deposition (NRD) samplers on 61 participants - 38 welders, 12 factory and 11 office controls - with ICP-OES metals and TEM/EDS morphology. Mild-steel fumes (n=31) were dominated by Mn and Fe, stainless (n=3) by Cr and Ni, aluminium (n=4) by Al; across the six metals, the nanoparticulate fraction carried 48.3% of the respirable metal mass on average, with substantial Mn, Cr and Ni in NPM. The stainless and aluminium arms are too small for inference, which the authors acknowledge. Relevance: if half the respirable metal sits below ~300 nm, optical sensors - which lose efficiency below that size - will systematically undercount the toxicologically dominant fraction in welding environments.

doi: 10.1016/j.jhazmat.2026.143540

Liu 2026 Atmosphere

Energy-Indoor Air Quality-Health Nexus in Rural Residential Retrofits: A Field-Calibrated Coupled Simulation and Multi-Criteria Evaluation for the Central Plains Transitional Climate of China

Solid-fuel brick-concrete rural homes in Anyang, Henan: 72-h paired winter and summer measurements of PM_{2.5}, PM₁₀, CO₂, formaldehyde and TVOC calibrate a two-way EnergyPlus-CONTAM model (indoor temperature NMBE -2.1%, CV(RMSE) 6.7%). An envelope-only retrofit cut thermal loads 52.2% but raised winter PM_{2.5} 25.8% and formaldehyde 32.4% via a 62.3% fall in infiltration; adding schedule-optimised natural ventilation kept 95.6% of savings while cutting the PM_{2.5} hazard quotient 20.3%. Calibration and out-of-sample validation are reported for temperature and CO₂, not for PM, so the PM results rest on unvalidated source and deposition terms; single author, one county. Relevance: tightening envelopes raises indoor PM from solid fuel - retrofit programmes need indoor sensors to catch it.

doi: 10.3390/atmos17100921

Liu et al. 2026 Environ Res

Spatiotemporal Contamination Characteristics, Bacterial Community Structure, and Airflow-Mediated Transport of Bacterial Aerosols in a Pediatric Emergency Consulting Room.

One paediatric emergency consulting room: culturable airborne and surface bacteria, high-throughput sequencing and CFD of aerosol transport and deposition. Airborne culturable counts differed little between locations (well-mixed small room), surface deposition was heterogeneous, and 73.47% of bacterial aerosol was below 3.3 µm; Staphylococcus was the core genus. CFD placed the consultation desk and floor at 79.71% of deposited particles, with high-risk zones around occupants. One room, culturable fraction only (typically <1% of total bacteria), and sampling duration and counts are not in the abstract. Relevance: marginal; bioaerosol is invisible to optical PM sensors, so ventilation decisions here need culture or fluorescence methods.

doi: 10.1016/j.envres.2026.125789

Liu et al. 2026 Atmosphere

Research on Comprehensive Evaluation of Air Quality in Underground Stations of Urban Rail Transit Systems

One island-platform Beijing Metro station: CFD of airflow for train absent, arriving and departing was used to place measurement points, then handheld instruments measured CO₂, PM_{2.5}, PM₁₀ and TVOC at morning and evening peaks, combined in an entropy-weighted index. Retention zones formed at the platform centre and escalator entrances under low-frequency ventilation; piston flow improved mixing but imported tunnel air; PM tracked outdoor inputs with some internal contribution. No concentrations, instrument models or sampling durations are in the abstract. Relevance: CFD-guided siting is directly transferable to placing fixed low-cost PM nodes in stations, where iron-rich wear particles bias optical readings.

doi: 10.3390/atmos17100938

Lv H et al. 2026 ACS Appl Mater Interfaces

Polyimide microsphere/nanofibre gradient aerogels for high-temperature filtration

A single-component polyimide gradient aerogel built to beat the dust-holding limit of uniform nanofibre mats, which cake at the surface and choke. Dual-layer structure: ≈730 nm coarse fibres under ≈300 nm medium fibres, with a composite functional layer of ≈3.34 µm microspheres and ≈60 nm ultrafine fibres on top. Performance: **99.996% PM_{0.3} efficiency at 45.08 Pa** initial pressure drop, **127 g m⁻² dust-holding capacity**, and **99.91% retained at 300 °C**. Single-component construction is claimed for recyclability and uniform thermal expansion. What is not reported is loaded pressure-drop evolution or efficiency after thermal cycling, which is what actually determines service life in a stack.

doi: 10.1021/acsami.6c10042

Lv et al. 2026 J Hazard Mater

Impact of biomass stove intervention on indoor air quality in rural homes of China: Attenuation of cooking contribution.

Advanced biomass stoves were introduced in Xuanwei households and real-time PM_{2.5} monitors logged kitchen, living room, bedroom and yard. Kitchen daily mean PM_{2.5} fell 29.3%, 55.8% and 30.7% with direct-fired, recirculating and gasifier stoves. Peak analysis showed gasifier stoves cut kitchen cooking-peak height by 59.3% and peak area by 60.2%, cooking's contribution to kitchen PM_{2.5} fell ~30%, and indoor/outdoor ratios in living rooms and bedrooms approached 1. The abstract gives no household count, no control arm and no sensor calibration - a pre/post design confounded by season and behaviour. Relevance: peak-resolved metrics from time-resolved sensors capture intervention effects that daily means understate - a methodological point for any stove or purifier evaluation.

doi: 10.1016/j.jhazmat.2026.143732

Ma L et al. 2026 Can J Urol

Surgical smoke from laser laparoscopic partial nephrectomy with an aspirating handle

Ex vivo porcine kidney comparison of electrocoagulation, ultrasonic, 980 nm diode and 532 nm KTP laser resection, with and without a novel aspirator-integrated handle. The KTP green laser produced the most smoke; coagulation-layer thickness ran diode (**4.16 ± 0.21 mm**) > green (1.55 ± 0.21) > electrocoagulation (0.35 ± 0.08) > ultrasonic (0.52 ± 0.08). With the aspirator, diode-laser formaldehyde fell **4.95 → 1.69 mg m⁻³**, TVOC **8.34 → 1.11 mg m⁻³** and CO **124 → 24 ppm**. Gases only — the study reports no particle-size distribution or number concentration, which is the fraction that reaches the alveoli and the reason surgical smoke is an occupational aerosol problem rather than a gas one. In vitro, no personal sampling.

doi: 10.32604/cju.2026.070096

Ma et al. 2026 Front Public Health

Association between combined occupational exposure to coal dust and noise and high-frequency hearing loss among workers: a cross-sectional study.

1,187 workers examined in Baiyin City in 2023: 276 noise-only, 911 noise plus coal dust. High-frequency hearing loss prevalence 35.5% vs 46.8% (P<0.001), with the characteristic 4 kHz notch and little speech-frequency involvement; adjusted OR for combined versus noise-alone exposure 1.620 (95% CI 1.198-2.190, P=0.002). The inference is weak in a specific way: exposure is a job label, not a measured dust concentration or a measured noise dose, so the combined group may simply work louder jobs and the OR then encodes noise intensity rather than a dust interaction. A cross-sectional design in a workforce also selects out the already-deafened. It is carried because dust-noise co-exposure is common and because it is a reminder that a sensor network characterising an industrial site is measuring only one of two correlated hazards.

doi: 10.3389/fpubh.2026.1929586

Mbazima et al. 2026 Int J Environ Health Res*Chalk dust puts teachers over a deposition threshold while staying under the occupational limit*

Sixty personal breathing-zone samples from 12 Johannesburg secondary-school teachers (7 female, 5 male), collected Monday–Friday 08:00–14:35 with GilAir-3 pumps, a 4 µm cut-point cyclone and 25 mm PVC filters at 1.85 L/min, then run through a multiple-path particle dosimetry model at low, moderate and high exertion for nasal and oronasal breathing. **PM₄ ranged 0.019–0.761 mg/m³, mean 0.126 ± 0.134 mg/m³**; chalk use ($\beta = 0.468$) and sampling duration ($\beta = 0.305$, both $p < 0.001$) predicted concentration. Breathing route dominated deposition, and alveolar clearance was slower than tracheobronchial in both sexes. The authors' conclusion — that risk persists below the occupational exposure limit — rests on modelled rather than measured dose, and 12 teachers in one region is a small and non-random base. Note the standard deviation exceeds the mean, so the distribution is heavily right-skewed and the mean is a poor summary.

doi: 10.1080/09603123.2026.2726420

Micallef & Scerri 2026 Build Environ*A flow-conditioned infiltration metric for traffic-related pollution ingress*

Ships on metadata only — Crossref carries no abstract and the record is not otherwise indexed on this run. The title indicates an outdoor-to-indoor infiltration metric for traffic-related air pollution in naturally ventilated buildings, conditioned on flow regime rather than assumed constant. Counted and listed; no numbers claimed. Worth retrying: a flow-dependent infiltration factor is exactly the term that makes outdoor-monitor-based exposure assignment wrong for naturally ventilated stock, and it is the term a paired indoor/outdoor sensor pair can actually estimate.

doi: 10.1016/j.buildenv.2026.115195

Occupational nanoparticle scoping 2026 Part Fibre Toxicol*Occupational nanoparticle exposure and its effects on pulmonary health, cardiovascular health, and systemic biomarkers: a scoping review*

MEDLINE, EMBASE and Web of Science to April 2024; 66 of 6,020 records included, mostly cross-sectional ($n = 37$) or panel ($n = 18$). Exposures are dominated by carbon nanotubes and fibres, titanium dioxide and metal oxides in manufacturing. Reported outcomes are reduced lung function and raised inflammatory and oxidative-stress markers, with cardiovascular and systemic biomarker changes. The review explicitly performed no quality appraisal, so the evidence map is a catalogue of what has been measured rather than a weighing of it, and the cross-sectional predominance means reverse causation is unaddressed across most of the base. Useful mainly for the biomarker inventory and for locating where UFP metrology in workplaces is missing

doi: 10.1186/s12989-026-00699-0

Oklahoma cannabis facility IAQ 2026 Ann Work Expo Health*Seven-instrument occupational air-quality characterisation of a cannabis grow room*

Continuous monitoring from 10 July to 1 August 2023 in a 186 m² HVAC-equipped medical cannabis cultivation room holding roughly 900 plants on a 12:12 photoperiod. Seven instruments across three spatial zones at 1.5 m breathing height — three Aranet4 NDIR CO₂ sensors, two TSI AirAssure multi-parameter monitors carrying size-resolved PM, and two PurpleAir PA-II-FLEX dual-channel counters — produced **~113,000 quality-controlled records**. CO₂ tracked the light cycle, with means of **979–1,086 ppm** and maxima above **2,200**; nighttime elevation is attributed to plant metabolism and enrichment rather than worker respiration, since workers leave at lights-out. **Median PM_{2.5} was low at 1.4–3.1 µg m⁻³** with episodic activity-driven spikes. One facility, one room, one summer month, and no personal sampling, so this is a determinants pilot rather than an exposure assessment — but it is the rare occupational study that runs consumer-grade and research-grade particle instruments side by side.

doi: 10.1093/annweh/wxag074

Opbroek et al. 2026 Sci Total Environ*Health effects of indoor environmental quality (IEQ): A systematic review of indoor environments in high-income settings.*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 22 Sep with no abstract, and none was recoverable from any leg. A systematic review of IEQ health effects in high-income settings would cover indoor PM alongside thermal, acoustic, lighting and ventilation parameters, but the deposit does not say which exposures, how many studies, or whether any pooling was done, and the author list is not evidence about content either. Carried as a placeholder because indoor PM is the dominant fraction of personal exposure in high-income settings; added to the retry list and to be re-screened rather than summarised from the title.

doi: 10.1016/j.scitotenv.2026.182158

PSH smoke-free home RCT 2026 JAMA Netw Open*A smoke-free-home trial that missed its primary outcome, reported plainly*

Cluster-randomised trial across 40 permanent-supportive-housing sites in the San Francisco Bay Area, 400 residents (mean age 54.5, 63% male) who smoked at home, January 2022 to March 2025, intervention sites receiving one-to-one coaching on adopting a smoke-free home and staff at all sites receiving brief cessation training. **The primary outcome failed**: 90-day smoke-free-home adoption was 6.8% versus 4.8%, **OR 1.45 (0.69–3.07)**. What moved was the softer secondary — any adoption of at least one day, 63.4% versus 36.8%, **adjusted OR 3.83 (2.63–5.57)** — and CO-verified 7-day abstinence, 6.3% versus 1.0%, **OR 6.94 (1.69–28.45)**, which rests on **14 events in total** and should not be read as an effect size. The honest reading is that brief coaching moves intention and not maintenance in a population with very high baseline exposure, and that a waiting-list control cannot rule out attention effects on the self-reported outcomes.

doi: 10.1001/jamanetworkopen.2026.32697

Perfilyeva et al. 2026 Genes (Basel)*Genetic Susceptibility to Impaired Lung Function in Underground Coal Miners from Kazakhstan.*

GWAS of six spirometric traits (FVC, FEV1, FEV1/FVC, MEF25/50/75) in 186 underground coal miners from the Karaganda basin, adjusted for age, mining exposure, BMI, smoking and population structure, with imputation. No variant reached genome-wide significance; recurrent suggestive signals at PRKCB, FANCE and UBTD1 across correlated traits. $n=186$ is grossly underpowered for GWAS, dust exposure is years-in-mining rather than measured, and there is no replication cohort. Relevance: low; underscores that occupational cohorts in Central Asia still lack any measured respirable-dust exposure.

doi: 10.3390/genes17091021

Pessoa Campos et al. 2026 J Toxicol Environ Health A*Salivary 8-hydroxy-2'-deoxyguanosine as a biomarker of oxidative DNA damage in urban outdoor workers exposed to traffic-related air pollution.*

Cross-sectional comparison of outdoor street vendors against indoor control workers, with workplace PM_{2.5} and noise logged by portable instruments and salivary 8-OHdG measured by ELISA. Exposed workers had higher salivary 8-OHdG; PM_{2.5} and BMI were independent predictors in multivariate models while income and education correlated inversely. What the record does not give is the whole quantitative content: no sample size, no effect estimate, no interval, and no city. Saliva sampled once cannot separate chronic burden from that morning's traffic, and the socioeconomic gradient is entangled with the exposure by construction. Carried because the design rests entirely on portable PM_{2.5} monitors at the workplace, and its credibility therefore rests on a calibration the abstract never describes.

doi: 10.1080/15287394.2026.2734428

Qian et al. 2026 Indoor Air*Impacts of Housing Retrofits on Indoor Environmental Quality and Energy Consumption: A Review*

A review of residential retrofit studies. Retrofits cut energy use by 10-84%; indoor temperature rises for winter-targeted and falls for summer-targeted measures, but effects on RH, CO₂ and pollutants including PM_{2.5} varied with ventilation, occupant behaviour and outdoor levels. The authors' main conclusion is critical of the evidence base: most studies lacked a causal framework, long observation, adequate samples, weather adjustment, matched controls and behaviour control. No pooled estimate and no search protocol in the abstract. Read together with the Anyang simulation in this issue, the gap is clear. Relevance: long-duration, matched-control indoor PM monitoring with low-cost sensors is the cheapest way to supply the causal designs this review says are missing.

doi: 10.1155/ina/3883472

Rouyn-Noranda citizen dust 2026 Environ Monit Assess*Citizen-science measurement of smelter-derived metals in snow and indoor dust*

Residents of **23 homes** 540–2,080 m from the Horne Smelter — Canada's only copper smelter — collected their own indoor dust and adjacent snow samples for Cu, Pb, Cd, As, Zn and Cr. Accounting for distance and wind direction, smelter location explained **73% of snow variance for Cu and Cr**, 58% for Cd and Pb, 56% for As and 44% for Zn ($p < 0.01$); for indoor dust the distance-only explanatory power was much lower at **36% (Cu) to 15% (As)**. Copper served as a smelter tracer and correlated strongly with the other metals in snow ($R^2 > 70\%$). Few residential characteristics predicted indoor dust once distance was controlled. Twenty-three homes is a small sample, snow and settled dust are deposition integrals rather than airborne concentrations, and participant-collected samples carry uncontrolled handling variance. The value is the demonstration itself: a participatory design produced a defensible source-attribution gradient in a community with a documented exposure dispute.

doi: 10.1007/s10661-026-15900-2

SAA wetting agent 2026 Environ Geochem Health*High-efficiency 'SAA' wetting agent for coal mine dust control: from preparation and mechanism to field application.*

A three-component surfactant blend (sodium gluconate, AEO-3, AES) lowers surface tension and contact angle on coal, raises oxygen-containing surface functional groups, and agglomerates micron-scale dust - D50 rose 22.58% over untreated. In a purpose-built simulation rig the suppression rate was 71.50%, 2.17x conventional water mist, and the formulation was then taken underground on a real working face. The mechanism-to-field arc is the strength; the evaluation is the weakness. A suppression rate measured in the authors' own custom rig has no external calibration, the field test reports no control shift or personal sampling, and no respirable-fraction or silica-specific result is given, which is what determines pneumoconiosis risk - agglomerating dust into coarser particles raises D50 partly by removing mass from exactly the respirable size range the health endpoint cares about, so the health-relevant efficacy is almost certainly different from 71.5% and is not reported. Relevant to occupational sensing as a reminder that a mass-based sensor reads an agglomeration intervention as a larger improvement than the lung sees.

doi: 10.1007/s10653-026-03491-3

Tam et al. 2026 Curr Environ Health Rep*Toxic air contaminant emissions from residential and commercial fossil gas appliances: a systematic review*

25 studies, the first systematic synthesis of emission factors across gas appliance types. Nonzero emissions documented for **at least 41 toxic air contaminants** via both leakage and combustion pathways. Only formaldehyde and benzene had enough data for quantitative characterisation, and their emission factors **spanned two orders of magnitude** across studies and operating conditions — a spread that makes any single-value inventory estimate indefensible. Coverage is badly skewed toward cooking stoves; heaters, water heaters and dryers are barely studied despite longer duty cycles. The operationally interesting claim, correctly stated as emerging rather than established, is that **methane and CO measurements may proxy for leakage- and combustion-related TAC emissions** — which is exactly the kind of cheap-channel inference target a low-cost indoor node can carry.

doi: 10.1007/s40572-026-00562-6

Tangshan CWP IGF1 2026 Front Public Health*Dust exposure, IGF1 and autophagy in coal workers' pneumoconiosis*

Retrospective analysis of 2,028 employees at one Tangshan coal enterprise (2023–2024) with a paired SiO₂-stimulated macrophage model. Logistic regression retained early initial working age, dust-exposure duration, smoking, white cell count and AST/ALT ratio; patients and stimulated macrophages both showed raised IGF1, altered LC3/P62 and increased PI3K/AKT/mTOR phosphorylation. Two weaknesses matter. Exposure is job history, not a measured respirable dust concentration, so no exposure–response can be estimated; and the cell model uses crystalline silica while the cohort's hazard is coal mine dust — these are different particles with different mechanisms, and the paper's own language stays correlative. The white-cell and liver-enzyme “risk factors” are as plausibly consequences of disease as causes.

doi: 10.3389/fpubh.2026.1899729

Tanzania house-design IAQ 2026 Research Square*A comparative study to investigate the impact of novel house design on indoor air quality in rural Tanzania*

PREPRINT. Magoda village, four novel-design houses against three local control houses over 77 days. Mean indoor PM_{2.5} was 2.8x lower and PM₁₀ 2.4x lower in the novel houses; the authors attribute this to integrated chimneys, increased envelope permeability, the lighting strategy and raised bedrooms, in a setting where ambient PM is low so indoor sources dominate. $n = 7$ houses is the limitation and it is a hard one: households are not randomised, occupancy and cooking behaviour differ, and the design elements are confounded with each other so no single feature is attributable. The secondary proposal - an hourly rather than 24-hour indoor guideline for homes empty during the day - is the more transferable idea

doi: 10.21203/rs.3.rs-10498557/v1

Taskin and Ozkal 2026 Indoor Air*Investigating Airborne Pollutant Dynamics in a Pathology Laboratory Using an Integrated High-Resolution Sensor Network and Operational-Context Framework.*

A university hospital pathology laboratory was logged at 1-min resolution for 42 days with a myRIO-based node (sensor-indicated HCHO, MOx eTVOC, eCO₂, PM, T, RH), aggregated to 123 eight-hour blocks. HCHO was higher in working than non-working blocks (median 313 vs 186 ppb) but the work-status coefficient fell from +0.430 to +0.019 ($p = 0.897$) once block position, temperature and humidity entered; 91.1% of blocks exceeded the 100 ppb ACGIH TLV regardless. Work days differed in their 15-min maxima (1,723 vs ~530–608 ppb). PM results are not reported in the abstract and the HCHO sensor is uncorrected against a reference. Relevance: the block-aggregation and confounding analysis is the right template for any occupational sensor deployment.

doi: 10.1155/ina/9352274

Turkish cigarette PM chamber 2026 Tob Use Insights*Particulate Matter Emissions of Turkish Cigarettes in Comparison to German Brands - An Aerosol Spectrometric Study*

Eleven Turkish brands, four German comparators and the 3R4F reference burned in a 2.88 m³ chamber. PM₁₀ 1044-1677 $\mu\text{g m}^{-3}$ and PM_{2.5} 1043-1675 $\mu\text{g m}^{-3}$, i.e. essentially all mass below 2.5 μm , with PM₁ 994-1548 $\mu\text{g m}^{-3}$ carrying the bulk. Five Turkish brands exceeded the reference cigarette on PM₁₀ and PM_{2.5}; all four German brands were lower. No correlation with tobacco weight, filling density, filter length or rod length, so the brand differences are not explained by the physical parameters measured. Limitation: a sealed unventilated chamber sized like a car cabin is a worst case, not a room, and the authors use it to argue for vehicle smoking bans rather than generalising to dwellings

doi: 10.1177/1179173x261473110

Wang et al. 2026 PLoS One*Research on dust control and ventilation parameter optimization in roadhead mining based on response surface methodology.*

ANSYS Fluent simulation of a fully mechanised heading face under long-forcing, short-exhaust ventilation, with a Box-Behnken design over exhaust-duct inlet angle, exhaust-to-forcing airflow ratio, exhaust inlet diameter and forcing outlet diameter. Inlet angle and exhaust diameter dominated the dust field; multi-objective optimisation cut dust mass concentration by 58% at the lower and 34% at the upper return-air-side corner, and a 1:4 scaled physical model reproduced the simulated field to within 5%. The validation is of the flow model, not of the dust reduction in an operating mine - no in-situ gravimetric comparison, no respirable size cut, no worker exposure. Relevance is direct nonetheless: corner accumulation zones are where fixed monitors are not, and a few nodes placed by this simulation would test it cheaply.

doi: 10.1371/journal.pone.0338763

Wang et al. 2026 J Hazard Mater*Aeration change from horizontal brush to fine bubble substantially reduces the emission of inhalable antibiotic resistance from wastewater to air.*

Seasonal metagenomics of PM_{2.5} and wastewater at one WWTP operating horizontal-brush and fine-bubble aeration under matched inflow and location. PM_{2.5}-borne ARGs showed higher occurrence, mobility and risk scores under brush aeration, peaking in winter; brush aeration transferred 74.5% of wastewater ARGs and 75.0% of pathogenic resistant bacteria to air versus 57.5% and 50.0% for fine bubble. Mobile ARGs were integrase-carried and mainly MLS-resistant; six pathogens including *K. pneumoniae* expressed mobility. One plant, and 'transfer percentages' are shared-gene fractions, not emission rates; no worker exposure or PM mass is reported. Relevance: low for sensing; PM_{2.5} is the carrier, but no optical sensor distinguishes bioaerosol.

doi: 10.1016/j.jhazmat.2026.143712

Wang et al. 2026 Toxics*Mixed Exposure to Underground Air Pollutants and Metabolic Syndrome in Coal Miners: A Cross-Sectional Study Integrating Multi-Pollutant Models and Urinary Metabolomics.*

Cross-sectional study of underground coal miners with logistic regression, BKMR and Qgcomp for coal dust, CO, CO₂ and NO₂, plus untargeted urinary metabolomics. Each pollutant was associated with metabolic syndrome; the mixture was positively associated in BKMR and Qgcomp, with coal dust and NO₂ contributing most - dust tracking central obesity and hypertension, NO₂ glucose and triglycerides. >113 metabolites differed; acylcarnitines tracked dust and leukotrienes NO₂. The abstract gives no n, no effect sizes and no exposure-measurement method; healthy-worker and shift-work confounding are unaddressed. Relevance: adds a metabolic endpoint to respirable coal dust, which is measured underground by personal dust monitors that low-cost sensors cannot yet replace.

doi: 10.3390/toxics14090746

Wei J et al. 2026 Front Immunol*Welding fume immunology, with the exposure-assessment gap named explicitly*

Mini review of metal-rich fine and ultrafine welding-fume particles as a stimulus to airway epithelium, alveolar macrophages, neutrophils, monocytes and circulating immune cells. Consistent signals across human exposure studies, controlled inhalation, cell models and workplace investigations: acute-phase proteins, cytokines, chemokines and neutrophilic responses, with some evidence of altered T-helper activity and blunted responsiveness to bacterial stimuli. TRPA1/TRPV1 sensory signalling is proposed as a rapid route to cough and airway reflex, and the review is careful to say that direct welding-fume-specific human evidence for that pathway is limited. The stated obstacle to synthesis is the one that matters here: **fume composition and particle metric vary so much between studies that biomarker results cannot be pooled.** No quantitative estimate, no pooling, no systematic search protocol reported.

doi: 10.3389/fimmu.2026.1882845

Wu et al. 2026 Int J Phytoremediation*Sphagnum palustre moss bags as sentinels to monitor indoor air PAHs in university libraries.*

Sphagnum palustre moss bags were exposed for 45 days in autumn (non-heating) and winter (heating) in study rooms, e-reading rooms, stacks and outdoors at three university libraries. Total PAHs were 487-1,050 $\mu\text{g/kg}$ moss in autumn and 735-3,167 $\mu\text{g/kg}$ in winter, dominated by phenanthrene, naphthalene, pyrene and fluoranthene; I/O ratios pointed to indoor sources in autumn and infiltration in winter, and diagnostic ratios to petrogenic then mixed sources. Moss uptake is not converted to an air concentration, gas and particle phases are not separated, and diagnostic PAH ratios are notoriously unreliable for apportionment. Relevance: low; a cheap integrating sampler, but no route to a $\mu\text{g m}^{-3}$ exposure metric.

doi: 10.1080/15226514.2026.2734570

Xing et al. 2026 Sci Bull (Beijing)*Household air pollution associated with different heating approaches in rural Northern China.*

Year-round indoor PM_{2.5} monitoring in 948 rural North China Plain households under the Clean Heating Initiative, with exposure modelling and cost-benefit analysis. Annual indoor PM_{2.5} was 109 $\pm$ 207 $\mu\text{g m}^{-3}$ and above the WHO guideline for >90% of the year. Pelletised-biofuel households had 20-30% lower indoor PM_{2.5} and ~20% lower population exposure than traditional solid-fuel users, and an estimated 8% lower attributable premature mortality; benefit-cost ratios were 4.39-8.64 for pellets versus 0.76-2.20 for full electrification. Fuel type was not randomised, the SD twice the mean signals heavy episodic tails, and the monitor type and its correction are not given in the abstract. Relevance: at n=948 year-round this is the scale at which household LCS networks become the exposure instrument of record.

doi: 10.1016/j.scib.2026.09.018

Yang et al. 2026 Atmosphere*Ventilation optimisation for a 300 m welding workshop with distributed dust sources*

CFD of airflow and welding-fume transport in a 300 $\times$ 28 $\times$ 21 m hall, varying exhaust-to-supply ratio and side-exhaust-port height, and contrasting low-density aluminium-alloy fume with high-density carbon-steel fume. Two air-velocity peaks appear at 0-2 m and 8-12 m above floor; the **top exhaust port outperforms the side port** throughout, higher ESR accelerates upward migration and suppresses lateral dispersion, and a badly set side-port height produces either short-circuiting or excess lateral spread. Optimal side-port height **5 m for aluminium, 6 m for carbon steel.** Simulation only — no measured personal exposure, and fume is treated as a passive density-differentiated tracer rather than a coagulating aerosol.

doi: 10.3390/atmos17090843

Yoshizaki et al. 2026 J Am Coll Surg*Differences in the effectiveness of smoke evacuator types for reducing surgical smoke: a randomized controlled trial*

Randomised, double-blind trial (UMIN000047033), 64 laparotomy patients at the University of Tokyo Hospital, Feb 2022 to May 2023, allocated 1:1:1 to electrocautery-incorporated evacuation at 8 mm tip distance ($n = 22$), 18 mm ($n = 23$) or hose-type ($n = 21$). The 8 mm system cut particle counts 96–97% across 0.3–5.0 μm against hose-type; **the 18 mm arm did not reach significance on particles**, so effectiveness collapses over a 10 mm change in capture distance — the practically important finding. Median acetaldehyde was 1.90, 2.70 and 7.90 $\mu\text{g}/\text{m}^3$ in the 8 mm, 18 mm and hose arms, both electrocautery arms below hose, with formaldehyde behaving similarly. Endpoints are area concentrations rather than personal breathing-zone exposure and no health outcome was measured; single centre, one procedure type. The distance sensitivity is the transferable lesson for any point-source capture or near-field sensing geometry.

doi: 10.1097/xcs.0000000000001943

Yu et al. 2026 Atmosphere*Development of a Qualitative Questionnaire for Indoor Air Quality Monitoring: A Scoping Review of Subjective Assessments in Studies Using Low-Cost Sensors*

Scopus-only scoping review (281 records, 35 included) of questionnaires used alongside low-cost IAQ sensors in homes and workplaces, grouping items into five domains - demographics, symptoms, perception, building/occupancy, monitored pollutants - and proposing a structured questionnaire. A single database and no appraisal of questionnaire validity limit it to an item inventory; no psychometric testing of the proposed tool is reported. Relevance: moderate for deployment design - pairing sensor time series with standardised perception items is how sensor data become usable for SBS-type outcomes.

doi: 10.3390/atmos17100943

Yuksel & Yildirim 2026 Front Public Health*An analysis of occupational exposure to particulate matter and erythrocyte morphological abnormalities among charcoal-grill restaurant workers*

25 charcoal-grill restaurant workers vs 25 office staff. Kitchen PM median 117.8 vs 0.23 $\mu\text{g m}^{-3}$ (the office value is implausibly low for any indoor space and suggests an instrument floor), with more wheeze (12/25 vs 0/25). After adjustment, four poikilocyte types were elevated - dacryocytes IRR 2.46 (95% CI 1.36–4.44), elliptocytes 2.33, sickle forms 4.47, schistocytes 1.56 - but total anomaly burden was not (IRR 1.57, 0.65–3.78). $n=25$ per arm, smears only for 19 controls, CO co-exposure unseparated and smoking unreported. Relevance: low-moderate - one more occupational combustion setting with nine-fold-plus PM gradients that cheap sensors could document over full shifts.

doi: 10.3389/fpubh.2026.1953388

ZCND aerogel filter 2026 J Colloid Interface Sci*Biomimetic dual-network nanofibrous aerogels with robust interfacial assembly for multifunctional air filtration.*

A zein/curcumin nanofibre plus down-fibre aerogel reached 99.51%, 99.74% and 99.93% efficiency for PM_{10} , $\text{PM}_{2.5}$ and PM_{10} at 27.39 Pa pressure drop, with 335.85 g/m² dust-holding capacity, and - the result that distinguishes it - retained 98.67% PM_{10} efficiency after 12 h at 90% relative humidity. It also adsorbs ammonia (104.95 cm³/g), shows visual NH_3 sensing, is antibacterial, and biodegrades in soil burial faster than commercial media. The target is livestock housing, where PM, NH_3 and bioaerosol co-occur and humidity is chronically high, so the multifunctionality is matched to a real setting rather than assembled for novelty. Standard caveats for this literature apply and are not minor: bench aerosol challenge under controlled conditions, no barn deployment, 12 h is not a service life, and nothing on mechanical durability under pulse cleaning or cost at scale. Noted here mainly because humidity-resistant PM_{10} capture is the same physical problem, inverted, that defeats nephelometric low-cost sensors in barns - both fail when water condensates on the particle, one by fouling and one by over-reading.

doi: 10.1016/j.jcis.2026.141640

Zigabe et al. 2026 Environ Pollut*Determinants of early-life household air pollution exposure (HAP): carbon monoxide (CO) and $\text{PM}_{2.5}$ concentrations in Bukavu, eastern Democratic Republic of Congo (DRC).*

MUTOTO pregnancy-cohort sub-study, 405 households, March 2025 to January 2026. Mean 24-h household $\text{PM}_{2.5}$ was 53.8 $\mu\text{g m}^{-3}$ (SD 34.2), maximum household mean 244.3; personal CO averaged 5.3 ppm (SD 8.7), maximum 65.1. Biomass fuel and cooking frequency drove both; 96% of kitchens had no chimney, and biomass use rose from 56% to 87% across CO exposure groups. The limitation is metrological: PurpleAir optical counts in dense biomass smoke are not reported as corrected, and uncorrected PMS sensors typically overestimate in smoke while underestimating coarse mass - the reported means could be biased in either direction without a gravimetric subsample. Single 24-h samples also cannot capture seasonal variability. For this watch it is the model deployment: low-cost PM plus CO loggers at cohort scale in a setting with no reference network.

doi: 10.1016/j.envpol.2026.129170

NOT CARRIED BY A DAILY ISSUE — EXTRACTIVE PRECIS OR METADATA ONLY, NOT AN ASSESSED SUMMARY (1 RECORDS)

Srinivasan et al. 2026 Natl Med J India*Occupational wood dust exposure and lung function impairment: a systematic review and meta-analysis*

No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.25259/nmj_668_2024

Burden, policy & mitigation**55 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (55 OF 55 RECORDS IN THIS SUBTOPIC)

Addis Ababa hybrid net 2026 Health Serv Insights *$\text{PM}_{2.5}$ spatiotemporal variation and monetised burden from a hybrid sensor network*

Two years of hourly $\text{PM}_{2.5}$ (2022–2023) from a **hybrid network of seven low-cost sensors and two beta attenuation monitors** — the design this project cares about, deployed in a city with almost no prior monitoring. Site means 15–33 $\mu\text{g}/\text{m}^3$, citywide 27, more than fivefold the WHO guideline of 5; concentrations were higher in the wet Kiremt season, with a **07:00 diurnal peak of 38**. Hazard quotients exceeded 1 in all age groups. The cancer-risk estimate is honestly reported as unstable: ELCR exceeded 10^{-4} under a slope factor of 0.008 but stayed acceptable at 0.0012. Monetised DALYs give **USD 652.95 M** over two years, **0.25% of GDP** in each year, dominated by ischaemic heart disease. The paper does not describe co-location or humidity correction for the seven low-cost nodes, and every health figure is transferred from external risk coefficients, so the burden estimate inherits both instrument bias and transfer error.

doi: 10.1177/11786329261487535

Aerosol Air Qual Res 2026 Aerosol Air Qual Res*Cross-national Evaluation of STEM-Based Indoor Air Quality (IAQ) Education for Sustainable Development.*

METADATA ONLY. A cross-national evaluation of STEM-delivered indoor air quality education; no sample, instrument or outcome is recoverable from the deposit. Logged at tier C because low-cost sensor deployments in schools are the standard vehicle for exactly this kind of programme, and whether the measurement component changes behaviour or only knowledge is an open question this watch has no evidence on either way. Nothing is claimed here. Deferred re-query.

doi: 10.1007/s44408-026-00168-5

Ann Glob Health 2026 Ann Glob Health*Disclosure without accountability: Air pollution reporting by major South African industrial companies.*

METADATA ONLY. Europe PMC held no abstract at screening. The title states the finding as a thesis - corporate air-pollution reporting that discloses without enabling accountability - which in the self-reported-emissions literature usually means figures are published at an aggregation or cadence that cannot be checked against any independent measurement. South Africa's Highveld is one of the few heavily industrialised regions where that gap is measurable, because ambient monitoring exists alongside the disclosures. This is precisely the gap independent low-cost networks are deployed to close, so the record is carried despite being unreadable. Deferred re-query.

doi: 10.5334/aogh.5437

Atmospheric infrastructures Czech activism 2026 Social Health Illn*Cities as Atmospheric Infrastructures: An Analysis of Activist Responses to Respiratory Illness in the Czech Republic*

Proposes 'atmospheric infrastructures' to hold together two things usually analysed apart: the city as a physical producer of the air residents breathe, and the city as a social infrastructure that makes collective response possible. Grounded in Czech air-pollution activism, where organisers frame respiratory health as a citizenship claim, fight for the health of the city rather than of individuals, and treat civic education as equal in weight to protest. The framing is directly useful to anyone deploying community sensor networks, because it names why data alone rarely mobilises and why the collective rather than individual framing carries. This is conceptual and interpretive work from a single national context; no exposure or health measurements are involved and no generalisability is claimed.

doi: 10.1111/1467-9566.70260

Australian air health risk index 2026 Environ Pollut*METADATA ONLY — community-specific weather and air health risk indices*

No abstract retrievable on the entry date, despite *Environ Pollut* being PubMed-indexed — the same Crossref-ahead-of-deposit lag seen elsewhere in this issue. The title commits to community- and outcome-specific weather-and-air health risk indices built with explainable machine learning across 66 Australian communities. The pollutant set, the health outcomes, the index construction and its skill are all unrecoverable and none is asserted here. Outcome-specific and community-specific index construction is the methodological direction that would replace one-size AQI banding, and 66 communities is enough to test whether the community-specific part earns its complexity; flagged for retrieval.

doi: 10.1016/j.envpol.2026.129117

Bao et al. 2026 J Environ Manage*Distribution-sensitive assessment of air-quality and health benefits from clean-heating pilots in northern China*

88 clean-heating pilot cities, 2013–23, evaluated with machine-learning weather normalisation and the augmented synthetic control method against city-specific no-policy counterfactuals. The distributional result is the point: the policy cut the heating-season mean only 3.01 $\mu\text{g}/\text{m}^3$ but P95 by 5.76, P99 by 5.96 and exceedance days by 7.92 per season — the tail moved roughly twice as far as the mean, which a mean-only evaluation misses entirely. Health impact assessment gave 12,454 fewer attributable all-cause deaths (95% CI 8,564–16,222), with 88.9% from the 2017–18 rollout cohorts and 56.2% from the top decile of cities; high-pollution days were 27.8% of improvement days but 48.8% of the mortality difference. Benefit is therefore concentrated in both time and space. ASCM counterfactuals depend on donor-pool comparability during a decade of nationwide decline, and the mortality figure is a transfer through an exposure-response function, not an observed death count.

doi: 10.1016/j.jenvman.2026.130960

Bouchriti et al. 2026 Environ Monit Assess*Spatial analysis of the burden of lung cancer attributable to long-term PM(2.5) exposure in Morocco: a national assessment using the WHO AirQ+ model.*

WHO AirQ+ applied to 2019 city-specific PM_{2.5} for 46 Moroccan urban and rural areas. PM_{2.5} ranged from 7.1 $\mu\text{g}/\text{m}^3$ in coastal El Jadida (RR 1.02; attributable proportion 2.35%) to 28.4 $\mu\text{g}/\text{m}^3$ in Errachidia (RR 1.30, 95% CI 1.17–1.42; AP 23.29%); Casablanca's moderate RR of 1.06 still yielded 24 attributable cases. The RRs are the model's concentration-response function evaluated at each city's mean, not observed associations, and a single year of concentration data - sources not specified, and Morocco's monitoring is sparse - drives everything. Southern desert areas likely carry a dust-dominated PM_{2.5} whose lung-cancer potency differs from the combustion-derived PM behind the CRF. Relevance: the uncertainty sits almost entirely in the exposure input, where a sensor network would change the answer most.

doi: 10.1007/s10661-026-15942-6

Busskamp et al. 2026 Atmos Environ*A virtual limit value for ultrafine particles: Frequency of threshold exceedances across Germany based on data from the German Ultrafine Aerosol Network*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 29 Sep with no abstract; PubMed fetch, Crossref, Europe PMC and the publisher landing page returned nothing. The title indicates a test of a hypothetical ultrafine-particle limit value against German Ultrafine Aerosol Network data. High relevance: the revised EU Ambient Air Quality Directive requires UFP monitoring without a limit value, and any future threshold will define what number-counting sensors must resolve. Retry first. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.atmosenv.2026.122390

CA wine-grape smoke 2026 (preprint) arXiv (preprint)*Quantifying impacts of wildfire smoke on crop yields in California*

Statistical regression of wine-grape yield on particulate matter alongside climate predictors at three Northern California sites. PM entered as a significant predictor jointly with temperature at two of the three, with final models reaching $R^2 = 0.79$ at Sonoma and 0.66 at Napa — read by the authors as a shift from climate-controlled to wildfire-and-climate-controlled yields. Three sites, yield as an annual aggregate, and no separation of smoke PM from background or from the heat and radiation anomalies that accompany fire weather; with that many candidate predictors, an R^2 of this size is weak evidence of a specific PM pathway. Included as the only record this window treating PM as an economic rather than a health exposure, and because the smoke-attribution problem it dodges is the same one source-resolved monitoring exists to solve. Not peer reviewed.

doi: 10.48550/arXiv.2609.04422

CLHLS exercise 2026 Front Public Health*Exercise, county industrial air pollution and all-cause mortality in older adults*

16,954 CLHLS participants aged 65+, 23 provinces, three waves over six years, Cox models with time-varying covariates and sequential adjustment. Habitual exercise **HR 0.835 (0.773–0.902)**; county industrial SO₂ **HR 1.001 (0.874–1.146)** after adjustment; soot indicators null; and **no significant interaction** between exercise and any of the four pollution indicators. The null is about the exposure metric as much as the hypothesis: annual county soot and SO₂ emission and removal tonnage is an administrative aggregate, not a concentration, and it carries neither PM_{2.5} nor any within-county gradient. Exercise is self-reported and reverse causation from subclinical illness is not addressed. Reported here as a bounded null, not as evidence against interaction.

doi: 10.3389/fpubh.2026.1894835

COVID air-quality lessons review 2026 Preprints.org*COVID-19 Pandemic as an Environmental Health Indicator: Lessons Learned from Air Pollution Changes and Climate Implications*

PREPRINT, single author. Reviews the lockdown natural experiment: short-term outdoor air-quality improvement and greenhouse-gas reduction across many regions, the reported interactions between meteorology, pollutant concentrations and respiratory viral transmission, and the argument that temporary activity reductions demonstrate the ceiling of behavioural mitigation rather than a strategy. No systematic search, no inclusion criteria, no quantitative synthesis, not peer reviewed, and the pollution-COVID-transmission literature it draws on is itself heavily confounded by the timing of interventions. Logged for completeness of the burden-and-policy subtopic; carries no estimate this archive should propagate

doi: 10.20944/preprints202609.0712.v1

Campania review 2026 Monaldi Arch Chest Dis*Air pollution and chronic respiratory diseases: a narrative review including a case study in Campania (Italy).*

Narrative review of mechanism (oxidative stress, chronic inflammation, epithelial damage, airway remodelling) attached to a Campania case study, one of the Italian regions most persistently above national average. Between 2015 and 2024 national PM₁₀ and PM_{2.5} fell slightly while Campania stayed high, with measurable respiratory impact and marked intra-regional disparity that the authors tie to socioeconomic gradient. Closing sections argue for machine learning and satellite-based forecasting to improve monitoring, and for environmental justice targeting. No systematic search, no pooled estimate, and the regional numbers are cited rather than re-analysed. Included for the inequality framing and because the monitoring-gap argument it makes is the case for dense low-cost networks in exactly such regions.

doi: 10.4081/monaldi.2026.4216

Liu et al. 2026 Environ Monit Assess*Spatiotemporal evolution of PM_{2.5} and its health and economic effects in Central China*

Henan, Hubei and Hunan analysed with global Moran's I, centroid migration and BenMAP, with Henan carried forward for valuation over 2017–2023. Spatial clustering is strong and stable (**Moran's I 0.524–0.738, all $p < 0.001$**), high-value clusters sit in Henan and around Wuhan and Changsha, and the concentration centroid moved only **2.86–24.37 km** per year. The authors explicitly *reject* a persistent southwestward migration because most displacements fall below their noise threshold — a welcome negative result in a literature prone to over-reading centroid drift. Attributable premature deaths declined with a late rebound, but a rising value of a statistical life meant monetised benefit grew more slowly than health benefit. BenMAP output inherits its CRF and baseline-rate assumptions entirely.

doi: 10.1007/s10661-026-15911-z

Cheong et al. 2026 J Expo Sci Environ Epidemiol*Community-level air quality and health implications of sustainable aviation fuels adoption: a scoping review*

METADATA ONLY. Metadata-only record: Springer Nature deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes a scoping review of near-airport air quality and health under sustainable aviation fuel adoption. Relevant to ultrafine-particle monitoring around airports. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1038/s41370-026-00971-7

China PM₁ burden projection 2026 Environ Pollut*Ambient PM₁-Related Health Burdens and Mortality Influencing Factors in China: Historical Trends to Future Projections*

METADATA ONLY. No abstract was available on the entry date from any of the four sources queried. Title establishes a China-wide PM₁ attributable-burden analysis running historical trends into projections with a decomposition of mortality drivers. No exposure product, no concentration-response function, no scenario definition and no burden figure are available, so nothing is asserted and the record is tier C. Worth re-reading: PM₁ burden work is scarce because almost no regulatory network measures PM₁, which makes whatever exposure surface this study used the interesting part

doi: 10.1016/j.envpol.2026.129144

China county ageing vs PM_{2.5} 2026 GeoHealth*Association of Environmental Factors and Aging Levels Under Different Urbanization Levels in China*

County-level Chinese data for 2010 and 2020, 22 candidate factors, fixed-effect and random-forest models against the 65+ and 85+ population shares, stratified by high, medium and low urbanisation. Ageing levels rose 1.5–2.1 fold over the decade. PM_{2.5} and resident savings rank among the important variables in low-urbanisation counties; in medium-urbanisation counties CO, SO₂, PM_{2.5} and O₃ all enter; in high-urbanisation counties January and July mean temperature dominate instead. The design cannot support a health claim: ageing share is a demographic composition driven chiefly by migration and fertility, and variable-importance rankings from random forests on two cross-sections are not effect estimates. Carried for the policy-facing exposure gradient, not for causation.

doi: 10.1029/2025gh001642

Covenant of Mayors co-benefit tool 2026 Open Res Eur*A tool to maximize air quality co-benefits while planning local climate actions: application to the Covenant of Mayors initiative.*

JRC tool translating residential-building and transport mitigation actions in city climate action plans into emission changes for five pollutants, applied to 531 plans from 494 European and neighbouring cities. Most greenhouse-gas actions are synergistic with air quality, but two trade-offs are quantified as risks: transport fuel switching can raise ammonia, and promoting solid biomass for residential heating raises primary PM_{2.5} – the failure mode that has repeatedly turned European heating decarbonisation into a local PM problem. The tool maps actions to emission factors and does not run dispersion, so nothing here is a concentration or an exposure; city-reported plans are also self-declared and not verified against realised emissions.

doi: 10.12688/openreseurope.22919.1

Ding et al. 2026 Environ Sci Technol*Which Roads Should Cities Control First? A Health-Risk Framework for Mobile-Source Air Pollution.*

METADATA ONLY. Metadata-only record: ACS deposited title, authors (Ding, Zhang, Li, Chen, Hu) and DOI on 22 Sep without an abstract, and none was recoverable from Crossref, Europe PMC or PubMed. The title promises a prioritisation framework that ranks road segments by population health risk rather than by emissions alone - squarely the question a near-road sensor network is built to answer. Nothing about pollutants, exposure model, city or validation can be stated from the deposit, and inferring a method from a title is not something this watch does. Placed on the retry list; re-screen on its merits when the abstract appears, and check whether the risk weights use measured or modelled concentrations.

doi: 10.1021/acs.est.6c12287

Dong et al. 2026 Nat Hum Behav*Impacts of PM_{2.5} air pollution on high-skilled worker productivity in China.*

Individual scholars' publication counts linked to annual PM_{2.5} at their city, with an instrumental-variable design. Each 1 $\mu\text{g m}^{-3}$ rise in annual PM_{2.5} reduced publications by 0.23% (95% CI -0.28% to -0.17%); scaled nationally, a 1% PM_{2.5} rise costs about 1,053 papers. Effects were larger in heavy-pollution years and among younger, more productive scholars; mechanism analyses point to fewer working hours, more hospital visits and weaker collaboration. The inference rests on the exclusion restriction, which the abstract does not name - common instruments (thermal inversions, upwind emissions) can also shift temperature and mobility. Annual city means are coarse against the indoor office environment where this work actually happens. Relevance: a cognitive-labour damage function in a high-exposure setting, and a case for workplace indoor monitoring rather than ambient assignment.

doi: 10.1038/s41562-026-02576-4

EOTBL cancer 2026 Adv Sci*Global Burden and Temporal Trends of Early-Onset Tracheal, Bronchial, and Lung Cancer: An Analysis of GLOBOCAN 2022 and GBD 2023 Data.*

Early-onset tracheal, bronchial and lung cancer caused 137,705 incident cases and 72,646 deaths globally in 2022, with the highest incidence in Eastern Asia and in High-HDI settings, but disproportionate mortality in Medium- and Low-HDI regions - the familiar detection-versus-survival split. The PM-relevant claim is that ambient particulate pollution emerges as a leading attributable risk in rapidly developing economies such as China, where active smoking dominates in countries like Russia. That attribution is GBD machinery, not new observation: it inherits GBD's exposure surfaces and its concentration-response assumptions wholesale, and no uncertainty interval accompanies the risk-factor ranking here. Registry coverage is also weakest in exactly the Low-HDI regions carrying the mortality burden.

doi: 10.1002/adv.77778

Flue-gas condensable PM removal 2026 J Hazard Mater*Removing the particulate mass that filterable-PM controls never see*

Condensable particulate matter leaves coal-fired and municipal-waste-incineration stacks as vapour and forms particles in the atmosphere downstream, which is why filterable-PM control devices and, in many jurisdictions, the compliance measurement itself miss it entirely. This bench study passes flue gas through a string-of-beads liquid and reports **29.56–65.27% overall CPM removal at 1.17–8.25% of the energy** of conventional condensation. The organic fraction is removed more readily than the inorganic throughout, with the sulfate fraction best at 78.34%. Mechanistically, condensation dominates at high flue-gas temperature and dissolution takes over as the gas cools, and a two-term model combining them reproduces the measured removal. It is bench scale with synthetic conditions — no long-duration fouling, no liquid regeneration economics, no full-scale pressure drop — so the efficiency range is an upper bound on what a retrofit would deliver.

doi: 10.1016/j.jhazmat.2026.143505

Fregni et al. 2026 J Am Heart Assoc*Stroke Burden Attributable to Risk Factors in the Americas, 1990 to 2021: A Temporal Trends Analysis From the GBD Study 2021.*

GBD 2021 stroke deaths and DALYs attributable to 23 modifiable risk factors across 39 countries and territories of the Americas, with joinpoint trends 1990–2021. The 23 factors together accounted for 78.8% of stroke deaths; high systolic pressure led (54.8% of deaths). Household air pollution from solid fuels showed the steepest decline of any factor (deaths AAPC -5.09%, 95% CI -5.17 to -5.02; DALYs -4.92%). Ambient PM_{2.5} is not reported in the abstract. GBD attributable fractions inherit modelled exposure and meta-analytic relative risks; this is a re-cut of existing estimates, not new evidence, and the narrow CIs are regression CIs on modelled series. Relevance: low; background showing HAP falling fastest while ambient PM is not separated.

doi: 10.1161/jaha.125.047114

Habit-locked household emissions 2026 Environ Sci Ecotechnol*Habit-locked-in carbon-air-pollutant emissions dominate China's household footprints.*

Consumption-based accounting decomposed into emissions locked in by behavioural inertia versus those driven by demand growth. Roughly 84% of China's household-driven CO₂, PM_{2.5}, NO_x and SO₂ is habit-locked, rising to about 97% for food and 84–86% for residential expenditure; per-capita habit-locked emissions peak in East and North China, with 32–53% originating in Nei Mongol and Shandong production systems. Under business-as-usual, habit-locked CO₂ peaks near 2030 while habit-locked air pollutants already decline. Pairing carbon-neutrality policy with attenuated consumption inertia yields a further 1285 Mt CO₂, 218 kt PM_{2.5}, 729 kt SO₂ and 126 kt NO_x of abatement. The locked-in share depends entirely on how inertia is parameterised, which the abstract does not expose.

doi: 10.1016/j.ese.2026.100757

Hohhot park PM removal 2026 Environ Pollut*Random-forest attribution of PM reduction by urban park green space*

Thirty-seven parks in Hohhot, summer daytime, reduction rates computed for **six PM size fractions**. Linear, GAM and random-forest models were compared under leave-one-park-out cross-validation and RF won, though the fit is modest and honestly reported — point-level R^2 **0.19–0.58**, park-level **0.18–0.72**. SHAP plus the derivatives of partial-dependence and ICE curves identified nine drivers and, more usefully, the **threshold ranges beyond which the marginal mitigation effect plateaus**: air temperature 22.08–33.26°C, RH 29.71–67.17%, wind speed 0.14–1.12 m/s, distance from centre 2.7–16.79 km, perimeter-area ratio 0.15–0.33, contiguity 0.24–0.48, aggregation index 78.46–86.85%. Summer daytime only, one semi-arid city, and the in-versus-out contrast cannot separate deposition from dispersion and shelter effects. The plateau structure is the transferable result; the specific thresholds are not.

doi: 10.1016/j.envpol.2026.129128

Intergenerational mortality equity 2026 Environ IntMETADATA ONLY — *intergenerational equity in acute and chronic mortality impacts*

No abstract retrievable on the entry date from Crossref per-DOI, Europe PMC or PubMed. What the record establishes: mortality impacts of *both* acute and chronic air pollution exposure have been assessed under an intergenerational health-equity framing — an unusual decomposition, since acute and chronic burden estimates are normally produced by separate literatures with incompatible baselines. What it does not establish: the populations, the pollutant set, whether PM is resolved, the discounting or life-table assumptions that an intergenerational comparison entirely depends on, the geography, or any result. None is asserted here. Flagged for retrieval as the highest-value of the four metadata-only records in this issue.

doi: 10.1016/j.envint.2026.110515

Jin Z. et al. 2026 Environ Sci Technol*Carbonaceous Components Underlying Wildfire Smoke PM_{2.5}-Associated Mortality in Older U.S. Adults*

77.5 million Medicare beneficiaries (2002–2019), annual high-resolution smoke and non-smoke PM_{2.5} and carbonaceous matter, stratified Cox models. Per 1 $\mu\text{g m}^{-3}$: HR 1.032 (95% CI 1.027–1.037) for smoke PM_{2.5} and 1.087 (1.076–1.098) for smoke carbonaceous matter, both above the non-smoke counterparts per unit mass; per IQR the ranking reverses for mass because smoke exposure varies little. Estimated 22,420 deaths/yr attributable to smoke PM_{2.5}, mostly its carbon. Stronger in Black and dual-eligible beneficiaries. Annual means smooth episodic smoke into a chronic metric, and smoke attribution is itself modelled. Relevance: high - composition-resolved smoke exposure is the case for adding absorption/BC channels to LCS networks in fire regions.

doi: 10.1021/acs.est.6c07219

Ju et al. 2026 Lancet Planet Health*Premature mortality and economic losses attributable to landscape fire-related PM_{2.5} in ten countries and territories*

1,366 communities, ten countries and territories, **47,844,851 deaths**, 2000–19, with community-specific associations estimated by instrumental-variable models and pooled by random-effects meta-analysis. Each 1 $\mu\text{g m}^{-3}$ of 2-day mean landscape-fire PM_{2.5} raised mortality **0.60% (95% CI 0.19–1.00)** and mortality-related economic losses **0.59% (0.21–0.97)**, yielding **906,869 deaths and US\$994.5 billion** attributable (attributable fractions 1.90% and 1.64%). The disproportion is the headline: fire PM_{2.5} is **19.0% of ambient mass but 30.6% of PM_{2.5}-attributable deaths**. IV estimates exceeded conventional association estimates by a **median 56.0% (2.2–111.3)**, so classical designs are biased toward the null here. The weak point is that the instrument is not named in the abstract and exclusion-restriction validity — against co-emitted gases and weather — carries the causal claim entirely; value-of-statistical-life transfer across ten economies adds uncertainty that is not propagated into the intervals.

doi: 10.1016/j.lanplh.2026.101506

Kan et al. 2026 Environ Pollut*From Emission-Dominated Mitigation to Meteorology-Modulated Secondary Pollution: Decadal Insights from Beijing's Short-Term Interventions.*

A natural experiment repeated a decade apart: aerosol composition and VOCs during the 2025 Beijing parade controls compared with the 2015 parade. Traffic controls in 2025 cut NO₂ by 41.6% and aromatic VOCs by 14.8%, yet PM_{2.5} rose 20.9%; in 2015 comparable controls took PM_{2.5} from 55.7 to 27.0 $\mu\text{g m}^{-3}$. ML attribution assigned ~90% of the 2015 drop to emission controls and the 2025 rise, led by sulfate and SOA, to warmer, more humid conditions, with biogenic, oxygenated and VCP emissions up. Two short control episodes are anecdotes with good instruments; the ML deweathering is only as good as its training-period representativeness. Relevance: short-term PM interventions evaluated by any network now need co-measured RH and composition to be interpretable.

doi: 10.1016/j.envpol.2026.129240

Kwon & Kim 2026 BMC Public Health*Mortality and morbidity consequences of particulate matter in South Korea: evidence from transboundary air pollution.*

Mortality 2015–2022, morbidity 2016–2019, national. The instrument is the useful part: local PM_{2.5} is endogenous to local economic activity, so the authors instrument it with transboundary transport - wind direction crossed with upwind Chinese monitors - which shifts Korean PM_{2.5} without plausibly acting on Korean health through any other channel. A 1 $\mu\text{g m}^{-3}$ increase raises 65+ mortality by 0.17% and respiratory hospitalisations by 5.1%; acute upper-respiratory visits rise 1.2%. The observed 5.58 $\mu\text{g m}^{-3}$ national decline over 2015–2022 is valued at USD 14.14 billion on avoided life-years alone. Two weaknesses: the exclusion restriction fails if the same synoptic conditions that import Chinese PM independently affect health (cold-air outbreaks, humidity), which the abstract does not address; and the monetisation counts only life-years, so it is a floor, not an estimate. No confidence intervals are reported for the elasticities, so none are plotted. The design is worth borrowing: for any dense sensor network sited in a downwind receptor region, upwind transport is a ready-made instrument for exposure.

doi: 10.1186/s12889-026-29307-8

Li Y et al. 2026 Environ Sci Technol*Multidimensional inequality of CO₂ and air-pollutant footprints in China*

Consumption-based per-capita footprint accounting for CO₂ and air pollutants across 2020–2060 under the dual-carbon policy. The top 10% carry **28.5 t CO₂ and 50.6 kg NO_x** per capita against **1.2 t and 4.7 kg** for the bottom 50%. The structural finding is that **over 90% of footprint inequality is intraprovincial**, not between provinces, and the gap widens to 2060 — which means province-level policy instruments are aimed at the wrong axis. Composition differs by stratum: the top 1% footprint is investment-driven (96% of SO₂), the bottom 50% consumption-driven (88% of NH₃). Input–output footprint accounting carries well-known allocation sensitivity, and the 2060 horizon is a scenario, not a projection.

doi: 10.1021/acs.est.6c04314

Li et al. 2026 Sci Bull (Beijing)*Clean air paradox: exacerbating urban-rural inequality in PM_{2.5} exposure.*

Income-stratified urban and rural PM_{2.5} exposure across mainland China, 2005–2019, combining modelled exposure with field-measured source-specific toxicity profiles. Despite the Clean Air Actions, the urban-rural exposure gap widened by 55.6%; rural residents had 3.2-fold higher exposure in 2019, rising to 15.0-fold once source-specific toxicity was applied, >85.2% of it from residential burning. A 20% residential-emission cut achieved 88.9% inequality mitigation in developed regions, with smaller gains from a 60% cut in low-income areas. The 15-fold figure rests entirely on the toxicity weighting, whose metric (oxidative potential or other) the abstract does not name. Relevance: rural indoor burning is where ambient networks are blind and where household sensors have the largest marginal value.

doi: 10.1016/j.scib.2026.09.015

Li et al. 2026b Environ Res*Contributions of local household fuel burning and regional transport to ambient PM_{2.5} in the Qinghai-Tibet Plateau.*

A residential emission inventory for the Qinghai-Tibet Plateau rebuilt from survey household-energy data for 1960-2019 and run through GEOS-Chem with a normalised marginal attribution. Residential emissions rose to about 1990 then declined with partial clean-energy transition, dominated by inefficient animal-dung combustion except for SO₂, which tracked coal. Residential sources are over 80% of primary PM_{2.5} emissions on the plateau, yet all local sources explain only 17% of ambient PM_{2.5} in Tibet against 46% in Qinghai, with South Asian transport supplying 65% of Tibet's annual ambient PM_{2.5}. Survey-derived inventories over six decades carry large unstated uncertainty and the attribution inherits the model's transport errors. The policy implication is sharp and testable with a sparse plateau sensor network: cleaning local stoves cannot fix Tibetan ambient PM_{2.5}.

doi: 10.1016/j.envres.2026.125734

Liu Z et al. 2026 Environ Sci Technol*Health co-benefits of China's 2035 NDC arrive before the climate benefits do*

Integrated assessment over 2025–2060 puts the NDC's net benefit at **+1,500.5 billion USD**, decomposing into **3,924.3 billion** in long-run climate benefits and **1,659.5 billion** in health co-benefits. The finding is the timing and the distribution, not the total: in the near term the NDC reduces climate benefit, especially in underdeveloped regions, because coal generation balances renewable intermittency and rapid afforestation releases soil organic carbon; long-run benefits reach all regions but stay concentrated in high-income ones. Air-quality health co-benefits are the component that accrues locally and immediately, which is the argument for treating PM monitoring as mitigation-policy infrastructure. Monetised co-benefits inherit every assumption in the concentration–response and valuation chain and should be read as scenario contrast, not as a forecast.

doi: 10.1021/acs.est.6c06024

Livestock ammonia review 2026 Preprints.org*Livestock Ammonia Emissions: Impacts on Air Quality, Animal and Human Health, and Mitigation Strategies*

PREPRINT, single author. Reviews NH₃ formation from urea and uric acid hydrolysis and the physical controls on volatilisation - pH, temperature, moisture, residence time, exposed area, airflow - and insists on the distinction between concentration and mass flux, which is the measurement point that matters for inventories. Mitigation is framed as layered barriers: precision protein nutrition, rapid excreta removal, litter and slurry stabilisation, ventilation with capture, covered storage, low-emission land application; warming is noted to erode mitigation performance. No systematic search strategy, no quantitative synthesis, not peer reviewed. Relevant as agricultural NH₃ is the precursor controlling ammonium nitrate PM_{2.5} across much of Europe and the US Midwest

doi: 10.20944/preprints202609.0590.v1

Lorences Noguera & Cardenas-Duran 2026 QJM*Disentangling Population Ageing from Epidemiological Change in the Rising Burden of PM_{2.5}-Attributable Non-Communicable Disease*

METADATA ONLY. Metadata-only record: Oxford University Press deposited title, authors and DOI on 28 Sep with no abstract; PubMed efetch, Crossref, Europe PMC and the publisher landing page returned nothing. The title indicates a decomposition of the rising PM_{2.5}-attributable NCD burden into population ageing versus epidemiological change; an author response to correspondence was deposited the same day. The ageing-versus-exposure split is the same point the 25 Sep Wan et al. ES&T projection made for China. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1093/qjmed/hcag257

Ma et al. 2026 Environ Sci Technol*Afforestation and Emission Cuts: Dual Drivers for China's Clean Air and Carbon Neutrality*

Emission inventories plus a chemical transport model test maximum-potential afforestation against deep anthropogenic cuts. Afforestation raises summer BVOC emissions by 3.20-3.82 Tg (20-24%), adding 0.39 μg m⁻³ PM_{2.5} (range 0.02-3.47) and 0.74 ppbv MDA8 O₃ in eastern China under current emissions; under carbon-neutral emissions SOA-related PM_{2.5} from afforestation falls 67% and the O₃ response turns slightly negative. Species choice alone can cut local O₃ impact to under a third. BVOC emission factors and SOA yields are the dominant, unquantified uncertainty. Relevance: low-moderate - the PM_{2.5} penalty is small relative to sensor error.

doi: 10.1021/acs.est.6c04722

Mairghani et al. 2026 Rev Environ Health*An umbrella review of public health interventions to mitigate air pollution in high-income countries*

Eight systematic reviews spanning city and transport policy, household exposure reduction and emission-reduction modelling. Transport interventions gave consistent cardiovascular benefit and fewer traffic injuries; portable air cleaners cut indoor PM_{2.5} with modest blood-pressure improvement; modelling projected a **median ~1.5% mortality reduction** under large-scale mitigation. The ordering is the useful output — structural policy beat individual and household measures, which were complementary rather than substitutive. Two limits constrain its use. **Equity was rarely assessed and inconsistent where it was**, so the review cannot say who benefits; and the 1.5% figure is modelled rather than measured, mixing evidence tiers inside one summary. Eight reviews, all high-income, so none of it transfers to the settings carrying the largest burden.

doi: 10.1515/reveh-2026-0061

Manchanda et al. 2026 Environ Sci Technol*Inverse Modeling Identifies Efficient Emission Control Strategy for Mitigating PM_{2.5} Pollution*

Recasts attainment planning as a constrained Bayesian inverse problem: find the smallest spatially explicit, sector- and precursor-resolved emission changes that bring modelled PM_{2.5} under the 9 μg m⁻³ annual NAAQS across the contiguous US. Strategies with similar total reductions ranged from full compliance to minimal progress depending on where and what was cut; aggregate tonnage does not predict attainment. No numeric results in the abstract; the solution inherits the transport model's linearised source-receptor relationships and inventory error. Relevance: moderate - attainment is judged at monitors, so where the monitors (and dense sensor networks) are determines which 'efficient' strategy appears to work.

doi: 10.1021/acs.est.6c06581

Massimi et al. 2026 Environ Sci Technol*Urban green walls under traffic stress: elemental composition of PM deposition and plant response*

A nine-species green wall at a high-traffic urban site, followed for a year. PCA on the insoluble fraction of Al, Cr, Cs, Fe, Mn, Mo, Rb, Sb, Sn, Ti and Zn identified *Photinia* × *fraseri* as the highest and most stable retainer of non-exhaust traffic particles and *Pteris vittata* as the lowest, with SEM-EDS showing **up to 100-fold higher retention** in the former. The methodological finding travels furthest: **only insoluble elements traced deposition reliably**, because soluble and whole-leaf fractions are dominated by endogenous pools of nutrients and of tracers such as Cs and Rb, so conventional whole-leaf digestion overestimates deposition. *Photinia* showed marked seasonal physiological and transcriptional modulation; *Pteris* did not. One wall at one site with no paired ambient PM mass measurement, so species are ranked against each other but no removal flux is quantified and no air-quality benefit can be inferred.

doi: 10.1021/acs.est.6c10086

Mundu et al. 2026 J Air Waste Manag Assoc*Fossil fuel use, urban air pollution and health in rapidly growing African cities: Implications for sustainable and low-emission urban development.*

PRISMA-reported critical review and policy synthesis on fossil-fuel sources, urban air quality, health impacts and governance in fast-growing African cities. Diesel road transport, backup generators and household energy emerge as the dominant PM, NO_x, black carbon and VOC sources; observed levels often exceed WHO guidelines; monitoring networks are sparse and unevenly distributed. No pooled estimates, no quality appraisal of the included studies is described in the abstract, and the recommendations (electrification, clean cooking, mini-grids) are generic. Its value is as a map, not as evidence. The monitoring-gap finding is the direct link to this watch: generator-dominated diurnal profiles and diesel BC are two conditions where uncorrected optical sensors are least reliable, so an African LCS rollout needs co-located BC or reference anchors from the start.

doi: 10.1080/10962247.2026.2737847

N fertilizer NH₃-PM_{2.5} 2026 Atmos Res*Integrating nitrogen fertilizer management for mitigating agricultural ammonia emissions and PM_{2.5} pollution.*

METADATA ONLY. Metadata-only record; no abstract deposited or recoverable from any leg. Agricultural ammonia is the control lever with the best cost-per-microgram in most secondary-inorganic-dominated airsheds and the one least often regulated, so a fertilizer-management framing is squarely in scope for the burden and mitigation track. Whether the paper is a modelling assessment, a field trial or a policy review is not determinable from the deposit. Retry list. Note it sits alongside the Athens new-particle-formation record in this same issue - two independent ammonia-centred papers deposited the same day, both illegible, which is a coverage gap rather than a coincidence worth reading into.

doi: 10.1016/j.atmosres.2026.109343

Jiang et al. 2026 Med Gas Res*Ambient particulate matter pollution and stroke burden*

GBD 2021 stroke deaths and DALYs attributable to ambient particulate matter, decomposed by region, sex and SDI over 1990–2021 with EAPC trends and concentration and slope indices of inequality. Absolute deaths rose **107%** and DALYs **101%** while age-standardised rates fell — the familiar decomposition of ageing and population growth against declining risk. Middle-SDI locations carry **46% of the global burden** and exceed high-SDI on both age-standardised measures. The concentration index suggests improving relative inequality while the slope index worsened against 1990, and the paper does not reconcile the two. Attributable burden is a modelled quantity resting on the exposure surface and the chosen CRF; this is not independent epidemiological evidence.

doi: 10.4103/mgr.MEDGASRES-D-25-00188

PPS antifouling filter 2026 Environ Sci Technol*Robust and Self-Cleaning Fluorine-Free Superhydrophobic PPS Filters for High-Humidity Flue Gas Filtration.*

Light-induced covalent grafting of a triple-cross-linked low-surface-energy network onto polyphenylene sulfide fibre membranes produces a fluorine-free antifouling bag-filter medium. Against the unmodified control it extends mean pulse-jet cleaning cycle to 550.84 s, lowers mean residual pressure drop to 280.69 Pa and raises quality factor to 0.044 /Pa under simulated humid dynamic filtration. The fluorine-free part is the policy-relevant part - PTFE-based antifouling treatments are the incumbent and are under PFAS restriction, so a non-fluorinated substitute that survives wet dust addresses a regulatory constraint on stationary-source PM control rather than just an engineering one. Bench-scale, simulated flue gas, no capture-efficiency-versus-size curve reported and no coating-lifetime data beyond the test campaign, so durability - the property the whole design targets - is demonstrated over hours rather than the months a baghouse runs. Tier C and upstream of this watch's usual scope; carried under mitigation because primary PM emission control determines the ambient field every downstream sensor measures.

doi: 10.1021/acs.est.6c08122

Puyang PM_{2.5}-O₃ decade 2026 Preprints.org*A decade of diverging PM_{2.5} and ozone in a Chinese oilfield city*

Preprint, not peer reviewed. Regulatory monitoring records for Puyang, one of the “2+26” channel cities, 2015–2025. **PM_{2.5} fell 79.6 to 39.3 μg m⁻³ while MDA8 ozone rose 123.0 to 189.6** — the canonical Chinese divergence, here in a city whose oil and gas VOC inventory makes it an unusually clean test of it. A positive PM_{2.5}–O₃ correlation appeared only in the compound-pollution corner (O₃ > 160, PM_{2.5} < 35), with seasonal *r* of 0.38 winter, 0.37 summer, 0.26 autumn, 0.21 spring — all weak. Ozone production efficiency declined over the record, correlated with PM_{2.5} at *R* = 0.46–0.65, and stayed above 7 monthly, which the authors read as NO_x-limited. OPE inferred from ambient monitoring rather than measured, no VOC speciation reported, and a single city; the control recommendation rests on that inference.

doi: 10.20944/preprints202609.0349.v1

Shi et al. 2026 Environ Sci Technol*Carbon-air pollution cobenefits and emission leakage in China's steel industry*

Unit-level emission database covering **2,866 enterprises and ~4,030 units**, relating regional energy-industrial configuration to mitigation priorities. Steel CO₂ peaked at 1.99 Gt in 2019 while SO₂ and PM fell **more than 80%** over 2011–2022 — the decoupling this watch keeps seeing. Over 2023–2050 the priorities diverge: production restraint gives the largest cumulative CO₂ reduction (22.1%) while **obsolete-capacity retirement dominates air-pollutant reduction (22.3%)**, so a carbon-optimal pathway is not a PM-optimal one. Regionally differentiated pathways induce **spatial emission leakage**, stronger for CO₂ (17.39% in 2030, falling to 2.38% by 2050) than for pollutants, and national mean pollutant concentrations fall only 5.4–12.3% by 2030 relative to 2022. Sector-specific and inventory-based.

doi: 10.1021/acs.est.6c08461

Simoes et al. 2026 J Xenobiot

Air Pollution, Temperature, and Demographic Vulnerability as Determinants of Cause-Specific Mortality in Portugal.

Nationwide monthly data for Portugal, 2010-2022, with GEE models of malignant, circulatory and respiratory mortality against NO₂, PM_{2.5} and temperature, exposures coded as concentration above WHO 2021 guideline values. The signal is NO₂, not PM: respiratory mortality RR 1.014 (95% CI 1.009-1.019) per 1 µg m⁻³ NO₂ above guideline, circulatory 1.005; PM_{2.5} appears mainly as a modifier attenuating the NO₂ association with tumour mortality, and pollutant findings were not robust in sensitivity analyses. A reported RR of 142.6 for males 65+ is a predicted risk ratio against a reference group, not an exposure effect. Monthly national aggregation is ecological and thresholding at guidelines discards most variance. Relevance: minimal; logged for the PM_{2.5}-modifies-NO₂ finding, which would need personal-scale data to test.

doi: 10.3390/jox16050172

TS-AQHI two-city validation 2026 Front Public Health

Development and two-city validation of a Temperature-Sensitive Air Quality Health Index incorporating synergistic effects of air pollution and thermal stress

Weighted quantile sum regression supplies pollutant weights to a GAM carrying temperature, humidity and two-day-lagged multiplicative interactions; validated against daily mortality in Ningbo and London, 2014-2017. For high-risk categories the temperature-sensitive index gave excess risk 22.9 (95% CI 21.1-24.7) vs 15.0 (13.4-16.7) for conventional AQHI in Ningbo, and 22.2 (20.7-23.8) vs 7.7 (6.3-9.0) in London. Limitation: index weights are estimated and validated on the same 1,461-day series in each city with no out-of-sample or out-of-period hold-out, so the margin over the benchmark is an in-sample fit advantage until someone tests it forward. Two cities on different continents is nonetheless a real generalisation test of the framework

doi: 10.3389/fpubh.2026.1806127

Turkatte et al. 2026 Am J Public Health

Implementing Community-Centered Air Quality Law: Success Stories From California's Assembly Bill 617.

Mixed-methods comparison of community emissions reduction programmes (CERPs) in nine California AB 617 communities: over 800 strategies coded into 25 groups and rated on capacity for health protection, adoption pathway and ease of implementation, with funding expenditure as the progress proxy. Success tracked creative programme development, interagency coordination and community capacity. Expenditure is a process measure, not an air-quality or health outcome, and 'success stories' framing invites selection. Relevance: AB 617 is the largest statutory deployment of community air monitoring in the US, yet the abstract evaluates strategies without any measured concentration change - the monitoring data that could close that loop exist and are not used here.

doi: 10.2105/ajph.2026.308696

Türkiye stroke GBD 2000-2023 2026 Neuroepidemiology

Ambient PM is the leading environmental risk for stroke death in Türkiye

GBD 2023 secondary analysis for Türkiye. Age-standardised stroke incidence fell from 141.47 to 103.05 per 100,000 between 2000 and 2023, deaths from 132.61 to 57.03, and DALYs from 2696.30 to 1073.51 — a genuine and large decline. In 2023, metabolic risks accounted for 38.24% of stroke deaths (high blood pressure alone 21.29%) and smoking 12.20% among behavioural risks; **ambient particulate matter led the environmental risks at 11.93%**, placing it within a percentage point of tobacco. Macroeconomic regressions gave mixed and partly counterintuitive results — universal health coverage service capacity correlated positively with death rates — which is a familiar ecological artefact of ascertainment improving with health-system capacity. The deeper limitation for this watch: GBD's PM exposure surface is itself a model, so the 11.93% is an attribution built on modelled exposure, not an independent measurement.

doi: 10.1159/med/adwag030

US net-zero multi-model 2026 Environ Sci Technol

A Multi-Model Assessment of the Air Quality and Health Impacts of U.S. Energy Decarbonization.

Three independent US energy system models (US-REGEN, GCAM-USA, REPEAT) run to 2050 agree on direction but not magnitude: PM_{2.5}-attributable mortality rates fall 28-73% against 2017 under net-zero, despite population growth and ageing. The spread is the result. It tracks how each model allocates CO₂ mitigation across sectors, with carbon dioxide removal deployment in 2050 varying threefold and setting how much residual fossil combustion - and therefore precursor emission - remains. A sensitivity analysis shows plausible increases in non-energy emissions, wildfire in particular, can offset the energy-sector benefit outright. Concentration-response functions are shared across the three models, so the reported spread understates total uncertainty: model structure varies, epidemiological input does not.

doi: 10.1021/acs.est.6c02914

Wan et al. 2026 Environ Sci Technol

Three-Phase Transition in China's Future Air Quality under Ambitious Mitigation Pathways Revealed by Large-Ensemble AI Projections.

A hybrid deep-learning surrogate (hyDL) of a chemical transport model cut compute by >99% while reproducing meteorology and emission sensitivities (R > 0.9), enabling daily ~50 km projections for China 2015-2060 under six emission pathways, each driven by 7-11 climate models. Megacities showed an exponential O₃-PM_{2.5} relationship implying three phases under ambitious mitigation: PM_{2.5} falls while O₃ persists (current), co-improvement, then O₃ gains with marginal PM_{2.5} returns. With population ageing, attributable mortality stays flat to ~2035 even on the cleanest pathway. A surrogate trained on the historical emission space is extrapolating when pathways leave it, and R > 0.9 on sensitivities is not validation of 2060 chemistry. Relevance: the surrogate approach is how sensor-network data could be assimilated at useful cost.

doi: 10.1021/acs.est.6c04494

Western Pacific stroke burden 2026 JACC Asia

Stroke Burden, Trends, and Inequalities in the Western Pacific Region, 1990-2021: A Population-Based Study.

GBD 2021 across 31 Western Pacific countries and areas: 34.56 million prevalent stroke cases (95% UI 32.11-37.21) and 64.53 million DALYs in 2021. Age-standardised DALY rates roughly halved from 4,155.11 to 2,326.92 per 100,000 while absolute DALYs rose, driven by ageing (219.51%) and population growth (70.27%); lower-SDI countries carried the larger absolute inequality. Environmental exposures accounted for 43.78% of stroke DALYs with ambient particulate matter the single largest contributor at 22.22%. That fraction is a modelled comparative-risk-assessment output resting on exposure-response functions imported from cohorts largely outside the region, and GBD uncertainty intervals do not propagate the uncertainty in those functions, so 22.22% should be read as an order of magnitude rather than an estimate.

doi: 10.1016/j.jacasi.2026.07.034

Yalali et al. 2026 Front Public Health

Geospatial modeling of environmental and socioeconomic drivers of health outcomes.

Census-tract prevalence of seven chronic conditions (hypertension, asthma, COPD, stroke, cancer, diabetes, mental distress) in Harris County, Texas, modelled with OLS, GWR and multiscale GWR on 2022 data and re-estimated for 2020-2024. MGWR raised adjusted R² (e.g. 0.57 to 0.75 for hypertension) and poverty and PM_{2.5} were the most consistent predictors, strongest near the eastern industrial corridors; patterns were stable across years (|dR²| ≤ 0.09). Outcomes are modelled small-area estimates (likely CDC PLACES), so the regression partly recovers the covariates used to make them; no effect size for PM_{2.5} is given. Ecological. Relevance: low; locates tracts where added sensors would coincide with high modelled burden.

doi: 10.3389/fpubh.2026.1790559

Yang et al. 2026 Thorax

PM_{2.5}, O₃ and survival gains from China's clean air actions among patients with lung cancer: a 20-year registry-based cohort study in Beijing.

172,845 lung cancer patients diagnosed 2003-2022 in the Beijing Cancer Registry, with three-year pre-diagnosis exposure from ChinaHighAir-Pollutants. Per 10 $\mu\text{g m}^{-3}$, PM_{2.5} gave HR 1.111 (95% CI 1.083-1.140) and O₃ HR 1.117 (1.070-1.166) for all-cause mortality, with a significant multiplicative interaction concentrated above roughly 105 $\mu\text{g m}^{-3}$ of O₃. Counterfactual simulation against 2013 levels attributed 13,229 avoided deaths over 2016-2022 and a 7.8-month gain in five-year extended expected lifespan - of which rising O₃ and the interaction offset about 40%. Exposure is a reanalysis product at residence at diagnosis with no mobility and no treatment covariates, and the survival gains are simulated, not observed. The interaction is the operational point: a PM_{2.5}-only network cannot rank policy benefit in an ozone-rising regime.

doi: 10.1136/thorax-2026-225199

Zhao N et al. 2026 J Hazard Mater

Attainment pressure under China's revised GB 3095-2026 standard in Shandong

Daily observations of six criteria pollutants across 16 Shandong cities, 2015–2025, condensed into a relative attainment pressure index combining concentration gap and additional non-attainment days, with random forest, SHAP and monotonicity-constrained XGBoost on socioeconomic predictors. PM_{2.5}, PM₁₀, SO₂, NO₂ and CO fell 41.0–79.6%; O₃ did not move, staying at 166–191 $\mu\text{g m}^{-3}$. Under the transition-stage limits a western high-pressure belt emerges — Heze 98.7, Dezhou 91.3, Liaocheng 89.1. Civilian vehicle ownership and secondary-industry share are the leading predictors for particulates and NO₂; the socioeconomic set explains PM_{2.5} far better than O₃. Even under the deep-transition pathway eight of sixteen cities remain above 30 $\mu\text{g m}^{-3}$ in 2030. One province, and the scenario arithmetic assumes the historical predictor–concentration relation holds forward.

doi: 10.1016/j.jhazmat.2026.143460

Other clinical endpoints

27 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (27 OF 27 RECORDS IN THIS SUBTOPIC)

12M student visual impairment 2026 Environ Int

Associations of long-term exposure to PM_{2.5} and its chemical constituents with incident visual impairment risk: A cohort study of 12 million school students

METADATA ONLY. No abstract was available on the entry date from Crossref per-DOI, Europe PMC, PubMed efetch or Semantic Scholar - the same same-day Elsevier deposit lag logged repeatedly since 2026-09-08. Title, journal and cohort scale establish that this is a constituent-resolved long-term PM_{2.5} cohort with an incident ocular endpoint in a 12-million-student population, which would make it the largest PM-vision cohort in the archive. No effect estimate, no exposure model description and no confounding structure are available, so nothing is asserted here and the record is tier C. Flag for re-reading once the abstract deposits

doi: 10.1016/j.envint.2026.110513

Aerosol Air Qual Res 2026 Aerosol Air Qual Res

Association Between Air Pollution and Psoriasis-Related Outcomes: A Systematic Review and Meta-analysis.

METADATA ONLY. Springer deposit of 19 September without an abstract; not yet indexed by Semantic Scholar. Carried on title and venue only. No pooled estimate, study count, heterogeneity statistic or pollutant metric is asserted, because none was deposited. Noted as the fourth systematic review or meta-analysis in a single day's window - alongside the venous thrombosis, climate-burden and ischaemic stroke reviews screened out or admitted above - which is itself a signal worth watching in the weekly rollup. Held for pickup.

doi: 10.1007/s44408-026-00166-7

Alhalwani et al. 2025 Saudi J Ophthalmol

Correlations of CO, SO₂, PM₁₀, NO₂, and O₃ with dry eye disease in Saudi Arabia.

492 dry eye disease diagnoses at King Abdulaziz Medical City, Jeddah, 2017-18, against three monitoring stations. Adjusted ORs reached significance only for SO₂ (1.182) and CO (1.162); PM₁₀, the particulate metric, was not associated. The reported SO₂ interval is 0.908-1.111 with P=0.021, which is internally inconsistent - an interval spanning unity cannot accompany that P value - and that error is in the published abstract, so the estimate should not be used. Three monitors for a city of four million gives essentially no exposure contrast within Jeddah, and a clinic case series with no non-DED comparison group cannot support an odds ratio at all as described. Carried at tier C as a record of the ocular-surface literature only; the finding that PM₁₀ was null here should not be read as evidence against a particulate effect.

doi: 10.4103/sjopt.sjopt_286_24

Ali et al. 2026 Toxics

The Impact of Polycyclic Aromatic Hydrocarbons Exposure from Seasonal Biomass Burning on Oxidative Stress and Genotoxicity Biomarkers in Patients with Diabetes and Hypertension.

152 adults in Chiang Mai (88 healthy, 35 diabetes, 15 hypertension, 14 both) sampled before (December) and after (May) the burning season. Urinary 1-OHP rose from 0.21 +/- 0.50 to 0.87 +/- 1.25 $\mu\text{mol/mol}$ creatinine and serum 8-OHdG from 51.04 +/- 24.90 to 66.08 +/- 27.82 pg/mL (both $p<0.001$), with the rise significant in diabetic and hypertensive groups; MDA did not change overall (6.09 to 6.00 nmol/mL). No ambient PM was measured, so season is the exposure and every other seasonal change is confounded; subgroups of 14-35 are small. Relevance: a reusable design - adding low-cost PM_{2.5} at participants' homes would turn the pre/post contrast into a dose-response.

doi: 10.3390/toxics14090771

Batisse et al. 2026 Environ Epidemiol

Prenatal ultrafine particles and childhood cancer in 965,000 children

Cohort of over 965,000 children born in Montreal or Toronto 1999–2020, with high-resolution outdoor UFP number concentration and mean UFP size assigned to six-digit residential postal codes from annual tax records, followed to age 14 (mean 9.1 years) against the Canadian Cancer Registry. 1,625 incident cancers. Per 10,000 particles/cm³ of prenatal exposure: acute lymphoblastic leukaemia HR 1.12 (0.94–1.34), neuroblastoma 1.15 (0.85–1.57); both intervals include the null, both persisted across exposure windows and sensitivity analyses, and other sites were null. Adjusting for UFP size as well as number is the methodological advance — number and size are correlated and separately toxicologically relevant. The honest summary is a consistent, imprecise positive signal, not a demonstrated association.

doi: 10.1097/ee9.0000000000000528

Chen K et al. 2026 Sichuan Da Xue Xue Bao Yi Xue Ban

PM_{2.5} components and incident frailty in CHARLS

CHARLS 2011–2020, $n=9,603$ middle-aged and older adults, frailty index outcome, exposures from the Tracking Air Pollution product, time-dependent Cox models. Per IQR of the preceding year: PM_{2.5} HR **1.100 (1.015–1.193)**; by component, organic matter **1.142 (1.055–1.236)**, black carbon **1.131 (1.041–1.229)**, ammonium **1.098 (1.010–1.194)**, sulfate 1.088 (0.996–1.187) and nitrate 1.081 (0.996–1.172). Carbonaceous components rank above bulk mass and above the secondary inorganic ions, and the two that cross the null are exactly the two secondary ions — an ordering consistent with combustion-source toxicity rather than with mass. The components co-vary strongly in the same exposure surface, so the ranking is suggestive and the six estimates are not six independent tests.

doi: 10.12182/20260760103

Ding E et al. 2026 Environ Sci Technol

Compound climate extremes and influenza, with a submultiplicative interaction

630,120 laboratory-confirmed influenza cases from hospitals in **335 Chinese cities**, 2005–2019, analysed by case-crossover with distributed lag non-linear models, machine-learning variable ranking, and an LSTM forecasting proof of concept for Beijing. Temperature, humidity and particulate matter rank as the primary contributors. The result worth keeping: co-exposure to multiple extreme events carries higher joint risk than single exposure, but the interaction is **submultiplicative on the multiplicative scale with no evidence of additive interaction** — so the joint effect is less than the product and the naive stacking of extreme-event risks overstates it. Adding extreme-event terms improved daily forecasting in the Beijing model. Caveats: the abstract reports no effect sizes at all, hospital-based case ascertainment varies across 335 cities and 15 years, and one-city forecasting is a demonstration, not a validation.

doi: 10.1021/acs.est.6c04755

Gyeltshen et al. 2026 Int J Cancer

Short-Term PM_{2.5} Exposure and Daily Cancer Mortality Across 47 Japanese Prefectures: A Nationwide Time-Stratified Case-Crossover Study, 2013–2022

Nationwide case-crossover, all 47 prefectures 2013–2022, prefecture-specific conditional quasi-Poisson pooled by random effects. Per 10 $\mu\text{g m}^{-3}$ PM_{2.5} at lag 0–2, all-cancer mortality rose 0.88% (95% CI 0.57–1.18), with breast 2.06%, colorectal 1.06%, stomach 0.81% and lung 0.77%, and no effect modification by sex, age or site. The association faded beyond lag 0–5 and lost significance after adjustment for NO₂ or SO₂ - the decisive weakness: an acute cancer-death effect is most plausibly harvesting of frail terminal patients via cardiorespiratory pathways, and co-pollutant sensitivity leaves PM specificity open. Prefecture-mean exposure. Relevance: moderate - prefecture-scale exposure is coarse; finer networks would test the co-pollutant question.

doi: 10.1002/ijc.70761

Hong et al. 2026 Front Public Health

Associations of temperature, particulate matter, and greenness with human papillomavirus positivity at screening: an analysis of Guangdong Province (2016–2022) using the treed distributed lag nonlinear model.

Screening records for 1,287,412 women in Guangdong, 2016–2022, analysed with treed distributed-lag nonlinear models against biweekly temperature, PM_{2.5} and NDVI. High PM_{2.5} (97.5th percentile, 38.0 $\mu\text{g m}^{-3}$ vs median 20.3) was associated with higher odds of HPV positivity over lag weeks 1–16 (cumulative OR 1.172, 95% CrI 1.051–1.283); low temperature and low NDVI also carried associations. The authors report that temporal sensitivity analyses attenuated the PM_{2.5} association. HPV positivity reflects acquisition months to years earlier and screening attendance is seasonal, so a weeks-scale lag structure is biologically implausible and likely confounded by attendance timing. Relevance: low; included as a caution on DLNM results with outcomes whose latency does not match the lag window.

doi: 10.3389/fpubh.2026.1842611

Huang et al. 2026 Medicine (Baltimore)

Secondhand smoke exposure in never-smokers: A cross-sectional and prospective cohort study of chronic kidney disease prevalence and all-cause mortality in affected patients.

13,276 never-smoking adults, NHANES 1999–2020, with 1,630 CKD cases followed for mortality. Moderate SHS exposure was associated with CKD prevalence (OR 1.37, 95% CI 1.07–1.76). Among those with CKD, moderate and heavy exposure carried all-cause mortality HRs of 2.03 (1.23–3.36) and 1.80 (1.11–2.93); each doubling of cotinine gave HR 1.04 (1.01–1.08). The non-monotonic ordering - moderate exceeding heavy for mortality, and heavy not associated with prevalence - undercuts a dose-response reading and points to residual confounding by household socioeconomic position. Cotinine is a good internal dose marker for the nicotine fraction and a poor one for the particulate fraction, which is the exposure of interest here; the two dissociate across ventilation conditions in a way an indoor PM sensor would resolve and a serum assay cannot.

doi: 10.1097/md.00000000000050614

Huang et al. 2026 Int J Epidemiol

Short-term health impacts of air pollution in Canada: a nonlinear multi-pollutant Bayesian case-crossover framework.

Cause-specific daily mortality 2001–2015 and hospitalisation 2001–2018 across 20 Census Divisions covering over half of Canada's population. A 10-unit contrast in the combined PM_{2.5}+NO₂ metric gave +1.5% non-accidental mortality (95% CrI 0.3 to 2.2), +0.6% hospitalisation (0.0 to 1.3) and +2.7% senior respiratory hospitalisation (0.9 to 4.5). Several PM_{2.5} and NO₂ curves were nonlinear; O₃ was near-linear. The methodological gain is real: exposure uncertainty from imputation is propagated rather than ignored, and CD-level curves are partially pooled. The threat is interpretive: a summed PM_{2.5}+NO₂ index has no single-pollutant counterpart, so the coefficient cannot be dropped into a PM_{2.5}-only burden model, and the CrIs for cause- and age-specific strata are wide. For sensing: imputed monitor gaps are treated as an exposure-error source, which is exactly where dense low-cost networks would shrink the posterior.

doi: 10.1093/ije/dyag194

Huang et al. 2026b Front Pediatr

The role and management of dietary and environmental factors in pediatric atopic dermatitis.

Stage 1: 535 paediatric atopic-dermatitis cases diagnosed 2022–2024 and 535 age- and sex-matched controls, conditional logistic regression. PM_{2.5} above 35 $\mu\text{g m}^{-3}$ carried an adjusted OR of 2.48 (95% CI 1.80 to 3.50), alongside early milk/egg sensitisation (3.18), winter indoor humidity below 40% (1.92) and *S. aureus* colonisation above 50% (4.08). Stage 2 was a 66-child non-randomised intervention comparison on diet and home care, with no air-quality component. The PM estimate is the weakest element: the exposure source, spatial resolution and averaging window are not given, a single dichotomy at 35 $\mu\text{g m}^{-3}$ discards the gradient, and matched hospital controls invite selection on healthcare use. Treat the OR as hypothesis-grade. Relevance to sensing is indirect - indoor humidity and PM_{2.5} are measured by the same low-cost units, so a home-sensor design could separate them properly.

doi: 10.3389/fped.2026.1862285

Kadoorie PM_{2.5} fracture/BMD 2026 J Hazard Mater*Associations of long-term exposure to ambient air pollution with fracture risk and bone mineral density: A cohort study of 0.5 million Chinese adults*

490,172 China Kadoorie Biobank participants, 15,614 incident fractures over a median 12.0 years, with calcaneal quantitative-ultrasound BMD in 32,545. PM_{2.5} gave HR 1.09 (95% CI 1.05-1.13) per 10 $\mu\text{g m}^{-3}$ and warm-season O₃ 1.08 (1.04-1.12); a one-unit pollutant principal-component increment gave 1.08 (1.03-1.13), with PM_{2.5} contributing 70% by quantile g-computation. BMD fell 0.26-0.31 SD per PC increment and mediated only 5.8-8.6% of the fracture association, so the dominant pathway is not bone density. Exposure is assigned from the recruitment clinic's community, not the residence, and is fixed in space - a large cohort cannot repair ecological exposure assignment, and the 1 km product has no local validation.

doi: 10.1016/j.jhazmat.2026.143551

Kelishadi et al. 2026 Int J Prev Med*The Effect of Synbiotic Supplementation on Malondialdehyde Levels in Adolescents Exposed to Air Pollution: A Randomized, Single-Blind, Controlled Clinical Trial.*

Seventy healthy 10-18-year-olds randomised, single-blind, to synbiotics (n = 38) or omega-3 (n = 32) for 8 weeks; outcome serum malondialdehyde. MDA fell within the synbiotic arm (148.8 to 51.5), but the between-arm difference was null (P = 0.75); only after covariate adjustment did synbiotic use associate with MDA reduction. That is the result that matters, and it is null on the pre-specified contrast. The comparator is an active antioxidant, so the trial cannot test synbiotics against nothing; there is no placebo, no measured exposure and the MDA standard deviations exceed the post-treatment mean. It cannot inform a PM dose-response. Carried because nutritional mitigation of PM oxidative stress keeps recurring in the trial registry; an exposure-stratified design using personal sensors is what this line of work lacks.

doi: 10.4103/ijpvm.ijpvm_246_25

Kim et al. 2026 J Am Geriatr Soc*Extreme Temperatures, Elevated PM_{2.5}, and Acute Care Events Among Dual-Eligible Nursing Facility Residents.*

24,784,081 facility-days from 24,760 nursing facilities in 47 states plus DC, Medicare-Medicaid linked claims 2017-20. On elevated-PM_{2.5} days ED visits rose 0.020 (95% CI 0.013-0.027) and hospitalisations 0.017 (0.012-0.022) per 1,000 resident-days; extreme heat gave 0.022 (0.003-0.041) and 0.025 (0.011-0.039), extreme cold 0.014 (0.001-0.027) hospitalisations. Effect sizes are tiny in absolute terms and the intervals for heat and cold nearly touch zero; what carries the paper is 24.8 million facility-days and a fixed-effects design that removes time-invariant facility confounding. The exposure is the weak link - an outdoor county-level value dichotomised at 9 $\mu\text{g m}^{-3}$, applied to a population that is almost entirely indoors, with no measurement of the HVAC infiltration the authors themselves identify as the mechanism. The policy asymmetry is the point: nursing facilities are regulated on indoor temperature and not at all on indoor PM.

doi: 10.1111/jgs.70720

King 2026 Br J Anaesth*Outdoor air pollution as a potentially modifiable perioperative risk factor.*

METADATA ONLY. Metadata-only record: PubMed/Elsevier deposited title, authors and DOI on 25 Sep with no abstract; Crossref, PubMed efetch and the publisher landing page returned nothing, and Europe PMC was down (HTTP 503) for the whole run. The title indicates a single-author piece framing ambient air pollution as a modifiable perioperative risk factor. Likely editorial; low priority unless it reports primary data. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.bja.2026.08.047

Leonforte et al. 2026 Toxics*Environmental Contaminants and Osteoporosis-Related Outcomes: A Systematic Review and Meta-Analysis.*

PRISMA review (PubMed, Scopus, WoS; 2016-Apr 2026; PROSPERO CRD420261460078) of 48 studies and 4,510,868 participants on environmental contaminants and BMD/osteoporosis, REML random-effects pooling where ≥ 3 studies existed. Air pollutants were the most consistent class. PM_{2.5} was associated with osteoporosis (OR 1.24, 95% CI 1.02-1.51) and PM₁₀ more weakly (OR 1.17, 0.96-1.42). The authors report extreme heterogeneity and sensitivity to influential studies; the abstract gives no exposure increment, so the ORs pool incommensurate contrasts. Relevance: low for sensing; a musculoskeletal endpoint that is starting to accumulate PM evidence.

doi: 10.3390/toxics14090790

Liu et al. 2026 Zhonghua Liu Xing Bing Xue Za Zhi*Association of exposure to fine particulate matter and its components with frailty in middle-aged and older adults in Beijing-Tianjin-Hebei region and surrounding areas.*

1,875 adults across ten cities, mean age 64.95 (SD 13.42), 49.4% men, deficit-accumulation frailty index mean 0.15 (SD 0.09). An IQR rise in 90-day PM_{2.5} was associated with a 1.30% higher frailty index (95% CI 0.29-2.31%), strongest at lag0-90. Components: OC 3.63% (1.17-6.09), Cu 1.50% (0.76-2.24), Ca 1.35% (0.51-2.20), EC 0.83% (0.45-1.21) per IQR. Two problems. Exposure is assigned from the nearest superstation, so within-city gradient is entirely unmodelled and the IQR is a between-city contrast wearing individual-level clothing - precisely the misassignment a dense network exists to fix. And the components are collinear combustion tracers; the OC-over-EC ordering is not evidence that organic carbon is the causal fraction. No subgroup modification was detected on any of six strata, which at n=1,875 is uninformative rather than reassuring.

doi: 10.3760/cma.j.cn112338-20251220-00911

Multimorbidity meta-analysis 2026 BMJ Public Health*Pooled risk of multimorbidity per 5 $\mu\text{g m}^{-3}$ of PM and NO₂*

Systematic review to 1 April 2026 across four databases (PROSPERO CRD42024613048); 22 studies included, 21 meta-analysed with random effects. Pooled adjusted risk ratios per 5 $\mu\text{g m}^{-3}$: PM_{2.5} 1.15 (1.08-1.23) from 4 studies at $I^2 = 93.0\%$; PM₁₀ 1.27 (1.11-1.47) from 3 at $I^2 = 99.5\%$; PM₁ 1.07 (1.05-1.09) from 2 at $I^2 = 0.0\%$; NO₂ 1.12 (1.00-1.26) from 4 at $I^2 = 96.6\%$. Nitrogen oxides, black carbon, SO₂, ozone and CO showed nothing. The limitation is structural, not incidental: harmonising to a per-5 $\mu\text{g m}^{-3}$ increment discarded most of the 22 studies, leaving two to four per pollutant, and at $I^2 = 93-99.5\%$ the PM_{2.5} and PM₁₀ pooled values are averages over non-exchangeable studies. **Zero eligible studies** on polypharmacy — half the stated aim — or on household air pollution, older adults, the Americas or Oceania.

doi: 10.1136/bmjph-2025-004782

Mtenga et al. 2026 Cancer Epidemiol*Ambient PM_{2.5} exposure and cancer incidence in Nigerian adults*

First prospective sub-Saharan African cohort on this question. 5,621 adults, median follow-up 4.7 years, 147 incident cancers confirmed histopathologically, mean PM_{2.5} 50.4 $\mu\text{g m}^{-3}$ — tenfold the WHO guideline. Each 10 $\mu\text{g m}^{-3}$ gives HR 1.12 (95% CI 1.06-1.19) for all-cancer incidence and the top quartile 1.82 (1.29-2.57); mediation analysis attributes 41.3% to oxidative stress and inflammation. Site-specific signals for lung, breast and haematopoietic cancer rest on very small strata at 147 total events, and biomarkers measured once in a subsample cannot establish the temporal ordering mediation requires. The 1 km satellite-land-use regression exposure surface has no ground reference network to anchor it.

doi: 10.1016/j.canep.2026.103232

Odiko et al. 2026 J Expo Sci Environ Epidemiol

Environmental lead exposure and clinical infectious respiratory disease in a sample of urban and disadvantaged children.

13,221 children aged 14 and under, Temple University Health System, Philadelphia, 2010–20; 28% above the CDC 3.5 ug/dL action level. Streptococcal pharyngitis was associated with BLC ≥ 3.5 ug/dL (aOR 2.74, 95% CI 1.92–3.90) and, almost identically, with BLC 1–3.4 ug/dL (2.45, 1.75–3.44); frequent strep showed the same pattern. Group was null. That the two lead strata give the same estimate is the problem – a genuine dose-response should separate them, and the flatness suggests both strata are marking the same underlying thing, most plausibly housing age and healthcare contact frequency, neither of which is adjusted for beyond insurance type. Outcome ascertainment is ICD-coded within one health system, so a child who is screened for lead is also a child who is seen more often. The ‘no safe level’ conclusion is not supported by a contrast this flat.

doi: 10.1038/s41370-026-00978-0

Park J et al. 2026 Int J Epidemiol

Ozone, disability and emergency department use in Korea

Time-stratified case-crossover on 3,484,877 ED visits from the Korean National Health Insurance Service (2015–19, 2022–23), exposure the 3-day moving average of daily 8-hour maximum O₃, adjusted for daily mean temperature and holidays. Per IQR (~22 ppb): +1.94% (1.64–2.25) in persons with disabilities during the warm season and +1.93% (1.62–2.25) year-round, against +1.36% (0.92–1.80) in persons without. A roughly 40% larger relative effect in a population defined administratively rather than clinically, with an interaction term tested across temperature strata. No particulate metric is included, which is both the limitation and, read against Gao et al. in this issue, part of the point: a disability register can identify a susceptible group that pollutant-specific monitoring networks were never designed to serve.

doi: 10.1093/ije/dyag189

Su et al. 2026 Lancet Oncol

Wildfire-related PM_{2.5} and overall survival of older lung cancer patients

414,016 SEER-Medicare patients aged 65+, 1,211,490 person-years, 2008–2019, time-varying Cox on 3-year moving averages of zip-code wildfire and non-wildfire PM_{2.5}. Per 1 $\mu\text{g m}^{-3}$: HR 1.0782 (95% CI 1.0641–1.0924) for wildfire against 1.0171 (1.0149–1.0192) for non-wildfire, a 4.6-fold difference in per-unit potency. The exposure contrast is the weakness — median wildfire PM_{2.5} was 0.28 $\mu\text{g m}^{-3}$ (IQR 0.21–0.39) against 8.11 for non-wildfire, so the per-unit hazard is extrapolated far outside the observed range, and the per-SD estimates (1.0229 versus 1.0384) reverse the ordering. Zip-code assignment with no residential history. Species-resolved rather than mass-only monitoring is the direct implication.

doi: 10.1016/s1470-2045(26)00295-0

Tianjin myopia 2026 Environ Pollut

Long-term PM_{2.5} components and childhood myopia

School-based study of 212,566 children in grades 4–6 in Tianjin, with three-year mean PM_{2.5}, its major components (SO₄²⁻, NO₃⁻, NH₄⁺, organic matter, black carbon) and ozone estimated by machine-learning exposure models and assigned to school locations. Restricted cubic splines and quartile-based modified Poisson models for single pollutants, quantile g-computation for the mixture. Sulfate carried the signal: top versus bottom quartile PR 1.10 (1.07–1.13), against a weak non-monotonic ozone association 1.03 (1.00–1.05) and a whole-mixture estimate of only 1.017 (1.007–1.027) per quartile increase. Cross-sectional, so myopia onset timing is unknown and reverse pathways through indoor time are not excluded; exposure at school ignores home and commute, which for a three-year window is substantial misclassification. The component contrast is the reason to read it — a speciated signal that unspiciated mass does not reproduce is the standing argument against mass-only networks.

doi: 10.1016/j.envpol.2026.129120

Wang C et al. 2026 Environ Sci Technol

Green space and cardiovascular health where the air-quality pathway is 75% of the model and the effects are fragile

Two-stage gradient-boosting plus structural equation modelling on a clinical cohort of 23,312 patients in Nanjing. Air-quality-related pathways account for more than 75% of the relative statistical contribution to model variance, exceeding psychological-restoration proxies — which, if it holds, says green-space cardiovascular benefit in a dense polluted city is mostly an air-quality mechanism. Distinct networks emerge for chronic ischaemic heart disease (vegetation vitality inverse to NO₂ and SO₂, linked to metabolic markers) and heart failure. The paper is unusually candid about its own weaknesses and they are disqualifying for the headline: the macroscale aerodynamic interpretation is called speculative for want of morphological or fluid-dynamics data, and the age-stratified canopy-density $\times$ PM_{2.5} slopes are nonsignificant with extremely small effect sizes. Relative variance contribution is not effect size.

doi: 10.1021/acs.est.6c08296

Zheng M et al. 2026 Ann Med

Occupational particulate exposure ranks with alcohol in an ecological cancer model

GBD 2021 incidence for adults ≥ 55 across 203 countries, 1990–2021, classified into quartile-based co-burden categories for oesophageal and lip/oral-cavity cancer, with 58 candidate exposures screened by random forest and SHAP to 36, then to 25 after variance-inflation screening, entering negative binomial models with population offsets. Countries split into consistent (73), oesophageal-dominant (61) and LOC-dominant (69). Occupational exposure to particulate matter, gases and fumes carries RR 1.29 (1.13–1.49) for oesophageal cancer, comparable to high alcohol use (1.32, 1.13–1.53) and above low nut/seed intake (1.16). The authors state the correct caveat themselves: these are country-level ecological associations and establish no individual-level causality. GBD exposure estimates for occupational PM are modelled, not measured, so the estimate is a model-on-model contrast.

doi: 10.1080/07853890.2026.2721134

Zhu et al. 2026 Environ Res

Long-term exposure to particulate matter and survival among patients with gastric cancer: A hospital-based cohort study.

4,441 stage I–III gastric cancer patients resected 2012–2022, median follow-up 3.83 years, 1,318 deaths (919 gastric-cancer-specific). Per 1 SD increase, all-cause mortality HRs were 1.15 (95% CI 1.06–1.25) for PM_{2.5}, 1.14 (1.06–1.24) for PM₁₀ and 1.12 (1.03–1.21) for the coarse fraction; gastric-cancer-specific subdistribution HRs were larger – 1.31 (1.20–1.44), 1.28 (1.17–1.40) and 1.17 (1.07–1.28) – with Fine-Gray models treating non-gastric deaths as competing events, which is the right handling and is often skipped in this literature. Counterfactual PAFs under lower historical exposure are reported. The inference is threatened most by exposure assignment: CHAP is a 1 km satellite-and-model reanalysis assigned at the residential address, so measurement error is spatially structured, and pre-index exposure is a proxy for decades of residence and occupation that the 36-month window does not capture. Reverse causation is limited by the pre-surgery window but not eliminated – sicker patients may relocate. That the cause-specific effect exceeds the all-cause effect is internally coherent but unusual enough to want replication outside Zhejiang.

doi: 10.1016/j.envres.2026.125689

Pooled corpus register — September 2026

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
45 studies, 29 pooled, 2010-2024	Environmental exposure and risk of death by suicide, attempted suicide and suicide ideation: a systematic review and meta-analysis.	<i>J Epidemiol Community Health</i>	Systematic review (PRISMA)	Ambient PM _{2.5} and PM ₁₀ , short- and long-term contrasts pooled together	Global / multi-region
4HAIE hair cortisol	Intensity-dependent associations of physical activity and cardiorespiratory fitness with hair cortisol concentrations in the context of lifetime air pollution exposure	<i>Stress</i>	Cross-sectional (cohort baseline)	Lifetime residential PM _{2.5} , PM ₁₀ and NO ₂ from historical models	Czech Republic
ABCD amygdala 2026	Air Pollution, Early Adversity, and Amygdala: Environmental Correlates of Psychopathology in Preadolescents	<i>bioRxiv (preprint)</i>	Prospective cohort	Annual residential PM _{2.5} total mass plus 15 components	United States
Barbalat et al.	Association of heat, cold, and ambient pollution in the first 1000 days of life with autism spectrum disorder risk indicators in two-year-olds: findings from the ELFE national birth cohort	<i>Environ Sci Technol</i>	Prospective birth cohort + DLNM	Weekly mean PM _{2.5} , PM ₁₀ , NO ₂ from 1 km daily outdoor predictions (200 m for NO ₂ /temperature in large urban areas)	France
CHARLS neurodegenerative composite	The association of long-term air pollution with a composite neurodegenerative risk score among Chinese elderly: evidence from the CHARLS.	<i>Environ Health</i>	Prospective cohort (CHARLS)	2012-2015 mean PM _{2.5} , PM ₁₀ , NO ₂ and O ₃ gridded to residence	China
Corrigendum	Corrigendum to 'Prenatal and childhood air pollution exposure, cellular immune biomarkers, and brain connectivity in early adolescence'.	<i>Brain Behav Immun Health</i>	Metadata only (no abstract)	Corrects 10.1016/j.bbih.2024.100799; (metadata only)	Not stated (metadata only)
Curzio et al.	The Exposome-Brain Axis: A Scoping Review of Biological Biomarkers in Environmental Neurotoxicity, Neuroinflammation, and Neurodegeneration.	<i>Toxics</i>	Scoping review	Ambient PM _{2.5} among other exposures (as reported by included studies)	Global / multi-region (review)
ENVIRONAGE eye-track	Residential early-life air pollution exposure and oculomotor markers of cognitive workload in children: a prospective birth cohort study	<i>EBioMedicine</i>	Prospective cohort	Modelled residential BC, NO ₂ and PM _{2.5} ; windowed, per IQR	Europe
Falkenstein et al.	Investigating effects of day-to-day variations in environmental exposure on the human brain: study protocol for the Day2Day Environment project.	<i>BMC Neurosci</i>	Study protocol	Wearable air-quality, noise and light sensors with GPS tracking over 25 sessions	Not stated (protocol)
Hong Kong momentary stress	An Explainable Multi-Modal Deep Learning Framework for Investigating the Influences of Dynamic Environmental Exposure on Momentary Stress.	<i>Environ Sci Technol</i>	Ecological momentary assessment with mobile sensing, multi-modal deep learning	Min-level mobile sensor PM _{2.5} alongside noise, temperature, humidity, greenery, street-view and remote-sensing imagery	Hong Kong
Huang et al.	Environmental Pollution, Biomarkers, and Dementia: An Evidence Review.	<i>Int J Mol Sci</i>	Narrative review	Long-term PM _{2.5} , PM ₁₀ , NO _x /NO ₂ , BC, O ₃ (as reported by included studies)	Global / multi-region (review)
Hubei Memory & Aging	Associations between compound heat exposure and cognitive impairment and the potential mediating role of plasma sphingolipid metabolism.	<i>Environ Pollut</i>	Repeated-measures cohort + metabolite mediation	ERA5-Land hourly temperature, CMDSC humidity; PM _{2.5} and O ₃ from the Tracking Air Pollution dataset as covariates	China

First author	Focus	Journal	Design	Metric	Region
India-Aldana et al.	The chemical exposome during peri-pregnancy, maternal sleep outcomes post-pregnancy, and the potential mediating role of body mass index.	<i>Environ Pollut</i>	Prospective cohort (mixtures)	Estimated air-pollutant mixture during pregnancy within a 52-marker chemical exposome	Mexico City, Mexico (PROGRESS cohort)
Januel et al.	Air pollution and the risk of multiple sclerosis relapses in the Paris region, the CONFISEP study	<i>J Neurol Neurosurg Psychiatry</i>	Self-controlled design	Weekly street-level PM ₁₀ , PM _{2.5} , NO ₂ , O ₃ (Airparif model)	France (Ile-de-France)
Kuznetsov et al.	Particulate matter air pollution and life satisfaction: links across and within twin pairs	<i>Environ Res</i>	Twin-pair longitudinal comparison	Residential PM _{2.5} and PM ₁₀ , repeated across six waves	Germany
Kwon	Fine particulate matter (PM _{2.5}) and suicide in South Korea: an instrumental variable analysis using thermal inversion.	<i>Front Public Health</i>	Instrumental-variable panel analysis	Regional PM _{2.5} instrumented by thermal inversions, 2015-2019	South Korea
Morrel et al.	Outdoor Air Pollution, Perivascular Space Morphology, and Cognition in Preadolescence.	<i>Biol Psychiatry Glob Open Sci</i>	Cross-sectional cohort imaging + weighted quantile sum regression	Annual-average PM _{2.5} , O ₃ , NO ₂ and 15 PM _{2.5} components, spatiotemporal models at residence	USA
Oh et al.	Association Between Long-Term Exposure to Air Pollution and Parkinson's Disease: Administrative Cohort of Older Adults in South Korea.	<i>Mov Disord</i>	Nationwide cohort	District-level annual PM _{2.5} , NO ₂ and O ₃ from CMAQ, 1-year moving average, time-varying	South Korea
Older-adult cognition review	Air pollution and cognitive outcomes in older adults: a systematic review	<i>Rev Environ Health</i>	Systematic review	Ambient PM _{2.5} and co-pollutants, long-term	Global
Passive smoking acute stress	Is passive smoking associated with acute stress response? An observational real-world study	<i>Front Public Health</i>	Cross-sectional survey	Self-reported secondhand smoke exposure, no measurement	China
Paulino et al.	Neural exposome burden stratifies cognitive trajectories in a subacute-to-chronic mild TBI outpatient cohort: an exploratory dimension-reduction study	<i>Int J Neurosci</i>	Prospective cohort + PCA	Residential PM _{2.5} as one of 13 exposome variables entering a principal-component burden score, alongside education, area deprivation and adverse childhood events	North America
Ren X et al.	Association between PM _{2.5} constituents and non-suicidal self-injury among Chinese adolescents: modifying role of biological rhythm	<i>Front Public Health</i>	Cross-sectional survey	PM _{2.5} total mass and constituents in logistic and weighted quantile sum regression, stratified by multidimensional biological-rhythm score	China
Rodulfo-Cardenas et al.	APOE genotype shapes persistent behavioral and molecular alterations following early postnatal particulate matter exposure in humanized apoE3 and apoE4 mice.	<i>Neurotoxicology</i>	Animal model	NIST SRM 2787 fine PM, whole-body inhalation 1 h/day PND4-14, nominal max 3789 µg/m ³ (not measured)	Laboratory (humanized apoE mice)
Sex modification dementia meta	Does Reported Sex Modifies the Association between Ambient Air Pollution exposure and Alzheimer's Disease and Related Dementia? A Systematic Review and Meta-Analysis	<i>Environ Pollut</i>	Systematic review + random-effects meta-analysis	Long-term PM _{2.5} and 11 further pollutants, F:M ratio	Global
Shupler et al.	Preconception, Prenatal and Postnatal PM _{2.5} Exposure and Neurodevelopmental Disorder Risk among Medicaid Recipients	<i>Environ Sci Technol</i>	Retrospective cohort	ZIP-code PM _{2.5} , preconception 12 wk, prenatal 37 wk, postnatal 3 y; DLM	USA (Medicaid, national)

First author	Focus	Journal	Design	Metric	Region
Temporao et al.	Daily associations between outdoor air pollution and sleep quality.	<i>medRxiv (preprint)</i>	Panel study with ambulatory EEG	Daytime outdoor PM _{2.5} , NO ₂ and O ₃ aggregated from city monitoring stations; same-day and previous-day	Hungary
WTC responder exposure mixture	Urban exposures, frailty, and mental illness in World Trade Center Health Program responders	<i>Nat Commun</i>	Prospective cohort (mixtures)	Annual PM _{2.5} , temperature, Green View Index	United States
Wang et al.	Environmental exposures and mental health: longitudinal results from the German National Cohort	<i>Environ Int</i>	Metadata only (no abstract)	Metadata only - environmental exposures and mental health in NAKO (by title)	Not recoverable from metadata
Zhang et al.	Associations between household solid fuel use, kitchen ventilation, and depressive symptoms among older adults in China: a national cross-sectional study.	<i>Front Public Health</i>	Cross-sectional (multistage)	Self-reported household fuel type and kitchen ventilation, no measured concentration	China
Zou et al.	Association of preconception to early childhood residential greenness with child neurodevelopmental outcomes and the mediating role of air pollution: a prospective cohort study.	<i>Environ Res</i>	Prospective birth cohort	PM _{2.5} , PM ₁₀ , NO ₂ , O ₃ as single-pollutant mediators of NDVI	China (Jiangsu)
Cardiovascular & metabolic					
Aysin et al.	Effects of Traffic-Related Diesel Exhaust Exposure and Moderate Exercise on Obesity, Metabolic Dysfunction, and Inflammatory Responses in Rats.	<i>Metabolites</i>	Animal model	Diesel exhaust, mean 300 µg/m ³ PM _{2.5} , 2 or 4 h/day for 8 weeks	Laboratory (Wistar rats)
Bikdeli et al.	Coated or chewable Aspirin and a hybrid air pollution mitigation strategy in patients with atherosclerotic disease: Rationale and Design of the COATED-AIR Randomized Trial.	<i>Eur Heart J Cardiovasc Pharmacother</i>	Trial / intervention	Real-time ambient air-quality alerts driving behavioural and KN95 responses	Not stated (design paper; sites not in abstract)
CHARLS coexposure	Association of coexposure to indoor solid fuels, outdoor air pollution, and heatwave with the risk of Cardiometabolic Multimorbidity among middle-aged and older Chinese adults	<i>Int Arch Occup Environ Health</i>	Prospective cohort	PM ₁ , PM _{2.5} and PM ₁₀ with solid-fuel use and heatwave, Qgcomp mixture	China
COPDB Beijing	Associations between air pollution and cardiorespiratory function and metabolic indicators: New insights into mixture exposures.	<i>J Hazard Mater</i>	Panel study	Continuously measured PM _{2.5} , UFP and accumulation-mode counts, SO ₂ /NO/NO ₂ /O ₃ , and PM _{2.5} carbon, metal and water-soluble ion components	China
California PM _{2.5} diabetes mortality	Associations of PM _{2.5} exposure with diabetes-related mortality for California residents	<i>Front Public Health</i>	Case-control (registry)	1-year rolling mean PM _{2.5} at geocoded residence	United States
Campagna et al.	Tracking biochemical indicators of air pollution exposure in acute cardiorespiratory emergencies: a scoping review.	<i>Front Public Health</i>	Scoping review	Short-term ambient PM _{2.5} , PM ₁₀ , NO ₂ (as reported by included studies)	Global / multi-region (review)
Campailla et al.	Ambient particulate matter exposure and acute type A aortic dissection: A retrospective time-stratified case-crossover study in northern Italy.	<i>Int J Cardiol Cardiovasc Risk Prev</i>	Case-crossover	Daily mean PM _{2.5} and PM ₁₀ from one urban-background station, same-day and 30-day windows	Italy (Piacenza)
Chen N et al.	Associations of PM _{2.5} components, residential greenness, and heatwaves with incident venous thromboembolism	<i>Eur J Prev Cardiol</i>	Prospective cohort (mixtures)	Elemental carbon, organic matter, sulfate, ammonium and nitrate at 3x3 km; NDVI in a 300 m buffer; heatwaves under several temperature thresholds	United Kingdom

First author	Focus	Journal	Design	Metric	Region
Chiang et al.	Ambient Carbon Monoxide as a Delayed Trigger of Acute Aortic Dissection: A Time-Stratified Case-Crossover Study in a Tropical Industrial Metropolis.	<i>J Am Heart Assoc</i>	Case-crossover	PM _{2.5} , PM ₁₀ , CO, NO ₂ , O ₃ , SO ₂ from 24 fixed stations; lags 0-3 d, per 1 ppm and per IQR	Taiwan
China WBC homeostasis	Long-term exposure to ambient air pollution and circulating white blood cell homeostasis among adults in China: Time-window-specific associations and the mediating role of fasting blood glucose	<i>Environ Pollut</i>	Retrospective cohort	1-, 6-, 12- and 24-month mean PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ , CO, O ₃	China
Constituent multiomics cardiopulmonary	PM _{2.5} Constituents and Cardiopulmonary Dysfunctions in Older Adults: A Multiomics Approach Unveiling Molecular Pathways	<i>Environ Sci Technol</i>	Prospective cohort (mixtures) + omics mediation	Personal PM _{2.5} monitoring with constituent speciation	China
Cui et al.	Differential Risks of Incident Heart Failure Subtypes Associated with PM _{2.5} Chemical Constituents in China between 2006 and 2021.	<i>Environ Sci Technol</i>	Prospective cohort	Long-term residential PM _{2.5} constituents (BC, OM, NH ₄ ⁺ , others)	China (Kailuan, Tangshan)
De Potter et al.	Influences of socioeconomic and environmental factors on STEMI incidence and related mortality: insights from the Belgian Environmental Myocardial Infarction Risk (EMIR) study	<i>Eur Heart J Acute Cardiovasc Care</i>	Time-stratified case-crossover + DLNM	Acute and chronic residential PM _{2.5} , NO ₂ , O ₃ , temperature, noise and greenspace from high-resolution models; per IQR	Belgium
Dzhambov et al.	Residential exposures associated with adiposity and proinflammatory markers in obese adults with metabolic syndrome: a cross-sectional study	<i>Arch Hig Rada Toksikol</i>	Observational - cross-sectional	Modelled residential NO ₂ , PM _{2.5} , PM ₁₀ ; traffic noise; solid-fuel use; greenness	Bulgaria
Fang et al.	Joint association of air pollutants and lifestyles on early-/late-onset heart failure: a population-based prospective cohort study	<i>Eur J Prev Cardiol</i>	Prospective cohort	Composite air-pollution score over NO ₂ , NO _x , PM ₁₀ , PM _{2.5} and PM _{2.5-10} , crossed with a six-component lifestyle score; multiplicative and additive interaction	United Kingdom
Fang et al.	Venous Thrombosis and Air Pollution: A Systematic Review and Meta-Analysis of 1.4 Million Adult Patients.	<i>J Thromb Haemost</i>	Systematic review + random-effects meta-analysis	Long-term (>=1 y average) and short-term PM _{2.5} , PM ₁₀ and gaseous pollutants, per 10 µg/m ³	Global
Fleming & Fleming	Tire Wear Pollution and Cardiovascular Risk.	<i>Am J Med</i>	Metadata only (no abstract)	Tire-wear (non-exhaust) particles; no exposure metric retrievable	Global
Hangzhou AAD PM components	Associations of Fine Particulate Matter Components With Acute Aortic Dissection: A Time-Stratified Case-Crossover Study	<i>J Am Heart Assoc</i>	Time-stratified case-crossover	Component-resolved PM _{2.5} : BC, OM, sulfate, nitrate	China
Hua et al.	Heterogeneous Risks of Chemical Components and Sources in Size-Resolved Particulate Matter on Ischemic and Hemorrhagic Stroke.	<i>Environ Health (Wash)</i>	Time-series	41 chemical components and 6 emission sources across 9 aerodynamic size fractions below 10 µm	China
Huang et al.	Association between air pollution exposure and metabolic dysfunction-associated steatotic liver disease in adults in China.	<i>Zhonghua Liu Xing Bing Xue Za Zhi</i>	Observational - cross-sectional	PM _{2.5} , PM ₁₀ , NO ₂ and warm-season O ₃ , per 1 SD; weighted environmental risk score for the mixture	China
Hyperuricemia low-concentration Korea	Long-term exposure to ambient air pollution and an increased risk of hyperuricemia.	<i>Clin Rheumatol</i>	Retrospective cross-sectional (health screening)	Annual mean PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ and CO from CMAQ at residential address	South Korea

First author	Focus	Journal	Design	Metric	Region
Jeong et al.	Joint effects of air pollution and temperature on glycemia: An exposome analysis in European cohorts.	<i>Environ Int</i>	Metadata only (no abstract)	Metadata only - air pollution and temperature jointly, metrics not recoverable	Europe (cohorts not recoverable from meta-data)
Jiang Z et al.	Long-term exposure to multiple air pollution and trajectory risk of cardio-renal-metabolic multimorbidity	<i>Atmos Environ</i>	Metadata only (no abstract)	Not retrievable: the publisher deposited no abstract to Crossref, Europe PMC or Semantic Scholar at index time	Not stated (metadata only)
Jin et al.	Ambient air pollution and stroke in two population-based cohorts: roles of systemic inflammation and polygenic risk.	<i>Neuroepidemiology</i>	Prospective cohort	Residential PM _{2.5} , PM ₁₀ (CHARLS; UK Biobank LUR)	China and UK (CHARLS, UK Biobank)
Kriti et al.	Ecological associations between ambient air pollution, gut microbiome composition, and metabolic markers in urban populations of central India.	<i>Front Public Health</i>	Ecological	Station PM _{2.5} , PM ₁₀ , NO _x , SO ₂ by residential locality	India (Bhopal)
Li et al.	Joint effects of short term exposure to heat and ozone on circulatory disease mortality in Beijing-Tianjin-Hebei region and surrounding areas.	<i>Zhonghua Liu Xing Bing Xue Za Zhi</i>	Time-series	Daily maximum 8-h O ₃ (mean 138.4 +/- 57.5 µg/m ³) and daily mean temperature, warm season	China
Lin et al.	Joint association of PM _{2.5} and AIP with advanced Cardiovascular-Kidney-Metabolic syndrome in middle-aged and older Chinese adults: a national cohort study.	<i>Research Square (preprint)</i>	Cross-sectional survey	Satellite-derived residential PM _{2.5} split at the median	China (national, CHARLS)
Lin et al.	Short-term lagged exposure to ambient air pollutants and hospitalization costs among AMI patients: health-economic evidence from a subtropical city	<i>Front Public Health</i>	Retrospective cohort	Lag 0-6 day PM _{2.5} , PM ₁₀ , SO ₂ , NO ₂ , CO, O ₃ (city monitors)	China (Jiangmen, Guangdong)
Pan SG et al.	Ozone and possible heat effects on out-of-hospital cardiac arrest: a time-series study	<i>Front Public Health</i>	Distributed-lag time-series	Daily mean temperature over lag 0-14 in a quasi-Poisson DLNM, plus lag0-1 PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ , CO and O ₃ in single- and two-pollutant models	China
Pellegrino et al.	A possible mediating role of SASP in the association between environmental air pollution and incident atherosclerotic cardiovascular disease	<i>Sci Total Environ</i>	Metadata only (no abstract)	Metadata only - air pollution, SASP mediation, incident ASCVD	Not recoverable from metadata
Pollutant mixtures atherosclerosis	A Perfect Storm of Pollutants: Environmental Mixtures in the Pathogenesis of Atherosclerosis	<i>Int J Mol Sci</i>	Narrative review	PM in mixtures with toxic metals, pesticides, PAHs and endocrine disruptors	Global / multi-region
Shahriary et al.	The Silent Intruders: Cardiovascular and Pregnancy-Related Health Effects of Ultrafine Particulate Matter.	<i>Toxicol Lett</i>	Mechanistic review	Ultrafine particulate matter, number-based rather than mass-based metrics	Global / multi-region
Sichuan Basin CAA	Temporal changes in short-term PM _{2.5} -cardiorespiratory mortality associations and implications for the health benefits of the clean air action in the Sichuan Basin, 2014-2022	<i>Int Arch Occup Environ Health</i>	Time-series	Ambient PM _{2.5} , county daily means; ER per 10 µg m ⁻³	China
Sun D et al.	Exploring China's Clean Air Act and associated cardiovascular disease risk: a prospective, quasi-experimental, and causal inference modelling study	<i>Lancet Planet Health</i>	Quasi-experimental (difference-in-differences)	Long-term PM _{2.5} , PM ₁₀ and O ₃ ; municipal PM-reduction target as the intervention assignment	China
Sun et al.	When air gets heavy: joint PM(2.5) and ambient humidity exposure, frailty, and stroke.	<i>Front Public Health</i>	Prospective cohort (CHARLS)	Annual PM _{2.5} and relative humidity at residence, dichotomised at median	China (nation-wide)

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Sun et al.	Solid Fuel Use and Risks of All-Cause and Cardiovascular Disease Mortality in Urban Northern China: A 12-Year Cohort Study	<i>Indoor Air</i>	Retrospective cohort	Household solid fuel use for heating/cooking (years); no PM measured	China (Tianjin, Shenyang, Taiyuan, Rizhao)
UK PH metabolomic mediation	Long-term ambient air pollution exposure, metabolomic signatures, and incident pulmonary hypertension: A large prospective cohort study	<i>Ecotoxical Environ Saf</i>	Prospective cohort (mixtures) + omics mediation	PM _{2.5} , PM ₁₀ , NO ₂ , NO _x and a joint air pollution score	UK
Wang et al.	Long-term exposure to fine particulate matter constituents and incident stroke risk among middle-aged and older adults in China: a cohort study.	<i>Front Public Health</i>	Prospective cohort (CHARLS)	TAP-dataset PM _{2.5} and five constituents (SO ₄ , NO ₃ , NH ₄ , OM, BC) at residence	China (nation-wide)
Wang et al.	Long-Term Air Pollution Exposure, Cardiovascular Disease, and Cardiometabolic Risk Factors in Middle-Aged and Older Chinese Adults: A Prospective Cohort Study.	<i>Bioengineering (Basel)</i>	Prospective cohort (CHARLS)	CHAP long-term PM _{2.5} , PM ₁₀ , NO ₂ , O ₃ at residence	China (CHARLS, national)
Wang et al.	Long-term PM _{2.5} exposure and incident heart failure in the UK Biobank: relative and absolute risk estimates.	<i>Front Public Health</i>	Prospective cohort	LUR annual PM _{2.5} , PM ₁₀ , NO ₂ (2010 ESCAPE surface)	UK
Yancheng, 2,312 admissions 2022-2024	Association between ambient fine particulate matter and hemorrhagic stroke in Yancheng, China.	<i>Front Public Health</i>	Time-stratified case-crossover	Station-average daily PM _{2.5} , lag 0 to lag 01	China
Yang et al.	Mitochondrial DNA methylation as a bridge: Linking indoor air purification to alleviated cardiovascular impairments in young adults.	<i>Environ Res</i>	Randomised crossover chamber	Personal PM _{2.5} under true vs sham indoor air purification (47.45% indoor reduction)	Not stated (young-adult panel)
Yu et al.	Effect modification of greenness on the associations between air pollutants and out-of-hospital cardiac arrest: Evidence from a multi-city study	<i>Public Health</i>	Time-stratified case-crossover	Short-term PM _{2.5} , PM ₁₀ , O ₃ , CO at onset location; NDVI/EVI at 250-1000 m	China (multi-city)
Zhang et al.	Cardiopulmonary hospitalization risks from wildfire-specific and non-wildfire PM _{2.5} in 20 US states.	<i>Nat Commun</i>	Self-controlled design	Two-year average wildfire-specific vs non-wildfire PM _{2.5} , modelled and source-separated, per 1 µg/m ³	USA
Zhang et al.	Air Pollution Exposure Induces Vascular Endothelial Cell Dysfunction via Elevated Circulating F2: Multiomics Insights from an Ex Vivo Biosensor Assay.	<i>Environ Health (Wash)</i>	Panel study	Ambient PM _{2.5} , PM ₁₀ , SO ₂ , NO ₂ and CO contrast between two cities, no personal sampling	China (Qingdao to Shijiazhuang)
Zhongshan AMI	Daily changes in fine particulate matter are associated with acute myocardial infarction hospitalizations: an individual-level time-stratified case-crossover study in Zhongshan, China	<i>Front Public Health</i>	Case-crossover	Day-to-day PM _{2.5} change, CHAP 1 km, residence-assigned	China
Reproductive & developmental					
Ammons et al.	Epidemiologic research on environmental chemical exposures and ovarian and endometrial cancers: a systematic review.	<i>Environ Res</i>	Systematic review + random-effects meta-analysis	Long-term ambient PM _{2.5} , per 5 µg/m ³ , alongside other environmental chemical exposures	North America
Bae et al.	Exposure to ambient air pollutants during early gestation and phenotypic variation in non-syndromic microtia.	<i>J Plast Reconstr Aesthet Surg</i>	Observational - cross-sectional	Modelled PM _{2.5} , PM ₁₀ , O ₃ , NO ₂ , SO ₂ , CO at gestational weeks 5-9 (address at registration)	South Korea (single centre)
Chen Y et al.	The impact of PM _{2.5} component exposure during early pregnancy on congenital heart disease: evidence from Guangdong, China	<i>Sci Rep</i>	Case-control (registry)	Satellite-derived PM _{2.5} components (OM, sulfate, nitrate, ammonium, BC) by week of early pregnancy; quantile g-computation for mixture weights	China

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ECHO prenatal wildfire PM _{2.5}	Prenatal exposure to wildfire fine particulate matter and fetal growth outcomes in the United States: Findings from the ECHO Cohort	<i>Environ Epidemiol</i>	Prospective birth cohort + DLNM	Daily wildfire-specific PM _{2.5} , census-tract resolution	United States
Eick et al.	Prenatal exposure to aromatic amines and associations with infant cognition in the ECHO.CA.IL cohorts	<i>J Expo Sci Environ Epidemiol</i>	Prospective birth cohort (mixtures)	Second-trimester urinary aniline, o/m-toluidine, 4-chloroaniline, o-anisidine, 4,4'-methylenedianiline, cotinine	Illinois and California, USA
France SUDI case-crossover	Short-Term Effect of Air Pollutants on Sudden Unexpected Death in Infancy: A Case-Crossover Study Based on the French National Registry	<i>Paediatr Perinat Epidemiol</i>	Time-stratified case-crossover	Modelled residential daily PM ₁₀ , PM _{2.5} , NO ₂ , O ₃	France
Gao et al.	Exposure biomarkers of airborne organic pollutants in pregnant women and developmental risks: evidence from epidemiology to in vitro assessment	<i>Environ Sci Technol</i>	Prospective cohort + in vitro screening	20 urinary biomarkers of airborne organics (6PPD, 6PPD-Q, nitro-PAHs, phenylguanidines, benzothiazoles, cotinine) by LC-MS/MS	China
Gielen et al.	The association of neighborhood socioeconomic status with birth weight is not confounded by ambient air pollution in twin pregnancies	<i>Neonatology</i>	Prospective birth cohort	Kriged PM ₁₀ and NO ₂ at residence (monitor + land cover)	Belgium (East Flanders)
Hu et al.	Real-world PM _{2.5} exposure triggers early testicular epigenetic responses in the absence of significant changes in measured reproductive endpoints.	<i>Reprod Toxicol</i>	Murine inhalation exposure	Real-world ambient PM _{2.5} , whole-body inhalation, 4, 8 and 12 weeks	China (Hangzhou)
Korean birth-cohort growth mixture	Prenatal bisphenols and ambient air pollutants are differentially associated with early childhood length and weight from birth to 36 months: a multi-domain mixture analysis in a Korean birth cohort	<i>Environ Pollut</i>	Metadata only (no abstract)	Prenatal ambient air pollutants in a multi-domain mixture	South Korea
Li et al.	Cohort profile: the Shenzhen Birth Cohort (SZBC) study.	<i>BMJ Open</i>	Prospective birth cohort	Prenatal and postnatal PM _{2.5} with component resolution (nitrate singled out); model not stated	China (Shenzhen)
Liaoning fertility clinics, n=1,349	Ambient air pollutant mixtures in relation to sonographic endometrial receptivity parameters during the peri-implantation window: a clinic-based observational study.	<i>Environ Res</i>	Observational exposure study	Monthly mean PM _{2.5} , PM ₁₀ , CO, NO ₂ , SO ₂ , O ₃ from a high-resolution reanalysis grid	China
Mbishi et al.	Early-life exposure to particulate matter and green spaces: Associations with infant growth	<i>Environ Pollut</i>	Prospective cohort	Perinatal PM ₁₀ and PM _{2.5} from the RIO detrended-kriging interpolation model; municipal vegetation split into low, high and total green space	Belgium
Nelin et al.	Associations of short-term changes in air pollution with spontaneous preterm birth: A time-stratified case-crossover analysis.	<i>Ann Epidemiol</i>	Time-stratified case-crossover + DLNM	Daily high-resolution modelled PM _{2.5} , week before delivery, three lag parameterisations	United States
Preeclampsia nationwide 2018-23	Association between air pollution and preeclampsia: a nationwide study from 2018 to 2023	<i>Eur J Intern Med</i>	Metadata only (no abstract)	Ambient air pollution, nationwide registry linkage	Not stated
Prenatal PM childhood NHL Spain	Exposure to Particulate Matter (PM) During Pregnancy and Non-Hodgkin Lymphomas in Children	<i>Cancers</i>	Case-control (registry)	Prenatal PM _{2.5} and PM ₁₀ from a random-forest exposure model, by trimester	Spain
Prenatal greenspace ASD window	Mid-to-Late Pregnancy: A Critical Window for the Protective Effect of Green Space Against Autism Spectrum Disorder	<i>Environ Pollut</i>	Matched case-control + DLNM	Monthly residential PM _{2.5} alongside NDVI, -10 to +3 months	China

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SIP prenatal Bayley	The effect of in utero exposure to air pollution and subsequent cognitive and behavioral development in children aged 1-3 years old	<i>Child Neuropsychol</i>	Prospective birth cohort	Prenatal PM _{2.5} and NO ₂ from public monitoring data	United States
Shriyan et al.	Ambient and Indoor Air Pollution in Pregnancy and the risk of Low birth weight and Ensuing Effects in Infants (APPLE): A cohort study in Bangalore, South India.	<i>Wellcome Open Res</i>	Study protocol	Paired indoor and ambient PM ₁₀ , PM _{2.5} and CO during pregnancy, nested in the MAASTHI cohort	India
Tamire et al.	Prenatal fine particulate matter (PM _{2.5}) exposure differentially impairs fetal skeletal and soft-tissue growth: a longitudinal study in rural Butajira, Ethiopia.	<i>J Expo Sci Environ Epidemiol</i>	Community-based longitudinal cohort, linear mixed-effects	Repeated 24-h personal PM _{2.5} , Atmotube Pro wearable monitor, at 20, 30 and 36 weeks gestation	Rural Butajira, Ethiopia
Thirumalazhagan et al.	A Comprehensive Meta-Analysis on the Impact of airborne Particulate Matter on Trophoblast Cell Function In Vitro.	<i>Reprod Toxicol</i>	Systematic review + random-effects meta-analysis	PM _{2.5} extracts applied to trophoblast cell lines across 12 in vitro studies	Laboratory (pooled in vitro studies)
Wang et al.	Associations of ASIC2, IKZF3, and GSDMB Variants with Gestational Diabetes Mellitus and Their Interactions with Environmental Exposures Among Pregnant Women.	<i>Biomolecules</i>	Case-control	Pregnancy PM _{2.5} and O ₃ (method not stated); urinary MIBP in a subset	China
Wang et al.	Long-term Exposure to PM _{2.5} and Risk of Adverse Pregnancy Events in Women with Pulmonary Hypertension: A Nationwide Multicenter Study	<i>J Heart Lung Transplant</i>	Retrospective cohort	Monthly PM _{2.5} , 3-year mean before end of pregnancy; trimester windows	China (24 tertiary hospitals)
Wanta et al.	Ultrastructural Changes in Human Placental Tissues and Evidence of Particulate Matter Transport into the Umbilical Vein Following PM _{2.5} Exposure in Chiang Rai.	<i>Cells Tissues Organs</i>	Biomonitoring + in vitro	Third-trimester high-PM _{2.5} season (Feb-Apr) vs no high exposure; SEM-EDS, TEM	Thailand (Mae Chan, Chiang Rai)
Wei Q et al.	Prenatal exposure to PM _{2.5} components and fetal growth: identifying critical gestational weeks and key toxic components in a Chinese birth cohort	<i>Int J Epidemiol</i>	Prospective birth cohort	Weekly black carbon, organic matter, nitrate, sulfate and ammonium from the TAP real-time exposure model, entered into distributed lag models with GEE plus ML mixture analysis	China
Xu ZL et al.	Timing matters: component-specific PM _{2.5} exposure windows and oocyte development in assisted reproduction	<i>Biomed Environ Sci</i>	Retrospective cohort + DLNM	Individual PM _{2.5} and its sulfate, nitrate, ammonium, organic matter and black carbon fractions over recent (0-3 mo), distal (4-12 mo) and cumulative (0-12 mo) windows	China
Zanini et al.	Maternal exposure to air pollution and fetoplacental Doppler markers during pregnancy	<i>Environ Pollut</i>	Metadata only	Maternal residential air-pollution exposure during pregnancy	Europe
Zhang et al.	Association between ambient PM(2.5) exposure during pregnancy and risk of preterm birth: a retrospective birth cohort study in Chengdu, Southwest China.	<i>Front Public Health</i>	Retrospective cohort	Satellite-derived vs monitor-based PM _{2.5} across gestational windows	China (Chengdu)
Zhang et al.	Second-trimester PM _{2.5} and PM ₁₀ exposure is associated with neonatal metabolome alterations and elevated risk of pregnancy loss	<i>Environ Pollut</i>	Prospective cohort	PM _{2.5} and PM ₁₀ in seven windows, pre-pregnancy to peripartum; neonatal metabolites	China

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Respiratory & allergic					
Amazon PM _{2.5} x heat	Interaction Effects of PM _{2.5} and High Temperatures on the Respiratory Health of Vulnerable Age Groups in the Brazilian Amazon	<i>Research Square</i>	Ecological panel	Monthly PM _{2.5} from biomass burning, temperature interaction	Brazil
Bengaluru COPD	Prevalence and Risk Factors of Chronic Obstructive Pulmonary Disease (COPD) in an Urban Poor Locality of Bengaluru, South India - A Cross-sectional Study	<i>Indian J Community Med</i>	Cross-sectional (multistage)	Self-reported cooking fuel, biomass and indoor smoke; no measurement	India
COPD anaemia metals	Air Pollution-Associated Intracellular Metals, Immune Gene Expression, and Erythrocyte Indices in COPD with Anemia	<i>Int J Chron Obstruct Pulmon Dis</i>	Repeated-measures biomarker	Annual PM ₁₀ , PM _{2.5} , NO ₂ , NO _x by land-use regression	Not stated
Cardiorespiratory review	Breathing in harm: current insights into the effects of air pollution on cardiorespiratory health	<i>Med Gas Res</i>	Narrative review	Ambient air pollution, no single exposure metric	Global
Carroll et al.	Global Tuberculosis Burden Attributable to Household and Outdoor Air Pollution	<i>Ann Am Thorac Soc</i>	Systematic review / meta-analysis	Household solid fuel; outdoor PM _{2.5} (short- and long-term)	Global / multi-region (review)
Chen et al.	Residential Green and Blue Space, the Natural Environment, Genetic Susceptibility, and Incident COPD.	<i>Int J Chron Obstruct Pulmon Dis</i>	Prospective cohort	Residential green space, blue space and natural environment at 300 m and 1000 m buffers; no PM term reported	UK
Cui L et al.	Epidemic thunderstorm asthma in China: a systematic review of an Artemisia-associated phenotype	<i>Clin Rev Allergy Immunol</i>	Systematic review / meta-analysis	Airborne pollen as the causal particle; most Chinese reports identify sensitisation only to genus-level Artemisia, without species-level aerobiological confirmation	China
Dinh et al.	Association of air pollution with chronic obstructive pulmonary disease: anemia-related outcomes and exploratory genetic susceptibility analyses.	<i>Front Med (Lausanne)</i>	Case-control (registry)	Residential 1-, 3-, 6- and 12-month PM ₁₀ , PM _{2.5} , NO ₂ , NO, NO _x , SO ₂ , CO and O ₃ at recruitment	Taiwan
Disaster respiratory review	Respiratory emergencies in disasters: a narrative review	<i>Thorac Res Pract</i>	Narrative review	No new measurement; wild-fire PM, silica and ash dust	Global / multi-region
Dou et al.	Risk factors for allergic rhinitis in children: an umbrella review of systematic review and meta-analysis.	<i>Front Immunol</i>	Umbrella review / meta-analysis	Air pollution as one of 75 pooled risk factors (metric not specified in abstract)	Global / multi-region
Gomez-Olarte et al.	Early-life exposure to ambient air pollution and respiratory outcomes: mixture and sex-stratified associations with inflammatory cytokine responses	<i>Environ Pollut</i>	Prospective birth cohort	Residential annual PM _{2.5} , PM ₁₀ , NO ₂ ; per IQR 2.0 / 2.8 / 9.1 ug m ⁻³	Germany
HEPA RCT synthesis	HEPA air filtration for asthma and COPD: lessons from randomized controlled trials.	<i>Expert Rev Respir Med</i>	Narrative synthesis of randomized controlled trials	Portable HEPA filtration as intervention; indoor PM _{2.5} as the mediating exposure	Multi-country trial evidence (PubMed and Scopus to 5 May 2026)
Heat-COPD meta, 36 studies	Heat exposure and its impact on COPD morbidity and mortality: a systematic review and meta-analysis.	<i>Int J Environ Health Res</i>	Systematic review and three-level random-effects meta-analysis	Co-exposure only: air-pollution adjustment tested as a moderator	Global / multi-region
Huang et al.	Global, regional, and 204 countries burden of disease attributed to second-hand smoke-related COPD, 1990-2021: A secondary data analysis of the Global Burden of Disease Study 2021.	<i>Tob Induc Dis</i>	Descriptive burden analysis, GBD 2021	Secondhand tobacco smoke as a fine-particulate exposure, GBD comparative risk assessment	Global, 204 countries
IPF acute PM _{2.5} hospitalisation	Short-term air pollution exposure and hospitalizations in idiopathic pulmonary fibrosis: data from the IPF-PRO Registry.	<i>Ann Am Thorac Soc</i>	Time-stratified case-crossover	Daily PM _{2.5} at the nearest regulatory monitor, lags 0-6	United States

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Johansson et al.	Associations Between Indoor Air Pollution and Asthma, Allergic Rhinitis and Eczema in a Cohort of Children in Durban, South Africa	<i>Indoor Air</i>	Birth-cohort sub-study with indoor measurement	Indoor PM _{2.5} measurements; prenatal PM _{2.5} ; ETS, biomass fuel, dampness by walkthrough	South Africa (Durban)
Kang et al.	Asthma Intervention with Residential Ventilation and Air Cleaner (AIR-VAC) Study: A 4-arm parallel-group randomized controlled trial protocol.	<i>PLoS One</i>	Trial / intervention	Indoor/outdoor PM, NO ₂ , CO, VOCs and mold by low-cost and research-grade sensors	USA (Dallas-Fort Worth, Texas)
Korea 8.25M asthma cohort	Cumulative Vulnerability to Particulate Matter and Incident Asthma: Joint Effects of Urbanicity, Poverty, and Age in a Nationwide Cohort	<i>Allergy</i>	Nationwide cohort	1-km long-term PM _{2.5} and PM ₁₀ estimates	South Korea
Korea paediatric severity IV	Air pollution drives concentrated clinical severity of respiratory illness in early childhood	<i>medRxiv</i>	Registry cohort, FE models	Monthly PM _{2.5} , thermal-inversion instrument	South Korea
Korean J Intern Med	Health effects of air pollution on chronic respiratory diseases.	<i>Korean J Intern Med</i>	Narrative review	Outdoor and indoor air pollutants, life-course framing; no single exposure metric	Not stated
Kulkarni et al.	Short-Term Exposure to PM _{2.5} and Emergency Department Visits for Asthma and Chronic Obstructive Pulmonary Disease.	<i>JAMA Netw Open</i>	Metadata only (no abstract)	Short-term ambient PM _{2.5} (details not deposited)	Singapore
Kuwait tobacco smoke morbidity	Exposure to tobacco smoke, concurrent short-term symptoms of respiratory, allergic, cardiovascular morbidities and health-services use among young adults in Kuwait	<i>BMC Health Serv Res</i>	Cross-sectional survey	Self-reported first-hand and second-hand tobacco smoke	Kuwait
Li H et al.	Multi-omics causal inference of childhood asthma triggered by ambient particulate matter	<i>ERJ Open Res</i>	Mendelian randomisation + TWAS	Genetically instrumented ambient PM _{2.5} from GWAS summary statistics	Europe
Maunakea et al.	Climate-amplified wildland-urban interface disasters create persistent internal metal mixtures associated with pulmonary impairment.	<i>Proc Natl Acad Sci U S A</i>	Observational - cross-sectional	24 urinary metals 6-18 months after the Aug 2023 Maui WUI fires (internal dose, not air)	USA (Maui, Hawaii)
Meng et al.	Exposure-response relationship between ambient PM _{2.5} and SO ₂ with COPD and small airway dysfunction prevalence among individuals at high risk for COPD in Yunnan Province, China: a threshold analysis.	<i>Front Public Health</i>	Cross-sectional (multistage)	Ambient annual PM _{2.5} and SO ₂ , residential assignment	China
MobiliSense ambulatory spirometry	Evaluation of the effects of air pollutants on lung function using ambulatory monitor data from the MobiliSense Project	<i>Research Square</i>	Personal exposure monitoring + panel spirometry	Continuous personal BC and PM _{2.5} with GPS, 15 min-6 h lags	France
Naka et al.	Impact of particulate matter 2.5 exposure on severity of pediatric chronic rhinitis: A prospective cohort study.	<i>Pediatr Allergy Immunol</i>	Prospective cohort	1-week mean ambient PM _{2.5} by inverse-distance weighting of monitors	Thailand (Bangkok)
Nanjing, 62,167 paediatric asthma visits	Extreme Cold Events and Pediatric Asthma Outpatient Visits: The Modifying Role of Ambient Carbon Monoxide and Sulfur Dioxide.	<i>J Asthma Allergy</i>	Time-series	Daily PM _{2.5} and PM ₁₀ as stratifying co-pollutants, not the exposure of interest	China
Narrative review, prenatal and early postnatal window	Early Life Exposure to Air Pollutants and Childhood Asthma.	<i>Curr Allergy Asthma Rep</i>	Narrative review	Prenatal and early postnatal ambient air pollutants, reviewed qualitatively	Global / multi-region
PM _{2.5} lung cancer never-smokers review	Breathing carcinogens: PM _{2.5} air pollution and lung cancer in never-smokers, a narrative review	<i>J Thorac Dis</i>	Narrative review	Narrative synthesis, PM _{2.5} exposure literature 2000-May 2026	Global / multi-region
Samdarshi et al.	Systematic misclassification of lung function in Indian adolescents using GLI-2012 reference equations: evidence from the multicity APEAL cohort.	<i>Thorax</i>	Cross-sectional (cohort baseline)	None at this analysis stage (APEAL air-pollution cohort baseline)	Delhi, Mumbai, Bengaluru and Mysuru, India

First author	Focus	Journal	Design	Metric	Region
San Juan-Reyes et al.	Short-term effects of ambient air pollution exposure on respiratory hospitalizations among children in the State of Mexico.	<i>Environ Sci Pollut Res Int</i>	Case-crossover	Daily PM _{2.5} , PM ₁₀ , CO, NO ₂ and O ₃ from fixed monitoring stations; primary window lag 1-7 cumulative	Mexico
Shi et al.	Hidden Hazards at Home: Indoor Air Pollution and Adolescent Asthma - A Systematic Review and Meta-Analysis	<i>Indoor Air</i>	Systematic review + random-effects meta-analysis	Indoor exposures incl. cooking/heating combustion and open fires	Global / multi-region (review)
Shijiazhuang AECOPD	Seasonal patterns and short-term environmental associations with AECOPD hospitalizations in Shijiazhuang, China, 2020-2024	<i>COPD</i>	Time-series	Daily PM _{2.5} / PM ₁₀ city means; lags 0-7 and 7-day mean	China
Song et al.	Air Pollution Effects on Influenza-like Illness Modified by Temperature and Humidity in Central China	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - air pollution x temperature/humidity, influenza-like illness	China (Central China)
South Africa, 59,017 IPD cases, 4,350 genomes	Pneumococcal population structure influences the effects of air pollution on invasive disease risk in South Africa.	<i>Nat Microbiol</i>	Bayesian spatiotemporal model	Ambient PM _{2.5} , lagged to 8 weeks, linked to serotype and lineage	Sub-Saharan Africa
Tang et al.	Analysis on epidemiology trends of chronic obstructive pulmonary disease in residents aged 40 years and above in Hubei Province, 2014-2024.	<i>Zhonghua Liu Xing Bing Xue Za Zhi</i>	Cross-sectional survey	Self-reported occupational dust and harmful-gas exposure; household coal heating fuel use	China
Thoracic tissue microplastics malignancy	Microplastics in Human Thoracic Tissues: Increased Burden in Patients with Lung Malignancy	<i>J Clin Med</i>	Prospective tissue biomonitoring series	Tissue microplastic particle counts by confocal micro-Raman spectroscopy	Greece
Umea wood smoke chamber	Acute controlled wood smoke exposure induces sensory airway irritation without detectable changes in small airway mechanics in healthy adults: a randomized crossover study	<i>Environ Res</i>	Randomised crossover chamber	Chamber PM ₁ 453 +/- 13 ug m-3, 2 h vs filtered air	Europe
Vance et al.	Greenspace, air pollution and respiratory health outcomes: a systematic review of cohort studies.	<i>Scand J Public Health</i>	Systematic review	Residential greenspace (NDVI and similar) with co-assigned ambient air pollution	Europe
Wangupadcha et al.	Short-term effects of ambient PM _{2.5} on respiratory healthcare utilisation in older adults: a time-series study in an industrialising area of Central Thailand	<i>J Glob Health</i>	Time-series	Daily PM _{2.5} from the nearest government monitor, single-lag and moving-average negative binomial models per 10 µg/m ³	Thailand
Wright et al.	Air pollution and heat impacts on respiratory morbidity and mortality outcomes in Africa: a systematic review towards a meta-analysis	<i>Curr Environ Health Rep</i>	Systematic review / meta-analysis	Reported indoor and ambient PM _{2.5} and nitrogen oxides across included case-control studies; no pooled exposure-response	Sub-Saharan Africa
Xia et al.	Short-term associations between air pollution and nocturnal cough frequency.	<i>Commun Health</i>	Multi-city time-series + random-effects meta-analysis	City-day ambient PM _{2.5} , daily mean	Global / multi-region
Zhang et al.	Associations between household particulate matter and inflammatory cytokines in cystic fibrosis.	<i>Respir Med</i>	Real-time sensor campaign	In-home real-time PM _{2.5} sensors, 4-11 days; mean and % time > 35 µg/m ³	Not stated (single-centre CF pilot)
Mechanistic toxicology					
6PPD-Q AC16 p53	Exploring the targets and mechanisms of 6PPD-Q-induced cardiotoxicity: a network toxicology and molecular docking study	<i>Ecotoxicol Environ Saf</i>	In silico + in vitro	No airborne metric; 6PPD-Q is a road-dust constituent	China

First author	Focus	Journal	Design	Metric	Region
AS-IV vs PM vascular senescence	Replicative and Air Pollution-Induced Senescence: Telomere Dysfunction, Mitochondrial Stress, and Modulatory Effects of Astragaloside IV in Endothelial and Vascular Smooth Muscle Cells	<i>Cardiovasc Toxicol</i>	Toxicological (in vitro)	Certified PM ₁₀ -like urban fine dust, 50 and 100 ug/mL	Computational (no site)
Ameli et al.	Acute co-exposure to urban particulate matter and aircraft noise elicits lung-centered immune-redox proteomic remodeling in mice.	<i>Environ Pollut</i>	Animal model	NIST SRM 1648a urban PM, 3 days, alone and with aircraft noise	Laboratory (C57BL/6J mice)
Arora et al.	Oxidative potential of fresh vs. O ₃ -aged PM _{2.5} across urban and rural sources in China	<i>J Environ Sci (China)</i>	Commentary / methods letter	PM _{2.5} oxidative potential by DTT assay, fresh vs ozone-aged, source-resolved	China
Berberine PM _{2.5} breast cancer	Berberine shows potential in mitigating PM _{2.5} -induced breast cancer progression by inducing DNA damage and inhibiting error-prone DNA repair pathways	<i>Apoptosis</i>	In vitro (cell lines)	Water-soluble extract of ambient PM _{2.5} (Delhi)	India
CSE HASMC isoforms	A comprehensive full-length transcriptome landscape of cigarette smoke-exposed HASMCs reveals extensive remodeling of mRNA isoforms and regulatory networks.	<i>PLoS One</i>	In vitro long-read transcriptomics, dose-response	Cigarette smoke extract 0.2-0.8% on human aortic smooth muscle cells (combustion-aerosol surrogate)	In vitro
Cai et al.	Prenatal PM _{2.5} exposure disrupts cortical excitatory/inhibitory balance in male offspring by suppressing Lhx8-regulated cholinergic homeostasis and cell adhesion	<i>Ecotoxicol Environ Saf</i>	Animal experiment	Prenatal PM _{2.5} (mouse model; dose not in abstract)	Not stated (laboratory)
Cao et al.	Interfacial chemistry and multiscale aging of atmospheric black carbon: from surface reactions to coating, morphology, and model parameterization	<i>Environ Sci Atmos</i>	Narrative / critical review	Black carbon particle state - surface reactivity, coating thickness, morphology and aging timescale - as the link between emission and optical, cloud and toxicological behaviour	Global / multi-region
Chen et al.	A Quasi-Michaelis-Menten Mechanism Explains the Nonlinear Relationship between Transition Metals and PM _{2.5} Oxidative Potential Measured by the Dithiothreitol (DTT) Assay.	<i>Environ Sci Technol</i>	Laboratory metrology	OP-DTT on laboratory single-metal and Cu-Mn mixture solutions and on 62 ambient PM _{2.5} filter samples	China
Cherriere et al.	Influence of lung overload and exposure scenarios on recovery following inhalation of an alumina nanoparticle and HCl gas mixture.	<i>Part Fibre Toxicol</i>	Animal inhalation exposure + dosimetry	Alumina nanoparticles 1-20 mg/m ³ co-exposed with 5 ppm HCl, nose-only inhalation	Laboratory (rat inhalation, France)
Driencourt et al.	The impact of post-exposure Streptococcus pneumoniae infections on pulmonary immunity: The role of air pollution and IL-20 cytokines in respiratory infections.	<i>Rev Mal Respir</i>	Narrative review	Traffic-derived PM _{2.5} (conceptual)	France
Dulong et al.	Effects of fine particulate matter on neuroblastoma cell line: metabolic and lipid deregulations and their consequences on the onset of neurological diseases.	<i>Biochimie</i>	Narrative review	PM _{2.5} , in vitro dosing; no harmonised mass concentration or duration reported	Global
Fetal macrophage FCER1G	Prenatal PM _{2.5} exposure drives epigenetic reprogramming of fetal macrophages linked to atopic dermatitis	<i>Nat Commun</i>	Birth cohort + multi-omics	First-trimester ambient PM _{2.5} , modelled residential	South Korea
Gao et al.	Integrative transcriptomic analysis identifies CCL22-associated immune signatures in air pollution-related atopic dermatitis.	<i>Front Public Health</i>	Integrative transcriptomics + murine validation	No measured PM; pollutant-associated gene sets taken from a toxicogenomic database and intersected with atopic-dermatitis transcriptomes	Not stated

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Ghio et al.	Iron, Tuberculosis, and Particle-Related Exposures Including Smoking.	<i>Pathogens</i>	Mechanistic review	Cigarette smoke, ambient PM, biomass smoke, silica (conceptual)	Global / multi-region (review)
Gradinaru et al.	Atmospheric Particulate Matter as a Carrier of Pb, Cd, and Ni: From Environmental Transfer and Bioaccessibility to Molecular Toxicity and Predictive Modeling.	<i>J Xenobiot</i>	Narrative review	PM-bound Pb, Cd and Ni across the source-to-effect continuum	Global / multi-region
Gulyaev et al.	Association between polyphenols and inhalation-induced oxidative stress in chronic lung diseases: A systematic review.	<i>Biomed Rep</i>	Systematic review	Inhalation-induced oxidative stress across in vitro, animal and clinical exposure models	Global / multi-region
Gulzar et al.	Mechanistic Insights into PM _{2.5} -Induced Cardiopulmonary Dysfunction: Bridging Pathways to Therapeutic Interventions	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - PM _{2.5} cardiopulmonary mechanisms (review by title)	Not recoverable from metadata
Han G et al. 2026	Picropodophyllotoxin attenuates PM _{2.5} -induced pulmonary injury via modulation of oxidative stress and mTOR-mediated autophagy	<i>Research Square (preprint)</i>	Experimental / toxicology	PM _{2.5} dosed in vitro at 25-100 ug/mL	East Asia (ex-China)
Heo et al.	PM ₁₀ exposure impairs differentiation and induces inflammation in human small airway epithelial cells cultured at the air-liquid interface	<i>Tuberc Respir Dis (Seoul)</i>	In vitro exposure	Repeated PM ₁₀ dosing of differentiated air-liquid-interface cultures; TEER, lineage markers, surfactant proteins, cytokines, Notch and NF-κB signalling	South Korea
IL23A / NF-κB lung tumour	IL23A acts as a pivotal mediator of PM _{2.5} -induced lung tumorigenesis through the NF-κB pathway activation	<i>Ecotoxicol Environ Saf</i>	Animal + in vitro	PM _{2.5} extract exposure, NSCLC lines + orthotopic mice	China
Jiang et al.	Nitration of Urban PM _{2.5} : A Hidden Driver of Enhanced Human Inflammation.	<i>Environ Sci Technol</i>	Exposome-wide association	Personal PM _{2.5} filters, 1000 nitrated aromatic compounds quantified by HRMS	China (Beijing)
Jiang et al.	Identification of shared and divergent molecular pathways linking ambient air pollutants to mortality and major chronic diseases through proteomic profiling	<i>Redox Biol</i>	Cohort proteomics + mediation + Mendelian randomization	Residential PM _{2.5} , PM ₁₀ , NO ₂ , NOx (LUR)	United Kingdom (UK Biobank)
Kamyabi et al.	A novel controlled wood smoke exposure model to evaluate physiological responses to smoke-derived air pollution in dairy calves.	<i>J Dairy Sci</i>	Animal experiment	6-h wood smoke in enclosures: PM _{2.5} 238.44 +/- 62.97 vs 9.94 +/- 4.50 µg/m ³	USA
Kou et al.	Pollutants regulate changes in pathological markers of neurodegenerative diseases: a new perspective in environmental toxicology.	<i>Front Toxicol</i>	Narrative review	Air pollutants treated as one class alongside metals and POPs; no exposure metric	Global
Lang et al.	Particulate matter selectively promotes macropinocytosis and malignant transformation-associated phenotypes in p53-impaired epithelial cells.	<i>Environ Toxicol Pharmacol</i>	In vitro (cell lines)	European certified reference PM _{2.5} , acute and repeated exposure	Laboratory (murine mammary, hamster respiratory cells)
Lin et al.	Air Pollution Triggers Metabolomic Signature of Mitochondrial Dysfunction in Mice and Humans: Implications for Cardiovascular Disease Development.	<i>Arterioscler Thromb Vasc Biol</i>	Animal + in vitro	Diesel exhaust (ApoE ^{-/-} mice, 2 wk) and a natural LA-to-Beijing move (PM _{2.5} 14.4 -> 67.6 µg/m ³ , 10 wk)	USA and China (Los Angeles-Beijing travel panel)
Ma et al.	Combined PM _{2.5} and microplastics exposure exacerbates allergic asthma via OPA1/SLC7A11-mediated airway epithelial ferroptosis.	<i>Ecotoxicol Environ Saf</i>	Animal + in vitro	PM _{2.5} and microplastic co-exposure, ovalbumin-sensitised murine model and MRE cells	China

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Meganathan et al.	Aryl hydrocarbon receptor (AhR) regulates organic dust and organic dust-associated bacterial extracellular vesicles-induced airway inflammation via ROS-NFkB signaling.	<i>Respir Res</i>	Animal + in vitro	Poultry-farm organic dust extract and dust-associated bacterial EVs (in vitro, murine)	Laboratory (Beas2B cells, AhR knockout mice)
Meldrum & Leonard	Pollutant particle priming amplifies airway compartment specific neutrophil and macrophage inflammatory programs following house dust mite acute exposure	<i>Front Immunol</i>	Murine inhalation exposure	Diesel exhaust particles, temporally separated priming dose before allergen challenge	United Kingdom
Messina et al.	Organ-specific apoptotic pathways induced by micro- and nanoplastics: a systematic review of in vitro toxicological evidence.	<i>Front Toxicol</i>	Systematic review (PRISMA)	Micro- and nanoplastic particles across cell and organoid models (airborne fraction not separated)	Not applicable (literature synthesis)
Monge et al.	Heart Function Changes During Environmentally Relevant Combined Exposure to Heat, Ultrafine Carbon Black, and Ozone: Role of NOD-like Receptor X1.	<i>Cells</i>	Animal model	Ultrafine carbon black 2.5 mg/m ³ + O ₃ 1 ppm, with or without 36 C heat, 3 h	Laboratory (Nlr1 knockout mice)
Na HW et al.	Adverse effects of airborne heavy metals on human skin cells: insights from sebum analysis of indoor and outdoor workers	<i>J Toxicol</i>	Biomonitoring + in vitro	24 metals quantified by ICP-MS in sebum, then reconstituted as sebum-mimicking metal mixtures at the measured indoor and outdoor profiles	China
Nasal PM _{2.5} testicular pyroptosis	Nasal Exposure to Environmental PM _{2.5} Drives Testicular Pyroptosis and Tissue Damage: Evidence from Inhibitor Intervention and Multi-Source Data	<i>Res Sq (preprint)</i>	Animal model	Intranasal PM _{2.5} , 10 mg/kg, 4 weeks	Computational (no site)
PM diabetic kidney MR transcriptomics	Particulate Matter Exposure and Diabetic Kidney Dysfunction: Insights from Integrated Transcriptomic and Bioinformatics Analyses	<i>Int J Mol Sci</i>	Mendelian randomization + TWAS	PM _{2.5} -10 coarse-fraction genetic instruments	Global / multi-region
PM skin AhR-Nrf2 framework	Particulate Matter-Induced Skin Injury: A Dual-Pathway AhR-Nrf2 Framework for Epidermal Homeostasis and Therapeutic Targeting	<i>Int J Mol Sci</i>	Mechanistic review	PM-bound reactive oxygen species, transition metals and PAHs	Taiwan
Pei et al.	Deciphering Allergenic Components Responsible for Differential Sensitization Potentials of Source-Specific Atmospheric Particulate Matter	<i>Environ Sci Technol</i>	In vitro exposure	NIST SRMs: urban PM 1648a, coal fly ash 2693, diesel exhaust particles 2975	Laboratory (THP-1, HaCaT co-culture)
Ren Y et al.	Unveiling particulate matter-lung interactions from static models to dynamic lung-on-chips	<i>Research (Wash D C)</i>	Method review	Compares 2D culture, 3D bioprinted constructs, organoids and lung-on-chip platforms for reproducing PM deposition and breath-mimicking mechanical strain	Global / multi-region
Retinal CRM	Environmental particulate reference materials increase DCFDA fluorescence and reduce ATP-dependent viability in primary mouse retinal cells	<i>Biochem Biophys Rep</i>	Experimental / toxicology	Two certified environmental particulate reference materials, dosed	Laboratory
Scrap tire valorization risks	Is recycling sustainable: The promise and risks of scrap tire valorization.	<i>iScience</i>	Narrative review, technical and toxicological	Process emissions: VOCs, PAHs, heavy metals, microplastics, 6PPD-quinone	Global / multi-region
Shen et al.	A Brake on Toxicity: Iron Suppresses Organic-Induced Superoxide Generation in Macrophages	<i>Environ Sci Technol</i>	In vitro exposure	Brake-wear particles, highway PM _{2.5} , Fe ²⁺ /Fe ³⁺ with isoprene SOA or 9,10-PQN	Laboratory (macrophage cell culture)

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Song et al.	Lung Toxicity Induced by Co-Exposure to Particulate Matter and Indoor Microorganisms in Mice.	<i>Toxicol Lett</i>	Animal experiment	PM plus dominant indoor microorganisms by intratracheal instillation, 2 weeks	Laboratory (C57BL/6 mice)
Sprague Dawley whole-body inhalation, GD5-19	Gestational inhalation of nanoparticles disrupts placental zone structure and induces vascular placentation in rats.	<i>Reprod Toxicol</i>	Animal model	nano-TiO ₂ aerosol, 9.23 +/- 0.39 mg/m ³ whole-body, as an ultrafine PM proxy	Laboratory
Tire wear particle phytotoxicity	Tire wear particles impact a roadside wild plant species	<i>Curr Biol</i>	Controlled exposure (ecotox)	Tire wear particles, non-exhaust traffic source	Japan
Tzagkaroulaki et al.	An A549 Cell-Based Approach Using Repeated Fluorescence Readouts for Assessing Reactive Oxygen Species Activity of Atmospheric Particulate Matter.	<i>Toxics</i>	Analytical method validation	NIST SRM 2584 and field PM _{2.5} (urban-background, high-altitude) on PTFE and quartz	Greece
UPM instillation reproduction null	Repeated Short-Term Intratracheal Exposure to Urban Particulate Matter Has No Apparent Adverse Effects on Female Reproductive Outcomes in Mice	<i>Cells</i>	Animal experiment	Intratracheal urban PM instillation, 40-200 mg/kg over one month	South Korea
Vancouver DE blood-lung concordance	Concordant transcriptomics analysis of the lungs and blood after exposure to diesel exhaust: a randomized, controlled human exposure study	<i>Environ Res</i>	Randomised crossover chamber	Controlled 2-h diesel exhaust vs filtered air	Canada
Vera-Colon et al.	Exposure to airborne fine particulate matter disrupts human embryonic stem cell osteogenic differentiation.	<i>Part Fibre Toxicol</i>	Metadata only (no abstract)	PM composition, dose and collection route not recoverable from metadata	Not stated
Wang Y et al.	Contrasting cellular reactive oxygen species modulation by alpha-dicarbonyl aqueous reaction products via interactions with redox-active particulate matter	<i>Environ Sci Technol Lett</i>	Bench / device experiment	Aqueous reaction products of glyoxal or methylglyoxal with ammonium sulfate or glycine, characterised by HRMS, then co-exposed with redox-active PM surrogates	Laboratory
Wang et al.	Ozone and PM _{2.5} co-exposure exacerbates allergic rhinitis and is associated with PPAR/TRPV1-related barrier damage: Insights from nasal transcriptomic and microbial analyses.	<i>Environ Pollut</i>	Animal model	Acute high-dose PM _{2.5} and O ₃ , single and combined, in OVA-sensitised mice	Laboratory (murine)
Wu et al.	Beta-carotene alleviates PM _{2.5} -induced pulmonary fibrosis via suppression of the Frizzled5-mediated Wnt5a/Ca ²⁺ pathway.	<i>Ecotoxicol Environ Saf</i>	Animal exposure study + in vitro receptor overexpression	Intratracheal PM _{2.5} instillation; dose, source and treatment concentration not reported	China
Yang et al.	The Human Microbiome in Airborne Particulate Matter-Associated Aging-Related Chronic Diseases: An Evidence-Stratified Review.	<i>Environ Health (Wash)</i>	Mechanistic review	Airborne PM (size fractions not separated across the reviewed literature)	Global / multi-region (literature synthesis)
Yang et al.	Molecular Mechanisms of Indoor Environmental Pollution tt-DDE-Induced Blood Pressure Elevation and the Protective Role of Human-Specific Gene CHRFAM7A.	<i>Environ Health (Wash)</i>	Animal + in vitro	trans,trans-2,4-decadienal, the dominant aldehyde in cooking-oil fumes, by inhalation	Laboratory (transgenic mice and endothelial cells, China)
Ye et al.	Mechanistic Study of SRSF1-Mediated Inhibition of Treg Cell Function via the mTORC1/ULK1 Signalling Axis Through Mitophagy in Asthma.	<i>Clin Exp Pharmacol Physiol</i>	Animal model	PM _{2.5} instillation arm added to OVA asthma model (dose not reported in abstract)	Laboratory (murine plus patient BALF)
e-waste children mucosal immunity	iNOS/Arg-1 balance and sIgA levels are associated with intestinal mucosal immune defense in children chronically exposed to air pollutants and lead.	<i>Front Immunol</i>	Cross-sectional + ML	Estimated average daily doses of PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ and CO plus blood Pb	China

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Occupational & indoor					
Albert et al.	Assessing home environmental hazards for older adults receiving in-home supportive services: the healthy homes assessment tool.	<i>Innov Aging</i>	Cross-sectional survey	Consumer-grade indoor PM _{2.5} and radon monitors vs care-coordinator hazard ratings	USA (older-adult households)
Assan et al.	Evaluation of Ambient Air Particulate Matter, Urine Arsenic Levels and the Effect on Residents Health in an Artisanal Mining Community and an Incorporated Mining Community in Ghana.	<i>Health Sci Rep</i>	Cross-sectional (comparative community)	Ambient PM _{2.5} and PM ₁₀ by Modular Air Quality System; urinary arsenic by atomic absorption spectrophotometry	Ghana
Auditorium IAQ + 16S bioaerosol	Indoor Air Quality (IAQ) analysis and 16S rRNA gene sequencing of indoor air pollutants in a 3000-capacity religious auditorium	<i>Sci Total Environ</i>	Metadata only (no abstract)	Metadata only (no abstract deposited)	Global / multi-region
Bi/Bi2Te3 workplace ICP-OES	Determination of bismuth and bismuth telluride in workplace air by ultra microwave digestion with inductively coupled plasma optical emission spectrometry	<i>Zhonghua Lao Dong Wei Sheng Zhi Ye Bing Za Zhi</i>	Analytical method validation	MCE filter, 75 L; Bi and Te by ICP-OES after digestion	China
Build Environ 2026 (cleanroom)	Multi-objective ventilation optimization in non-unidirectional cleanrooms: an integrated RSM-NSGA-II-TOPSIS approach	<i>Build Environ</i>	Metadata only (no abstract)	Not recoverable from metadata	Not recoverable from metadata
Build Environ 2026 (subway)	Spatiotemporal distribution characteristics of multiple air pollutants within subway stations	<i>Build Environ</i>	Metadata only (no abstract)	Not recoverable from metadata	Not recoverable from metadata
CFRP inhalation risk review	Inhalation risks of carbon fiber-reinforced plastic-derived emissions: Occupational exposure and regulatory perspectives	<i>Regul Toxicol Pharmacol</i>	Critical review	Respirable CFRP fibre fragments and thermal ultrafines	Global
Campiotti et al.	Beyond leaf traits: context dependent particulate matter accumulation across outdoor, indoor and active green walls	<i>Build Environ</i>	Metadata only (no abstract)	Metadata only - PM deposition on outdoor, indoor and active green walls (by title)	Not recoverable from metadata
Chen Y et al.	Household solid fuel combustion-related PAH exposure in rural northern China: associations of phenanthrene metabolites with DNA damage and systemic inflammation	<i>Atmos Environ</i>	Metadata only	Household solid-fuel PAH exposure indexed by urinary phenanthrene metabolites	China
China 2021 sandstorm, Baidu search DID	Indoor Air Protection Under Extreme Air Pollution Risk: Evidence From a Sandstorm in China.	<i>Indoor Air</i>	Ecological panel	Sandstorm-driven outdoor PM episode as an exogenous shock; no PM measured directly	China
China gym co-exposure	Association of indoor multiple airborne pollutants co-exposure and typical respiratory symptoms among gym employees: A nationwide observational study in China	<i>Environ Pollut</i>	Cross-sectional survey	Indoor PM _{2.5} , PM ₁₀ , total fungal and total plate count	China
Dennis & Ball	Strategies to Improve Compliance With Smoke Safety.	<i>AORN J</i>	Narrative review	Surgical smoke (particles, gases) from energy-based devices	USA
Derwein et al.	Combined effects of face mask blocking, filtration, and ventilation strategy on exhaled air transmission and infection risk	<i>Build Environ</i>	Metadata only (no abstract)	Not retrievable: the publisher deposited no abstract to Crossref, Europe PMC or Semantic Scholar at index time	Not stated (metadata only)
Dutch school air cleaner RCT	Evaluating air cleaner effectiveness in schools: a comprehensive review and protocol of a cluster-randomized controlled trial of physicochemical and microbial markers	<i>Front Public Health</i>	Cluster-randomised trial protocol	Continuous PM ₁₀ , PM ₄ , PM _{2.5} , PM ₁ plus CO ₂ , T, RH, VOC	Netherlands
Eghomwanre & Akharam	Spatio-temporal assessment of classroom air quality and thermal comfort as determinants of respiratory health risks in urban pre-tertiary institutions.	<i>Environ Sci Pollut Res Int</i>	Comparative field monitoring (built environment)	Classroom PM _{2.5} , PM ₁₀ , CO, CO ₂ and thermal indices, six schools, 10 weeks	Nigeria (Benin City)

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Environ Int	Portable HEPA air filtration effects on particulate matter, radioactivity, and chemical constituents in occupied buildings.	<i>Environ Int</i>	Metadata only (no abstract)	Not recoverable from the deposited record	Not stated
Environ Int 2026 (button mfg)	Plastic button manufacturing as an under-recognized source of airborne multi-pollutant mixtures: process-resolved characterization	<i>Environ Int</i>	Metadata only (no abstract)	Not recoverable from metadata	Not recoverable from metadata
Ethiopia dormitory ventilation	Occupant-driven ventilation and indoor exposure dynamics in naturally ventilated Ethiopian university dormitories	<i>Indoor Environments</i>	Metadata only (no abstract)	Indoor exposure under occupant-driven natural ventilation	Ethiopia
Europe indoor PAH children	Polycyclic aromatic hydrocarbon (PAH) levels in indoor air and their associated health risks for children across Europe	<i>Environ Res</i>	Multi-city indoor survey	Active indoor air sampling plus deposited dust, 12 PAHs	Europe
Feng R et al.	Clean heating renovation differentially reduces nitro-PAH and nitrated phenol exposure: implications from real-world personal monitoring and NF-kB-based toxicity	<i>Environ Sci Technol Lett</i>	Personal exposure monitoring + in vitro	Size-resolved personal exposure to particulate nitrated phenols and nitro-PAHs among users of clean coal, raw coal chunk and biomass	China
Figueiredo et al.	Chemical and toxicological assessment of particulate matter in a charcoal barbecue restaurant.	<i>Environ Toxicol Pharmacol</i>	Measurement campaign	Simultaneous indoor/outdoor PM ₁₀ filters: ions, OC/EC, organic speciation, OP (AA, DTT)	Portugal
Fire-station diesel, 3 stations	Characterization of diesel-related pollutant migration in fire stations using low-cost direct-reading monitors.	<i>J Occup Environ Hyg</i>	Repeated-measures monitoring, mixed-effects models	Minute-level PM size fractions, NO ₂ , VOCs and CO ₂ from direct-reading low-cost monitors, 7 days per station	United States
Gaillard et al.	Potential Link Between Indoor Air Quality and Coronary Artery Disease in Amsterdam, a Field Study	<i>Preprints.org (preprint)</i>	Long-term indoor monitoring campaign	Multi-week in-home PM _{2.5} , PM ₁₀ and CO ₂ monitoring	Netherlands (Amsterdam)
Gulzar et al.	PAHs Exposure, Health Risk Evaluation, and Renal Associations by Household Fuel Type in Rural Residents of Northwestern China	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - household-fuel PAH exposure and renal indicators (by title)	China (north-western, rural)
Hospital IAQ, 1.79M records 2023-2025	Toward Integrated Hospital IAQ Monitoring: Continuous Sensing and Targeted Chemical Characterization.	<i>Atmosphere</i>	Long-term continuous sensing plus targeted chemical sampling	CO ₂ , PM _{2.5} , PM ₁₀ , formaldehyde, T, RH, TVOC from low-cost sensors; June 2023 - December 2025	Europe
Hu et al.	Analysis of carbon fiber dust exposure status and related occupational health risk factors among workers during the molding process of carbon fiber products.	<i>Front Public Health</i>	Observational - cross-sectional	Workplace carbon fibre dust measurement at molding workstations, ICMM quantitative risk tiering	China
Ibrahim	From complaint-driven response to preventive management: Strengthening indoor air quality management in Malaysian buildings	<i>Qeios (preprint)</i>	Commentary / methods letter	Argues for routine indoor monitoring of particulate matter and outdoor-to-indoor transport alongside CO ₂ , rather than complaint-triggered investigation under ICOP IAQ 2010	Malaysia
Ikeda et al.	Characterizing layout-dependent breathing-zone contaminant concentration in office spaces with mixing ventilation using an occupant-to-exhaust distance metric.	<i>Build Environ</i>	Metadata only (no abstract)	Breathing-zone contaminant concentration under mixing ventilation; no quantities retrievable	Japan

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Indoor OCEC hazard synthesis	Organic Contaminants of Emerging Concern (OCECs) in Indoor Air: A Perspective on Global Distribution and Hazard Burden.	<i>Environ Sci Technol</i>	Systematic synthesis with inhalation hazard prioritisation	Airborne concentrations of 130+ compounds across seven OCEC classes	Global / multi-region
JECS indoor PM Kawasaki	Household-specific indoor PM _{2.5} and PM _{10-2.5} concentrations and Kawasaki disease onset by age six: the Japan Environment and Children's Study	<i>Int J Hyg Environ Health</i>	Birth-cohort sub-study with indoor measurement	Household indoor PM _{2.5} and PM _{10-2.5} , 1-week gravimetric at ages 1.5 and 3.0 y	Japan
JP laboratory exposure	Assessment of personal exposure and spatial concentration variability in a small-scale chemical laboratory.	<i>Ann Work Expo Health</i>	Repeated field measurement with CFD simulation	Paired fixed-point area sampling and personal passive sampling, plus computational fluid dynamics	Japan (university teaching laboratory, 2024 and 2025 campaigns)
Jaiswal et al.	Using nature-based solutions in mitigating indoor air quality: a case study of EDM emission from a gear manufacturing unit, India	<i>Research Square (preprint)</i>	Trial / intervention	Indoor PM _{2.5} and CO before and after five months of NBS filtration plus ventilation redesign	India
Jamali et al.	Wooden-framed all-cellulose HEPA nanofilter for sustainable air purification	<i>Int J Biol Macromol</i>	Laboratory instrument development	PM0.3 removal efficiency and pressure drop of a fog-sprayed, freeze-dried bacterial-cellulose nanofibre layer on a cellulose-microfibre substrate	Laboratory
Khan et al.	Exposure to cement industry emissions and respiratory health outcomes: a global systematic review and meta-analysis from 2000 to 2025.	<i>Front Public Health</i>	Systematic review + random-effects meta-analysis	Cement dust and cement-plant emissions, occupational and residential; no harmonised gravimetric metric across studies	Global / multi-region
Kusi et al.	Mechanistic pathways of biochemical and toxicological responses to waste tyre combustion emissions: A systematic review	<i>Research Square (preprint)</i>	Systematic review (PRISMA)	Particulate matter, PAHs, VOCs and heavy metals from open waste-tyre burning, mapped to organ-specific biochemical and toxicological endpoints	Global / multi-region
Lee et al.	Comparative analysis of metal concentrations in respirable particles and nanoparticulate matter (NPM) from mild steel, aluminum, and stainless steel welding fumes.	<i>J Hazard Mater</i>	Field measurement	Breathing-zone respirable and NRD nanoparticle samplers; Mn, Fe, Cr, Ni, Cu, Al by ICP-OES	Not stated (manufacturing facilities)
Liu	Energy-Indoor Air Quality-Health Nexus in Rural Residential Retrofits: A Field-Calibrated Coupled Simulation and Multi-Criteria Evaluation for the Central Plains Transitional Climate of China	<i>Atmosphere</i>	Repeated field measurement with CFD simulation	72-h paired before/after indoor PM _{2.5} , PM ₁₀ , CO ₂ , HCHO, TVOC; EnergyPlus-CONTAM co-simulation	China (Anyang, Henan)
Liu et al.	Spatiotemporal Contamination Characteristics, Bacterial Community Structure, and Airflow-Mediated Transport of Bacterial Aerosols in a Pediatric Emergency Consulting Room.	<i>Environ Res</i>	Repeated field measurement with CFD simulation	Size-resolved culturable bacterial aerosol (Andersen-type stages), CFD deposition	China (paediatric hospital)
Liu et al.	Research on Comprehensive Evaluation of Air Quality in Underground Stations of Urban Rail Transit Systems	<i>Atmosphere</i>	Repeated field measurement with CFD simulation	Handheld PM _{2.5} , PM ₁₀ , CO ₂ , TVOC at CFD-selected points; one island-platform station	China (Beijing Metro)

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Lv H et al.	Structurally engineered polyimide microsphere/nanofibrous gradient aerogels with enhanced dust-holding capacity for high-temperature air filtration	<i>ACS Appl Mater Interfaces</i>	Laboratory instrument development	PM0.3 filtration efficiency, pressure drop and dust-holding capacity of a single-component polyimide gradient aerogel	Laboratory
Lv et al.	Impact of biomass stove intervention on indoor air quality in rural homes of China: Attenuation of cooking contribution.	<i>J Hazard Mater</i>	Trial / intervention	Real-time PM _{2.5} monitors in kitchen, living room, bedroom and yard; pre/post stove change	China (Xuanwei, Yunnan)
Ma L et al.	Evaluating surgical smoke in laser-assisted laparoscopic partial nephrectomy with a novel handle: an in vitro study	<i>Can J Urol</i>	Bench / device experiment	Surgical-smoke formaldehyde, TVOC and CO with and without an integrated aspirator handle	Laboratory
Ma et al.	Association between combined occupational exposure to coal dust and noise and high-frequency hearing loss among workers: a cross-sectional study.	<i>Front Public Health</i>	Cross-sectional survey	Job-category coal dust exposure, dichotomous, combined with occupational noise; no gravimetric metric	China
Mbazima et al.	Exposure and respiratory tract deposition of particulate matter among teachers using chalkboards in selected public secondary schools in the Johannesburg region	<i>Int J Environ Health Res</i>	Measurement campaign	Respirable PM4, personal breathing-zone sampling + MPPD deposition model	Sub-Saharan Africa
Micallef & Scerri	A flow conditioned infiltration metric for traffic-related air pollution ingress and indoor air quality in naturally ventilated buildings	<i>Build Environ</i>	Metadata only	Outdoor-to-indoor infiltration of traffic-related air pollution, conditioned on flow regime	Europe
Occupational nanoparticle scoping	Occupational nanoparticle exposure and its effects on pulmonary health, cardiovascular health, and systemic biomarkers: a scoping review	<i>Part Fibre Toxicol</i>	Scoping review	Engineered nanomaterials and workplace ultrafine particles	Global
Oklahoma cannabis facility IAQ	Multi-parameter characterization of indoor air quality in a medical marijuana cultivation facility in Oklahoma: a pilot study of occupational exposure determinants	<i>Ann Work Expo Health</i>	Multi-instrument field evaluation	Size-resolved PM by TSI AirAssure + 2 PurpleAir PA-II-FLEX	United States
Opbroek et al.	Health effects of indoor environmental quality (IEQ): A systematic review of indoor environments in high-income settings.	<i>Sci Total Environ</i>	Metadata only (no abstract)	Metadata only - indoor environmental quality parameters (scope not recoverable)	Not recoverable from metadata
PSH smoke-free home RCT	Smoke-Free Home Intervention in Permanent Supportive Housing: A Cluster Randomized Clinical Trial	<i>JAMA Netw Open</i>	Cluster-randomised controlled trial	Secondhand smoke in multiunit housing, CO-verified	USA
Perfilyeva et al.	Genetic Susceptibility to Impaired Lung Function in Underground Coal Miners from Kazakhstan.	<i>Genes (Basel)</i>	Cross-sectional (cohort baseline)	Occupational coal mine dust (mining exposure as covariate)	Kazakhstan (Karaganda coal basin)
Pessoa Campos et al.	Salivary 8-hydroxy-2'-deoxyguanosine as a biomarker of oxidative DNA damage in urban outdoor workers exposed to traffic-related air pollution.	<i>J Toxicol Environ Health A</i>	Cross-sectional (comparative community)	Workplace PM _{2.5} and noise measured with portable instruments; no personal sampling	Not stated (urban street-vending and indoor control workplaces)
Qian et al.	Impacts of Housing Retrofits on Indoor Environmental Quality and Energy Consumption: A Review	<i>Indoor Air</i>	Narrative review	Indoor PM _{2.5} , VOCs, NO ₂ , HCHO, radon, CO ₂ , T and RH before and after retrofit	Global / multi-region
Rouyn-Noranda citizen dust	Metal contamination in residential snow and indoor dust near the Horne Smelter: a citizen-science study in Rouyn-Noranda, Quebec	<i>Environ Monit Assess</i>	Citizen-science sampling	Cu, Pb, Cd, As, Zn, Cr in snow and settled indoor dust	Canada

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SAA wetting agent	High-efficiency 'SAA' wetting agent for coal mine dust control: from preparation and mechanism to field application.	<i>Environ Geochem Health</i>	Laboratory formulation and mechanism study with underground field trial	Respirable coal dust; median particle size D50 and dust suppression rate	China (3-1802 working face, deep coal seam)
Srinivasan et al.	Occupational wood dust exposure and lung function impairment: a systematic review and meta-analysis	<i>Natl Med J India</i>	Systematic review / meta-analysis	Occupational wood dust exposure, categorical (exposed vs unexposed); no quantitative dust concentration pooled	Global / multi-region
Tam et al.	Toxic Air Contaminant Emissions from Residential and Commercial Fossil Gas Appliances: A Systematic Review.	<i>Curr Environ Health Rep</i>	Systematic review (PRISMA)	Appliance emission factors for 41 toxic air contaminants, leakage and combustion pathways	Global / multi-region
Tangshan CWP IGF1	Effects of dust exposure on IGF1 and autophagy related to coal workers' pneumoconiosis	<i>Front Public Health</i>	Experimental / toxicology	Cumulative occupational coal dust by job history; SiO ₂ in vitro	China
Tanzania house-design IAQ	A comparative study to investigate the impact of novel house design on indoor air quality in rural Tanzania	<i>Research Square</i>	Comparative field monitoring (built environment)	Continuous indoor PM _{2.5} and PM ₁₀ , 77-day deployment	Tanzania
Taskin and Ozkal	Investigating Airborne Pollutant Dynamics in a Pathology Laboratory Using an Integrated High-Resolution Sensor Network and Operational-Context Framework.	<i>Indoor Air</i>	Real-time sensor campaign	NI myRIO multi-sensor node, 1-min HCHO, eTVOC, eCO ₂ , PM, T, RH; 42 days	Not stated (university hospital pathology laboratory)
Turkish cigarette PM chamber	Particulate Matter Emissions of Turkish Cigarettes in Comparison to German Brands - An Aerosol Spectrometric Study	<i>Tob Use Insights</i>	Chamber emission characterisation	PM ₁₀ , PM _{2.5} , PM1 by laser aerosol spectroscopy, 2.88 m ³ chamber	Turkiye
Wang et al.	Research on dust control and ventilation parameter optimization in roadhead mining based on response surface methodology.	<i>PLoS One</i>	CFD simulation + scaled-model validation	Simulated dust mass concentration at return-air-side corners of a heading face	China (underground coal heading face)
Wang et al.	Aeration change from horizontal brush to fine bubble substantially reduces the emission of inhalable antibiotic resistome from wastewater to air.	<i>J Hazard Mater</i>	Field measurement	PM _{2.5} metagenomics (ARGs, pathogens) under two aeration modes, seasonal	Not stated (single WWTP)
Wang et al.	Mixed Exposure to Underground Air Pollutants and Metabolic Syndrome in Coal Miners: A Cross-Sectional Study Integrating Multi-Pollutant Models and Urinary Metabolomics.	<i>Toxics</i>	Observational - cross-sectional	Underground coal dust, CO, CO ₂ , NO ₂ (job-exposure assignment; method not in abstract)	China (coal mines)
Wei J et al.	Welding fume exposure and respiratory innate immune activation: metal-rich particles, inflammatory biomarkers, and immune-informed occupational prevention	<i>Front Immunol</i>	Mechanistic review	Metal-rich fine and ultrafine welding-fume particles as a stimulus to airway epithelium, alveolar macrophages, neutrophils and circulating immune cells	Global / multi-region
Wu et al.	Sphagnum palustre moss bags as sentinels to monitor indoor air PAHs in university libraries.	<i>Int J Phytoremediation</i>	Field measurement	Sphagnum moss bags, 45-day deployments, total PAHs (ug/kg moss), indoor and outdoor	Not stated (three university libraries)
Xing et al.	Household air pollution associated with different heating approaches in rural Northern China.	<i>Sci Bull (Beijing)</i>	Long-term field measurement campaign	Year-round indoor PM _{2.5} monitoring in 948 households, plus exposure modelling	China (North China Plain, rural)

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Yang et al.	Optimization of ventilation systems in large welding workshops with multiple dust sources: a case study of a 300 m long welding workshop	<i>Atmosphere</i>	Chemical transport modelling	Welding fume transport by CFD; low-density aluminium vs high-density carbon-steel fume	China
Yoshizaki et al.	Differences in the Effectiveness of Smoke Evacuators Types for Reducing Surgical Smoke: A Randomized Controlled Trial.	<i>J Am Coll Surg</i>	Randomised controlled trial	Optical particle counts at 0.3, 0.5, 1.0 and 5.0 µm plus acetaldehyde and formaldehyde, 10-min sampling	Japan
Yu et al.	Development of a Qualitative Questionnaire for Indoor Air Quality Monitoring: A Scoping Review of Subjective Assessments in Studies Using Low-Cost Sensors	<i>Atmosphere</i>	Scoping review	Studies pairing low-cost IAQ sensors with occupant questionnaires	Global / multi-region (review)
Yuksel & Yildirim	An analysis of occupational exposure to particulate matter and erythrocyte morphological abnormalities among charcoal-grill restaurant workers	<i>Front Public Health</i>	Cross-sectional survey	Kitchen PM mass (spot), CO, CO ₂	Turkiye (Kocaeli)
ZCND aerogel filter	Biomimetic dual-network nanofibrous aerogels with robust interfacial assembly for multifunctional air filtration.	<i>J Colloid Interface Sci</i>	Materials synthesis and bench filtration characterization	PM ₁ , PM _{2.5} and PM ₁₀ filtration efficiency, pressure drop, dust-holding capacity under controlled humidity	Bench (live-stock housing application)
Zigabe et al.	Determinants of early-life household air pollution exposure (HAP): carbon monoxide (CO) and PM _{2.5} concentrations in Bukavu, eastern Democratic Republic of Congo (DRC).	<i>Environ Pollut</i>	Cross-sectional (cohort baseline)	24-h household PM _{2.5} by PurpleAir Zen; 24-h personal CO by Lascar EL-USB-CO	Bukavu, Democratic Republic of Congo
Sensing, forecasting & instrumentation					
3D-LSS particle shape	Enhanced three-dimensional light scattering sensor (3D-LSS) for online characterization of aerosol particle shape: Development and validation with test aerosols	<i>J Aerosol Sci</i>	Metadata only (no abstract)	Angularly resolved elastic scattering, single particle	Not stated
AAQR	A Sensor-Driven State-Space Framework for PM _{2.5} -Informed Natural Ventilation Opportunities Under Coupled Pollution-Thermal-Humidity Constraints.	<i>Aerosol Air Qual Res</i>	Monitoring-record analysis	Trailing 24-h mean PM _{2.5} from AirGradient low-cost nodes via OpenAQ, threshold 15 µg/m ³ (WHO 2021 24-h AQG)	Southeast Asia
AIRWISE Delhi	Improving aerosol-radiation interactions in the operational forecasting system - AIRWISE.	<i>Atmos Chem Phys</i>	Operational NWP sensitivity experiments with observational evaluation	Species-resolved complex refractive indices driving aerosol-radiation interaction in an operational forecast	Delhi, India (October 2023 - January 2024)
Aeolus transatlantic smoke, Sep	Characterization of aerosol properties during a huge transatlantic smoke transport event using Aeolus observations in synergy with multi-platform data.	<i>Atmos Chem Phys</i>	Spaceborne HSRL case study with trajectory modelling	Aerosol extinction and backscatter from ALADIN; AOD	Global / multi-region
Aerosol core/morphology optics	Coupled Effects of Core Material and Aggregate Morphology on the Optical Properties of Atmospheric Aerosols	<i>J Aerosol Sci</i>	Metadata only (no abstract)	Metadata only (no abstract deposited)	Computational (no site)
Alderotti et al.	Raman spectroscopy resolves nitrogen-driven metabolic acclimation to urban air pollution in <i>Quercus ilex</i>	<i>Sci Total Environ</i>	Bench / device experiment	Leaf Raman band intensities (chlorophyll, carotenoid, flavonoid) as a non-destructive biomonitoring index along an NO ₂ and PM ₁₀ gradient	Europe

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Alkylamine IC-MS/MS method	An accurate and sensitive method for identifying atmospheric alkylamines: selective enrichment by alkaline purging-acidic absorption coupled with ion chromatography-tandem mass spectrometry	<i>J Chromatogr A</i>	Analytical method validation	Speciated gas and PM-phase alkylamines by IC-MS/MS	China
Alraddadi et al.	Explainable one-hour-ahead air quality index forecasting across Saudi cities using pollutant and meteorological data	<i>Atmosphere</i>	Algorithm development	Hourly AQI (PM-dominated) with and without surface meteorological predictors	Saudi Arabia
Anwer et al.	Artificial Intelligence in Sand and Dust Storm Research: A Systematic and Bibliometric Review	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - AI methods for sand and dust storm detection/forecasting (by title)	Not recoverable from metadata
BC coating optics	Effects of V2O5, NiO, and Fe2O3 surface coatings on the optical properties of black carbon aggregates	<i>J Aerosol Sci</i>	Metadata only (no abstract)	Optical properties of coated BC aggregates	Laboratory
Bai et al.	Role of low-level jet evolution in vertical aerosol redistribution: a Doppler wind lidar study over East China.	<i>Atmos Meas Tech</i>	Remote sensing / lidar retrieval	Coherent Doppler wind lidar backscatter, spectral width and shear profiles against surface PM ₁₀	China
Baltic ship plume OFR aging	In Situ Collection and Photochemical Aging of Individual Ship Plumes Captured at Sea	<i>Environ Sci Technol Lett</i>	Field campaign / shipborne	Shipborne oxidation flow reactor on individual vessel plumes	Europe
Barth et al.	Online characterisation of oxidative potential and reactive oxygen species of fresh and aged aircraft turbine and ship engine particulate emissions.	<i>Environ Sci Atmos</i>	Chamber emission characterisation	Online OP and particle-bound ROS of fresh and photochemically aged aircraft-turbine and ship-engine exhaust	Chamber / laboratory (engine test facility)
Bogota BiLSTM Huber	BiLSTM with asymmetric Huber loss for 24 h PM _{2.5} forecasting and early exceedance detection at an industrial monitoring station in Bogota, Colombia	<i>Atmosphere</i>	Deep-learning forecasting	Hourly PM _{2.5} , Puente Aranda industrial corridor	Colombia
Boston sparse network	Air quality in cities: what do sparse sensor networks miss?	<i>Environ Sci Technol</i>	Mobile vs fixed co-measurement	Fixed-station PM _{2.5} vs mobile within a 250 m radius	United States
Broda et al.	A new balloon-borne system for measuring the vertical variability of aerosol optical properties using low-cost sensors.	<i>Atmos Meas Tech</i>	Instrument development	Sensirion SPS30 low-cost OPC, calibrated to scattering coefficient, Angstrom exponent and effective radius	Poland (southeastern)
CALIOP Shanghai aerosol stratification	Long-term CALIOP observations of vertical aerosol stratification over Shanghai: Meteorological modulation and height-resolved transport	<i>Atmos Res</i>	Metadata only (no abstract)	Metadata only (no abstract deposited); CALIOP spaceborne lidar	China
Cai J et al.	Scale-dependent drivers shape airborne microbiomes across the Shanghai subway network	<i>Environ Sci Technol</i>	Environmental surveillance campaign	Paired TSP (<=100 um) and PM _{2.5} sampling with microbial taxonomy, transcriptional potential, PM _{2.5} chemical composition, microclimate and network topology	China
Central-eastern China, 14-day lead time	14-Day PM _{2.5} and O3 Forecasting over Eastern China Using a Hybrid CNN-Transformer Architecture Driven by Multilevel Meteorological Fields.	<i>Environ Sci Technol</i>	Deep-learning model	Gridded surface PM _{2.5} and O3 forecast from multilevel meteorology plus emissions	China

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Chaparro et al.	Monitoring airborne fine particles with biological and technological sensors for preventive conservation of cultural heritage sites	<i>Sci Total Environ</i>	Metadata only	Airborne fine particles measured in parallel by biomonitors and instrumental sensors	Latin America
Chen C et al.	Characteristics and gas-particle partitioning of amine-containing particles from firework emissions: new insights from machine learning and single-particle mass spectrometry	<i>J Hazard Mater</i>	Real-time sensor campaign	Single-particle aerosol mass spectrometry of amine-containing particles; modified aerosol acidity Rra'	China
Clean air advocacy data approaches	Examining Data-Driven Approaches for Clean Air Advocacy in a Changing Climate: a Scoping Review.	<i>Curr Environ Health Rep</i>	Scoping review	Low-cost sensor, satellite and modelled air quality data used for advocacy	Global / multi-region
Climate-stratified PM _{2.5} LightGBM	Climate-stratified assessment of PM _{2.5} using machine learning: Geographic controls dominate meteorological factors	<i>PLoS One</i>	Machine-learning model	MERRA-2 satellite-assimilated monthly PM _{2.5}	Global / multi-region
Contini et al.	10-year trends of particle size distributions and equivalent black carbon at the Environmental-Climate Observatory (ECO) in Lecce (Italy).	<i>Environ Pollut</i>	Long-term field measurement campaign	Online PM _{2.5} , PM _{10-2.5} , PNSD 10-10,000 nm, eBC; offline OC/EC for MAC, 2015-2024	Italy (Lecce)
Cooking aerosol FIGAERO partitioning	Unravelling gas-particle partitioning dynamics in cooking aerosol oxidation through FIGAERO-CIMS analysis	<i>Atmos Chem Phys</i>	Aerosol flow reactor + model implementation	FIGAERO-I-CIMS speciation, 2-D VBS volatility	China
Coriolis sampler inactivation	Sampler-induced loss of respiratory virus infectivity revealed by spiked collection-liquid experiments	<i>Indoor Air</i>	Laboratory sampler validation	Coriolis micro liquid impinger, spiked collection liquid	Laboratory
Cui C et al.	Explainable machine learning prediction of long-term variations in cloud condensation nuclei at a high mountain site in the North China Plain	<i>Environ Sci Technol</i>	Data-fusion / ML surrogate model	CCN number concentration reconstructed from particle number size distribution and chemistry; bulk activation ratio and hygroscopicity parameter	China
Cyprus dust typing + DRE	Characterization of Saharan and Middle Eastern dust aerosols and their associated shortwave direct radiative effects over Cyprus	<i>Atmos Meas Tech</i>	Field campaign / remote sensing	AERONET optical properties, LIDAR, MODIS, HYSPLIT typing	Cyprus
Delkash	Receptor-scale imprint of barometric pressure variability on community air monitoring near an elevated-temperature landfill: a multi-method analysis across two mitigation interventions	<i>Environ Pollut</i>	Metadata only	Community-monitor receptor concentrations against barometric pressure variability, across two landfill mitigation interventions	United States
Embassy monitor shutdown, 46 cities	Air pollution after the shutdown of US embassy monitoring.	<i>Proc Natl Acad Sci U S A</i>	Matched difference-in-differences, satellite AOD	Real-time satellite aerosol optical depth, calibrated to pre-shutdown embassy PM _{2.5}	Global / multi-region
Europe BLH regression kriging	Modelling monthly Boundary Layer Height maps combining radiosonde, satellite, and reanalysis over Europe	<i>Atmos Meas Tech</i>	Regression kriging map product	Gridded monthly PBL height, 25 km, 00 and 12 UTC	Europe
FY-4B/AGRI AOD evaluation	Evaluation of the FY-4B/AGRI Land Aerosol Optical Depth Product and Comparison with Geostationary Satellite Products over Asia	<i>Atmos Environ</i>	Metadata only (no abstract)	Geostationary land AOD retrieval, FY-4B/AGRI vs peer products	Global
Feature engineering without co-location	Machine Learning-Based Calibration of Low-Cost PM _{2.5} Sensors Using Location-Specific Environmental Covariates and Feature Engineering Strategies.	<i>Chemosensors</i>	Ablation study, time-based validation	Low-cost PM _{2.5} sensor in an outdoor residential setting	Turkiye

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Front Bioeng	Bionic breathing-type airway exposure system for inhalation exposure studies.	<i>Front Bioeng Biotechnol</i>	Laboratory instrument development	Branch-level particle deposition in a CT-derived airway replica under programmed breathing waveforms	Laboratory
Galioto et al.	Physics-informed machine learning for urban nitrogen dioxide forecasting in Palermo, Italy	<i>Atmosphere</i>	Chemical transport modelling	Hourly NO ₂ at 11 m resolution: COPERT-Italy emission factors dispersed through the SIRANE street-network model, then fused with measurements in an ML stage	Europe
Genoa absorption intercomparison	Multi-Wavelength Optical Characterization of Carbonaceous Aerosols at a Coastal Urban Site in Genoa: Intercomparison of Real-Time and Filter-Based Techniques	<i>Atmosphere</i>	Multi-instrument field evaluation	AE33 aethalometer vs MWA and BLAnCA filter analysis	Italy
Giang et al.	Cumulative and real-time performance of consumer-grade electronic radon monitors in a Canadian single-family home over three months	<i>J Radiol Prot</i>	Sensor co-location / calibration	Radon activity concentration from 45 consumer monitors against an AlphaGuard PQ2000 and two alpha-track detectors; PM _{2.5} , CO ₂ , T and RH co-recorded	North America
Guth et al.	Exploring the variability in emissions for broadcast and pile prescribed burns using low-cost sensors	<i>Atmosphere</i>	Field emission-factor campaign	Low-cost-sensor emission factors for CO and PM _{2.5} , with filter-based EC/OC and speciated organics, resolved by modified combustion efficiency	North America
Hall et al.	Indoor Air Quality Forecasting With Machine Learning: A Systematic Review	<i>Indoor Air</i>	Systematic review (PRISMA)	Indoor PM and other IAQ variables; forecasting models	Global / multi-region (review)
He J et al.	Molecular packing engineering enables a high performance ppb-level room-temperature NO ₂ sensor	<i>ACS Sens</i>	Laboratory instrument development	Gas-phase NO ₂ (not particulate): chemiresistive response of cyano-substituted distyrylbenzene films with tuned alkyl-chain length and molecular packing	Laboratory
Herman et al.	An Advanced, Multifunctional, and Extreme-Condition Test Chamber for Performance Evaluation of Air Quality Sensors	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - sensor test chamber (by title)	Not recoverable from metadata
Hong et al.	Predicting fungal and bacterial bioaerosol concentrations across diverse building types: a framework integrating generalized additive models and machine learning	<i>Indoor Air</i>	Cross-sectional + prediction model	Culturable bacterial and fungal bioaerosol concentrations vs indoor temperature and relative humidity	South Korea
Hu et al.	Advanced insights into biomass burning aerosols during the 2023 Canadian wildfires from dual-site Raman and fluorescence lidar observations.	<i>Atmos Chem Phys</i>	Remote sensing / lidar retrieval	Multi-wavelength Raman, depolarisation and fluorescence lidar profiles of transported smoke	Global / multi-region

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Huang et al.	Interpretable Spatiotemporal Coupling Network for Robust Emission Prediction and Implicit Mechanism Mining in Sludge Gasification.	<i>Environ Sci Technol</i>	Deep-learning forecasting	Continuous industrial stack monitoring of PM, SO ₂ and NO _x from a sludge-gasification plant, with feature-level optimal time lags	Not stated
Ichikawa et al.	Tower- and drone-based BVOC observations in a suburban Tokyo forest: methodological insights and MEGAN comparison	<i>Atmos Chem Phys</i>	Multi-instrument field evaluation	Isoprene and monoterpene mixing ratios and emission fluxes at multiple heights on a 30 m flux tower, cross-checked by drone sampling and against MEGAN	Japan
India winter PM _{2.5} seasonal forecast	Seasonal predictability of winter PM _{2.5} pollution severity in India through a strong pollution-extratropical storm connection	<i>Sci Adv</i>	Seasonal prediction framework	Government reference PM _{2.5} network, >10 winters	India
J Aerosol Sci	High-efficiency plasma-near-passage unipolar aerosol charger for sub-10 nm particles.	<i>J Aerosol Sci</i>	Metadata only (no abstract)	Not recoverable from the deposited record	Not stated
Ji et al.	Quantifying the aging of dust aerosols using Mie-polarization-fluorescence lidar: linking optical evolution to chemical transformations	<i>Atmos Res</i>	Metadata only (no abstract)	Dust optical properties by Mie-polarization-fluorescence lidar	China
Jonsson et al.	Tracing biological, anthropogenic, and inorganic sources of coarse aerosols via single-particle fluorescence and optical morphology	<i>Atmos Chem Phys</i>	Laboratory instrument development	Coarse-mode single particles, Multiparameter Bioaerosol Spectrometer (fluorescence + morphology)	Norway (Svalbard, Zeppelin Observatory)
Kalluri et al.	Long-term (2008-2024) analysis of black carbon aerosols over a semi-arid rain-shadow region in southern peninsular India: trends, wavelet analysis, machine learning approaches and source regions.	<i>Atmos Environ</i>	Metadata only (no abstract)	Black carbon, presumably aethalometer-derived; instrument, wavelength set and trend magnitudes not recoverable	India
Kaur et al.	Performance evaluation of Plantower's PMSX003N: laboratory and field analysis	<i>Aerosol Sci Technol</i>	Sensor co-location / calibration	PM _{2.5} and PM ₁₀ from three PMSX003N, three PMS5003T and three Alphasense OPC-N3 against FEM reference; laboratory size-bin response to monodisperse DOS aerosol	North America
Khalili Hollo et al.	Validation and Correction of PM _{2.5} Mass Concentrations from Consumer-Grade Personal Exposure and Outdoor Fixed Monitors: A Two-Year Colocation Study.	<i>ACS Environ Au</i>	Sensor co-location / calibration	Atmotube Pro (n=50) and QuantAQ Modulair-PM (n=5) vs GRIMM EDM180, 1-h PM _{2.5}	USA (Denver, Colorado)
Kim C-h et al.	OAV-based source characterization of designated odor compounds in long-term continuous monitoring data with severe missingness	<i>Atmosphere</i>	Monitoring-record analysis	22 designated odour compounds at 20 min resolution; block-holdout imputation with BiLSTM, BiGRU, TCN, Transformer and HistGradient-Boosting, then odour activity value and odour contribution	South Korea

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Krajnc Pogacnik et al.	Selective real-time monitoring of respiratory droplets for indoor air safety in educational environments.	<i>Build Environ</i>	Metadata only (no abstract)	Respiratory-droplet detection; sensing principle, size range and selectivity mechanism not recoverable	Not stated
LCS field calibration correction	Correction: Long-term performance evaluation and seasonal field calibration of a low-cost particulate matter sensor	<i>J Environ Health Sci Eng</i>	Metadata only (no abstract)	Low-cost PM sensor, seasonal field calibration	Iran
LLM-compiled adaptive sampling policies	Agentic LLM Compilation of Adaptive Sampling Policies for Low-Power IoT Sensing	<i>Sensors</i>	Algorithm development	Beijing PM _{2.5} series used as the benchmark signal for duty-cycled sampling	China
Lawson et al.	Humidity-dependent photoacoustic absorption biases and effective water mass accommodation coefficients for multicomponent aerosol particles	<i>Aerosol Sci Technol</i>	Metadata only (no abstract)	Metadata only - photoacoustic absorption bias vs RH, multicomponent aerosol	Laboratory (not recoverable from metadata)
Le et al.	Real-time monitoring of ultrafine aerosol size distributions using parallel non-bouncing impactors coupled with corona charger.	<i>Meas Sci Technol</i>	Instrument development	Particle number size distribution 10-560 nm; total PNC and UFP (PM _{0.1})	Single urban rooftop (location not stated)
Li et al.	A multi-angle and polarization-based retrieval algorithm for aerosol layer height of smoke and dust.	<i>Atmos Meas Tech</i>	Satellite retrieval evaluation	Aerosol layer height from HARP2/PACE multi-angle polarimetry, validated against EarthCARE ATLID lidar	Global / multi-region
Liu et al.	Morphology Change of Diesel Exhaust Black Carbon Particles during Humidification and Evaporation.	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - black carbon particle morphology through humidification and evaporation cycles	China (laboratory, site not recoverable)
Low-cutoff cyclone method	An improved method for measuring the penetration efficiency of low cutoff cyclones.	<i>Aerosol Sci Technol</i>	Metadata only (no abstract)	Metadata only - penetration efficiency measurement for low cut-point cyclone inlets	Not recoverable
Mathissen et al.	Development of the Worldwide Harmonized Heavy Duty Brake Cycle (WHBC) for Brake Wear Particle Measurement	<i>Atmosphere</i>	Test-cycle development + dynamometer testing	Brake-wear particles (PM ₁₀ /PM _{2.5} , PN), heavy-duty vehicles	Global / multi-region (dataset)
Medellin MABI eBC	Multi-wavelength spectral attribution of equivalent black carbon in Medellin, Colombia (2010-2023): Disentangling fossil fuel and biomass burning using the MABI instrument.	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - Multi-wavelength Absorption Black carbon Instrument, spectral source attribution, 14-year record	Medellin, Colombia
Mohd Nadzir et al.	Short-term variability of trace gases and particulate matter in the Antarctic Peninsula atmosphere using a low-cost air sensor	<i>Atmosphere</i>	Real-time sensor campaign	Minute-scale NO ₂ , O ₃ , CO, SO ₂ , PM _{2.5} , PM ₁₀ (plus TVOC in May) from a single low-cost multi-sensor node	Antarctica
Morel et al.	Evaluation of Two Cyclonic Bioaerosol Samplers for Hospital Air Monitoring in a Calibrated Test Chamber Integrating Aerodynamics and Biological Endpoints	<i>Indoor Air</i>	Laboratory sampler validation	Bioaerosols (SARS-CoV-2, <i>S. aureus</i> , <i>A. fumigatus</i>); cyclonic samplers	France (test chamber)
NaCl aerosol turbulent size redistribution	Turbulence-Induced Size Redistribution of Submicron NaCl Aerosols in Ribbed Confined Flows Prior to Electrostatic Collection	<i>J Aerosol Sci</i>	Metadata only (no abstract)	Metadata only (no abstract deposited)	Computational (no site)

First author	Focus	Journal	Design	Metric	Region
Ni et al.	An AI-based algorithm for retrieving aerosol optical depth and single scattering albedo using All-Sky Imager observations.	<i>Atmos Meas Tech</i>	Algorithm development	AOD and single scattering albedo retrieved from all-sky camera imagery, sun-photometer targets	Global / multi-region
Open-pit mine dust stacking model	Multi-Point Prediction and Interpretable Analysis of Dust Concentration in Open-Pit Coal Mine During the Cold Season Using a Stacking Learning Framework	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Multi-point cold-season mine dust concentration	Not stated
Organic aerosol self-diffusion MD	Predicting the Self-Diffusion Coefficient of Atmospheric Organic Aerosol Components via Molecular Dynamics Simulations.	<i>ACS Omega</i>	Modelling / inventory	Particle-phase self-diffusion of atmospheric organic components	Not applicable
PMS5003 and Sniffer4D, concrete drilling	Non-Cumulative, Size-Specific Calibration of Low-Cost Particulate Matter Sensors Under Simulated Construction Drilling Events.	<i>Atmosphere (preprint of record)</i>	Chamber / laboratory	Non-cumulative PM ₁ , PM _{2.5} , PM ₁₀ fractions under simulated concrete-slab drilling	Not stated
PPDRTL leaf-image AQI	PPDRTL: A novel framework for predicting pollutant deposition on roadside tree leaves using linear regression	<i>PLoS One</i>	Image-feature regression proxy	AQI regressed on leaf-surface image texture features	India
Papathanasiou et al.	Evaluating Personal Air Quality Monitors for PM _{2.5} and CO ₂ : Insights from a Year-Long Pilot Study.	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - personal PM _{2.5} / CO ₂ monitors, year-long pilot (by title)	Not recoverable from metadata
Parks et al.	Evaluating performance of real-time monitors for quantifying elemental carbon in diesel particulate matter.	<i>Aerosol Sci Technol</i>	Metadata only (no abstract)	Metadata only - real-time elemental carbon monitors against diesel particulate matter	Not stated (abstract not deposited)
Partector 2 Pro vs SMPS and CPC, 4 filaments	Evaluation of the Partector 2 Pro for measuring ultrafine particles emitted from 3D printing.	<i>J Occup Environ Hyg</i>	Multi-instrument field evaluation	UFP number concentration and D10-D95 mobility diameter percentiles	Laboratory
Peng et al.	Inhibition of condensable particulate matter by co-firing biomass and high sulfur coal on a 50 kW self-sustaining combustion furnace.	<i>Environ Pollut</i>	Chamber emission characterisation	Condensable particulate matter mass and speciation, mg/MJ, dilution sampling	China
PurpleAir Raleigh and Delhi	Evaluating and Calibrating Low-Cost Air Quality Sensors in Contrasting Aerosol Regimes.	<i>Aerosol Air Qual Res</i>	Co-location calibration campaign	PurpleAir PA-II PM _{2.5} against Federal Equivalent Method reference	United States; India
RI-URBANS urban AQ mapping	Enhancing Urban Air Quality Mapping through Novel Measurement and Modelling approaches, and Citizen Science: Actionable Insights from the RI-URBANS Project	<i>Atmos Meas Tech</i>	Multi-platform observation synthesis	Low-cost sensor networks, mobile platforms, citizen data	Europe
Raju et al.	Investigations of black carbon aerosol sources and strengths over a rural-valley location in Haryana, India.	<i>J Environ Sci Health A</i>	Multi-year aethalometer monitoring + source apportionment	Equivalent black carbon, seven-wavelength AE33, 880 nm; annual and seasonal means in $\mu\text{g}/\text{m}^3$	India
Raso et al.	Influence of temporal resolution on the characterization of episodic air pollution events: experimental evidence from New Year's Eve fireworks	<i>Atmosphere</i>	Real-time sensor campaign	PM ₁₀ , NO ₂ , CO, O ₃ at minute, hourly and daily resolution; certified reference instruments alongside calibrated low-cost sensors	Italy

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Rosas et al.	Biomass burning and African dust influence on seasonal aerosol chemistry over the Western Caribbean (Part I): Atmospheric variability and human health	<i>Environ Sci Pollut Res Int</i>	Field campaign / remote sensing	Continuous PM _{2.5} , PM ₁₀ and particle-bound PAH with the PM _{2.5} / PM ₁₀ ratio as a regime discriminator, joined to FIRMS fire detections and HYSPLIT back-trajectories	Mexico
Rosca & Stancu	Indirect Machine Learning Prediction of a Composite Indoor Air Quality Index in an Airport Passenger Transit Area	<i>Indoor Air</i>	Machine-learning model	Indoor PM _{2.5} , PM ₁₀ and CO ₂ sensor data; composite IAQ index	Italy (Brindisi Airport)
Rural wintertime SIA metal catalysis	Important role of acidity-regulated metal catalysis in rural wintertime secondary inorganic aerosol formation in the North China Plain	<i>Atmos Res</i>	Metadata only (no abstract)	Rural wintertime secondary inorganic aerosol in PM _{2.5}	China
SAQIN	Long-short-term decoupled residual learning for high-resolution spatio-temporal air quality inference	<i>Neural Netw</i>	Machine-learning model	Criteria-pollutant fields inferred from sparse reference stations	Not stated
SE India monsoon aerosol	Monsoon-Regulated Aerosol Variability and Radiative Effects over a Rural Receptor Site in Southeast India	<i>Atmosphere</i>	Long-term field measurement campaign	MICROTOPS-II sunphotometer AOD500 and Angstrom exponent	India
SIBS inexpensive single-particle	Feasibility of an inexpensive single-particle SIBS instrument	<i>Aerosol Sci Technol</i>	Metadata only (no abstract)	Single-particle spark-induced breakdown spectroscopy	Not stated
Sahin et al.	Validation Design Governs the Apparent Performance of Machine Learning Calibration for Low-Cost Optical Particle Counters: A PM1 Study	<i>Atmosphere</i>	Multi-instrument field evaluation	Indoor PM1 from three low-cost OPCs vs reference nephelometers; six ML calibrators	Not stated (indoor co-location)
Seong & Hoque	Influence of boundary layer conditions and surface properties on bacterial interaction with indoor built surfaces	<i>Front Pediatr</i>	Laboratory metrology	Attachment fraction and centrifugal detachment of <i>Corynebacterium</i> sp. on carpet, wood, metal and glass, with near-surface velocity fields by particle image velocimetry	Laboratory
Shafi & Scafetta	Management of missing air pollution data within urban environments using machine learning regressions: a case study for Delhi, India	<i>Environ Sci Pollut Res Int</i>	Data-fusion / ML surrogate model	Iterative reconstruction of missing daily PM _{2.5} , PM ₁₀ , O ₃ , NO ₂ , SO ₂ and CO from the four most-correlated neighbouring stations; 35 candidate regressors per gap	South Asia
Shen et al.	Apportioning light absorption of ambient aerosols to black carbon, brown carbon, and lensing effect using a PAX-ISS hybrid method: insights into absorption enhancement.	<i>Atmos Chem Phys</i>	Multi-instrument field evaluation	Photoacoustic absorption (PAX) plus solvent-mediated integrating-sphere absorption of ambient filter samples	Beijing, China
Shenzhen tower vertical PM _{2.5} DL	Leveraging multi-source observations for vertical PM _{2.5} estimation: A deep learning fusion model	<i>J Hazard Mater</i>	Data-fusion / ML surrogate model	Lidar extinction, microwave radiometer profiles, tower PM _{2.5} at 200/300 m	China
Shi S et al.	Potential aerosol tradeoffs of vessel speed reduction for fresh ship plume composition and tracer coverage	<i>npj Clim Atmos Sci</i>	Measurement campaign	Ultrafine and coarse modes of fresh ship plumes; V/VO and sulfate markers	China

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Shin et al.	A Continuous Virus Aerosol-into-Liquid Sampler (VAiLS) Integrating Twin-Vortex Condensation Growth and Wet-Cyclone Enrichment.	<i>Environ Sci Technol</i>	Laboratory sampler validation	Number-weighted collection efficiency for KCl nanoparticles and MS2/phi6 bioaerosols at 16 L/min	Laboratory (bioaerosol test rig, South Korea)
Song	Relocation-associated discontinuities in Korean air-quality monitoring: a Bayesian structural time-series analysis of PM ₁₀ and PM _{2.5} .	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - station relocation step changes in PM ₁₀ /PM _{2.5} records (by title)	South Korea (national monitoring network, by title)
TROPOMI SO2 EAKF assimilation	Assimilation of SO2 TROPOMI Retrievals at the European Scale with EAKF Implemented in MINNI Through DART	<i>Atmosphere</i>	Satellite + chemical transport model	TROPOMI SO2-COBRA total column into MINNI CTM	Europe
Thomas et al.	Procedural Challenges in the Analysis of Toxic Metals in Electronic Cannabinoid Delivery System Aerosols Produced for Vaping Cannabinoids.	<i>J Test Eval</i>	Method review	Aerosol trapping media for ICP-MS determination of metals in e-cigarette and cannabis vaping aerosol	United States (ASTM D37.08 work group)
Tu et al.	Assessing the Need for Site-Specific Calibration of ATP Bioluminescence Sensors for Airborne Bacteria Monitoring in Indoor Environments	<i>Indoor Environ</i>	Metadata only (no abstract)	Metadata only - ATP bioluminescence sensors for airborne bacteria (by title)	Not recoverable from metadata
UAV single-particle CCSEM	Temporal evolution and vertical distribution of size-resolved aerosol composition and mixing state within the boundary layer using an unmanned aerial vehicle	<i>Environ Sci Technol</i>	UAV instrument deployment	Microscopy sampler on a UAV, 30 m and 1 min resolution	Global / multi-region
US precipitation washout DoubleML	Particulate matter removal by precipitation under normal and wildfire conditions across the United States using double machine learning	<i>Research Square</i>	Causal ML on monitoring records	Hourly EPA PM _{2.5} and PM ₁₀ vs NLDAS-2 precipitation	USA
Urban OOM flux	Urban surface-atmosphere fluxes of selected pptv-level oxygenated organic molecules (OOMs) from eddy covariance observations.	<i>Atmos Chem Phys</i>	Eddy covariance flux measurement, method development	Iodide-adduct CIMS eddy covariance fluxes of 16 oxygenated organic molecules (SOA precursors)	Central China megacity, hottest month
Veselovskii et al.	Evaluation of smoke mass concentration within the PBL based on observations of fluorescence lidar with several discrete channels	<i>Atmos Meas Tech</i>	Remote sensing / lidar retrieval	Smoke mass concentration retrieved from five-channel fluorescence lidar spectra, separating smoke from urban aerosol; detection threshold 0.1 µg/m ³	Europe
Vishakhapatnam long-term co-location	Exploring Environmental and Temporal Performances of Machine Learning Models for Calibration of a Low-Cost PM _{2.5} Sensor.	<i>Aerosol Air Qual Res</i>	Sensor co-location	Low-cost PM _{2.5} against reference, binned by temperature, RH, hour and month	India
W Sub-Saharan Africa PM monitoring	Monitoring and estimating PM _{2.5} and PM ₁₀ concentrations in Western Sub-Saharan Africa: Challenges and perspectives	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Regional PM _{2.5} /PM ₁₀ monitoring and estimation capacity	West Africa
Wang S et al.	Impact of fuel moisture on chaparral emissions under flaming and smoldering conditions	<i>Fire Saf J</i>	Chamber / laboratory	PM and gas emission factors by FTIR, flaming vs smoldering	North America
Wang et al.	Artificial intelligence for atmospheric chemistry: from observation to process understanding	<i>npj Clim Atmos Sci</i>	Metadata only (no abstract)	Metadata only - AI in atmospheric chemistry observation and process studies (by title)	Not recoverable from metadata
Wildfire CHON PM _{2.5}	Integrating FT-ICR MS with Rule-Based Prediction and QSAR to Resolve Nitrogen-Containing Organic Compounds and Their Toxicity in Wildfire-Derived PM _{2.5}	<i>Environ Res</i>	Measurement campaign	Wildfire PM _{2.5} by GC-MS and FT-ICR MS; nitrated PAC in ng m ⁻³	South Korea

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Wu et al.	Spatiotemporal Prediction of Sand-storm Utilizing Multi-Source Data and ConvLSTM Model: A Case Study in Northwest China	<i>Atmosphere</i>	Deep-learning forecasting	Surface PM ₁₀ with Himawari-8/9 AOD assimilated by EnKF into ConvLSTM	China (Gansu)
Xing et al.	Multisource observation-constrained tuning for air quality forecasting in a machine learning-enhanced NOAA Unified Forecast System	<i>Environ Sci Technol</i>	Data-fusion / ML surrogate model	Surface PM _{2.5} and O ₃ fields reconstructed from VIIRS AOD, TEMPO NO ₂ and EPA AirNow, used as initial conditions for 72 h UFS-AQM forecasts	North America
Xu et al.	Integrating Fragmented Mobile Monitoring Data into a High-Resolution Spatiotemporal Exposure Model for Black Carbon	<i>Environ Sci Technol</i>	Mobile monitoring campaign	Equivalent black carbon (eBC), vehicle-platform aethalometry	China (Beijing)
Zeng et al.	Decoupling between PM _{2.5} oxidative potential and light extinction: Roles of chemical composition and aerosol thermodynamic state	<i>J Hazard Mater</i>	Source apportionment	Volume-normalised oxidative potential and light extinction placed on a common quadrant framework, with water-soluble organic carbon, aerosol liquid water content, aerosol pH and PMF source factors as explanatory terms	Taiwan
Zhang Z et al.	A microfluidic multispectral diffraction platform for quantitative analysis of atmospheric black carbon aerosols	<i>Anal Chem</i>	Laboratory instrument development	Black carbon mass concentration from lensless multispectral (400-800 nm) diffraction fingerprints after microfluidic inertial separation and electrostatic capture	Laboratory
Zhang et al.	Adaptive robotic sensing for exposure-oriented mapping of hazardous airborne particle dispersion in indoor environments.	<i>J Hazard Mater</i>	Adaptive mobile-robot sampling + CFD validation	PM _{2.5} mapped by a mobile optical sensor on a robot; Kernel DM+V field reconstruction	Laboratory
Zhang et al.	Are expensive sensors necessary for assessing indoor environmental quality? A data-driven approach using minimal sensing.	<i>Build Environ</i>	Metadata only (no abstract)	Metadata only - minimal-sensing IEQ assessment (by title)	Not recoverable from metadata
Zhao et al.	A novel deep learning prediction method for PM _{2.5} integrating improved MSTL decomposition and FATA optimization algorithm.	<i>PLoS One</i>	Deep-learning forecasting	PM _{2.5} time series with meteorological and pollutant covariates, station-based	China
Zhou et al.	Effects of geometrical parameters on eccentric cylindrical differential mobility analyzer	<i>Aerosol Sci Technol</i>	Metadata only (no abstract)	Metadata only - eccentric cylindrical DMA geometry (by title)	Not recoverable from metadata
cCPC/OPC/SPS30 field evaluation	Field Evaluation of Community Condensation Particle Counters, Optical Particle Counters, and a Low-Cost Particle Sensor	<i>Aerosol Sci Technol</i>	Multi-instrument field evaluation	cCPC 0.005-2.5 µm number; OPC 0.3-10 µm; Sensirion SPS30 mass	United States
sMCPC supersaturation model	Performance modeling of a small mixing-type condensation particle counter.	<i>Atmos Meas Tech</i>	Numerical / CFD instrument model	Particle number, critical activation diameter D _{p,50}	Not applicable (laboratory / numerical)
Exposure assessment & modelling					
118 CMAQ scenarios, 17 regions x 7 sectors	A LASSO-based reduced-form CMAQ model for predicting ozone and PM _{2.5} responses to emission changes in South Korea.	<i>PLoS One</i>	Chemical transport modelling	CMAQ-simulated surface PM _{2.5} and O ₃ response to sectoral emission perturbation	South Korea

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13 Java coal plants, 41,119 daily obs 2015-2025	Predictive modelling of fine-mode combustion aerosol events near Indonesian coal-fired power plants using AE-filtered Himawari-8 satellite data.	<i>Environ Monit Assess</i>	Machine-learning model	Himawari-8 AOD filtered by Angstrom exponent > 1.2 as a fine-mode combustion proxy	Not stated
33 microwave radiometers, China, 2024-2025	Surface-Based Temperature Inversions Climatology Across China from Microwave Radiometer Network and Their Relationship to PM _{2.5} Pollution.	<i>Environ Pollut</i>	Field measurement	PM _{2.5} at hotspot stations against hourly MWR-derived inversion and mixing-layer metrics	China
8 Korean agricultural sites, 547 days	Relationships Between PM _{2.5} and Its Precursor Gases in Agricultural Areas of South Korea Using a Loading-Rate-Based Analysis.	<i>Atmosphere</i>	Field measurement	Hourly PM _{2.5} and NH ₃ , NO ₂ , SO ₂ , expressed as concentration and as wind-scaled loading rate	South Korea
Accra traffic intersections	Meteorological drivers of ambient PM _{2.5} pollution at urban traffic intersections in Accra, Ghana	<i>Environ Monit Assess</i>	Portable sensor monitoring campaign	Aeroqual Ranger portable dust monitor, 9 traffic hotspots	Ghana
Ahmad et al.	Characterizing Ambient Fine Particulate Matter Pollution in Jordan: National Monitoring Evidence of Spatial Inequality, PM _{2.5} Hotspots, and Temporal Trends	<i>Preprints.org (preprint)</i>	Monitoring-record analysis	Annual PM _{2.5} from Jordan's national monitoring network, 2018-2024	Jordan
Arpornwichanop & Chuersuwan	Life-stage-resolved screening of inhalation exposure to PM _{2.5} -bound elements in Nonthaburi, Greater Bangkok, Thailand	<i>Environ Geochem Health</i>	Exposure model (secondary data)	PM _{2.5} -bound elements; respiratory-deposited intake adjusted by simulated-lung-fluid solubility	Thailand
Arunachalam et al.	Residential gas combustion drives building-sector air pollution mortality in the U.S.	<i>Research Square (preprint)</i>	Health impact model	CMAQ Decoupled Direct Method sensitivities of NO ₂ , O ₃ and PM _{2.5} to building-sector precursor emissions, propagated through BenMAPR to premature deaths	United States
Athens NPF	The influence of air mass origin and ammonia emissions on new particle formation at a Mediterranean urban site (Athens, Greece).	<i>Atmos Res</i>	Metadata only (no abstract)	Metadata only - new particle formation events, ammonia, back-trajectory air-mass origin	Athens, Greece
Atik et al.	Inter-annual trends of PM _{2.5} -bound heavy metals in Dhaka, Bangladesh: a comparative study for source apportionment and health risk assessment	<i>Environ Sci Atmos</i>	Source apportionment	Elevated-altitude PM _{2.5} filter sampling, heavy metals, PMF	Bangladesh (Dhaka)
Atmos Pollut Res	Canyons and Canopies: Effects of building configuration and green infrastructure on urban air quality.	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Not recoverable from the deposited record	Not stated
Benin PM _{2.5} -water grid screening	Grid-based screening of fine particulate matter and water-optical conditions in Southern Benin: the Cotonou, Lake Nokoue, and Porto-Novo Corridor	<i>Environ Monit Assess</i>	Satellite remote sensing	Satellite-derived annual PM _{2.5} , 1 km grid; Sentinel-2 water optics	Benin
Bhattarai et al.	Spatio-temporal assessment of PM _{2.5} and PM ₁₀ pollution in core and peripheral municipalities of the Kathmandu valley using regulatory and low-cost sensor data	<i>Research Square (preprint)</i>	Network monitoring + burden	Regulatory monitors integrated with a high-resolution low-cost sensor network, daily PM _{2.5} /PM ₁₀ , 2023-2024	Nepal (Kathmandu Valley)
Bi et al.	Associations of street morphology and local wind environment with annual cumulative ambient PM _{2.5} exceedance: Evidence from Wuhan	<i>J Environ Manage</i>	Machine-learning model	Annual cumulative excess PM _{2.5} exposure per street; CatBoost-SHAP + PLS-PM	China (Wuhan)

First author	Focus	Journal	Design	Metric	Region
Blond et al.	Boundary-Layer Height Outweighs Wind Speed in the Meteorological Control of Urban Air Pollution: Unique-Variance Attribution and a City Typology for Ten French Cities	<i>Preprints.org (preprint)</i>	Statistical model	Station PM _{2.5} , PM ₁₀ , NO ₂ , O ₃ vs boundary-layer height and wind	France (10 cities)
Bloomberg et al.	Associations of individual exposure to particulate matter with next-day physical activity and sedentary behaviour in older adults: a micro-longitudinal study	<i>medRxiv (preprint)</i>	Panel study	Personal air-quality monitors (waking hours, 5 days) vs area-level PM _{2.5} / PM ₁₀	UK
Boyle et al.	Creating outdoor air pollution exposure metrics for assessing breast and lung cancer risks in the United States	<i>Cancer Epidemiol Biomarkers Prev</i>	Metric development + retrospective application	Census-tract criteria-pollutant concentrations (CACES land-use regression) plus RSEI industrial-release model, weighted by cancer-specific meta-analytic risk estimates	United States
Brake wear SO ₂ uptake	Unexpectedly rapid SO ₂ heterogeneous reaction on the surface of vehicular brake wear particles	<i>Nat Commun</i>	Heterogeneous kinetics experiment	Brake wear particles, SO ₂ uptake coefficient	China
Bravo et al.	No one is immune: neighborhood trends and disparities in air pollution exposure in the city of Chicago during summer 2023 wildfires	<i>J Expo Sci Environ Epidemiol</i>	Exposure model (secondary data)	Daily 24 h tract-level PM _{2.5} from hourly 500 m gridded estimates, split by wildfire-smoke and non-smoke days	North America
Brisbane Aitken-mode PNC	Long-term changes in Aitken-mode particle number concentration in Brisbane: roles of meteorology, traffic, and biomass burning	<i>Environ Int</i>	Metadata only (no abstract)	Long-term Aitken-mode particle number concentration	Australia
Broda et al.	Vertical and Zonal Characteristics of Transatlantic Biomass Burning Aerosol Transport During the Extreme 2023 Canadian Wildfire Season.	<i>Atmosphere</i>	Multi-platform observation synthesis	NAAPS reanalysis smoke AOD (550 nm), MODIS fire and AOD, modelled plume injection heights	North America to Europe (transatlantic)
Chen et al.	Chemical Characteristics, Source Apportionment and Toxic Potential of PM _{2.5} in Southern Taiwan: Insights from the Kao-Ping Experiment (KPEX).	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - speciated PM _{2.5} with source apportionment and a toxicity metric	Taiwan (Kao-Ping region)
Chen et al.	Formation mechanism of rapid daytime nitrate production in a typical semi-arid city near dust source regions	<i>Atmos Res</i>	Metadata only (no abstract)	Metadata only - daytime particulate nitrate formation near dust sources	Not recoverable from metadata
Chen et al.	Solid Waste Burning: An Overlooked Atmospheric Source of p-Phenylenediamine Antioxidants and Their Quinones	<i>Environ Sci Technol Lett</i>	Field measurement	PPDs and PPD-quinones in PM _{2.5} from open burning and incineration	China
China NO _x isotopes	Driving factors of atmospheric nitrogen oxides emissions and pollution in China.	<i>Environ Pollut</i>	Source apportionment	delta15N of particulate and precipitation nitrate across eight Chinese regions, 2001-2022	China
China aerosol vertical	Spatiotemporal Dynamics and Vertical Distribution of Aerosols in China: Insights from Multi-Source Evidence from 2000 to 2022.	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - column and vertically resolved aerosol from multiple platforms, 2000-2022	China (national)
Construction equipment PM-bound EPFR	Real-world emissions of PM-bound environmentally persistent free radicals and reactive radical species generation from non-road construction equipment	<i>Atmos Environ</i>	Metadata only (no abstract)	PM-bound environmentally persistent free radicals	Not stated

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Delhi N2O5	Enhanced nocturnal N2O5-CINO2 chemistry in Delhi.	<i>Atmos Chem Phys</i>	Field campaign	Nocturnal N2O5 and CINO2 mixing ratios; particulate chloride 4.7 $\mu\text{g}/\text{m}^3$ nighttime mean	India
Dou et al.	Volatile organic compounds in Chengdu: Functional zone disparities and four-year dynamics in a basin city of Southwest China.	<i>Environ Pollut</i>	Metadata only (no abstract)	Ambient VOCs by functional zone, four-year record; no quantities retrievable	China
Douala households	What is the impact of the concentration of particulate matter on the respiratory health of households in urban Cameroon?	<i>Spat Spatiotemporal Epidemiol</i>	Observational exposure study	PM ₁₀ , PM ₅ , PM _{2.5} by OC 300 Dust Particle Laser, low-cost portable	Sub-Saharan Africa
EROD-HR/SOILHD Iberia	Impact of high-resolution soil erodibility datasets on dust simulations in WRF-Chem with the GOCART scheme	<i>Atmos Chem Phys</i>	Regional CTM evaluation	Modelled PM ₁₀ vs SINCLAIR network, 5 dust episodes	Spain
Engeroff et al.	Air pollution exposure during urban running: a randomized crossover comparison of particulate matter exposure on park, roadside, and residential routes with consideration of rush-hour timing	<i>Research Square (preprint)</i>	Mobile monitoring campaign	Mobile personal PM ₁₀ , PM _{2.5} and PM ₁ during 2 km runs, referenced to local urban background PM _{2.5} /PM ₁₀	Germany
Environ Pollut	Underlying Surface Heterogeneity Modulates Meteorological Drivers of PM _{2.5} and O ₃ Pollution in the Sichuan Basin.	<i>Environ Pollut</i>	Metadata only (no abstract)	Not recoverable from the deposited record	China
Fireman et al.	Biomonitoring of urban ultrafine particle exposure and associated health effects.	<i>Int J Public Health</i>	Biomonitoring + in vitro	Ultrafine particle counts in exhaled breath condensate, industrial vs control urban setting	Israel
Firozabad size-segregated lung deposition	Seasonal variation of particulate matter deposition in human lungs through different exposure pathways in an industrial city.	<i>Environ Monit Assess</i>	Deposition sampling	Size-segregated PM _{1.0} , PM _{2.5} and PM ₁₀ with metal speciation	India
Frueh et al.	Source apportionment of volatile organic compounds in a Philadelphia fenceline community: results from THRIVEair.	<i>J Expo Sci Environ Epidemiol</i>	Community passive-sampler campaign + PMF source apportionment	38 VOCs, one-week integrated passive thermal-desorption tubes, 20 sites, 12 months	USA
GPS+LCS schoolchild exposure model	Integrating GPS and Low-Cost Sensors to Enhance Personal PM _{2.5} Exposure Modeling for Schoolchildren	<i>Environ Pollut</i>	Personal monitoring + microenvironment model	Portable PM _{2.5} sensor with GPS, 918 person-days; BAM-1020 reference	Taiwan
Gao J et al.	Understanding unexpected ozone-health associations: are secondary oxidative pollutants key?	<i>Environ Sci Technol</i>	Narrative / critical review	Measured O ₃ as a proxy for an unmeasured oxidant field; argues low O ₃ can coincide with active oxidative chemistry producing toxic unmeasured secondary products	Global / multi-region
Garcia-Vara et al.	Expanding the chemical space of urban air: Identification and semiquantification of polar emerging contaminants in PM _{2.5} through LC-HRMS non-target analysis.	<i>J Hazard Mater</i>	Field measurement	Filter PM _{2.5} at 8 stations (traffic, urban background, port); LC-HRMS NTA	Spain (Barcelona)
Godri Pollitt	Beyond Inhalation: Air as an Integrative Matrix for Exposome Research.	<i>Environ Sci Technol</i>	Metadata only (no abstract)	No measured PM; air framed as an exposome sampling matrix	Global / multi-region
Gonzalez-Mendez et al.	Morpho-chemical fingerprinting and distance-dependent attenuation of airborne Particulate Matter from abandoned mine tailings in Northwestern Mexico.	<i>Environ Pollut</i>	Source apportionment	Sigma-2 passive samplers, 9 sites; automated SEM-EDX single-particle (3,017 particles)	Mexico (northwestern)

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Guo & Sudo	Global simulation of polycyclic aromatic hydrocarbons (PAHs) with the CHASER chemistry-climate model	<i>Atmosphere</i>	Chemical transport modelling	Gas-particle partitioning of NAP, PHE, PYR and BaP under a coupled soot-adsorption / organic-matter-absorption scheme	Global
Guo et al.	Seasonal Formation, Sources, and Long-Term Evolution of PM _{2.5} -Bound Water-Soluble Inorganic Ions in Nanchang, China: The Increasing Importance of Wintertime Nitrate.	<i>Toxics</i>	Field observations + box model + thermodynamic equilibrium	Day/night PM _{2.5} filters, nine WSIs; 2009-2026 historical records	China (Nanchang, Jiangxi)
Guo et al.	Atmospheric Environment of the Northern Tianshan Urban Agglomeration, Xinjiang: A Comprehensive Review of Pollutant Characteristics, Photochemical Processes, and Health Implications	<i>Atmosphere</i>	Narrative review	PM _{2.5} , coarse dust, BC/BrC, O ₃ , VOCs (review)	China (Xinjiang)
Gurler Akyuz & Akyuz	Seasonal and size-resolved elemental composition and source apportionment of atmospheric particulate matter in a coastal urban-industrial region of Zonguldak, Turkiye	<i>J Environ Sci Health A</i>	Field measurement	PM _{2.5} and PM _{2.5} -10 elemental composition; enrichment factor, PCA and discriminant analysis	Turkiye
Han et al.	Environmental Mixture-Health Associations: Current Analytical Practice and Reporting Recommendations.	<i>China CDC Wkly</i>	Review / synthesis	Not applicable; the unit of analysis is the mixture-modelling study	Global
He B. et al.	Contrasting seasonal behavior and inhalation risks of atmospheric PAHs, PAEs, and OPEs in a heating-influenced urban atmosphere	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - particle/gas PAHs, phthalates, organophosphate esters (by title)	Not recoverable from metadata
He et al.	Interpretable machine learning for spatially differentiated nonlinear relationships between PM _{2.5} and landscape patterns in an urban-rural integrated zone	<i>J Environ Manage</i>	Machine-learning model	Gridded PM _{2.5} vs landscape metrics (MSPA, FRAGSTATS, building height, NDVI)	China (Xuchang, Henan)
Hien et al.	Long-term Observations of Fine Particulate Sulfate in Hanoi during 1999-2023: Impact of SO ₂ Regional Emissions and Monsoon Circulations.	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - fine particulate sulfate time series, 1999-2023	Vietnam (Hanoi)
Hong Kong GEMA PM _{2.5} mobility	Spatiotemporal associations between PM _{2.5} , greenspace, physical activity, and stress: a geographic ecological momentary assessment study in Hong Kong	<i>Environ Res</i>	Geographic ecological momentary assessment	Mobile personal PM _{2.5} , 5-600 min cumulative windows	China
Hu et al.	Long-term spatiotemporal evolution, relative hotspot persistence, and predictive attribution of PM _{2.5} across China.	<i>Environ Pollut</i>	Metadata only (no abstract)	Metadata only - gridded long-term PM _{2.5} with hotspot persistence and attribution	China
Hulme et al.	Exposure of children to multiple long-term air pollutants (PM _{2.5} , BC, NO ₂) and noise at home and school in Accra, Ghana	<i>Environ Res Lett</i>	Exposure model (secondary data)	Annual mean PM _{2.5} , NO ₂ and BC from land-use regression at geocoded home and school points, combined into a time-location-weighted total; Lday/Lnight/Lden noise alongside	Sub-Saharan Africa
Hurynovich et al.	Transferability of dust-related PM _{2.5} models under contrasting arid environmental regimes.	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - dust PM _{2.5} model transferability (by title)	Not recoverable from metadata

First author	Focus	Journal	Design	Metric	Region
Ireland Census linkage	Ambient air pollution at home, work and school: Evidence from Ireland.	<i>Health Place</i>	Population-wide administrative linkage, cross-sectional	EPA modelled annual ambient PM _{2.5} , PM ₁₀ , NO ₂ and O ₃ assigned to residential and to work/school address	Republic of Ireland (2022 Census of Population)
Kagawa et al.	Seasonal variations of aerosol selenium concentrations and oxidation states in central Japan.	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - aerosol selenium concentration and speciation by oxidation state	Japan (central)
Kaur et al.	Long-term PM _{2.5} source apportionment at 12 sites in the western United States from 2000-2020.	<i>Atmos Pollut Res</i>	Source apportionment	Speciated PM _{2.5} (CSN/IMPROVE), Positive Matrix Factorization and Chemical Mass Balance jointly	Utah, California, Nevada, Idaho, Colorado (USA)
Kim et al.	Residential indoor air monitoring to assess long-term exposure to air pollution in susceptible populations for epidemiology: overview and early findings.	<i>Environ Health</i>	Long-term indoor monitoring campaign	Indoor and matched outdoor PM _{2.5} plus 36 chemical components, RTI MicroPEM, 5-day runs in each of four seasons, hourly resolution	South Korea
Klebach et al.	Aircraft observations suggest an important contribution of methanesulfonic and sulfuric acids to tropical Indo-Pacific aerosol	<i>Atmos Chem Phys</i>	Aircraft measurement campaign	Gas- and particle-phase MSA and H ₂ SO ₄ by nitrate chemical-ionisation mass spectrometry	Indo-Pacific
Kong J et al.	Investigating the photosensitized chemistry of proxies for tropospheric brown carbon	<i>J Environ Sci (China)</i>	Bench / device experiment	Excited triplet-state brown carbon proxy (1,4-naphthoquinone); aqueous-phase oxidation kinetics	Laboratory
Konon et al.	Seasonal Variation in Volatile Organic Compound Emissions and Secondary Organic Aerosol Formation from Unconventional Oil and Gas Development.	<i>Atmos Environ</i>	Metadata only (no abstract)	VOC emissions and SOA formation potential; no quantities retrievable	USA
Kuntz et al.	Dynamics of tire-related chemicals in roadside aerosol over one year in Leipzig, Germany-influencing factors, potential source materials and comparison with rural aerosol	<i>J Hazard Mater</i>	Measurement campaign	Roadside PM ₁₀ filters; 40 tire-related chemicals by LC-MS/MS	Germany (Leipzig)
Lahore, 30 filters, Mar-Apr	Carbonaceous Composition and Multi-Wavelength Optical Properties of Particulate Matter in Lahore, Pakistan.	<i>Atmosphere</i>	Measurement campaign	Gravimetric PM _{2.5} , OC/EC, five-wavelength absorption (375-850 nm) by MWAA	Not stated
Lai et al.	What are the sources of unclassified organosulfates in the atmosphere?	<i>Environ Sci Technol</i>	Narrative / critical review	Molecular formulas of ambient particulate organosulfates, which contribute up to 30% of aerosol mass, mapped to known precursors and formation routes	Global / multi-region
Lei et al.	Source transition and health implication of PM _{2.5} -bound metals: multi-year high-resolution observation in a coastal megacity	<i>Atmos Environ</i>	Metadata only	Hourly PM _{2.5} -bound metals, multi-year record	China

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Leung et al.	Investigating the impacts of abandoned oil and gas wells on indoor air quality and health symptoms in Ontario, Canada: A pilot study	<i>Indoor Air</i>	Observational exposure study	13 passively sampled indoor VOCs (gas phase, not particulate) against inverse-distance-squared-weighted abandoned-well exposure within 3 km, with self-reported acute symptoms	Canada
Li et al.	A Theoretical Model for Predicting Visibility During Dust Events.	<i>Atmosphere</i>	Theoretical / physical model	Horizontal visibility derived from saltation-controlled dust emission and near-surface concentration	Global / multi-region
Liu S et al.	Quantifying the combined effects of semi-closed terrain and monsoon on PM _{2.5} and O ₃ pollution aggregation in the North China Plain	<i>J Environ Sci (China)</i>	Spatial analysis / GIS	Gridded PM _{2.5} and O ₃ stratified by landform and altitude; Terrain-Wind Close Index	China
Liu S et al.	Wind shear structures and their association with the vertical distribution of PM _{2.5} in Taiyuan Basin	<i>Environ Pollut</i>	Multi-platform observation synthesis	Vertical PM _{2.5} profiles from aerosol lidar and UAV, against vertical wind shear regimes from wind profiler radar, microwave radiometer, radiosonde and WRF	China
Liu et al.	Airborne Bacterial and Eukaryotic Communities in Temuco, Chile, a City Dominated by Residential Wood Combustion.	<i>Life (Basel)</i>	Bioaerosol monitoring campaign	PM _{2.5} and PM ₁₀ filters, WSIs; 16S and 18S rRNA sequencing	Chile (Temuco)
Liu et al.	Effect of Skin Types and Microenvironment of Dermal Exposure to Polycyclic Aromatic Hydrocarbon Derivatives and Heavy Metals.	<i>Environ Health (Wash)</i>	Bench / device experiment	Particle-bound NPAHs, OPAHs and 11 metals adsorbed onto glycerol-trioleate-coated surrogate skin	China (campus indoor and outdoor sites, including dust events)
Liupan tethered eBC	Measurement report: impacts of thermodynamic and dynamic processes on the vertical distribution of carbonaceous aerosols: lessons from in-situ observations at the eastern foothills of Liupan Mountains, Loess Plateau	<i>Atmos Chem Phys</i>	Tethered-balloon profiling	eBC and UVP profiles, tethered balloon, July 23 and 24	China
Ma M. et al.	Atmospheric Redistribution of Microplastics Reshapes Environmental Risk Across China	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - airborne microplastics (by title)	China (national, by title)
Mainka & Mainka	Modelled Physical Activity Benefits and PM _{2.5} Risks in Two Urban Cyclists.	<i>Toxics</i>	Health impact model	Trip-resolved PM _{2.5} from portable optical monitors (162 trips) vs fixed reference station	Poland (Gliwice)
Melbourne, 3 acellular OP assays, 2 sites	Source Apportionment of PM _{2.5} Oxidative Potential in metropolitan Melbourne: integrating PMF with multiple linear regression across three acellular assays.	<i>Environ Pollut</i>	Source apportionment	PM _{2.5} oxidative potential by DTT, AA and DCFH, regressed on PMF source contributions	Australia
Melssen Pfahl et al.	Renoxification via Reduction of Inorganic Nitrate by Photoacids in Aerosols.	<i>Environ Sci Technol</i>	Chamber / laboratory	Nebulised gallic acid plus nitrate aerosol, UV-visible irradiation, NOy and HONO yields across pH	Laboratory
Milan pizza-oven dispersion	Assessment of Air Quality Impacts from Wood-Fired Pizza Ovens: Comparison with Major Urban Emission Sources	<i>Preprints.org</i>	CTM simulation + source attribution	Measured emission factors + dispersion modelling vs ARPA network	Italy

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Momen et al.	Explainable Machine Learning for Predicting Thermal Discomfort from Multipollutant Exposure on Dhaka Roadways	<i>Research Square (preprint)</i>	Field measurement	Roadside PM _{2.5} with noise and thermal sensation votes, monsoon	Bangladesh (Dhaka)
Monte Carlo wind erosion PM, Sardinia	Refinement and Quantitative Evaluation of a Monte Carlo Model for Wind-Driven PM Emissions from Industrial Granular Materials.	<i>Atmosphere</i>	Physical emission model, numerical sensitivity analysis	PM emission mass from saltation-driven sandblasting of granular surfaces	Europe
Montreal smoke advisory mobility	Urban mobility responses to wildfire smoke-related air quality advisories in Montreal, Canada.	<i>Health Place</i>	Retrospective time series	Wildfire-smoke air quality advisory days, 2022-2025	Canada
Multi-Tree	Multi-Tree Model for Precision Environmental Health with Longitudinally Assessed Mixture Exposure.	<i>J R Stat Soc Ser C Appl Stat</i>	Method / algorithm	Weekly-average gestational PM _{2.5} and maximum daily temperature, distributed-lag structure	United States
Mutic et al.	Characterizing preschool children's multi-matrix air pollutant exposures across home and early childhood education settings: a paired silicone wristband study protocol	<i>medRxiv (preprint)</i>	Cross-sectional time-activity survey	Paired home- and centre-designated silicone wristbands plus continuous indoor air-quality monitoring in two classrooms per site	United States
N China indoor Finf 48 homes	Multivariable Determinants of Indoor PM _{2.5} and Infiltration Factors in 48 Residential Buildings Across Eight Northern Chinese Cities: A Seasonal Monitoring and Linear Mixed-Effects Modelling Study	<i>Atmosphere</i>	Seasonal indoor-outdoor monitoring + LMM	Paired indoor/ outdoor PM _{2.5} at 5 min, hourly aggregate	China
NZ tractor fleet PM inventory	A calibrated, uncertainty-aware, and spatially explicit emission inventory of New Zealand's agricultural diesel tractor fleet	<i>Sci Total Environ</i>	Emission inventory	Non-road machinery PM _{2.5} , Monte Carlo uncertainty	New Zealand
Na et al.	Satellite-based estimation of PM _{2.5} chemical composition under sparse observations in Seoul metropolitan area	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - satellite-derived PM _{2.5} composition (by title)	South Korea (Seoul, by title)
Niamien et al.	Spatio-temporal analysis of the physical properties of aerosols in urban and rural environments in Cote d'Ivoire	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - aerosol physical properties (by title)	Cote d'Ivoire (by title)
Ningombam et al.	Assessment of global aerosol burden over land and oceans using ground-based, satellite, and reanalysis data	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - global AOD from ground network, satellite and reanalysis (by title)	Not recoverable from metadata
ORACLES 4STAR	Subseasonal and spatial variability of biomass burning aerosol radiative properties observed over the Southeast Atlantic during ORACLES.	<i>Atmos Chem Phys</i>	Aircraft measurement campaign	4STAR sky-scan retrievals of single scattering albedo and absorption Angstrom exponent for biomass burning aerosol above cloud	Sub-Saharan Africa
Odu-Onikosi et al.	Characteristics, sources, and health risks of PM _{2.5} -bound potentially toxic elements (PTEs) in Lagos, Nigeria	<i>Atmos Pollut Res</i>	Metadata only	PM _{2.5} -bound potentially toxic elements with source apportionment and health-risk characterisation	Sub-Saharan Africa
Oradea Metropolitan Area, 4 reference stations	Spatiotemporal assessment of urban PM ₁₀ and population exposure using integrated satellite, meteorological, and socio-spatial data.	<i>Front Public Health</i>	Field campaign / remote sensing	MODIS-MAIAC AOD regressed onto station PM ₁₀ , plus a socio-spatial exposure index	Europe

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PRD limona ketone	Ground-gridded observations of high concentrations of ambient limona ketone suggest overlooked limonene oxidation in subtropical megacities.	<i>Environ Pollut</i>	Kinetic model + field validation	Limona ketone as an observational constraint on limonene oxidation; secondary organic aerosol precursor chemistry	China
Paches et al.	Estimation of background PM _{2.5} concentrations in the Valencian Community applying clustering technique	<i>Atmos Pollut Res</i>	Metadata only	Regional background PM _{2.5} separated from local increment by clustering of monitoring-network records	Europe
Paediatric exposome	Assessing environmental exposures in children: toward a comprehensive exposome approach	<i>Pediatr Res</i>	Narrative review	Survey of questionnaires, biomarkers, sensors and geospatial indicators	Global
Papetta et al.	Source-dependent optical and mineral signatures of dust outbreaks over the Mediterranean	<i>Atmos Chem Phys</i>	Multi-platform observation synthesis	Dust optical depth, AOD, single-scattering albedo and asymmetry factor; AERONET + IASI + MODIS MIDAS + HYSPLIT	Mediterranean
Paris Saharan dust heatwave	Numerical study of a Saharan dust plume impact on radiation, urban climate and thermal comfort in the Paris region during an heatwave event	<i>Atmos Chem Phys</i>	Mesoscale simulation + campaign validation	CAMS dust aerosol in Meso-NH, AOD-validated	France
Paygude et al.	Heavy metal exposure and source profiling based on leaf-deposited particulate matter in Pune city.	<i>Environ Monit Assess</i>	Biomonitoring + source apportionment	Leaf-deposited dust, ten elements, enrichment factors and PCA source profiles	India
Petersen et al.	Direct photolysis and photosensitized reactions of sulfate ions in aerosol water form radical oxidants, secondary brown carbon, and organosulfates	<i>Environ Sci Technol</i>	Bench / device experiment	Sulfate-radical (SO ₄ anion radical) production in concentrated aerosol water; transient absorption plus radical trapping with LC-HRMS	Laboratory
Portland segregation typologies, 342 tracts	Beyond redlining: Gentrification, displacement, disadvantages, and exclusivity predict urban environmental and health inequities.	<i>Proc Natl Acad Sci U S A</i>	Cross-sectional typology construction, PCA and rank correlation	Tract-level pollution exposure among 104 environmental, hazard and health variables	United States
Qin et al.	Seasonal Variation of Airborne Bacteria and Potential Pathogens Adhere to Fine and Coarse Particles in a Coastal City.	<i>Microorganisms</i>	Bioaerosol monitoring campaign	Size-segregated PM _{2.5} and PM ₁₀ filter samples, 16S rRNA sequencing	China (Qingdao)
Rana & Guzman	Heterogeneous Nitrate Radical Oxidation of Catechol Produces Nitrocatechols, Oligomers, and Brown Carbon.	<i>ACS Earth Space Chem</i>	Chamber / laboratory	Condensed-phase products of NO ₃ oxidation of catechol thin films (FTIR, UV-vis, ESI-MS, UHPLC-MS)	Laboratory (thin-film reactor)
Rawat et al.	Size-resolved aerosols from residential fuel combustion in the northwestern Himalayas: potential effects on light absorption and human health	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - size-resolved residential-combustion aerosol (by title)	India (northwestern Himalayas, by title)
Savannah and boreal fire vapour ageing	Organic vapors from Savannah and European Boreal fire emissions: insights from photochemical and dark aging.	<i>Atmos Chem Phys</i>	Controlled burn plus 29 m ³ chamber ageing, PTR-MS	Organic vapour emission factors, SOA precursors	Global / multi-region
Sawan et al.	Uncoupling aviation and roadway contributions to ultrafine particle and black carbon exposure in a marginalized neighborhood	<i>Environ Sci Technol</i>	Source apportionment	Ultrafine particle number and black carbon, wind-sector stratified	North America

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Scalabrin et al.	Chemical tracers for assessing the contribution of tire wear particles in atmospheric aerosol: analytical approaches, challenges and future perspectives	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Non-exhaust tire wear particles via chemical tracers	Europe
Schaefer et al.	Upward transport and segregation of ice-nucleating particles in deep convective clouds	<i>Atmos Chem Phys</i>	Aircraft measurement campaign	Immersion-mode INP concentration-temperature spectra from filter samples of convective inflow, outflow and cloud-particle residuals	Europe
Seibert et al.	National-scale NMF source apportionment reveals component-specific west-to-east asymmetries in PM _{2.5} concentrations associated with trans-boundary transport.	<i>Environ Pollut</i>	Metadata only (no abstract)	Metadata only - national NMF source apportionment of PM _{2.5} (by title)	Not recoverable from metadata
Sen et al.	Genomic and Phenotypic Classification of Staphylococcus Species Isolated From Delhi Air.	<i>Environ Microbiol Rep</i>	Bioaerosol monitoring campaign	Culturable bioaerosol sampling (Merck MAS Eco-100), six sites, peak pollution season	India
Shahadat et al.	Spatiotemporal variability, trends, and health risk assessment of ambient PM _{2.5} and PM ₁₀ in Bangladesh during 2018-2025.	<i>Environ Geochem Health</i>	Monitoring-record analysis	16 DoE continuous monitoring stations, PM _{2.5} and PM ₁₀ , 2018-2025	Bangladesh (national, 16 stations)
Shen et al.	A Hybrid Deep Learning-Geostatistical Framework for PM _{2.5} Spatial Estimation with SHAP Interpretation	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - hybrid DL + geostatistical PM _{2.5} surface with SHAP	Not recoverable from metadata
Shenzhen, 2014-2024 filter series	Unraveling decadal changes in PM _{2.5} sources and secondary aerosol formation drivers using integrated receptor model and machine learning.	<i>J Environ Manage</i>	Source apportionment	Speciated PM _{2.5} filter samples, PMF factors plus ML driver attribution	China
Sichuan Basin	Topographic Modulation of High-Temperature Ozone Suppression: Core-Periphery Threshold Structure in the Sichuan Basin.	<i>Environ Pollut</i>	Observational + ML	Hourly O ₃ from 22 city monitoring stations, 2020-2024, with ERA5 meteorology	China
Singh et al.	Aerosol Absorption Coefficients Across Global Sites: Impacts of Chemical Composition, Emissions, and Meteorology	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - aerosol absorption coefficients at global sites (by title)	Not recoverable from metadata
Solaiman et al.	Secondary organic aerosol formation from the gas-phase oxidation of myrtenal by NO ₃ radicals: SOA yields, chemical composition, and highly oxygenated molecules	<i>Environ Sci Atmos</i>	Smog chamber experiment	SOA mass yields and HOM composition from a biogenic aldehyde oxidised by NO ₃ at night	Laboratory
Sonipat ACSM campaign	Chemical characterization and source apportionment of carbonaceous aerosols during post-monsoon biomass burning and Diwali at an upwind site of Delhi.	<i>Atmos Chem Phys</i>	Intensive field campaign, PMF source apportionment	Composition-based PM _{2.5} (ToF-ACSM non-refractory PM _{2.5} plus multi-wavelength Aethalometer eBC)	Sonipat, Haryana, India (upwind receptor for Delhi-NCR)
Southern Hemisphere fire trend 2003-2022	Characteristics and trend of the wild-fire activity in the Southern Hemisphere (2003-2022).	<i>Sci Total Environ</i>	Satellite time-series analysis	MODIS Active Fire detections as a proxy for biomass-burning PM source strength	Global / multi-region
St Elias proglacial dust	Modelling mineral dust emissions from proglacial valleys of the St. Elias Mountains, Canada	<i>Atmos Chem Phys</i>	Sub-km CTM development	Modelled dust emission vs in situ, camera and Doppler	Canada
Stergiou et al.	A spatiotemporal graph neural network framework for retrospective bias correction of CMAQ air quality simulations over the continental United States	<i>Atmos Pollut Res</i>	Algorithm development	CMAQ-simulated criteria pollutants including PM _{2.5} , monitor-anchored bias field	United States

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Sun et al.	Nonlinear Nitrate Formation Governed by Aerosol Water and VOC-Sensitive Oxidation in a Humid Urban Basin.	<i>Environ Sci Technol</i>	Field observations + box model + thermodynamic equilibrium	Wintertime PM _{2.5} composition, NO ₂ , O ₃ and aerosol liquid water content	China
Sunyer-Caldu	Beyond Blood and Urine: Decoding the Human Chemical Exposome across Matrices and Time.	<i>Environ Sci Technol</i>	Perspective / framework proposal	No measured PM; air positioned as one external matrix within a multi-matrix exposome design	Global / multi-region
Surdu et al.	Summertime Aerosol Composition in Southern Greenlandic Fjords: Impacts of Sources and New Particle Formation Events	<i>Environ Sci Technol</i>	Field campaign / shipborne	NEMS-FTIR aerosol composition from two research vessels; PMF; number size distributions	Greenland (Narsaq fjords)
Taiwan LRT bioaerosol	Impacts of long-range transport on the concentration, composition, and persistence of airborne fungi in northern Taiwan	<i>Environ Res</i>	Event-based monitoring	Daily culturable/spore fungal counts under ADS and FP	Taiwan
Taiwan PM _{2.5} -bound metals 71 sites	PM _{2.5} -Bound Metals in a High-Tech Industrial Area of Northern Taiwan: Spatial Distribution, Sources, and Health Risks	<i>Res Sq (preprint)</i>	Source apportionment	71-site PM _{2.5} filters, ICP-MS for 30 elements, PMF	Taiwan
Ullah et al.	Transboundary biomass-burning influence on springtime PM _{2.5} population exposure across southern China during 2015-2019	<i>Environ Pollut</i>	CTM simulation + source attribution	Population-weighted share of days above the WHO 2021 24 h guideline (15 µg/m ³) and China GB 3095-2012 Grade II (75 µg/m ³), with the Indo-China Peninsula fire contribution isolated by a paired GEOS-Chem run	China
Vilnius grey-green PM _{2.5} SHAP	Unraveling associations between grey-green spatial trade-offs and PM _{2.5} concentrations using explainable machine learning	<i>J Environ Manage</i>	Machine-learning model	Gridded PM _{2.5} 2016-2024; GSCI, NDVI, nighttime lights	Lithuania
W. India BC-dust	Seasonal Variability and Interactions of Black Carbon and Mineral Dust over Semi-Arid Western India: Satellite and Ground-Based Observations.	<i>Environ Sci Atmos</i>	Multi-year satellite and ground observational synthesis	Columnar and surface black carbon with mineral dust loading, satellite retrievals against ground stations	Semi-arid western India
Wang J et al.	Water-soluble matter mediated phototransformation of guaiacol leading to formation of aqueous secondary organic aerosol: mechanism and kinetics	<i>J Environ Sci (China)</i>	Bench / device experiment	Aqueous SOA from guaiacol photosensitised by water-soluble aerosol extracts; ROS and triplet WSOM	Laboratory
Wang T et al.	An integrated multiscale air quality modelling framework for industrial park pollution: linking local emissions to regional transport	<i>J Environ Sci (China)</i>	Chemical transport modelling	PM _{2.5} at 100 m resolution from NAQPMS coupled to a city-scale CTM with explicit Gaussian point-source plumes	China
Wang X et al.	Interactive migration of heavy metals between dust and soil demonstrated by the coupling coordination analysis	<i>Environ Monit Assess</i>	Source-apportionment method	Cr, Mn, Cu, Zn, As and Pb in paired surface dust and soil, scored with Igeo and the Nemerow index and coupled across the two media	China

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Wu Q et al.	Molecular characterization of night-time new particle formation in a coastal urban environment: season-dependent roles of sulfuric acid and oxygenated organics	<i>Environ Sci Technol</i>	Long-term field measurement campaign	Nocturnal H ₂ SO ₄ , its dimers and oxygenated organic molecules measured alongside nascent particle number, by nitrate- and Vocus-type mass spectrometry	China
Xiamen chlorine NO ₂ uptake	Chlorine enhances nocturnal heterogeneous uptake of NO ₂ in coastal atmosphere under sea-land breeze circulation	<i>Atmos Chem Phys</i>	Kinetic model + field validation	Nocturnal particulate nitrate and HONO under sea-land breeze	China
Xiao S et al.	Pollution characteristics and health risks of PAHs and OPAHs in PM _{2.5} of an inland basin city in China under the influence of the clean air action plan	<i>Environ Geochem Health</i>	Source-apportionment method	PM _{2.5} -bound parent PAHs and oxygenated PAHs, PMF source resolution, toxic-equivalent concentration	China
Xu L et al.	Marine air promotes structural compaction and coating growth of soot aerosols after long-range transport from East Asia	<i>J Environ Sci (China)</i>	Field measurement	Single-particle soot morphology and mixing state by TEM; fractal dimension and D _p /D _{core}	China
Xu X et al.	Enhancement of secondary organic aerosol formation from isoprene photooxidation by ammonia	<i>J Environ Sci (China)</i>	Smog chamber experiment	Isoprene SOA mass and composition; nitrate-CIMS gas phase plus HR-TOF-AMS particle phase	Laboratory
Yang & Li	From percentiles to distributions: nationwide ZCTA-level modeling of indoor radon exposure across the contiguous United States, 2000-2021	<i>Environ Sci Technol</i>	Exposure model (secondary data)	Within-area distributions of indoor radon concentration modelled at ZCTA level, rather than area means or medians	North America
Yang et al.	Complexation strength between organic carbon and transition metal ions dominates the photochemical conversion of SO ₂ to sulfates.	<i>Atmos Chem Phys</i>	Heterogeneous kinetics experiment	SO ₂ uptake coefficients and sulfate mass on coal-combustion organic carbon under irradiation	Laboratory (irradiated OC films)
Youn et al.	Seasonal distributions and source identification of water-soluble arsenic species of PM _{2.5} in the urban atmosphere of Northeast Asia	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - water-soluble As speciation in PM _{2.5}	Northeast Asia (not further recoverable)
Yu H et al.	Machine learning attribution of atmospheric black carbon in a coastal city: source variability and the sea-land breeze recirculation mechanism	<i>Environ Sci Technol</i>	Source-apportionment method	Online BC split into fossil-fuel and biomass-burning fractions by the Aethalometer model, then attributed to meteorology by random forest and trajectory analysis	China
Yu et al.	Continental pollutants modulate organic nitrogen and light absorption of marine organic aerosols over East Asian marginal seas.	<i>Atmos Chem Phys</i>	Field campaign / shipborne	Shipboard aerosol organic nitrogen, water-soluble and water-insoluble fractions, ON/OC ratio and light absorption over three seasons	China
Zahid & Morris	Demand or Congestion? Disentangling Event-Related Emissions Using a Directional Control on I-15.	<i>Preprints.org (preprint)</i>	Method / algorithm	Link-level CO ₂ , NO _x and PM _{2.5} emissions estimated from one-minute free-way detector data	Las Vegas, Nevada, USA (I-15)

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Zero-filling DLM bias	Gestational misalignment with fixed exposure windows: the potential dangers of zero-filling in distributed lag models	<i>Int J Epidemiol</i>	Method / algorithm	Weekly gestational PM _{2.5} exposure histories of unequal length	United States
Zhang	Within-Year PM _{2.5} Exposure Structure Provides Mortality-Predictive Information Beyond the Annual Mean.	<i>medRxiv (preprint)</i>	Metric development + retrospective application	Daily modelled PM _{2.5} summarised as the magnitude-rank index (PMRI) alongside the annual mean	United States (3,064 counties)
Zhang N et al.	Effects of aerosol aging on composition and light-absorbance of nitrogen-containing organic compounds: evidences from ultra-high-resolution mass spectrometry analysis	<i>J Environ Sci (China)</i>	Field measurement	PM1 nitrogen-containing organics by UHPLC-Orbitrap MS; mass absorption efficiency at 365 nm	China
Zhao C et al.	Incomplete reduction of particle-bound PAH toxicity under lower PM _{2.5} conditions: Evidence from Kunming, China	<i>Environ Pollut</i>	Source apportionment	16 priority PM _{2.5} -bound PAHs, BaP-equivalent burden, particle-normalised toxicity intensity (BaPeq/PM _{2.5}) and incremental lifetime cancer risk, split by lower- and higher-PM _{2.5} stage	China
Zhao H et al.	Lead isotope characteristics and source analysis of dust fall in different functional areas of a typical city in northern China	<i>Environ Monit Assess</i>	Source-apportionment method	Dustfall trace elements (V, Mn, Co, Cr, Ni, Cu, Zn, As, Cd, Pb) with Pb isotope source fingerprinting	China
Zhao et al.	Unveiling the Importance of Organic Hydroxy Acids-Related Autoxidation Pathway in Isoprene Secondary Organic Aerosol Formation.	<i>Environ Sci Technol</i>	Year-long regional tracer measurements + box model	PM _{2.5} -phase isoprene SOA tracers, six iHOM species, ng/m3, nine sites	China
Zhao et al.	Spatial-Temporal Variability of PM _{2.5} and Ozone Pollution in the Huaihai Economic Zone of China.	<i>Atmosphere (Basel)</i>	Monitoring-record analysis	PM _{2.5} and O3 concentration data 2000-2023 against MEIC emissions 2000-2020	China (Huaihai Economic Zone)
Zhou et al.	Can Electric Vehicles Reduce Summertime Air Pollution Exposure Inequalities in Metro Atlanta? A High-Resolution Analysis	<i>GeoHealth</i>	Chemical transport modelling	CMAQ at 1 km vs 12 km, NO2, MDA8 O3 and PM _{2.5} under 30% EV adoption	USA (Atlanta)
Zhu et al.	Revisiting the critical role of stabilized Criegee intermediates in sulfuric acid formation: coupling mechanistic updates with interpretable machine learning	<i>Atmos Chem Phys</i>	Chemical transport modelling	Gas-phase SO2 oxidation rate via stabilized Criegee intermediates; sCI fraction of total SO2 loss	China
da Silva et al.	Chemical Regimes and PMF-Resolved Sources Linked to Acute PM ₁₀ Ecotoxicity in Luanda, Angola.	<i>Toxics</i>	Source apportionment	116 daily PM ₁₀ filter samples, speciated; PMF; Microtox aqueous-extract toxicity	Angola (Luanda)
Burden, policy & mitigation					
Addis Ababa hybrid net	Spatiotemporal variation of PM _{2.5} concentration and health impacts in Addis Ababa, Ethiopia: trends and patterns	<i>Health Serv Insights</i>	Network monitoring + burden	Hourly PM _{2.5} , 7 low-cost sensors plus 2 BAM, 2022-23	Ethiopia
Aerosol Air Qual Res 2026 (IAQ education)	Cross-national Evaluation of STEM-Based Indoor Air Quality (IAQ) Education for Sustainable Development	<i>Aerosol Air Qual Res</i>	Metadata only (no abstract)	Not recoverable from metadata	Cross-national (countries not recoverable from metadata)

First author	Focus	Journal	Design	Metric	Region
Ann Glob Health 2026 (disclosure)	Disclosure without accountability: Air pollution reporting by major South African industrial companies.	<i>Ann Glob Health</i>	Metadata only (no abstract)	Not recoverable from metadata	South Africa
Atmospheric infrastructures Czech activism	Cities as Atmospheric Infrastructures: An Analysis of Activist Responses to Respiratory Illness in the Czech Republic	<i>Sociol Health Illn</i>	Qualitative / ethnographic study	Urban air pollution as the object of civic mobilisation	Czech Republic
Australian air health risk index	Constructing Community- and Outcome-specific Weather and Air Health Risk Indices Using Explainable Machine Learning: A Multicentre Study in 66 Australian Communities	<i>Environ Pollut</i>	Metadata only (no abstract)	Community- and outcome-specific air health risk index	Australia
Bao et al.	Distribution-sensitive assessment of air-quality and health benefits from clean-heating pilots in northern China.	<i>J Environ Manage</i>	Augmented synthetic control + weather normalisation	City-daily PM _{2.5} from 1-km gridded product, ML weather-normalised, heating-season mean/P95/P99	China
Bouchriti et al.	Spatial analysis of the burden of lung cancer attributable to long-term PM(2.5) exposure in Morocco: a national assessment using the WHO AirQ+ model.	<i>Environ Monit Assess</i>	Health impact model	City-specific annual PM _{2.5} (2019) for 46 areas into WHO AirQ+	Morocco
Busskamp et al.	A virtual limit value for ultrafine particles: Frequency of threshold exceedances across Germany based on data from the German Ultrafine Aerosol Network	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - UFP number-concentration threshold exceedances, GUAN (by title)	Germany
CA wine-grape smoke 2026	Quantifying Impacts of Wildfire Smoke on Crop Yields in California	<i>arXiv (preprint)</i>	Modelling / inventory	Regulatory PM as a yield predictor alongside temperature covariates	United States
CLHLS exercise	Exercise is associated with lower all-cause mortality among older adults: a national cohort survival analysis with county-level industrial air pollution indicators	<i>Front Public Health</i>	Prospective cohort	County annual industrial soot and SO ₂ emissions and removals	China
COVID air-quality lessons review	COVID-19 Pandemic as an Environmental Health Indicator: Lessons Learned from Air Pollution Changes and Climate Implications	<i>Preprints.org</i>	Narrative review	Lockdown-period ambient PM and co-pollutant changes	Global
Campania review	Air pollution and chronic respiratory diseases: a narrative review including a case study in Campania (Italy).	<i>Monaldi Arch Chest Dis</i>	Narrative review	Regional PM _{2.5} , PM ₁₀ , NO ₂ and O ₃ concentrations 2015-2024 against national averages	Italy
Central China BenMAP 2017-2024	Assessment of the spatiotemporal evolution of PM _{2.5} and its health and economic effects in Central China based on spatial statistics and BenMAP.	<i>Environ Monit Assess</i>	Spatial statistics plus BenMAP health impact assessment	Ground monitors and the Tracking Air Pollution in China (TAP) reanalysis, 2017-2024	China
Cheong et al.	Community-level air quality and health implications of sustainable aviation fuels adoption: a scoping review	<i>J Expo Sci Environ Epidemiol</i>	Metadata only (no abstract)	Metadata only - aviation-fuel emissions incl. ultrafine particles (by title)	Not recoverable from metadata
China PM1 burden projection	Ambient PM1-Related Health Burdens and Mortality Influencing Factors in China: Historical Trends to Future Projections	<i>Environ Pollut</i>	Metadata only (no abstract)	Ambient PM1, historical reconstruction and projection	China
China county ageing vs PM _{2.5}	Association of Environmental Factors and Aging Levels Under Different Urbanization Levels in China	<i>GeoHealth</i>	Ecological panel	County-level annual PM _{2.5} , CO, SO ₂ , O ₃ ; 2010 and 2020	China
Covenant of Mayors co-benefit tool	A tool to maximize air quality co-benefits while planning local climate actions: application to the Covenant of Mayors initiative.	<i>Open Res Eur</i>	Tool / software	PM _{2.5} , NO _x , SO ₂ , NH ₃ and VOC emission changes from city climate action plans	Europe
Ding et al.	Which Roads Should Cities Control First? A Health-Risk Framework for Mobile-Source Air Pollution.	<i>Environ Sci Technol</i>	Metadata only (no abstract)	Metadata only - road-segment mobile-source pollution with a health-risk weighting	Not recoverable from metadata

First author	Focus	Journal	Design	Metric	Region
Dong et al.	Impacts of PM _{2.5} air pollution on high-skilled worker productivity in China.	<i>Nat Hum Behav</i>	Ecological panel	Annual city-level ambient PM _{2.5} (mean 45.9 µg/m ³), instrumented	China (national)
EOTBL cancer	Global Burden and Temporal Trends of Early-Onset Tracheal, Bronchial, and Lung Cancer: An Analysis of GLOBOCAN 2022 and GBD 2023 Data.	<i>Adv Sci</i>	Ecological (GLOBOCAN secondary analysis)	Ambient particulate matter pollution as a GBD comparative-risk-assessment attributable risk factor	Global / multi-region
Flue-gas condensable PM removal	Study on the removal of condensable particulate matter from flue gas using string-of-beads liquid	<i>J Hazard Mater</i>	Bench / device experiment	Condensable PM from coal and MSW incineration flue gas	China
Fregni et al.	Stroke Burden Attributable to Risk Factors in the Americas, 1990 to 2021: A Temporal Trends Analysis From the GBD Study 2021.	<i>J Am Heart Assoc</i>	GBD secondary analysis	GBD 2021 household air pollution from solid fuels (one of 23 risk factors)	Western Hemisphere, 39 countries and territories (GBD 2021)
Habit-locked household emissions	Habit-locked-in carbon-air-pollutant emissions dominate China's household footprints.	<i>Environ Sci Ecotechnol</i>	Emission inventory / decomposition	Household-driven PM _{2.5} , NO _x , SO ₂ and CO ₂ from consumption-based accounting	China
Hohhot park PM removal	Quantifying the reduction in particulate matter by urban park green spaces using a random forest model: a case study of Hohhot, China	<i>Environ Pollut</i>	Cross-sectional + ML	Six PM size fractions, summer daytime, in/out of park	China
Intergenerational mortality equity	Assessing intergenerational health equity in the mortality impacts of acute and chronic air pollution exposure	<i>Environ Int</i>	Metadata only (no abstract)	Acute and chronic exposure, mortality impact attribution	Not stated
Jin Z. et al.	Carbonaceous Components Underlying Wildfire Smoke PM _{2.5} -Associated Mortality in Older U.S. Adults	<i>Environ Sci Technol</i>	Cohort	Annual smoke vs non-smoke PM _{2.5} mass and carbonaceous fraction	USA (contiguous)
Ju et al.	Premature mortality and economic losses attributable to landscape fire-related PM _{2.5} in ten countries and territories: a causal analysis.	<i>Lancet Planet Health</i>	Instrumental-variable time-series + random-effects meta-analysis	Population-weighted daily landscape-fire PM _{2.5} vs non-fire PM _{2.5} , 2-day moving average, modelled	Global / multi-region
Kan et al.	From Emission-Dominated Mitigation to Meteorology-Modulated Secondary Pollution: Decadal Insights from Beijing's Short-Term Interventions.	<i>Environ Pollut</i>	Measurement campaign	Aerosol composition and VOCs, 2025 vs 2015 Victory Day Parade controls; ML attribution	China (Beijing)
Kwon & Kim	Mortality and morbidity consequences of particulate matter in South Korea: evidence from transboundary air pollution.	<i>BMC Public Health</i>	Instrumental-variable panel analysis	Local monitored PM _{2.5} , instrumented by wind direction interacted with upwind Chinese monitor levels	South Korea
Li Y et al.	Multidimensional inequality of CO ₂ and air pollutant footprints in China under dual-carbon policy	<i>Environ Sci Technol</i>	Modelling / inventory	Per-capita consumption-based footprints of CO ₂ , SO ₂ , NO _x , NH ₃ and primary PM	China
Li et al.	Clean air paradox: exacerbating urban-rural inequality in PM _{2.5} exposure.	<i>Sci Bull (Beijing)</i>	Exposure model (secondary data)	Modelled PM _{2.5} exposure, weighted by field-measured source-specific toxicity	China (mainland)
Li et al.	Contributions of local household fuel burning and regional transport to ambient PM _{2.5} in the Qinghai-Tibet Plateau.	<i>Environ Res</i>	CTM simulation + source attribution	Survey-based residential emission inventory 1960-2019 with GEOS-Chem normalised marginal attribution	China (Qinghai-Tibet Plateau)

First author	Focus	Journal	Design	Metric	Region
Liu Z et al.	Significant near-term inequity under China's 2035 NDC due to a time lag in mitigation effects across regions	<i>Environ Sci Technol</i>	Integrated assessment modelling	Air-quality health co-benefits monetised alongside climate benefits under the 2035 NDC pathway, resolved by region and by decade	China
Livestock ammonia review	Livestock Ammonia Emissions: Impacts on Air Quality, Animal and Human Health, and Mitigation Strategies	<i>Preprints.org</i>	Narrative review	NH3 flux as secondary inorganic PM precursor	Global
Lorences Noguera & Cardenas-Duran	Disentangling Population Ageing from Epidemiological Change in the Rising Burden of PM _{2.5} -Attributable Non-Communicable Disease	<i>QJM</i>	Metadata only (no abstract)	Metadata only - PM _{2.5} -attributable NCD burden decomposition (by title)	Not recoverable from metadata
Ma et al.	Afforestation and Emission Cuts: Dual Drivers for China's Clean Air and Carbon Neutrality	<i>Environ Sci Technol</i>	Chemical transport model	Summer PM _{2.5} (SOA) and MDA8 O3 from BVOC and anthropogenic scenarios	China (national)
Mairghani et al.	An umbrella review of public health interventions to mitigate air pollution and improve health outcomes in high-income countries.	<i>Rev Environ Health</i>	Umbrella review / meta-analysis	Intervention-level exposure change (traffic policy, portable air cleaners, net-zero pathways)	Global / multi-region
Manchanda et al.	Inverse Modeling Identifies Efficient Emission Control Strategy for Mitigating PM _{2.5} Pollution	<i>Environ Sci Technol</i>	Chemical transport model	Annual PM _{2.5} vs precursor emissions by sector and location	USA (contiguous)
Massimi et al.	Urban Green Walls under Traffic Stress: Linking the Elemental Composition of Particulate Matter Deposition to Plant Physiological and Gene Expression Responses.	<i>Environ Sci Technol</i>	Biomonitoring + source apportionment	Multi-elemental fractionation of PM deposited on green-wall foliage - insoluble, soluble and whole-leaf fractions - with SEM-EDS, over one year	Italy
Mundu et al.	Fossil fuel use, urban air pollution and health in rapidly growing African cities: Implications for sustainable and low-emission urban development.	<i>J Air Waste Manag Assoc</i>	Systematic review (PRISMA)	Review of urban PM, NOx, BC and VOC evidence; source-sector attribution	Africa (urban, multi-country)
N fertilizer NH3-PM _{2.5}	Integrating nitrogen fertilizer management for mitigating agricultural ammonia emissions and PM _{2.5} pollution.	<i>Atmos Res</i>	Metadata only (no abstract)	Metadata only - agricultural NH3 emissions and secondary inorganic PM _{2.5} response	Not recoverable
PM stroke burden GBD 1990-2021	Ambient particulate matter pollution and stroke burden.	<i>Med Gas Res</i>	Descriptive burden analysis, GBD 2021	GBD ambient particulate matter pollution risk-factor exposure	Global / multi-region
PPS antifouling filter	Robust and Self-Cleaning Fluorine-Free Superhydrophobic PPS Filters for High-Humidity Flue Gas Filtration.	<i>Environ Sci Technol</i>	Materials synthesis with simulated dynamic filtration testing	Industrial flue-gas dust; cleaning cycle, residual pressure drop, quality factor under humid conditions	Bench (industrial flue gas application)
Puyang PM _{2.5} -O3 decade	Long-Term Trends and Nonlinear Interactions of PM _{2.5} -O3 Compound Pollution in a Typical Oilfield City of the 2+26 Region	<i>Preprints.org</i>	Monitoring-record analysis	Regulatory PM _{2.5} and MDA8 O3, Puyang, 2015-2025	China
Shi et al.	Coupled energy-industrial transitions shape heterogeneous carbon-air pollution cobenefits and spatial emission leakage in China's steel industry	<i>Environ Sci Technol</i>	Modelling / inventory	Unit-level SO2 and primary PM emissions alongside CO2, with modelled ambient response	China

First author	Focus	Journal	Design	Metric	Region
Simoes et al.	Air Pollution, Temperature, and Demographic Vulnerability as Determinants of Cause-Specific Mortality in Portugal.	<i>J Xenobiot</i>	Ecological panel	Monthly NO ₂ and PM _{2.5} exceedance above WHO 2021 guideline values	Portugal
TS-AQHI two-city validation	Development and two-city validation of a Temperature-Sensitive Air Quality Health Index incorporating synergistic effects of air pollution and thermal stress	<i>Front Public Health</i>	Metric development + retrospective application	Multi-pollutant WQS-weighted index with temperature interaction	Global
Turkatte et al.	Implementing Community-Centered Air Quality Law: Success Stories From California's Assembly Bill 617.	<i>Am J Public Health</i>	Qualitative / ethnographic study	Community emissions reduction programme strategies across nine AB 617 communities	USA (California)
Turkiye stroke GBD 2000-2023	Trends and Determinants of Stroke Burden in Turkiye, 2000-2023: An Analysis Using Global Burden of Disease Data	<i>Neuroepidemiology</i>	GBD secondary analysis	GBD ambient particulate matter risk-attributable fraction	Turkiye
US net-zero multi-model	A Multi-Model Assessment of the Air Quality and Health Impacts of U.S. Energy Decarbonization.	<i>Environ Sci Technol</i>	Model intercomparison	Modelled ambient PM _{2.5} from energy-sector precursor emissions under net-zero scenarios to 2050	United States
Wan et al.	Three-Phase Transition in China's Future Air Quality under Ambitious Mitigation Pathways Revealed by Large-Ensemble AI Projections.	<i>Environ Sci Technol</i>	Data-fusion / ML surrogate model	Hybrid deep-learning CTM surrogate, daily PM _{2.5} and O ₃ , 50 km, 2015-2060	China (national)
Western Pacific stroke burden	Stroke Burden, Trends, and Inequalities in the Western Pacific Region, 1990-2021: A Population-Based Study.	<i>JACC Asia</i>	GBD secondary analysis	Ambient particulate matter attributable fraction of stroke DALYs	Global / multi-region
Yalali et al.	Geospatial modeling of environmental and socioeconomic drivers of health outcomes.	<i>Front Public Health</i>	Spatial regression (GWR)	Census-tract PM _{2.5} among environmental and socioeconomic determinants	USA (Harris County, Texas)
Yang et al.	PM _{2.5} , O ₃ and survival gains from China's clean air actions among patients with lung cancer: a 20-year registry-based cohort study in Beijing.	<i>Thorax</i>	Prospective cohort (registry linkage)	Three-year pre-diagnosis mean PM _{2.5} and O ₃ from the ChinaHigh-AirPollutants reanalysis	China (Beijing)
Zhao N et al.	Uneven attainment pressure under China's tightened ambient air quality standard: socioeconomic predictors and differentiated control pathways for PM _{2.5} and O ₃ in Shandong	<i>J Hazard Mater</i>	Spatial regression (GWR)	Daily PM _{2.5} , PM ₁₀ , SO ₂ , NO ₂ , CO and O ₃ , 2015-2025; relative attainment pressure index against GB 3095-2026	China
Other clinical endpoints					
12M student visual impairment	Associations of long-term exposure to PM _{2.5} and its chemical constituents with incident visual impairment risk: A cohort study of 12 million school students	<i>Environ Int</i>	Metadata only (no abstract)	Long-term PM _{2.5} and specified constituents	China
Aerosol Air Qual Res	Association Between Air Pollution and Psoriasis-Related Outcomes: A Systematic Review and Meta-analysis.	<i>Aerosol Air Qual Res</i>	Metadata only (no abstract)	Not recoverable from the deposited record	Not stated
Alhalwani et al.	Correlations of CO, SO ₂ , PM ₁₀ , NO ₂ , and O ₃ with dry eye disease in Saudi Arabia.	<i>Saudi J Ophthalmol</i>	Observational exposure study	Daily PM ₁₀ , SO ₂ , NO ₂ , O ₃ and CO from three fixed monitoring stations in Jeddah	Saudi Arabia
Ali et al.	The Impact of Polycyclic Aromatic Hydrocarbons Exposure from Seasonal Biomass Burning on Oxidative Stress and Genotoxicity Biomarkers in Patients with Diabetes and Hypertension.	<i>Toxics</i>	Repeated-measures biomarker	Biomass-burning season (pre Dec vs post May); urinary 1-OHP as PAH dose	Thailand (Chiang Mai)

First author	Focus	Journal	Design	Metric	Region
Batisse et al.	Exposure to outdoor ultrafine particle number concentrations and incidence of childhood cancer: a population-based cohort study	<i>Environ Epidemiol</i>	Prospective cohort (registry linkage)	High-resolution outdoor UFP number concentration and mean UFP size at six-digit residential postal codes from annual tax records; prenatal and postnatal windows	North America
Chen K et al.	Effect of long-term exposure to PM _{2.5} and its chemical components on frailty incidence in middle-aged and older adults in China: a prospective cohort study	<i>Sichuan Da Xue Xue Bao Yi Xue Ban</i>	Prospective cohort (registry linkage)	One-year mean PM _{2.5} and its black carbon, organic matter, sulfate, nitrate and ammonium fractions from the Tracking Air Pollution dataset, per IQR	China
Ding E et al.	Climate factors, air pollutants, and extreme events shape influenza risk in China: a hospital-based case-crossover and machine-learning study	<i>Environ Sci Technol</i>	Time-series / case-crossover	Exposure-lag-response for meteorology, air pollution and compound extreme events by DLNM, variable ranking by ML, and an LSTM forecasting proof of concept for Beijing	China
Gyeltshen et al.	Short-Term PM _{2.5} Exposure and Daily Cancer Mortality Across 47 Japanese Prefectures: A Nationwide Time-Stratified Case-Crossover Study, 2013-2022	<i>Int J Cancer</i>	Time-stratified case-crossover	Daily prefecture PM _{2.5} (lag 0-2)	Japan (47 prefectures)
Hong et al.	Associations of temperature, particulate matter, and greenness with human papillomavirus positivity at screening: an analysis of Guangdong Province (2016-2022) using the treed distributed lag nonlinear model.	<i>Front Public Health</i>	Retrospective cross-sectional (health screening)	Biweekly mean PM _{2.5} at examination date, treed DLNM over 24 weeks	China (Guangdong)
Huang et al.	Secondhand smoke exposure in never-smokers: A cross-sectional and prospective cohort study of chronic kidney disease prevalence and all-cause mortality in affected patients.	<i>Medicine (Baltimore)</i>	Retrospective cohort	Serum cotinine as a biomarker of secondhand smoke particulate exposure, graded moderate/heavy	USA
Huang et al.	Short-term health impacts of air pollution in Canada: a nonlinear multi-pollutant Bayesian case-crossover framework.	<i>Int J Epidemiol</i>	Case-crossover	Daily PM _{2.5} , NO ₂ and O ₃ from imputed fixed-site monitoring, jointly modelled with nonlinear exposure-response	Canada (20 Census Divisions)
Huang et al.	The role and management of dietary and environmental factors in pediatric atopic dermatitis.	<i>Front Pediatr</i>	Case-control	Ambient PM _{2.5} dichotomised at 35 µg/m ³ (source and averaging period not stated)	Not stated (single-centre hospital series)
Kadoorie PM _{2.5} fracture/BMD	Associations of long-term exposure to ambient air pollution with fracture risk and bone mineral density: A cohort study of 0.5 million Chinese adults	<i>J Hazard Mater</i>	Prospective cohort (mixtures)	Annual PM _{2.5} , PM ₁₀ , NO ₂ , warm-season O ₃ at 1 km	China
Kelishadi et al.	The Effect of Synbiotic Supplementation on Malondialdehyde Levels in Adolescents Exposed to Air Pollution: A Randomized, Single-Blind, Controlled Clinical Trial.	<i>Int J Prev Med</i>	Trial / intervention	Residence in a polluted setting; no individual PM measurement reported	Not stated (single urban setting)
Kim et al.	Extreme Temperatures, Elevated PM _{2.5} , and Acute Care Events Among Dual-Eligible Nursing Facility Residents.	<i>J Am Geriatr Soc</i>	Panel study	Daily outdoor PM _{2.5} dichotomised at >9 µg/m ³ ; extreme heat and extreme cold day indicators	USA

First author	Focus	Journal	Design	Metric	Region
King	Outdoor air pollution as a potentially modifiable perioperative risk factor.	<i>Br J Anaesth</i>	Metadata only (no abstract)	Metadata only - ambient air pollution and perioperative risk (commentary / review by title)	Not recoverable from metadata
Leonforte et al.	Environmental Contaminants and Osteoporosis-Related Outcomes: A Systematic Review and Meta-Analysis.	<i>Toxics</i>	Systematic review + random-effects meta-analysis	Ambient PM _{2.5} , PM ₁₀ , PM1 and gases among ten contaminant classes	Global / multi-region (review)
Liu et al.	Association of exposure to fine particulate matter and its components with frailty in middle-aged and older adults in Beijing-Tianjin-Hebei region and surrounding areas.	<i>Zhonghua Liu Xing Bing Xue Za Zhi</i>	Observational - cross-sectional	Superstation PM _{2.5} and speciated components, lag0-90 to lag0-180 day moving averages; 90-day mean 53.92 +/- 10.51 µg/m ³	China
Multimorbidity meta-analysis	Association between air pollution, multimorbidity and polypharmacy: a systematic review and meta-analysis	<i>BMJ Public Health</i>	Systematic review + random-effects meta-analysis	Long-term ambient PM _{2.5} / PM ₁₀ / PM1 / NO ₂ , per 5 µg m-3	Global
Nigeria 3-state cohort, n=5,621	Ambient PM _{2.5} exposure and cancer incidence in Nigerian adults: A prospective cohort study.	<i>Cancer Epidemiol</i>	Prospective cohort	Hybrid satellite + land-use regression, 1 km annual PM _{2.5} , PM ₁₀ , NO ₂	Sub-Saharan Africa
Odiko et al.	Environmental lead exposure and clinical infectious respiratory disease in a sample of urban and disadvantaged children.	<i>J Expo Sci Environ Epidemiol</i>	Retrospective cohort	Blood lead concentration; lead is a PM-borne metal but no airborne measurement is made	USA
Park J et al.	Association between ozone and emergency department visits in people with disabilities in South Korea: a nationwide case-crossover study	<i>Int J Epidemiol</i>	Time-stratified case-crossover	Daily 8 h maximum ozone, 3-day moving average (lag 0-2), per IQR (22 ppb), adjusted for daily mean temperature; no particulate metric	South Korea
SEER-Medicare wildfire PM _{2.5} , 414,016 lung cancers	Long-term exposure to wildfire-related PM _{2.5} and overall survival of older patients with lung cancer: a retrospective, registry-based cohort study.	<i>Lancet Oncol</i>	Retrospective cohort	3-year moving average wildfire vs non-wildfire PM _{2.5} , zip-code, model-derived	United States
Tianjin myopia	Association of long-term exposure to ambient air pollution and myopia in Chinese children	<i>Environ Pollut</i>	School-based cross-sectional	3-yr PM _{2.5} and components, ML model at school point	China
Wang C et al.	Age-dependent heterogeneity trends in the associations between urban green space and cardiovascular health in a humid-subtropical megacity	<i>Environ Sci Technol</i>	Cross-sectional + prediction model	Two-stage gradient-boosting plus structural equation model; air-quality pathways (NO ₂ , SO ₂ , PM _{2.5}) carry >75% of the relative contribution to model variance	China
Zheng M et al.	Spatiotemporal co-occurrence patterns and related risk factors of esophageal and lip and oral cavity cancers among older adults across 203 countries and territories	<i>Ann Med</i>	GBD secondary analysis	Occupational exposure to particulate matter, gases and fumes as one of 25 country-level exposures in negative binomial models with population offsets	Global / multi-region

First author	Focus	Journal	Design	Metric	Region
Zhu et al.	Long-term exposure to particulate matter and survival among patients with gastric cancer: A hospital-based cohort study.	<i>Environ Res</i>	Retrospective cohort	36-month pre-surgery residential PM _{2.5} , PM ₁₀ and PM _{2.5-10} from the ChinaHigh-AirPollutants (CHAP) reanalysis	Zhejiang, China

Provenance and method

Pure aggregation over `state/corpus/2026-09-*.json` (30 daily issues, entry windows 1–30 September 2026, contiguous), `state/metrics.csv` and `state/trials.json`. No API was re-queried and no DOI re-resolved; every record passed the DOI gate at daily-issue time. Figures are pooled re-renders of the daily set; the forest panels catalogue estimates and do not pool them. Abstracts remain the copyright of their respective publishers.